Normalization, Optimization and Replay Schemes

Table of Contents

Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift

Sergey Ioffe
Google Inc., *sioffe@google.com*

Christian Szegedy
Google Inc., *szegedy@google.com*

Abstract

Training Deep Neural Networks is complicated by the fact that the distribution of each layer's inputs changes during training, as the parameters of the previous layers change. This slows down the training by requiring lower learning rates and careful parameter initialization, and makes it notoriously hard to train models with saturating nonlinearities. We refer to this phenomenon as *internal covariate shift*, and address the problem by normalizing layer inputs. Our method draws its strength from making normalization a part of the model architecture and performing the normalization *for each training mini-batch*. Batch Normalization allows us to use much higher learning rates and be less careful about initialization. It also acts as a regularizer, in some cases eliminating the need for Dropout. Applied to a state-of-the-art image classification model, Batch Normalization achieves the same accuracy with 14 times fewer training steps, and beats the original model by a significant margin. Using an ensemble of batch-normalized networks, we improve upon the best published result on ImageNet classification: reaching 4.9% top-5 validation error (and 4.8% test error), exceeding the accuracy of human raters.

1 Introduction

Deep learning has dramatically advanced the state of the art in vision, speech, and many other areas. Stochastic gradient descent (SGD) has proved to be an effective way of training deep networks, and SGD variants such as momentum (Sutskever et al., 2013) and Adagrad (Duchi et al., 2011) have been used to achieve state of the art performance. SGD optimizes the parameters Θ of the network, so as to minimize the loss

$$\Theta = \arg\min_{\Theta} \frac{1}{N} \sum_{i=1}^{N} \ell(\mathrm{x}_i, \Theta)$$

where $\mathrm{x}_{1\ldots N}$ is the training data set. With SGD, the training proceeds in steps, and at each step we consider a *mini-batch* $\mathrm{x}_{1\ldots m}$ of size m. The mini-batch is used to approximate the gradient of the loss function with respect to the parameters, by computing

$$\frac{1}{m} \frac{\partial \ell(\mathrm{x}_i, \Theta)}{\partial \Theta}.$$

Using mini-batches of examples, as opposed to one example at a time, is helpful in several ways. First, the gradient of the loss over a mini-batch is an estimate of the gradient over the training set, whose quality improves as the batch size increases. Second, computation over a batch can be much more efficient than m computations for individual examples, due to the parallelism afforded by the modern computing platforms.

While stochastic gradient is simple and effective, it requires careful tuning of the model hyper-parameters, specifically the learning rate used in optimization, as well as the initial values for the model parameters. The training is complicated by the fact that the inputs to each layer are affected by the parameters of all preceding layers – so that small changes to the network parameters amplify as the network becomes deeper.

The change in the distributions of layers' inputs presents a problem because the layers need to continuously adapt to the new distribution. When the input distribution to a learning system changes, it is said to experience *covariate shift* (Shimodaira, 2000). This is typically handled via domain adaptation (Jiang, 2008). However, the notion of covariate shift can be extended beyond the learning system as a whole, to apply to its parts, such as a sub-network or a layer. Consider a network computing

$$\ell = F_2(F_1(\mathrm{u}, \Theta_1), \Theta_2)$$

where F_1 and F_2 are arbitrary transformations, and the parameters Θ_1, Θ_2 are to be learned so as to minimize the loss ℓ. Learning Θ_2 can be viewed as if the inputs $\mathrm{x} = F_1(\mathrm{u}, \Theta_1)$ are fed into the sub-network

$$\ell = F_2(\mathrm{x}, \Theta_2).$$

For example, a gradient descent step

$$\Theta_2 \leftarrow \Theta_2 - \frac{\alpha}{m} \sum_{i=1}^{m} \frac{\partial F_2(\mathrm{x}_i, \Theta_2)}{\partial \Theta_2}$$

(for batch size m and learning rate α) is exactly equivalent to that for a stand-alone network F_2 with input x. Therefore, the input distribution properties that make training more efficient – such as having the same distribution between the training and test data – apply to training the sub-network as well. As such it is advantageous for the distribution of x to remain fixed over time. Then, Θ_2 does

not have to readjust to compensate for the change in the distribution of x.

Fixed distribution of inputs to a sub-network would have positive consequences for the layers *outside* the subnetwork, as well. Consider a layer with a sigmoid activation function $\mathrm{z} = g(W\mathrm{u} + \mathrm{b})$ where u is the layer input, the weight matrix W and bias vector b are the layer parameters to be learned, and $g(x) = \frac{1}{1+\exp(-x)}$. As $|x|$ increases, $g'(x)$ tends to zero. This means that for all dimensions of $\mathrm{x} = W\mathrm{u}+\mathrm{b}$ except those with small absolute values, the gradient flowing down to u will vanish and the model will train slowly. However, since x is affected by W, b and the parameters of all the layers below, changes to those parameters during training will likely move many dimensions of x into the saturated regime of the nonlinearity and slow down the convergence. This effect is amplified as the network depth increases. In practice, the saturation problem and the resulting vanishing gradients are usually addressed by using Rectified Linear Units (Nair & Hinton, 2010) $ReLU(x) = \max(x, 0)$, careful initialization (Bengio & Glorot, 2010; Saxe et al., 2013), and small learning rates. If, however, we could ensure that the distribution of nonlinearity inputs remains more stable as the network trains, then the optimizer would be less likely to get stuck in the saturated regime, and the training would accelerate.

We refer to the change in the distributions of internal nodes of a deep network, in the course of training, as *Internal Covariate Shift*. Eliminating it offers a promise of faster training. We propose a new mechanism, which we call *Batch Normalization*, that takes a step towards reducing internal covariate shift, and in doing so dramatically accelerates the training of deep neural nets. It accomplishes this via a normalization step that fixes the means and variances of layer inputs. Batch Normalization also has a beneficial effect on the gradient flow through the network, by reducing the dependence of gradients on the scale of the parameters or of their initial values. This allows us to use much higher learning rates without the risk of divergence. Furthermore, batch normalization regularizes the model and reduces the need for Dropout (Srivastava et al., 2014). Finally, Batch Normalization makes it possible to use saturating nonlinearities by preventing the network from getting stuck in the saturated modes.

In Sec. 4.2, we apply Batch Normalization to the best-performing ImageNet classification network, and show that we can match its performance using only 7% of the training steps, and can further exceed its accuracy by a substantial margin. Using an ensemble of such networks trained with Batch Normalization, we achieve the top-5 error rate that improves upon the best known results on ImageNet classification.

2 Towards Reducing Internal Covariate Shift

We define *Internal Covariate Shift* as the change in the distribution of network activations due to the change in network parameters during training. To improve the training, we seek to reduce the internal covariate shift. By fixing the distribution of the layer inputs x as the training progresses, we expect to improve the training speed. It has been long known (LeCun et al., 1998b; Wiesler & Ney, 2011) that the network training converges faster if its inputs are whitened – i.e., linearly transformed to have zero means and unit variances, and decorrelated. As each layer observes the inputs produced by the layers below, it would be advantageous to achieve the same whitening of the inputs of each layer. By whitening the inputs to each layer, we would take a step towards achieving the fixed distributions of inputs that would remove the ill effects of the internal covariate shift.

We could consider whitening activations at every training step or at some interval, either by modifying the network directly or by changing the parameters of the optimization algorithm to depend on the network activation values (Wiesler et al., 2014; Raiko et al., 2012; Povey et al., 2014; Desjardins & Kavukcuoglu). However, if these modifications are interspersed with the optimization steps, then the gradient descent step may attempt to update the parameters in a way that requires the normalization to be updated, which reduces the effect of the gradient step. For example, consider a layer with the input u that adds the learned bias b, and normalizes the result by subtracting the mean of the activation computed over the training data: $\widehat{x} = x - E[x]$ where $x = u + b$, $\mathcal{X} = \{x_{1\ldots N}\}$ is the set of values of x over the training set, and $\mathrm{E}[x] = \frac{1}{N}\sum_{i=1}^{N} x_i$. If a gradient descent step ignores the dependence of $\mathrm{E}[x]$ on b, then it will update $b \leftarrow b + \Delta b$, where $\Delta b \propto -\partial\ell/\partial\widehat{x}$. Then $u + (b + \Delta b) - \mathrm{E}[u + (b + \Delta b)] = u + b - \mathrm{E}[u + b]$. Thus, the combination of the update to b and subsequent change in normalization led to no change in the output of the layer nor, consequently, the loss. As the training continues, b will grow indefinitely while the loss remains fixed. This problem can get worse if the normalization not only centers but also scales the activations. We have observed this empirically in initial experiments, where the model blows up when the normalization parameters are computed outside the gradient descent step.

The issue with the above approach is that the gradient descent optimization does not take into account the fact that the normalization takes place. To address this issue, we would like to ensure that, for any parameter values, the network *always* produces activations with the desired distribution. Doing so would allow the gradient of the loss with respect to the model parameters to account for the normalization, and for its dependence on the model parameters Θ. Let again x be a layer input, treated as a

vector, and $\mathcal{X}$ be the set of these inputs over the training data set. The normalization can then be written as a transformation

$$\hat{\mathrm{x}} = \mathrm{Norm}(\mathrm{x}, \mathcal{X})$$

which depends not only on the given training example x but on all examples $\mathcal{X}$ – each of which depends on Θ if x is generated by another layer. For backpropagation, we would need to compute the Jacobians

$$\frac{\partial \mathrm{Norm}(\mathrm{x}, \mathcal{X})}{\partial \mathrm{x}} \text{ and } \frac{\partial \mathrm{Norm}(\mathrm{x}, \mathcal{X})}{\partial \mathcal{X}};$$

ignoring the latter term would lead to the explosion described above. Within this framework, whitening the layer inputs is expensive, as it requires computing the covariance matrix $\mathrm{Cov}[\mathrm{x}] = \mathrm{E}_{\mathrm{x}\in\mathcal{X}}[\mathrm{x}\mathrm{x}^T] - \mathrm{E}[\mathrm{x}]\mathrm{E}[\mathrm{x}]^T$ and its inverse square root, to produce the whitened activations $\mathrm{Cov}[\mathrm{x}]^{-1/2}(\mathrm{x} - \mathrm{E}[\mathrm{x}])$, as well as the derivatives of these transforms for backpropagation. This motivates us to seek an alternative that performs input normalization in a way that is differentiable and does not require the analysis of the entire training set after every parameter update.

Some of the previous approaches (e.g. (Lyu & Simoncelli, 2008)) use statistics computed over a single training example, or, in the case of image networks, over different feature maps at a given location. However, this changes the representation ability of a network by discarding the absolute scale of activations. We want to a preserve the information in the network, by normalizing the activations in a training example relative to the statistics of the entire training data.

3 Normalization via Mini-Batch Statistics

Since the full whitening of each layer's inputs is costly and not everywhere differentiable, we make two necessary simplifications. The first is that instead of whitening the features in layer inputs and outputs jointly, we will normalize each scalar feature independently, by making it have the mean of zero and the variance of 1. For a layer with d-dimensional input $\mathrm{x} = (x^{(1)} \ldots x^{(d)})$, we will normalize each dimension

$$\hat{x}^{(k)} = \frac{x^{(k)} - \mathrm{E}[x^{(k)}]}{\sqrt{\mathrm{Var}[x^{(k)}]}}$$

where the expectation and variance are computed over the training data set. As shown in (LeCun et al., 1998b), such normalization speeds up convergence, even when the features are not decorrelated.

Note that simply normalizing each input of a layer may change what the layer can represent. For instance, normalizing the inputs of a sigmoid would constrain them to the linear regime of the nonlinearity. To address this, we make sure that *the transformation inserted in the network can represent the identity transform*. To accomplish this, we introduce, for each activation $x^{(k)}$, a pair of parameters $\gamma^{(k)}, \beta^{(k)}$, which scale and shift the normalized value:

$$y^{(k)} = \gamma^{(k)}\hat{x}^{(k)} + \beta^{(k)}.$$

These parameters are learned along with the original model parameters, and restore the representation power of the network. Indeed, by setting $\gamma^{(k)} = \sqrt{\mathrm{Var}[x^{(k)}]}$ and $\beta^{(k)} = \mathrm{E}[x^{(k)}]$, we could recover the original activations, if that were the optimal thing to do.

In the batch setting where each training step is based on the entire training set, we would use the whole set to normalize activations. However, this is impractical when using stochastic optimization. Therefore, we make the second simplification: since we use mini-batches in stochastic gradient training, *each mini-batch produces estimates of the mean and variance* of each activation. This way, the statistics used for normalization can fully participate in the gradient backpropagation. Note that the use of mini-batches is enabled by computation of per-dimension variances rather than joint covariances; in the joint case, regularization would be required since the mini-batch size is likely to be smaller than the number of activations being whitened, resulting in singular covariance matrices.

Consider a mini-batch $\mathcal{B}$ of size m. Since the normalization is applied to each activation independently, let us focus on a particular activation $x^{(k)}$ and omit k for clarity. We have m values of this activation in the mini-batch,

$$\mathcal{B} = \{x_{1\ldots m}\}.$$

Let the normalized values be $\hat{x}_{1\ldots m}$, and their linear transformations be $y_{1\ldots m}$. We refer to the transform

$$\mathrm{BN}_{\gamma,\beta} : x_{1\ldots m} \to y_{1\ldots m}$$

as the *Batch Normalizing Transform*. We present the BN Transform in Algorithm 1. In the algorithm, ϵ is a constant added to the mini-batch variance for numerical stability.

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1\ldots m}\}$;
Parameters to be learned: γ, β
Output: $\{y_i = \mathrm{BN}_{\gamma,\beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m} x_i \qquad \text{// mini-batch mean}$$

$$\sigma^2_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_{\mathcal{B}})^2 \qquad \text{// mini-batch variance}$$

$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma^2_{\mathcal{B}} + \epsilon}} \qquad \text{// normalize}$$

$$y_i \leftarrow \gamma\hat{x}_i + \beta \equiv \mathrm{BN}_{\gamma,\beta}(x_i) \qquad \text{// scale and shift}$$

Algorithm 1: Batch Normalizing Transform, applied to activation x over a mini-batch.

The BN transform can be added to a network to manipulate any activation. In the notation $y = \mathrm{BN}_{\gamma,\beta}(x)$, we

indicate that the parameters γ and β are to be learned, but it should be noted that the BN transform does not independently process the activation in each training example. Rather, $\mathrm{BN}_{\gamma,\beta}(x)$ depends both on the training example *and the other examples in the mini-batch*. The scaled and shifted values y are passed to other network layers. The normalized activations $\widehat{x}$ are internal to our transformation, but their presence is crucial. The distributions of values of any $\widehat{x}$ has the expected value of 0 and the variance of 1, as long as the elements of each mini-batch are sampled from the same distribution, and if we neglect ϵ. This can be seen by observing that $\sum_{i=1}^{m} \widehat{x}_i = 0$ and $\frac{1}{m}\sum_{i=1}^{m} \widehat{x}_i^2 = 1$, and taking expectations. Each normalized activation $\widehat{x}^{(k)}$ can be viewed as an input to a sub-network composed of the linear transform $y^{(k)} = \gamma^{(k)}\widehat{x}^{(k)} + \beta^{(k)}$, followed by the other processing done by the original network. These sub-network inputs all have fixed means and variances, and although the joint distribution of these normalized $\widehat{x}^{(k)}$ can change over the course of training, we expect that the introduction of normalized inputs accelerates the training of the sub-network and, consequently, the network as a whole.

During training we need to backpropagate the gradient of loss ℓ through this transformation, as well as compute the gradients with respect to the parameters of the BN transform. We use chain rule, as follows (before simplification):

$$\begin{aligned}
\frac{\partial \ell}{\partial \widehat{x}_i} &= \frac{\partial \ell}{\partial y_i} \cdot \gamma \\
\frac{\partial \ell}{\partial \sigma_{\mathcal{B}}^2} &= \sum_{i=1}^{m} \frac{\partial \ell}{\partial \widehat{x}_i} \cdot (x_i - \mu_{\mathcal{B}}) \cdot \frac{-1}{2}(\sigma_{\mathcal{B}}^2 + \epsilon)^{-3/2} \\
\frac{\partial \ell}{\partial \mu_{\mathcal{B}}} &= \left(\sum_{i=1}^{m} \frac{\partial \ell}{\partial \widehat{x}_i} \cdot \frac{-1}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \right) + \frac{\partial \ell}{\partial \sigma_{\mathcal{B}}^2} \cdot \frac{\sum_{i=1}^{m} -2(x_i - \mu_{\mathcal{B}})}{m} \\
\frac{\partial \ell}{\partial x_i} &= \frac{\partial \ell}{\partial \widehat{x}_i} \cdot \frac{1}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} + \frac{\partial \ell}{\partial \sigma_{\mathcal{B}}^2} \cdot \frac{2(x_i - \mu_{\mathcal{B}})}{m} + \frac{\partial \ell}{\partial \mu_{\mathcal{B}}} \cdot \frac{1}{m} \\
\frac{\partial \ell}{\partial \gamma} &= \sum_{i=1}^{m} \frac{\partial \ell}{\partial y_i} \cdot \widehat{x}_i \\
\frac{\partial \ell}{\partial \beta} &= \sum_{i=1}^{m} \frac{\partial \ell}{\partial y_i}
\end{aligned}$$

Thus, BN transform is a differentiable transformation that introduces normalized activations into the network. This ensures that as the model is training, layers can continue learning on input distributions that exhibit less internal covariate shift, thus accelerating the training. Furthermore, the learned affine transform applied to these normalized activations allows the BN transform to represent the identity transformation and preserves the network capacity.

3.1 Training and Inference with Batch-Normalized Networks

To *Batch-Normalize* a network, we specify a subset of activations and insert the BN transform for each of them, according to Alg. 1. Any layer that previously received x as the input, now receives $\mathrm{BN}(x)$. A model employing Batch Normalization can be trained using batch gradient descent, or Stochastic Gradient Descent with a mini-batch size $m > 1$, or with any of its variants such as Adagrad (Duchi et al., 2011). The normalization of activations that depends on the mini-batch allows efficient training, but is neither necessary nor desirable during inference; we want the output to depend only on the input, deterministically. For this, once the network has been trained, we use the normalization

$$\widehat{x} = \frac{x - \mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}}$$

using the population, rather than mini-batch, statistics. Neglecting ϵ, these normalized activations have the same mean 0 and variance 1 as during training. We use the unbiased variance estimate $\mathrm{Var}[x] = \frac{m}{m-1} \cdot \mathrm{E}_{\mathcal{B}}[\sigma_{\mathcal{B}}^2]$, where the expectation is over training mini-batches of size m and $\sigma_{\mathcal{B}}^2$ are their sample variances. Using moving averages instead, we can track the accuracy of a model as it trains. Since the means and variances are fixed during inference, the normalization is simply a linear transform applied to each activation. It may further be composed with the scaling by γ and shift by β, to yield a single linear transform that replaces $\mathrm{BN}(x)$. Algorithm 2 summarizes the procedure for training batch-normalized networks.

Input: Network N with trainable parameters Θ; subset of activations $\{x^{(k)}\}_{k=1}^{K}$

Output: Batch-normalized network for inference, $N_{\mathrm{BN}}^{\mathrm{inf}}$

1: $N_{\mathrm{BN}}^{\mathrm{tr}} \leftarrow N$ // Training BN network
2: **for** $k = 1 \ldots K$ **do**
3: Add transformation $y^{(k)} = \mathrm{BN}_{\gamma^{(k)},\beta^{(k)}}(x^{(k)})$ to $N_{\mathrm{BN}}^{\mathrm{tr}}$ (Alg. 1)
4: Modify each layer in $N_{\mathrm{BN}}^{\mathrm{tr}}$ with input $x^{(k)}$ to take $y^{(k)}$ instead
5: **end for**
6: Train $N_{\mathrm{BN}}^{\mathrm{tr}}$ to optimize the parameters $\Theta \cup \{\gamma^{(k)}, \beta^{(k)}\}_{k=1}^{K}$
7: $N_{\mathrm{BN}}^{\mathrm{inf}} \leftarrow N_{\mathrm{BN}}^{\mathrm{tr}}$ // Inference BN network with frozen // parameters
8: **for** $k = 1 \ldots K$ **do**
9: // For clarity, $x \equiv x^{(k)}, \gamma \equiv \gamma^{(k)}, \mu_{\mathcal{B}} \equiv \mu_{\mathcal{B}}^{(k)}$, etc.
10: Process multiple training mini-batches $\mathcal{B}$, each of size m, and average over them:

$$\mathrm{E}[x] \leftarrow \mathrm{E}_{\mathcal{B}}[\mu_{\mathcal{B}}]$$
$$\mathrm{Var}[x] \leftarrow \frac{m}{m-1}\mathrm{E}_{\mathcal{B}}[\sigma_{\mathcal{B}}^2]$$

11: In $N_{\mathrm{BN}}^{\mathrm{inf}}$, replace the transform $y = \mathrm{BN}_{\gamma,\beta}(x)$ with $y = \frac{\gamma}{\sqrt{\mathrm{Var}[x] + \epsilon}} \cdot x + \left(\beta - \frac{\gamma\,\mathrm{E}[x]}{\sqrt{\mathrm{Var}[x] + \epsilon}}\right)$
12: **end for**

Algorithm 2: Training a Batch-Normalized Network

3.2 Batch-Normalized Convolutional Networks

Batch Normalization can be applied to any set of activations in the network. Here, we focus on transforms

that consist of an affine transformation followed by an element-wise nonlinearity:

$$\mathrm{z} = g(W\mathrm{u} + \mathrm{b})$$

where W and b are learned parameters of the model, and $g(\cdot)$ is the nonlinearity such as sigmoid or ReLU. This formulation covers both fully-connected and convolutional layers. We add the BN transform immediately before the nonlinearity, by normalizing $\mathrm{x} = W\mathrm{u}+\mathrm{b}$. We could have also normalized the layer inputs u, but since u is likely the output of another nonlinearity, the shape of its distribution is likely to change during training, and constraining its first and second moments would not eliminate the covariate shift. In contrast, $W\mathrm{u} + \mathrm{b}$ is more likely to have a symmetric, non-sparse distribution, that is "more Gaussian" (Hyvärinen & Oja, 2000); normalizing it is likely to produce activations with a stable distribution.

Note that, since we normalize $W\mathrm{u}+\mathrm{b}$, the bias b can be ignored since its effect will be canceled by the subsequent mean subtraction (the role of the bias is subsumed by β in Alg. 1). Thus, $\mathrm{z} = g(W\mathrm{u} + \mathrm{b})$ is replaced with

$$\mathrm{z} = g(\mathrm{BN}(W\mathrm{u}))$$

where the BN transform is applied independently to each dimension of $\mathrm{x} = W\mathrm{u}$, with a separate pair of learned parameters $\gamma^{(k)}$, $\beta^{(k)}$ per dimension.

For convolutional layers, we additionally want the normalization to obey the convolutional property – so that different elements of the same feature map, at different locations, are normalized in the same way. To achieve this, we jointly normalize all the activations in a mini-batch, over all locations. In Alg. 1, we let $\mathcal{B}$ be the set of all values in a feature map across both the elements of a mini-batch and spatial locations – so for a mini-batch of size m and feature maps of size $p \times q$, we use the effective mini-batch of size $m' = |\mathcal{B}| = m \cdot p\,q$. We learn a pair of parameters $\gamma^{(k)}$ and $\beta^{(k)}$ per feature map, rather than per activation. Alg. 2 is modified similarly, so that during inference the BN transform applies the same linear transformation to each activation in a given feature map.

3.3 Batch Normalization enables higher learning rates

In traditional deep networks, too-high learning rate may result in the gradients that explode or vanish, as well as getting stuck in poor local minima. Batch Normalization helps address these issues. By normalizing activations throughout the network, it prevents small changes to the parameters from amplifying into larger and suboptimal changes in activations in gradients; for instance, it prevents the training from getting stuck in the saturated regimes of nonlinearities.

Batch Normalization also makes training more resilient to the parameter scale. Normally, large learning rates may increase the scale of layer parameters, which then amplify the gradient during backpropagation and lead to the model explosion. However, with Batch Normalization, backpropagation through a layer is unaffected by the scale of its parameters. Indeed, for a scalar a,

$$\mathrm{BN}(W\mathrm{u}) = \mathrm{BN}((aW)\mathrm{u})$$

and we can show that

$$\frac{\partial \mathrm{BN}((aW)\mathrm{u})}{\partial \mathrm{u}} = \frac{\partial \mathrm{BN}(W\mathrm{u})}{\partial \mathrm{u}}$$
$$\frac{\partial \mathrm{BN}((aW)\mathrm{u})}{\partial (aW)} = \frac{1}{a} \cdot \frac{\partial \mathrm{BN}(W\mathrm{u})}{\partial W}$$

The scale does not affect the layer Jacobian nor, consequently, the gradient propagation. Moreover, larger weights lead to *smaller* gradients, and Batch Normalization will stabilize the parameter growth.

We further conjecture that Batch Normalization may lead the layer Jacobians to have singular values close to 1, which is known to be beneficial for training (Saxe et al., 2013). Consider two consecutive layers with normalized inputs, and the transformation between these normalized vectors: $\hat{\mathrm{z}} = F(\hat{\mathrm{x}})$. If we assume that $\hat{\mathrm{x}}$ and $\hat{\mathrm{z}}$ are Gaussian and uncorrelated, and that $F(\hat{\mathrm{x}}) \approx J\hat{\mathrm{x}}$ is a linear transformation for the given model parameters, then both $\hat{\mathrm{x}}$ and $\hat{\mathrm{z}}$ have unit covariances, and $I = \mathrm{Cov}[\hat{\mathrm{z}}] = J\mathrm{Cov}[\hat{\mathrm{x}}]J^T = JJ^T$. Thus, $JJ^T = I$, and so all singular values of J are equal to 1, which preserves the gradient magnitudes during backpropagation. In reality, the transformation is not linear, and the normalized values are not guaranteed to be Gaussian nor independent, but we nevertheless expect Batch Normalization to help make gradient propagation better behaved. The precise effect of Batch Normalization on gradient propagation remains an area of further study.

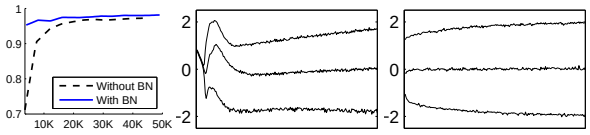
3.4 Batch Normalization regularizes the model

When training with Batch Normalization, a training example is seen in conjunction with other examples in the mini-batch, and the training network no longer producing deterministic values for a given training example. In our experiments, we found this effect to be advantageous to the generalization of the network. Whereas Dropout (Srivastava et al., 2014) is typically used to reduce overfitting, in a batch-normalized network we found that it can be either removed or reduced in strength.

4 Experiments

4.1 Activations over time

To verify the effects of internal covariate shift on training, and the ability of Batch Normalization to combat it, we considered the problem of predicting the digit class on the MNIST dataset (LeCun et al., 1998a). We used a very simple network, with a 28x28 binary image as input, and



(a) (b) Without BN (c) With BN

Figure 1: (a) *The test accuracy of the MNIST network trained with and without Batch Normalization, vs. the number of training steps. Batch Normalization helps the network train faster and achieve higher accuracy.* (b, c) *The evolution of input distributions to a typical sigmoid, over the course of training, shown as* $\{15, 50, 85\}$*th percentiles. Batch Normalization makes the distribution more stable and reduces the internal covariate shift.*

3 fully-connected hidden layers with 100 activations each. Each hidden layer computes $\mathrm{y} = g(W\mathrm{u}+\mathrm{b})$ with sigmoid nonlinearity, and the weights W initialized to small random Gaussian values. The last hidden layer is followed by a fully-connected layer with 10 activations (one per class) and cross-entropy loss. We trained the network for 50000 steps, with 60 examples per mini-batch. We added Batch Normalization to each hidden layer of the network, as in Sec. 3.1. We were interested in the comparison between the baseline and batch-normalized networks, rather than achieving the state of the art performance on MNIST (which the described architecture does not).

Figure 1(a) shows the fraction of correct predictions by the two networks on held-out test data, as training progresses. The batch-normalized network enjoys the higher test accuracy. To investigate why, we studied inputs to the sigmoid, in the original network N and batch-normalized network $N^{\mathrm{tr}}_{\mathrm{BN}}$ (Alg. 2) over the course of training. In Fig. 1(b,c) we show, for one typical activation from the last hidden layer of each network, how its distribution evolves. The distributions in the original network change significantly over time, both in their mean and the variance, which complicates the training of the subsequent layers. In contrast, the distributions in the batch-normalized network are much more stable as training progresses, which aids the training.

4.2 ImageNet classification

We applied Batch Normalization to a new variant of the Inception network (Szegedy et al., 2014), trained on the ImageNet classification task (Russakovsky et al., 2014). The network has a large number of convolutional and pooling layers, with a softmax layer to predict the image class, out of 1000 possibilities. Convolutional layers use ReLU as the nonlinearity. The main difference to the network described in (Szegedy et al., 2014) is that the 5×5 convolutional layers are replaced by two consecutive layers of 3×3 convolutions with up to 128 filters. The network contains $13.6 \cdot 10^6$ parameters, and, other than the top softmax layer, has no fully-connected layers. More details are given in the Appendix. We refer to this model as *Inception* in the rest of the text. The model was trained using a version of Stochastic Gradient Descent with momentum (Sutskever et al., 2013), using the mini-batch size of 32. The training was performed using a large-scale, distributed architecture (similar to (Dean et al., 2012)). All networks are evaluated as training progresses by computing the validation accuracy @1, i.e. the probability of predicting the correct label out of 1000 possibilities, on a held-out set, using a single crop per image.

In our experiments, we evaluated several modifications of Inception with Batch Normalization. In all cases, Batch Normalization was applied to the input of each nonlinearity, in a convolutional way, as described in section 3.2, while keeping the rest of the architecture constant.

4.2.1 Accelerating BN Networks

Simply adding Batch Normalization to a network does not take full advantage of our method. To do so, we further changed the network and its training parameters, as follows:

Increase learning rate. In a batch-normalized model, we have been able to achieve a training speedup from higher learning rates, with no ill side effects (Sec. 3.3).

Remove Dropout. As described in Sec. 3.4, Batch Normalization fulfills some of the same goals as Dropout. Removing Dropout from Modified BN-Inception speeds up training, without increasing overfitting.

Reduce the L_2 weight regularization. While in Inception an L_2 loss on the model parameters controls overfitting, in Modified BN-Inception the weight of this loss is reduced by a factor of 5. We find that this improves the accuracy on the held-out validation data.

Accelerate the learning rate decay. In training Inception, learning rate was decayed exponentially. Because our network trains faster than Inception, we lower the learning rate 6 times faster.

Remove Local Response Normalization While Inception and other networks (Srivastava et al., 2014) benefit from it, we found that with Batch Normalization it is not necessary.

Shuffle training examples more thoroughly. We enabled within-shard shuffling of the training data, which prevents the same examples from always appearing in a mini-batch together. This led to about 1% improvements in the validation accuracy, which is consistent with the view of Batch Normalization as a regularizer (Sec. 3.4): the randomization inherent in our method should be most beneficial when it affects an example differently each time it is seen.

Reduce the photometric distortions. Because batch-normalized networks train faster and observe each training example fewer times, we let the trainer focus on more "real" images by distorting them less.

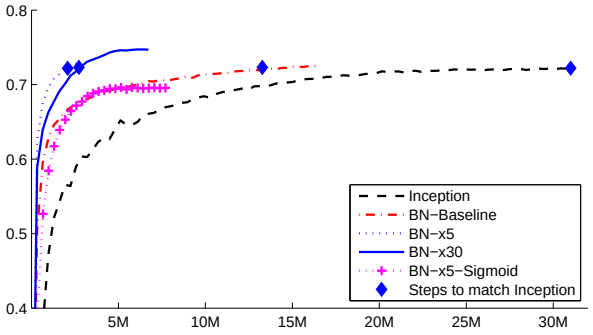


Figure 2: *Single crop validation accuracy of Inception and its batch-normalized variants, vs. the number of training steps.*

Model	Steps to 72.2%	Max accuracy
Inception	$31.0 \cdot 10^6$	72.2%
BN-Baseline	$13.3 \cdot 10^6$	72.7%
BN-x5	$2.1 \cdot 10^6$	73.0%
BN-x30	$2.7 \cdot 10^6$	74.8%
BN-x5-Sigmoid		69.8%

Figure 3: *For Inception and the batch-normalized variants, the number of training steps required to reach the maximum accuracy of Inception (72.2%), and the maximum accuracy achieved by the network.*

4.2.2 Single-Network Classification

We evaluated the following networks, all trained on the LSVRC2012 training data, and tested on the validation data:

Inception: the network described at the beginning of Section 4.2, trained with the initial learning rate of 0.0015.

BN-Baseline: Same as Inception with Batch Normalization before each nonlinearity.

BN-x5: Inception with Batch Normalization and the modifications in Sec. 4.2.1. The initial learning rate was increased by a factor of 5, to 0.0075. The same learning rate increase with original Inception caused the model parameters to reach machine infinity.

BN-x30: Like *BN-x5*, but with the initial learning rate 0.045 (30 times that of Inception).

BN-x5-Sigmoid: Like *BN-x5*, but with sigmoid nonlinearity $g(t) = \frac{1}{1+\exp(-x)}$ instead of ReLU. We also attempted to train the original Inception with sigmoid, but the model remained at the accuracy equivalent to chance.

In Figure 2, we show the validation accuracy of the networks, as a function of the number of training steps. Inception reached the accuracy of 72.2% after $31 \cdot 10^6$ training steps. The Figure 3 shows, for each network, the number of training steps required to reach the same 72.2% accuracy, as well as the maximum validation accuracy reached by the network and the number of steps to reach it.

By only using Batch Normalization (*BN-Baseline*), we match the accuracy of Inception in less than half the number of training steps. By applying the modifications in Sec. 4.2.1, we significantly increase the training speed of the network. *BN-x5* needs 14 times fewer steps than Inception to reach the 72.2% accuracy. Interestingly, increasing the learning rate further (*BN-x30*) causes the model to train somewhat *slower* initially, but allows it to reach a higher final accuracy. It reaches 74.8% after $6 \cdot 10^6$ steps, i.e. 5 times fewer steps than required by Inception to reach 72.2%.

We also verified that the reduction in internal covariate shift allows deep networks with Batch Normalization to be trained when sigmoid is used as the nonlinearity, despite the well-known difficulty of training such networks. Indeed, *BN-x5-Sigmoid* achieves the accuracy of 69.8%. Without Batch Normalization, Inception with sigmoid never achieves better than $1/1000$ accuracy.

4.2.3 Ensemble Classification

The current reported best results on the ImageNet Large Scale Visual Recognition Competition are reached by the Deep Image ensemble of traditional models (Wu et al., 2015) and the ensemble model of (He et al., 2015). The latter reports the top-5 error of 4.94%, as evaluated by the ILSVRC server. Here we report a top-5 validation error of 4.9%, and test error of 4.82% (according to the ILSVRC server). This improves upon the previous best result, and exceeds the estimated accuracy of human raters according to (Russakovsky et al., 2014).

For our ensemble, we used 6 networks. Each was based on *BN-x30*, modified via some of the following: increased initial weights in the convolutional layers; using Dropout (with the Dropout probability of 5% or 10%, vs. 40% for the original Inception); and using non-convolutional, per-activation Batch Normalization with last hidden layers of the model. Each network achieved its maximum accuracy after about $6 \cdot 10^6$ training steps. The ensemble prediction was based on the arithmetic average of class probabilities predicted by the constituent networks. The details of ensemble and multicrop inference are similar to (Szegedy et al., 2014).

We demonstrate in Fig. 4 that batch normalization allows us to set new state-of-the-art by a healthy margin on the ImageNet classification challenge benchmarks.

5 Conclusion

We have presented a novel mechanism for dramatically accelerating the training of deep networks. It is based on the premise that covariate shift, which is known to complicate the training of machine learning systems, also ap-

Model	Resolution	Crops	Models	Top-1 error	Top-5 error
GoogLeNet ensemble	224	144	7	-	6.67%
Deep Image low-res	256	-	1	-	7.96%
Deep Image high-res	512	-	1	24.88	7.42%
Deep Image ensemble	variable	-	-	-	5.98%
BN-Inception single crop	224	1	1	25.2%	7.82%
BN-Inception multicrop	224	144	1	21.99%	5.82%
BN-Inception ensemble	224	144	6	20.1%	**4.9%***

Figure 4: *Batch-Normalized Inception comparison with previous state of the art on the provided validation set comprising 50000 images. *BN-Inception ensemble has reached 4.82% top-5 error on the 100000 images of the test set of the ImageNet as reported by the test server.*

plies to sub-networks and layers, and removing it from internal activations of the network may aid in training. Our proposed method draws its power from normalizing activations, and from incorporating this normalization in the network architecture itself. This ensures that the normalization is appropriately handled by any optimization method that is being used to train the network. To enable stochastic optimization methods commonly used in deep network training, we perform the normalization for each mini-batch, and backpropagate the gradients through the normalization parameters. Batch Normalization adds only two extra parameters per activation, and in doing so preserves the representation ability of the network. We presented an algorithm for constructing, training, and performing inference with batch-normalized networks. The resulting networks can be trained with saturating nonlinearities, are more tolerant to increased training rates, and often do not require Dropout for regularization.

Merely adding Batch Normalization to a state-of-the-art image classification model yields a substantial speedup in training. By further increasing the learning rates, removing Dropout, and applying other modifications afforded by Batch Normalization, we reach the previous state of the art with only a small fraction of training steps – and then beat the state of the art in single-network image classification. Furthermore, by combining multiple models trained with Batch Normalization, we perform better than the best known system on ImageNet, by a significant margin.

Interestingly, our method bears similarity to the standardization layer of (Gülçehre & Bengio, 2013), though the two methods stem from very different goals, and perform different tasks. The goal of Batch Normalization is to achieve a stable distribution of activation values throughout training, and in our experiments we apply it before the nonlinearity since that is where matching the first and second moments is more likely to result in a stable distribution. On the contrary, (Gülçehre & Bengio, 2013) apply the standardization layer to the *output* of the nonlinearity, which results in sparser activations. In our large-scale image classification experiments, we have not observed the nonlinearity *inputs* to be sparse, neither with nor without Batch Normalization. Other notable differentiating characteristics of Batch Normalization include the learned scale and shift that allow the BN transform to represent identity (the standardization layer did not require this since it was followed by the learned linear transform that, conceptually, absorbs the necessary scale and shift), handling of convolutional layers, deterministic inference that does not depend on the mini-batch, and batch-normalizing each convolutional layer in the network.

In this work, we have not explored the full range of possibilities that Batch Normalization potentially enables. Our future work includes applications of our method to Recurrent Neural Networks (Pascanu et al., 2013), where the internal covariate shift and the vanishing or exploding gradients may be especially severe, and which would allow us to more thoroughly test the hypothesis that normalization improves gradient propagation (Sec. 3.3). We plan to investigate whether Batch Normalization can help with domain adaptation, in its traditional sense – i.e. whether the normalization performed by the network would allow it to more easily generalize to new data distributions, perhaps with just a recomputation of the population means and variances (Alg. 2). Finally, we believe that further theoretical analysis of the algorithm would allow still more improvements and applications.

References

Bengio, Yoshua and Glorot, Xavier. Understanding the difficulty of training deep feedforward neural networks. In *Proceedings of AISTATS 2010*, volume 9, pp. 249–256, May 2010.

Dean, Jeffrey, Corrado, Greg S., Monga, Rajat, Chen, Kai, Devin, Matthieu, Le, Quoc V., Mao, Mark Z., Ranzato, Marc'Aurelio, Senior, Andrew, Tucker, Paul, Yang, Ke, and Ng, Andrew Y. Large scale distributed deep networks. In *NIPS*, 2012.

Desjardins, Guillaume and Kavukcuoglu, Koray. Natural neural networks. (unpublished).

Duchi, John, Hazan, Elad, and Singer, Yoram. Adaptive subgradient methods for online learning and stochastic

type	**patch size/ stride**	**output size**	**depth**	#1×1	**#3×3 reduce**	#3×3	**double #3×3 reduce**	**double #3×3**	**Pool +proj**
convolution*	7×7/2	112×112×64	1						
max pool	3×3/2	56×56×64	0						
convolution	3×3/1	56×56×192	1		64	192			
max pool	3×3/2	28×28×192	0						
inception (3a)		28×28×256	3	64	64	64	64	96	avg + 32
inception (3b)		28×28×320	3	64	64	96	64	96	avg + 64
inception (3c)	stride 2	28×28×576	3	0	128	160	64	96	max + pass through
inception (4a)		14×14×576	3	224	64	96	96	128	avg + 128
inception (4b)		14×14×576	3	192	96	128	96	128	avg + 128
inception (4c)		14×14×576	3	160	128	160	128	160	avg + 128
inception (4d)		14×14×576	3	96	128	192	160	192	avg + 128
inception (4e)	stride 2	14×14×1024	3	0	128	192	192	256	max + pass through
inception (5a)		7×7×1024	3	352	192	320	160	224	avg + 128
inception (5b)		7×7×1024	3	352	192	320	192	224	max + 128
avg pool	7×7/1	1×1×1024	0						

Figure 5: Inception architecture

Layer Normalization

Jimmy Lei Ba
University of Toronto
jimmy@psi.toronto.edu

Jamie Ryan Kiros
University of Toronto
rkiros@cs.toronto.edu

Geoffrey E. Hinton
University of Toronto
and Google Inc.
hinton@cs.toronto.edu

Abstract

Training state-of-the-art, deep neural networks is computationally expensive. One way to reduce the training time is to normalize the activities of the neurons. A recently introduced technique called batch normalization uses the distribution of the summed input to a neuron over a mini-batch of training cases to compute a mean and variance which are then used to normalize the summed input to that neuron on each training case. This significantly reduces the training time in feed-forward neural networks. However, the effect of batch normalization is dependent on the mini-batch size and it is not obvious how to apply it to recurrent neural networks. In this paper, we transpose batch normalization into layer normalization by computing the mean and variance used for normalization from all of the summed inputs to the neurons in a layer on a *single* training case. Like batch normalization, we also give each neuron its own adaptive bias and gain which are applied after the normalization but before the non-linearity. Unlike batch normalization, layer normalization performs exactly the same computation at training and test times. It is also straightforward to apply to recurrent neural networks by computing the normalization statistics separately at each time step. Layer normalization is very effective at stabilizing the hidden state dynamics in recurrent networks. Empirically, we show that layer normalization can substantially reduce the training time compared with previously published techniques.

1 Introduction

Deep neural networks trained with some version of Stochastic Gradient Descent have been shown to substantially outperform previous approaches on various supervised learning tasks in computer vision [Krizhevsky et al., 2012] and speech processing [Hinton et al., 2012]. But state-of-the-art deep neural networks often require many days of training. It is possible to speed-up the learning by computing gradients for different subsets of the training cases on different machines or splitting the neural network itself over many machines [Dean et al., 2012], but this can require a lot of communication and complex software. It also tends to lead to rapidly diminishing returns as the degree of parallelization increases. An orthogonal approach is to modify the computations performed in the forward pass of the neural net to make learning easier. Recently, batch normalization [Ioffe and Szegedy, 2015] has been proposed to reduce training time by including additional normalization stages in deep neural networks. The normalization standardizes each summed input using its mean and its standard deviation across the training data. Feedforward neural networks trained using batch normalization converge faster even with simple SGD. In addition to training time improvement, the stochasticity from the batch statistics serves as a regularizer during training.

Despite its simplicity, batch normalization requires running averages of the summed input statistics. In feed-forward networks with fixed depth, it is straightforward to store the statistics separately for each hidden layer. However, the summed inputs to the recurrent neurons in a recurrent neural network (RNN) often vary with the length of the sequence so applying batch normalization to RNNs appears to require different statistics for different time-steps. Furthermore, batch normaliza-

tion cannot be applied to online learning tasks or to extremely large distributed models where the minibatches have to be small.

This paper introduces layer normalization, a simple normalization method to improve the training speed for various neural network models. Unlike batch normalization, the proposed method directly estimates the normalization statistics from the summed inputs to the neurons within a hidden layer so the normalization does not introduce any new dependencies between training cases. We show that layer normalization works well for RNNs and improves both the training time and the generalization performance of several existing RNN models.

2 Background

A feed-forward neural network is a non-linear mapping from a input pattern $\mathbf{x}$ to an output vector y. Consider the l^{th} hidden layer in a deep feed-forward, neural network, and let a^l be the vector representation of the summed inputs to the neurons in that layer. The summed inputs are computed through a linear projection with the weight matrix W^l and the bottom-up inputs h^l given as follows:

$$a_i^l = {w_i^l}^\top h^l \qquad h_i^{l+1} = f(a_i^l + b_i^l) \tag{1}$$

where $f(\cdot)$ is an element-wise non-linear function and w_i^l is the incoming weights to the i^{th} hidden units and b_i^l is the scalar bias parameter. The parameters in the neural network are learnt using gradient-based optimization algorithms with the gradients being computed by back-propagation.

One of the challenges of deep learning is that the gradients with respect to the weights in one layer are highly dependent on the outputs of the neurons in the previous layer especially if these outputs change in a highly correlated way. Batch normalization [Ioffe and Szegedy, 2015] was proposed to reduce such undesirable "covariate shift". The method normalizes the summed inputs to each hidden unit over the training cases. Specifically, for the i^{th} summed input in the l^{th} layer, the batch normalization method rescales the summed inputs according to their variances under the distribution of the data

$$\bar{a}_i^l = \frac{g_i^l}{\sigma_i^l}\left(a_i^l - \mu_i^l\right) \qquad \mu_i^l = \mathop{\mathbb{E}}_{\mathbf{x}\sim P(\mathbf{x})}\left[a_i^l\right] \qquad \sigma_i^l = \sqrt{\mathop{\mathbb{E}}_{\mathbf{x}\sim P(\mathbf{x})}\left[\left(a_i^l - \mu_i^l\right)^2\right]} \tag{2}$$

where $\bar{a}_i^l$ is normalized summed inputs to the i^{th} hidden unit in the l^{th} layer and g_i is a gain parameter scaling the normalized activation before the non-linear activation function. Note the expectation is under the whole training data distribution. It is typically impractical to compute the expectations in Eq. (2) exactly, since it would require forward passes through the whole training dataset with the current set of weights. Instead, μ and σ are estimated using the empirical samples from the current mini-batch. This puts constraints on the size of a mini-batch and it is hard to apply to recurrent neural networks.

3 Layer normalization

We now consider the layer normalization method which is designed to overcome the drawbacks of batch normalization.

Notice that changes in the output of one layer will tend to cause highly correlated changes in the summed inputs to the next layer, especially with ReLU units whose outputs can change by a lot. This suggests the "covariate shift" problem can be reduced by fixing the mean and the variance of the summed inputs within each layer. We, thus, compute the layer normalization statistics over all the hidden units in the same layer as follows:

$$\mu^l = \frac{1}{H}\sum_{i=1}^{H} a_i^l \qquad \sigma^l = \sqrt{\frac{1}{H}\sum_{i=1}^{H}\left(a_i^l - \mu^l\right)^2} \tag{3}$$

where H denotes the number of hidden units in a layer. The difference between Eq. (2) and Eq. (3) is that under layer normalization, all the hidden units in a layer share the same normalization terms μ and σ, but different training cases have different normalization terms. Unlike batch normalization, layer normaliztion does not impose any constraint on the size of a mini-batch and it can be used in the pure online regime with batch size 1.

3.1 Layer normalized recurrent neural networks

The recent sequence to sequence models [Sutskever et al., 2014] utilize compact recurrent neural networks to solve sequential prediction problems in natural language processing. It is common among the NLP tasks to have different sentence lengths for different training cases. This is easy to deal with in an RNN because the same weights are used at every time-step. But when we apply batch normalization to an RNN in the obvious way, we need to to compute and store separate statistics for each time step in a sequence. This is problematic if a test sequence is longer than any of the training sequences. Layer normalization does not have such problem because its normalization terms depend only on the summed inputs to a layer at the current time-step. It also has only one set of gain and bias parameters shared over all time-steps.

In a standard RNN, the summed inputs in the recurrent layer are computed from the current input $\mathbf{x}^t$ and previous vector of hidden states $\mathbf{h}^{t-1}$ which are computed as $\mathbf{a}^t = W_{hh}h^{t-1} + W_{xh}\mathbf{x}^t$. The layer normalized recurrent layer re-centers and re-scales its activations using the extra normalization terms similar to Eq. (3):

$$\mathbf{h}^t = f\left[\frac{\mathbf{g}}{\sigma^t} \odot \left(\mathbf{a}^t - \mu^t\right) + \mathbf{b}\right] \qquad \mu^t = \frac{1}{H}\sum_{i=1}^{H} a_i^t \qquad \sigma^t = \sqrt{\frac{1}{H}\sum_{i=1}^{H}\left(a_i^t - \mu^t\right)^2} \tag{4}$$

where W_{hh} is the recurrent hidden to hidden weights and W_{xh} are the bottom up input to hidden weights. $\odot$ is the element-wise multiplication between two vectors. $\mathbf{b}$ and $\mathbf{g}$ are defined as the bias and gain parameters of the same dimension as $\mathbf{h}^t$.

In a standard RNN, there is a tendency for the average magnitude of the summed inputs to the recurrent units to either grow or shrink at every time-step, leading to exploding or vanishing gradients. In a layer normalized RNN, the normalization terms make it invariant to re-scaling all of the summed inputs to a layer, which results in much more stable hidden-to-hidden dynamics.

4 Related work

Batch normalization has been previously extended to recurrent neural networks [Laurent et al., 2015, Amodei et al., 2015, Cooijmans et al., 2016]. The previous work [Cooijmans et al., 2016] suggests the best performance of recurrent batch normalization is obtained by keeping independent normalization statistics for each time step. The authors show that initializing the gain parameter in the recurrent batch normalization layer to 0.1 makes significant difference in the final performance of the model. Our work is also related to weight normalization [Salimans and Kingma, 2016]. In weight normalization, instead of the variance, the L2 norm of the incoming weights is used to normalize the summed inputs to a neuron. Applying either weight normalization or batch normalization using expected statistics is equivalent to have a different parameterization of the original feed-forward neural network. Re-parameterization in the ReLU network was studied in the Path-normalized SGD [Neyshabur et al., 2015]. Our proposed layer normalization method, however, is not a re-parameterization of the original neural network. The layer normalized model, thus, has different invariance properties than the other methods, that we will study in the following section.

5 Analysis

In this section, we investigate the invariance properties of different normalization schemes.

5.1 Invariance under weights and data transformations

The proposed layer normalization is related to batch normalization and weight normalization. Although, their normalization scalars are computed differently, these methods can be summarized as normalizing the summed inputs a_i to a neuron through the two scalars μ and σ. They also learn an adaptive bias b and gain g for each neuron after the normalization.

$$h_i = f(\frac{g_i}{\sigma_i}\left(a_i - \mu_i\right) + b_i) \tag{5}$$

Note that for layer normalization and batch normalization, μ and σ is computed according to Eq. 2 and 3. In weight normalization, μ is 0, and $\sigma = \|w\|_2$.

	Weight matrix re-scaling	Weight matrix re-centering	Weight vector re-scaling	Dataset re-scaling	Dataset re-centering	Single training case re-scaling
Batch norm	Invariant	No	Invariant	Invariant	Invariant	No
Weight norm	Invariant	No	Invariant	No	No	No
Layer norm	Invariant	Invariant	No	Invariant	No	Invariant

Table 1: Invariance properties under the normalization methods.

Table 1 highlights the following invariance results for three normalization methods.

Weight re-scaling and re-centering: First, observe that under batch normalization and weight normalization, any re-scaling to the incoming weights w_i of a single neuron has no effect on the normalized summed inputs to a neuron. To be precise, under batch and weight normalization, if the weight vector is scaled by δ, the two scalar μ and σ will also be scaled by δ. The normalized summed inputs stays the same before and after scaling. So the batch and weight normalization are invariant to the re-scaling of the weights. Layer normalization, on the other hand, is not invariant to the individual scaling of the single weight vectors. Instead, layer normalization is invariant to scaling of the entire weight matrix and invariant to a shift to all of the incoming weights in the weight matrix. Let there be two sets of model parameters θ, θ' whose weight matrices W and W' differ by a scaling factor δ and all of the incoming weights in W' are also shifted by a constant vector $\boldsymbol{\gamma}$, that is $W' = \delta W + \mathbf{1}\boldsymbol{\gamma}^\top$. Under layer normalization, the two models effectively compute the same output:

$$\begin{aligned}\mathbf{h}' =& f(\frac{\mathbf{g}}{\sigma'}(W'\mathbf{x} - \mu') + \mathbf{b}) = f(\frac{\mathbf{g}}{\sigma'}\left((\delta W + \mathbf{1}\boldsymbol{\gamma}^\top)\mathbf{x} - \mu'\right) + \mathbf{b}) \\ =& f(\frac{\mathbf{g}}{\sigma}(W\mathbf{x} - \mu) + \mathbf{b}) = \mathbf{h}. \end{aligned} \tag{6}$$

Notice that if normalization is only applied to the input before the weights, the model will not be invariant to re-scaling and re-centering of the weights.

Data re-scaling and re-centering: We can show that all the normalization methods are invariant to re-scaling the dataset by verifying that the summed inputs of neurons stays constant under the changes. Furthermore, layer normalization is invariant to re-scaling of individual training cases, because the normalization scalars μ and σ in Eq. (3) only depend on the current input data. Let $\mathbf{x}'$ be a new data point obtained by re-scaling $\mathbf{x}$ by δ. Then we have,

$$h_i' = f(\frac{g_i}{\sigma'}\left(w_i^\top \mathbf{x}' - \mu'\right) + b_i) = f(\frac{g_i}{\delta\sigma}\left(\delta w_i^\top \mathbf{x} - \delta\mu\right) + b_i) = h_i. \tag{7}$$

It is easy to see re-scaling individual data points does not change the model's prediction under layer normalization. Similar to the re-centering of the weight matrix in layer normalization, we can also show that batch normalization is invariant to re-centering of the dataset.

5.2 Geometry of parameter space during learning

We have investigated the invariance of the model's prediction under re-centering and re-scaling of the parameters. Learning, however, can behave very differently under different parameterizations, even though the models express the same underlying function. In this section, we analyze learning behavior through the geometry and the manifold of the parameter space. We show that the normalization scalar σ can implicitly reduce learning rate and makes learning more stable.

5.2.1 Riemannian metric

The learnable parameters in a statistical model form a smooth manifold that consists of all possible input-output relations of the model. For models whose output is a probability distribution, a natural way to measure the separation of two points on this manifold is the Kullback-Leibler divergence between their model output distributions. Under the KL divergence metric, the parameter space is a Riemannian manifold.

The curvature of a Riemannian manifold is entirely captured by its Riemannian metric, whose quadratic form is denoted as ds^2. That is the infinitesimal distance in the tangent space at a point in the parameter space. Intuitively, it measures the changes in the model output from the parameter space along a tangent direction. The Riemannian metric under KL was previously studied [Amari, 1998] and was shown to be well approximated under second order Taylor expansion using the Fisher

information matrix:

$$ds^2 = \mathrm{D}_{\mathrm{KL}}\big[P(y \,|\, \mathbf{x};\ \theta) \| P(y \,|\, \mathbf{x};\ \theta + \delta)\big] \approx \frac{1}{2}\delta^\top F(\theta)\delta, \tag{8}$$

$$F(\theta) = \mathop{\mathbb{E}}_{\mathbf{x}\sim P(\mathbf{x}), y\sim P(y\,|\,\mathbf{x})}\left[\frac{\partial \log P(y \,|\, \mathbf{x};\ \theta)}{\partial \theta}\frac{\partial \log P(y \,|\, \mathbf{x};\ \theta)}{\partial \theta}^\top\right], \tag{9}$$

where, δ is a small change to the parameters. The Riemannian metric above presents a geometric view of parameter spaces. The following analysis of the Riemannian metric provides some insight into how normalization methods could help in training neural networks.

5.2.2 The geometry of normalized generalized linear models

We focus our geometric analysis on the generalized linear model. The results from the following analysis can be easily applied to understand deep neural networks with block-diagonal approximation to the Fisher information matrix, where each block corresponds to the parameters for a single neuron.

A generalized linear model (GLM) can be regarded as parameterizing an output distribution from the exponential family using a weight vector w and bias scalar b. To be consistent with the previous sections, the log likelihood of the GLM can be written using the summed inputs a as the following:

$$\log P(y \,|\, \mathbf{x};\ w, b) = \frac{(a+b)y - \eta(a+b)}{\phi} + c(y, \phi), \tag{10}$$

$$\mathbb{E}[y \,|\, \mathbf{x}] = f(a+b) = f(w^\top \mathbf{x} + b),\ \ \mathrm{Var}[y \,|\, \mathbf{x}] = \phi f'(a+b), \tag{11}$$

where, $f(\cdot)$ is the transfer function that is the analog of the non-linearity in neural networks, $f'(\cdot)$ is the derivative of the transfer function, $\eta(\cdot)$ is a real valued function and $c(\cdot)$ is the log partition function. ϕ is a constant that scales the output variance. Assume a H-dimensional output vector $\mathbf{y} = [y_1, y_2, \cdots, y_H]$ is modeled using H independent GLMs and $\log P(\mathbf{y} \,|\, \mathbf{x};\ W, \mathbf{b}) = \sum_{i=1}^{H} \log P(y_i \,|\, \mathbf{x};\ w_i, b_i)$. Let W be the weight matrix whose rows are the weight vectors of the individual GLMs, $\mathbf{b}$ denote the bias vector of length H and $\mathrm{vec}(\cdot)$ denote the Kronecker vector operator. The Fisher information matrix for the multi-dimensional GLM with respect to its parameters $\theta = [w_1^\top, b_1, \cdots, w_H^\top, b_H]^\top = \mathrm{vec}([W, \mathbf{b}]^\top)$ is simply the expected Kronecker product of the data features and the output covariance matrix:

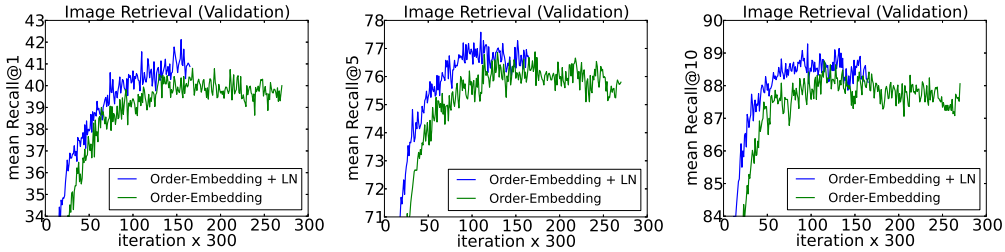
$$F(\theta) = \mathop{\mathbb{E}}_{\mathbf{x}\sim P(\mathbf{x})}\left[\frac{\mathrm{Cov}[\mathbf{y} \,|\, \mathbf{x}]}{\phi^2} \otimes \begin{bmatrix}\mathbf{x}\mathbf{x}^\top & \mathbf{x} \\ \mathbf{x}^\top & 1\end{bmatrix}\right]. \tag{12}$$

We obtain normalized GLMs by applying the normalization methods to the summed inputs a in the original model through μ and σ. Without loss of generality, we denote $\bar{F}$ as the Fisher information matrix under the normalized multi-dimensional GLM with the additional gain parameters $\theta = \mathrm{vec}([W, \mathbf{b}, \mathbf{g}]^\top)$:

$$\bar{F}(\theta) = \begin{bmatrix}\bar{F}_{11} & \cdots & \bar{F}_{1H} \\ \vdots & \ddots & \vdots \\ \bar{F}_{H1} & \cdots & \bar{F}_{HH}\end{bmatrix}, \quad \bar{F}_{ij} = \mathop{\mathbb{E}}_{\mathbf{x}\sim P(\mathbf{x})}\left[\frac{\mathrm{Cov}[y_i,\ y_j\ \,|\, \mathbf{x}]}{\phi^2}\begin{bmatrix}\frac{g_i g_j}{\sigma_i\sigma_j}\chi_i\chi_j^\top & \chi_i\frac{g_i}{\sigma_i} & \chi_i\frac{g_i(a_j-\mu_j)}{\sigma_i\sigma_j} \\ \chi_j^\top\frac{g_j}{\sigma_j} & 1 & \frac{a_j-\mu_j}{\sigma_j} \\ \chi_j^\top\frac{g_j(a_i-\mu_i)}{\sigma_i\sigma_j} & \frac{a_i-\mu_i}{\sigma_i} & \frac{(a_i-\mu_i)(a_j-\mu_j)}{\sigma_i\sigma_j}\end{bmatrix}\right] \tag{13}$$

$$\chi_i = \mathbf{x} - \frac{\partial \mu_i}{\partial w_i} - \frac{a_i - \mu_i}{\sigma_i}\frac{\partial \sigma_i}{\partial w_i}. \tag{14}$$

Implicit learning rate reduction through the growth of the weight vector: Notice that, comparing to standard GLM, the block $\bar{F}_{ij}$ along the weight vector w_i direction is scaled by the gain parameters and the normalization scalar σ_i. If the norm of the weight vector w_i grows twice as large, even though the model's output remains the same, the Fisher information matrix will be different. The curvature along the w_i direction will change by a factor of $\frac{1}{2}$ because the σ_i will also be twice as large. As a result, for the same parameter update in the normalized model, the norm of the weight vector effectively controls the learning rate for the weight vector. During learning, it is harder to change the orientation of the weight vector with large norm. The normalization methods, therefore,



(a) Recall@1 (b) Recall@5 (c) Recall@10

Figure 1: Recall@K curves using order-embeddings with and without layer normalization.

	MSCOCO							
	Caption Retrieval				Image Retrieval			
Model	**R@1**	**R@5**	**R@10**	**Mean *r***	**R@1**	**R@5**	**R@10**	**Mean *r***
Sym [Vendrov et al., 2016]	45.4		88.7	5.8	36.3		85.8	9.0
OE [Vendrov et al., 2016]	46.7		88.9	5.7	37.9		85.9	8.1
OE (ours)	46.6	79.3	89.1	5.2	37.8	73.6	85.7	7.9
OE + LN	**48.5**	**80.6**	**89.8**	**5.1**	**38.9**	**74.3**	**86.3**	**7.6**

Table 2: Average results across 5 test splits for caption and image retrieval. **R@K** is Recall@K (high is good). **Mean *r*** is the mean rank (low is good). Sym corresponds to the symmetric baseline while OE indicates order-embeddings.

have an implicit "early stopping" effect on the weight vectors and help to stabilize learning towards convergence.

Learning the magnitude of incoming weights: In normalized models, the magnitude of the incoming weights is explicitly parameterized by the gain parameters. We compare how the model output changes between updating the gain parameters in the normalized GLM and updating the magnitude of the equivalent weights under original parameterization during learning. The direction along the gain parameters in $\bar{F}$ captures the geometry for the magnitude of the incoming weights. We show that Riemannian metric along the magnitude of the incoming weights for the standard GLM is scaled by the norm of its input, whereas learning the gain parameters for the batch normalized and layer normalized models depends only on the magnitude of the prediction error. Learning the magnitude of incoming weights in the normalized model is therefore, more robust to the scaling of the input and its parameters than in the standard model. See Appendix for detailed derivations.

6 Experimental results

We perform experiments with layer normalization on 6 tasks, with a focus on recurrent neural networks: image-sentence ranking, question-answering, contextual language modelling, generative modelling, handwriting sequence generation and MNIST classification. Unless otherwise noted, the default initialization of layer normalization is to set the adaptive gains to 1 and the biases to 0 in the experiments.

6.1 Order embeddings of images and language

In this experiment, we apply layer normalization to the recently proposed order-embeddings model of Vendrov et al. [2016] for learning a joint embedding space of images and sentences. We follow the same experimental protocol as Vendrov et al. [2016] and modify their publicly available code to incorporate layer normalization [1] which utilizes Theano [Team et al., 2016]. Images and sentences from the Microsoft COCO dataset [Lin et al., 2014] are embedded into a common vector space, where a GRU [Cho et al., 2014] is used to encode sentences and the outputs of a pre-trained VGG ConvNet [Simonyan and Zisserman, 2015] (10-crop) are used to encode images. The order-embedding model represents images and sentences as a 2-level partial ordering and replaces the cosine similarity scoring function used in Kiros et al. [2014] with an asymmetric one.

[1] https://github.com/ivendrov/order-embedding

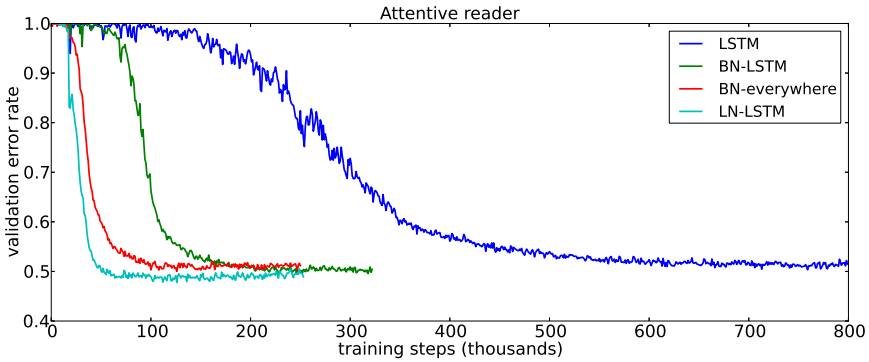


Figure 2: Validation curves for the attentive reader model. BN results are taken from [Cooijmans et al., 2016].

We trained two models: the baseline order-embedding model as well as the same model with layer normalization applied to the GRU. After every 300 iterations, we compute Recall@K (R@K) values on a held out validation set and save the model whenever R@K improves. The best performing models are then evaluated on 5 separate test sets, each containing 1000 images and 5000 captions, for which the mean results are reported. Both models use Adam [Kingma and Ba, 2014] with the same initial hyperparameters and both models are trained using the same architectural choices as used in Vendrov et al. [2016]. We refer the reader to the appendix for a description of how layer normalization is applied to GRU.

Figure 1 illustrates the validation curves of the models, with and without layer normalization. We plot R@1, R@5 and R@10 for the image retrieval task. We observe that layer normalization offers a per-iteration speedup across all metrics and converges to its best validation model in 60% of the time it takes the baseline model to do so. In Table 2, the test set results are reported from which we observe that layer normalization also results in improved generalization over the original model. The results we report are state-of-the-art for RNN embedding models, with only the structure-preserving model of Wang et al. [2016] reporting better results on this task. However, they evaluate under different conditions (1 test set instead of the mean over 5) and are thus not directly comparable.

6.2 Teaching machines to read and comprehend

In order to compare layer normalization to the recently proposed recurrent batch normalization [Cooijmans et al., 2016], we train an unidirectional attentive reader model on the CNN corpus both introduced by Hermann et al. [2015]. This is a question-answering task where a query description about a passage must be answered by filling in a blank. The data is anonymized such that entities are given randomized tokens to prevent degenerate solutions, which are consistently permuted during training and evaluation. We follow the same experimental protocol as Cooijmans et al. [2016] and modify their public code to incorporate layer normalization [2] which uses Theano [Team et al., 2016]. We obtained the pre-processed dataset used by Cooijmans et al. [2016] which differs from the original experiments of Hermann et al. [2015] in that each passage is limited to 4 sentences. In Cooijmans et al. [2016], two variants of recurrent batch normalization are used: one where BN is only applied to the LSTM while the other applies BN everywhere throughout the model. In our experiment, we only apply layer normalization within the LSTM.

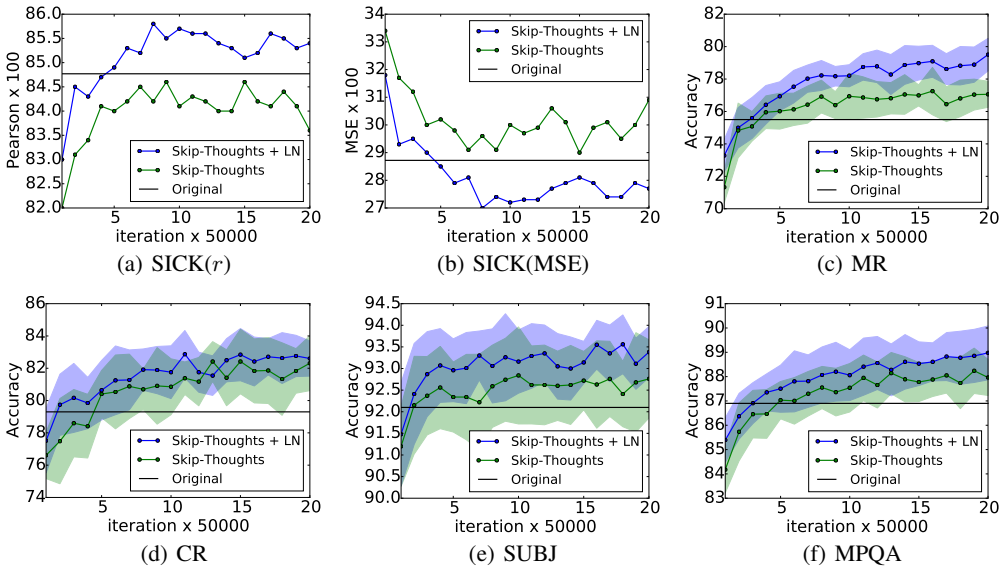
The results of this experiment are shown in Figure 2. We observe that layer normalization not only trains faster but converges to a better validation result over both the baseline and BN variants. In Cooijmans et al. [2016], it is argued that the scale parameter in BN must be carefully chosen and is set to 0.1 in their experiments. We experimented with layer normalization for both 1.0 and 0.1 scale initialization and found that the former model performed significantly better. This demonstrates that layer normalization is not sensitive to the initial scale in the same way that recurrent BN is. [3]

6.3 Skip-thought vectors

Skip-thoughts [Kiros et al., 2015] is a generalization of the skip-gram model [Mikolov et al., 2013] for learning unsupervised distributed sentence representations. Given contiguous text, a sentence is

[2] https://github.com/cooijmanstim/Attentive_reader/tree/bn

[3] We only produce results on the validation set, as in the case of Cooijmans et al. [2016]



(a) SICK(r) (b) SICK(MSE) (c) MR

(d) CR (e) SUBJ (f) MPQA

Figure 3: Performance of skip-thought vectors with and without layer normalization on downstream tasks as a function of training iterations. The original lines are the reported results in [Kiros et al., 2015]. Plots with error use 10-fold cross validation. Best seen in color.

Method	**SICK**(r)	**SICK**(ρ)	**SICK**(MSE)	**MR**	**CR**	**SUBJ**	**MPQA**
Original [Kiros et al., 2015]	0.848	0.778	0.287	75.5	79.3	92.1	86.9
Ours	0.842	0.767	0.298	77.3	81.8	92.6	87.9
Ours + LN	0.854	0.785	0.277	**79.5**	82.6	93.4	89.0
Ours + LN †	**0.858**	**0.788**	**0.270**	79.4	**83.1**	**93.7**	**89.3**

Table 3: Skip-thoughts results. The first two evaluation columns indicate Pearson and Spearman correlation, the third is mean squared error and the remaining indicate classification accuracy. Higher is better for all evaluations except MSE. Our models were trained for 1M iterations with the exception of (†) which was trained for 1 month (approximately 1.7M iterations)

encoded with a encoder RNN and decoder RNNs are used to predict the surrounding sentences. Kiros et al. [2015] showed that this model could produce generic sentence representations that perform well on several tasks without being fine-tuned. However, training this model is time-consuming, requiring several days of training in order to produce meaningful results.

In this experiment we determine to what effect layer normalization can speed up training. Using the publicly available code of Kiros et al. [2015] [4], we train two models on the BookCorpus dataset [Zhu et al., 2015]: one with and one without layer normalization. These experiments are performed with Theano [Team et al., 2016]. We adhere to the experimental setup used in Kiros et al. [2015], training a 2400-dimensional sentence encoder with the same hyperparameters. Given the size of the states used, it is conceivable layer normalization would produce slower per-iteration updates than without. However, we found that provided CNMeM [5] is used, there was no significant difference between the two models. We checkpoint both models after every 50,000 iterations and evaluate their performance on five tasks: semantic-relatedness (SICK) [Marelli et al., 2014], movie review sentiment (MR) [Pang and Lee, 2005], customer product reviews (CR) [Hu and Liu, 2004], subjectivity/objectivity classification (SUBJ) [Pang and Lee, 2004] and opinion polarity (MPQA) [Wiebe et al., 2005]. We plot the performance of both models for each checkpoint on all tasks to determine whether the performance rate can be improved with LN.

The experimental results are illustrated in Figure 3. We observe that applying layer normalization results both in speedup over the baseline as well as better final results after 1M iterations are performed as shown in Table 3. We also let the model with layer normalization train for a total of a month, resulting in further performance gains across all but one task. We note that the performance

[4] https://github.com/ryankiros/skip-thoughts

[5] https://github.com/NVIDIA/cnmem

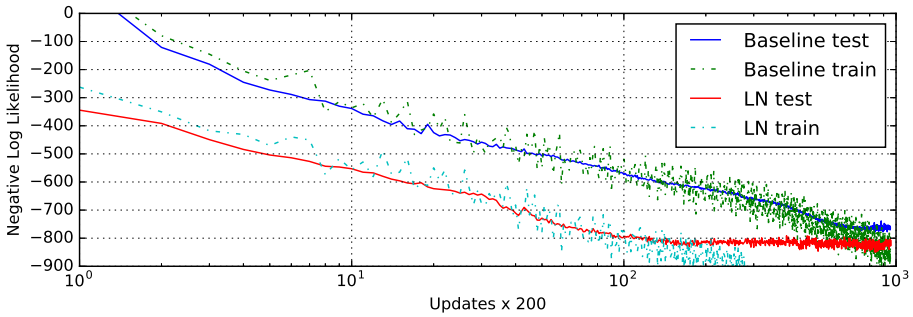


Figure 5: Handwriting sequence generation model negative log likelihood with and without layer normalization. The models are trained with mini-batch size of 8 and sequence length of 500,

differences between the original reported results and ours are likely due to the fact that the publicly available code does not condition at each timestep of the decoder, where the original model does.

6.4 Modeling binarized MNIST using DRAW

We also experimented with the generative modeling on the MNIST dataset. Deep Recurrent Attention Writer (DRAW) [Gregor et al., 2015] has previously achieved the state-of-the-art performance on modeling the distribution of MNIST digits. The model uses a differential attention mechanism and a recurrent neural network to sequentially generate pieces of an image. We evaluate the effect of layer normalization on a DRAW model using 64 glimpses and 256 LSTM hidden units. The model is trained with the default setting of Adam [Kingma and Ba, 2014] optimizer and the minibatch size of 128. Previous publications on binarized MNIST have used various training protocols to generate their datasets. In this experiment, we used the fixed binarization from Larochelle and Murray [2011]. The dataset has been split into 50,000 training, 10,000 validation and 10,000 test images.

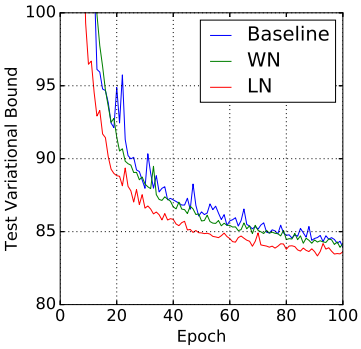


Figure 4: DRAW model test negative log likelihood with and without layer normalization.

Figure 4 shows the test variational bound for the first 100 epoch. It highlights the speedup benefit of applying layer normalization that the layer normalized DRAW converges almost twice as fast than the baseline model. After 200 epoches, the baseline model converges to a variational log likelihood of 82.36 nats on the test data and the layer normalization model obtains 82.09 nats.

6.5 Handwriting sequence generation

The previous experiments mostly examine RNNs on NLP tasks whose lengths are in the range of 10 to 40. To show the effectiveness of layer normalization on longer sequences, we performed handwriting generation tasks using the IAM Online Handwriting Database [Liwicki and Bunke, 2005]. IAM-OnDB consists of handwritten lines collected from 221 different writers. When given the input character string, the goal is to predict a sequence of x and y pen co-ordinates of the corresponding handwriting line on the whiteboard. There are, in total, 12179 handwriting line sequences. The input string is typically more than 25 characters and the average handwriting line has a length around 700.

We used the same model architecture as in Section (5.2) of Graves [2013]. The model architecture consists of three hidden layers of 400 LSTM cells, which produce 20 bivariate Gaussian mixture components at the output layer, and a size 3 input layer. The character sequence was encoded with one-hot vectors, and hence the window vectors were size 57. A mixture of 10 Gaussian functions was used for the window parameters, requiring a size 30 parameter vector. The total number of weights was increased to approximately 3.7M. The model is trained using mini-batches of size 8 and the Adam [Kingma and Ba, 2014] optimizer.

The combination of small mini-batch size and very long sequences makes it important to have very stable hidden dynamics. Figure 5 shows that layer normalization converges to a comparable log likelihood as the baseline model but is much faster.

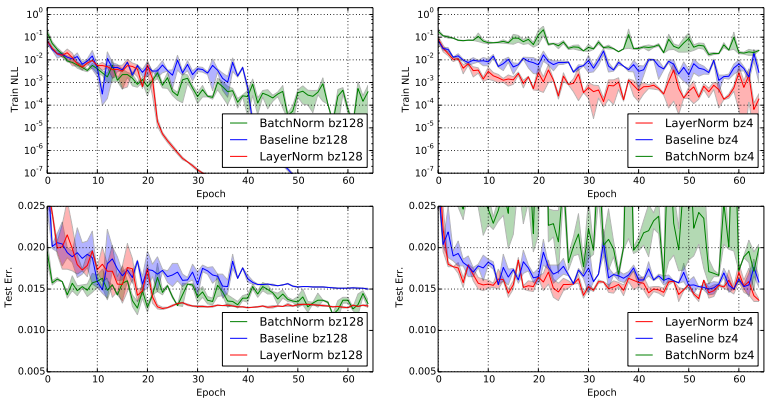


Figure 6: Permutation invariant MNIST 784-1000-1000-10 model negative log likelihood and test error with layer normalization and batch normalization. (Left) The models are trained with batch-size of 128. (Right) The models are trained with batch-size of 4.

6.6 Permutation invariant MNIST

In addition to RNNs, we investigated layer normalization in feed-forward networks. We show how layer normalization compares with batch normalization on the well-studied permutation invariant MNIST classification problem. From the previous analysis, layer normalization is invariant to input re-scaling which is desirable for the internal hidden layers. But this is unnecessary for the logit outputs where the prediction confidence is determined by the scale of the logits. We only apply layer normalization to the fully-connected hidden layers that excludes the last softmax layer.

All the models were trained using 55000 training data points and the Adam [Kingma and Ba, 2014] optimizer. For the smaller batch-size, the variance term for batch normalization is computed using the unbiased estimator. The experimental results from Figure 6 highlight that layer normalization is robust to the batch-sizes and exhibits a faster training convergence comparing to batch normalization that is applied to all layers.

6.7 Convolutional Networks

We have also experimented with convolutional neural networks. In our preliminary experiments, we observed that layer normalization offers a speedup over the baseline model without normalization, but batch normalization outperforms the other methods. With fully connected layers, all the hidden units in a layer tend to make similar contributions to the final prediction and re-centering and re-scaling the summed inputs to a layer works well. However, the assumption of similar contributions is no longer true for convolutional neural networks. The large number of the hidden units whose receptive fields lie near the boundary of the image are rarely turned on and thus have very different statistics from the rest of the hidden units within the same layer. We think further research is needed to make layer normalization work well in ConvNets.

7 Conclusion

In this paper, we introduced layer normalization to speed-up the training of neural networks. We provided a theoretical analysis that compared the invariance properties of layer normalization with batch normalization and weight normalization. We showed that layer normalization is invariant to per training-case feature shifting and scaling.

Empirically, we showed that recurrent neural networks benefit the most from the proposed method especially for long sequences and small mini-batches.

Acknowledgments

This research was funded by grants from NSERC, CFI, and Google.

Supplementary Material

Application of layer normalization to each experiment

This section describes how layer normalization is applied to each of the papers' experiments. For notation convenience, we define layer normalization as a function mapping $LN : \mathbb{R}^D \rightarrow \mathbb{R}^D$ with two set of adaptive parameters, gains $\boldsymbol{\alpha}$ and biases $\boldsymbol{\beta}$:

$$LN(\mathbf{z}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{(\mathbf{z} - \mu)}{\sigma} \odot \boldsymbol{\alpha} + \boldsymbol{\beta}, \tag{15}$$

$$\mu = \frac{1}{D}\sum_{i=1}^{D} z_i, \quad \sigma = \sqrt{\frac{1}{D}\sum_{i=1}^{D}(z_i - \mu)^2}, \tag{16}$$

where, z_i is the i^{th} element of the vector $\mathbf{z}$.

Teaching machines to read and comprehend and handwriting sequence generation

The basic LSTM equations used for these experiment are given by:

$$\begin{pmatrix} \mathbf{f}_t \\ \mathbf{i}_t \\ \mathbf{o}_t \\ \mathbf{g}_t \end{pmatrix} = \mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + b \tag{17}$$

$$\mathbf{c}_t = \sigma(\mathbf{f}_t) \odot \mathbf{c}_{t-1} + \sigma(\mathbf{i}_t) \odot \tanh(\mathbf{g}_t) \tag{18}$$

$$\mathbf{h}_t = \sigma(\mathbf{o}_t) \odot \tanh(\mathbf{c}_t) \tag{19}$$

The version that incorporates layer normalization is modified as follows:

$$\begin{pmatrix} \mathbf{f}_t \\ \mathbf{i}_t \\ \mathbf{o}_t \\ \mathbf{g}_t \end{pmatrix} = LN(\mathbf{W}_h \mathbf{h}_{t-1}; \boldsymbol{\alpha}_1, \boldsymbol{\beta}_1) + LN(\mathbf{W}_x \mathbf{x}_t; \boldsymbol{\alpha}_2, \boldsymbol{\beta}_2) + b \tag{20}$$

$$\mathbf{c}_t = \sigma(\mathbf{f}_t) \odot \mathbf{c}_{t-1} + \sigma(\mathbf{i}_t) \odot \tanh(\mathbf{g}_t) \tag{21}$$

$$\mathbf{h}_t = \sigma(\mathbf{o}_t) \odot \tanh(LN(\mathbf{c}_t; \boldsymbol{\alpha}_3, \boldsymbol{\beta}_3)) \tag{22}$$

where $\boldsymbol{\alpha}_i, \boldsymbol{\beta}_i$ are the additive and multiplicative parameters, respectively. Each $\boldsymbol{\alpha}_i$ is initialized to a vector of zeros and each $\boldsymbol{\beta}_i$ is initialized to a vector of ones.

Order embeddings and skip-thoughts

These experiments utilize a variant of gated recurrent unit which is defined as follows:

$$\begin{pmatrix} \mathbf{z}_t \\ \mathbf{r}_t \end{pmatrix} = \mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t \tag{23}$$

$$\hat{\mathbf{h}}_t = \tanh(\mathbf{W}\mathbf{x}_t + \sigma(\mathbf{r}_t) \odot (\mathbf{U}\mathbf{h}_{t-1})) \tag{24}$$

$$\mathbf{h}_t = (1 - \sigma(\mathbf{z}_t))\mathbf{h}_{t-1} + \sigma(\mathbf{z}_t)\hat{\mathbf{h}}_t \tag{25}$$

Layer normalization is applied as follows:

$$\begin{pmatrix} \mathbf{z}_t \\ \mathbf{r}_t \end{pmatrix} = LN(\mathbf{W}_h \mathbf{h}_{t-1}; \boldsymbol{\alpha}_1, \boldsymbol{\beta}_1) + LN(\mathbf{W}_x \mathbf{x}_t; \boldsymbol{\alpha}_2, \boldsymbol{\beta}_2) \tag{26}$$

$$\hat{\mathbf{h}}_t = \tanh(LN(\mathbf{W}\mathbf{x}_t; \boldsymbol{\alpha}_3, \boldsymbol{\beta}_3) + \sigma(\mathbf{r}_t) \odot LN(\mathbf{U}\mathbf{h}_{t-1}; \boldsymbol{\alpha}_4, \boldsymbol{\beta}_4)) \tag{27}$$

$$\mathbf{h}_t = (1 - \sigma(\mathbf{z}_t))\mathbf{h}_{t-1} + \sigma(\mathbf{z}_t)\hat{\mathbf{h}}_t \tag{28}$$

just as before, $\boldsymbol{\alpha}_i$ is initialized to a vector of zeros and each $\boldsymbol{\beta}_i$ is initialized to a vector of ones.

Modeling binarized MNIST using DRAW

The layer norm is only applied to the output of the LSTM hidden states in this experiment:

The version that incorporates layer normalization is modified as follows:

$$\begin{pmatrix} \mathbf{f}_t \\ \mathbf{i}_t \\ \mathbf{o}_t \\ \mathbf{g}_t \end{pmatrix} = \mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + b \tag{29}$$

$$\mathbf{c}_t = \sigma(\mathbf{f}_t) \odot \mathbf{c}_{t-1} + \sigma(\mathbf{i}_t) \odot \tanh(\mathbf{g}_t) \tag{30}$$

$$\mathbf{h}_t = \sigma(\mathbf{o}_t) \odot \tanh(LN(\mathbf{c}_t; \boldsymbol{\alpha}, \boldsymbol{\beta})) \tag{31}$$

where $\boldsymbol{\alpha}, \boldsymbol{\beta}$ are the additive and multiplicative parameters, respectively. $\boldsymbol{\alpha}$ is initialized to a vector of zeros and $\boldsymbol{\beta}$ is initialized to a vector of ones.

Learning the magnitude of incoming weights

We now compare how gradient descent updates changing magnitude of the equivalent weights between the normalized GLM and original parameterization. The magnitude of the weights are explicitly parameterized using the gain parameter in the normalized model. Assume there is a gradient update that changes norm of the weight vectors by δ_g. We can project the gradient updates to the weight vector for the normal GLM. The KL metric, ie how much the gradient update changes the model prediction, for the normalized model depends only on the magnitude of the prediction error. Specifically,

under batch normalization:

$$ds^2 = \frac{1}{2} \operatorname{vec}([0, 0, \delta_g]^\top)^\top \bar{F} (\operatorname{vec}([W, \mathbf{b}, \mathbf{g}]^\top) \operatorname{vec}([0, 0, \delta_g]^\top) = \frac{1}{2} \delta_g^\top \mathop{\mathbb{E}}_{\mathbf{x} \sim P(\mathbf{x})} \left[\frac{\operatorname{Cov}[\mathbf{y} \mid \mathbf{x}]}{\phi^2} \right] \delta_g. \tag{32}$$

Under layer normalization:

$$\begin{aligned} ds^2 =& \frac{1}{2} \operatorname{vec}([0, 0, \delta_g]^\top)^\top \bar{F} (\operatorname{vec}([W, \mathbf{b}, \mathbf{g}]^\top) \operatorname{vec}([0, 0, \delta_g]^\top) \\ =& \frac{1}{2} \delta_g^\top \frac{1}{\phi^2} \mathop{\mathbb{E}}_{\mathbf{x} \sim P(\mathbf{x})} \begin{bmatrix} \operatorname{Cov}(y_1, y_1 \mid \mathbf{x}) \frac{(a_1 - \mu)^2}{\sigma^2} & \cdots & \operatorname{Cov}(y_1, y_H \mid \mathbf{x}) \frac{(a_1 - \mu)(a_H - \mu)}{\sigma^2} \\ \vdots & \ddots & \vdots \\ \operatorname{Cov}(y_H, y_1 \mid \mathbf{x}) \frac{(a_H - \mu)(a_1 - \mu)}{\sigma^2} & \cdots & \operatorname{Cov}(y_H, y_H \mid \mathbf{x}) \frac{(a_H - \mu)^2}{\sigma^2} \end{bmatrix} \delta_g \end{aligned} \tag{33}$$

Under weight normalization:

$$\begin{aligned} ds^2 =& \frac{1}{2} \operatorname{vec}([0, 0, \delta_g]^\top)^\top \bar{F} (\operatorname{vec}([W, \mathbf{b}, \mathbf{g}]^\top) \operatorname{vec}([0, 0, \delta_g]^\top) \\ =& \frac{1}{2} \delta_g^\top \frac{1}{\phi^2} \mathop{\mathbb{E}}_{\mathbf{x} \sim P(\mathbf{x})} \begin{bmatrix} \operatorname{Cov}(y_1, y_1 \mid \mathbf{x}) \frac{a_1^2}{\|w_1\|_2^2} & \cdots & \operatorname{Cov}(y_1, y_H \mid \mathbf{x}) \frac{a_1 a_H}{\|w_1\|_2 \|w_H\|_2} \\ \vdots & \ddots & \vdots \\ \operatorname{Cov}(y_H, y_1 \mid \mathbf{x}) \frac{a_H a_1}{\|w_H\|_2 \|w_1\|_2} & \cdots & \operatorname{Cov}(y_H, y_H \mid \mathbf{x}) \frac{a_H^2}{\|w_H\|_2^2} \end{bmatrix} \delta_g. \end{aligned} \tag{34}$$

Whereas, the KL metric in the standard GLM is related to its activities $a_i = w_i^\top \mathbf{x}$, that is depended on both its current weights and input data. We project the gradient updates to the gain parameter $\delta_g i$ of the i^{th} neuron to its weight vector as $\delta_{gi} \frac{w_i}{\|w_i\|_2}$ in the standard GLM model:

$$\begin{aligned} & \frac{1}{2} \operatorname{vec}([\delta_{gi} \frac{w_i}{\|w_i\|_2}, 0, \delta_{gj} \frac{w_j}{\|w_i\|_2}, 0]^\top)^\top F([w_i^\top, b_i, w_j^\top, b_j]^\top) \operatorname{vec}([\delta_{gi} \frac{w_i}{\|w_i\|_2}, 0, \delta_{gj} \frac{w_j}{\|w_j\|_2}, 0]^\top) \\ =& \frac{\delta_{gi} \delta_{gj}}{2\phi^2} \mathop{\mathbb{E}}_{\mathbf{x} \sim P(\mathbf{x})} \left[\operatorname{Cov}(y_i, y_j \mid \mathbf{x}) \frac{a_i a_j}{\|w_i\|_2 \|w_j\|_2} \right] \end{aligned} \tag{35}$$

The batch normalized and layer normalized models are therefore more robust to the scaling of the input and its parameters than the standard model.

Normalization and effective learning rates in reinforcement learning

Clare Lyle[†] **Zeyu Zheng**[†] **Khimya Khetarpal**[†] **James Martens**[†] **Hado van Hasselt**[†]

Razvan Pascanu[†] **Will Dabney**[†]

Abstract

Normalization layers have recently experienced a renaissance in the deep reinforcement learning and continual learning literature, with several works highlighting diverse benefits such as improving loss landscape conditioning and combatting overestimation bias. However, normalization brings with it a subtle but important side effect: an equivalence between growth in the norm of the network parameters and decay in the effective learning rate. This becomes problematic in continual learning settings, where the resulting effective learning rate schedule may decay to near zero too quickly relative to the timescale of the learning problem. We propose to make the learning rate schedule explicit with a simple re-parameterization which we call Normalize-and-Project (NaP), which couples the insertion of normalization layers with weight projection, ensuring that the effective learning rate remains constant throughout training. This technique reveals itself as a powerful analytical tool to better understand learning rate schedules in deep reinforcement learning, and as a means of improving robustness to nonstationarity in synthetic plasticity loss benchmarks along with both the single-task and sequential variants of the Arcade Learning Environment. We also show that our approach can be easily applied to popular architectures such as ResNets and transformers while recovering and in some cases even slightly improving the performance of the base model in common stationary benchmarks.

1 Introduction

Many of the most promising application areas of deep learning, in particular reinforcement learning (RL), require training on a problem which is in some way nonstationary. In order for this type of training to be effective, the neural network must maintain its ability to adapt to new information as it becomes available, i.e. it must remain *plastic*. Several recent works have shown that loss of plasticity can present a major barrier to performance improvement in RL and in continual learning [Dohare et al., 2021, Lyle et al., 2021, Nikishin et al., 2022]. These works have proposed a variety of explanations for plasticity loss such as the accumulation of saturated ReLU unit and increased sharpness of the loss landscape [Lyle et al., 2023], along with mitigation strategies, such as resetting dead units [Sokar et al., 2023] and regularizing the parameters towards their initial values [Kumar et al., 2023]. Many of these explanations and their corresponding mitigation strategies center around reducing drift in the distribution of pre-activations [Lyle et al., 2024], a problem which has historically been resolved in the supervised learning setting by incorporating normalization layers into the network architecture. Indeed, normalization layers have been shown to be highly effective at stabilizing optimization in both continual learning and RL [Hussing et al., 2024, Ball et al., 2023].

[†]Google DeepMind. Correspondence to clarelyle@google.com

Preprint. Under review.

While effective, normalization on its own is insufficient to avoid loss of plasticity [Lee et al., 2023]. Part of the reason for this lies in a subtle property of normalization: a normalization layer causes the subnetwork preceding it to become scale-invariant, which means that the layer's *effective learning rate* (ELR) now depends on the norm of its parameters [Van Laarhoven, 2017]. In particular, when the norm of the parameters grows, as it typically does in neural networks trained without regularization, the effective learning rate shrinks. While this property has been studied extensively in idealized supervised learning settings [Arora et al., 2018], its practical implications on optimization in nonstationary problems have previously only been noted in the form of 'saturation' of normalization layers [Lyle et al., 2024].

This work begins with an analysis of the effect of layer normalization on a network's plasticity, revealing two surprising benefits. We first show that when coupled with adaptive optimizers such as Adam and RMSProp whose step size is independent of gradient norm, saturated units still receive sufficient gradient signal to make non-trivial changes to their parameters, providing the opportunity for them to recover as a natural byproduct of optimization. We go on to study the implicit learning rate schedule that layer normalization induces, arriving at the striking observation that even without an explicit learning rate schedule, the implicit learning rate decay induced by parameter norm growth in value-based deep RL algorithms is in fact critical to the optimization process. This observation yields an explanation for the dramatic performance degradations often induced by weight decay in value-based deep RL agents: certain components of the value function require a sufficiently small ELR in order to be learned, and an optimization process which does not reach this value will therefore underfit the value function in ways that can inhibit performance improvement. We further show that, while often beneficial in single-task settings, this implicit schedule can harm performance in nonstationary regimes, where the natural effective learning rate schedule drives the learning rate to trivial values too quickly relative to the task duration.

Leveraging this insight, we propose Normalize-and-Project (NaP), a simple protocol which ensures that the learning rate used by the optimizer reflects the true effective learning rate on the network. NaP avoids implicit learning rate decay, allowing us to attain impressive performance in a variety of nonstationary learning problems even without explicit regularization intended to prevent loss of plasticity. We conduct an empirical evaluation of NaP, confirming that it can be applied to a variety of architectures and datasets without interfering with (indeed, in some cases even improving) performance, including 400M transformer models trained on the C4 dataset and vision models trained on CIFAR-10 and ImageNet. We conclude with a study on the Sequential Arcade Learning Environment, where NaP demonstrates remarkable robustness to task changes and outperforms a baseline Rainbow agent with freshly initialized parameters after 400M training frames (100M optimizer steps).

2 Background and related work

We begin by providing background on trainability and its loss in nonstationary learning problems. We additionally give an overview of neural network training dynamics and effective learning rates.

2.1 Training dynamics and plasticity in neural networks

Early work on neural network initialization centered around the idea of controlling the norm of the activation vectors [LeCun et al., 2002, Glorot and Bengio, 2010, He et al., 2015] using informal arguments. More recently, this perspective has been formalized and expanded [Poole et al., 2016, Daniely et al., 2016, Martens et al., 2021] to include the inner-products between pairs of activation vectors (for different inputs to the network). The function that describes the evolution of these inner-products determines the network's gradients at initialization up to rotation, and this in turn determines trainability (which was shown formally in the Neural Tangent Kernel regime by Xiao et al. [2020] and Martens et al. [2021]). A variety of initialization methods have been developed to ensure the network avoids "shattering" [Balduzzi et al., 2017] or collapsing gradients [Poole et al., 2016, Martens et al., 2021, Zhang et al., 2021b].

Once training begins, learning dynamics can be well-characterized in the infinite-width limit by the neural tangent kernel and related quantities [Jacot et al., 2018, Yang, 2019], although in practice optimization dynamics diverge significantly from the infinite-width limit [Fort et al., 2020]. A complementary line of work has empirically and theoretically characterized self-stabilization properties

[Lewkowycz et al., 2020, Cohen et al., 2021, Agarwala et al., 2022], along with implicit regularization effects [Barrett and Dherin, 2020, Smith et al., 2020] from using non-infinitesimal learning rate. A variety of architectural choices can accelerate the training of extremely deep networks, including residual connections [He et al., 2016] and normalization layers [Ioffe and Szegedy, 2015, Ba et al., 2016]. Some additional works have aimed to replicate the benefits of normalization layers via normalization of the network *parameters* [Salimans and Kingma, 2016, Arpit et al., 2016], though layer normalization remains standard practice.

Maintaining trainability not only at initialization, but also, over the course of optimization is of critical importance in reinforcement and continual learning, and failure to do so has been extensively documented throughout the literature under the term *loss of plasticity* [Abbas et al., 2023, Lyle et al., 2021, Dohare et al., 2021, Sodhani et al., 2020, Nikishin et al., 2022]. This phenomenon has been shown to present a limiting factor to performance in a number of RL tasks [Nikishin et al., 2023], along with continual learning and warm-starting neural network training [Berariu et al., 2021, Ash and Adams, 2020]. Plasticity loss can be further decomposed into two distinct components [Lee et al., 2023]: loss of trainability and reduced generalization performance, of which we focus on the former.

2.2 Effective learning rates

As noted by several prior works [Van Laarhoven, 2017, Li and Arora, 2020, Li et al., 2020b], normalization introduces scale-invariance into the layers to which it is applied, where by a scale-invariant function f we mean $f(c\theta, \mathbf{x}) = f(\theta, \mathbf{x})$ for any positive scalar $c > 0$. This leads to the gradient scaling inversely with the parameter norm, whereby $\nabla f(c\theta) = \frac{1}{c}\nabla f(\theta)$. The intuition behind this property is simple: changing the direction of a large vector requires a greater perturbation than changing the direction of a small vector. This motivates the concept of an 'effective learning rate', which provides a scale-invariant notion of optimizer step size. In the following definition, we take the approach of Kodryan et al. [2022] and assume an implicit 'reference norm' of size 1 for the parameters.

Definition 1 (Effective learning rate)**.** *Consider a scale-invariant function f, parameters θ and update function $\theta_{t+1} \leftarrow \theta_t + \eta g(\theta_t)$. Letting $\rho = \frac{1}{\|\theta\|}$, we then define the* effective learning rate $\tilde{\eta}$ *as follows:*

$$\tilde{\eta} = \begin{cases} \eta\rho^2, & \text{if } g(\theta_t) = \nabla_\theta f(\theta_t) \\ \eta\rho, & \text{if } g(\theta_t) = \frac{\nabla_\theta f(\theta_t)}{\|\nabla_\theta f(\theta_t)\|} \end{cases} \tag{1}$$

where, letting $\tilde{\theta} = \theta\frac{1}{\|\theta\|}$ we then have $f(\tilde{\theta} + \tilde{\eta} g(\tilde{\theta})) = f(\theta + \eta g(\theta))$

This suggests that regularization of the network norm can have the dual effect of increasing the effective learning rate, as noted by prior work [Van Laarhoven, 2017, Hoffer et al., 2018]. Several works have since offered more fine-grained theoretical analyses of such implicit learning rate schedules [Arora et al., 2018], and suggested means of translating these implicit schedules into explicit ones [Li and Arora, 2020, Li et al., 2020a]. More recently, the work of Lobacheva et al. [2021] and Kodryan et al. [2022] has studied the training properties of scale-invariant networks trained with parameters constrained to the unit sphere, a training regime we expand upon in this work.

3 Analysis of normalization layers and plasticity

Although widely used and studied, the precise reasons behind the effectiveness of layer normalization remain mysterious. In this section, we provide some new insights into how normalization can help neural networks to maintain plasticity by facilitating the recovery of saturated nonlinearities, and highlight the importance of controlling the parameter norm in networks which incorporate normalization layers. We leverage these insights to propose Normalize-and-Project, a simple training protocol to maintain important statistics of the layers and gradients throughout training.

3.1 Layer normalization

It is widely accepted that achieving approximately mean-zero, unit-variance pre-activations (assuming suitable choices of activation functions) is useful to ensure a network is trainable at initialization [e.g Martens et al., 2021], and many neural network initialization schemes aim to maintain this property

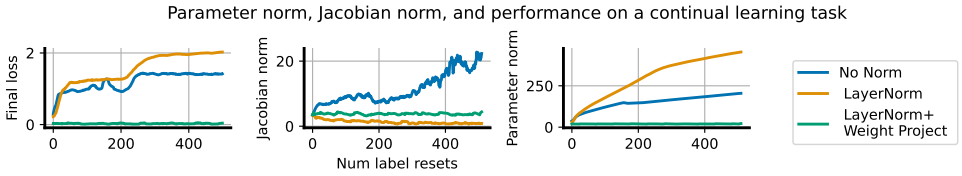


Figure 1: Continual random-labels CIFAR training: simple feedforward network architecture (No Norm) exhibits rapid growth in its parameter norm and the norm of its gradients, whereas the otherwise-identical network with layer normalization sees parameter norm growth coupled with a *reduction* in the norm of its gradients and reduced performance on later tasks. Constraining the parameter norm of this network maintains the performance of a random initialization.

as the depth of the network grows. Indeed, it is easy to show that in extreme cases large deviations of these statistics from their initial values can lead to a variety of network pathologies including saturated units and low numerical rank of the empirical neural tangent kernel [Xiao et al., 2020]. Layer normalization not only guarantees that activations are unit-norm, mean-zero at initialization, but also that they stay that way over the course of training even if the data distribution changes, assuming no scale or offset parameters. This property not only improves robustness to a variety of pathologies that cause loss of plasticity [Lyle et al., 2024], but also helps to improve the conditioning of the network's gradients in RL [Ball et al., 2023].

Beyond re-normalizing the pre-activation statistics, layer normalization also introduces a dependency between units in a given layer via the mean subtraction and division by standard deviation transformations, which translates to correlations in the gradients of their corresponding weights. This mixing step allows gradients to propagate through a pre-activation even if the unit is saturated, provided layer normalization is applied prior to the nonlinearity, a property we highlight in Proposition 1. We will use the notation f_{RMS} to refer to the transform $h \mapsto \frac{h}{\|h\|}$, and $f'(x) \equiv \frac{\partial}{\partial x} f(x)$ to refer to the scalar derivative of any f at scalar x.

Proposition 1. *Consider two indices i and j of a feature embedding $\phi(f_{\text{RMS}}(h))$ such that $\phi'(f_{\text{RMS}}(h)_j) \neq 0$, and $h_i, h_j \neq 0$. Then we have*

$$\frac{d}{dh_i}\phi(f_{\text{RMS}}(h))_j = -\phi'(f_{\text{RMS}}(h)_j)\frac{1}{\|h\|^3}h_i h_j \neq 0\ .$$

In contrast, for post-activation normalization the gradient is zero whenever $\phi'(h_i) = 0$, taking the form $\partial_{h_i} f_{\text{RMS}}(\phi(h))_j = -\phi'(h_i)\frac{1}{\|\phi(h)\|^3}\phi(h_i)\phi(h_j)$.

In other words, normalization effectively gives dead ReLU units a second chance at life – rather than immediately decaying to zero, the gradients flowing through a dead ReLU unit will instead take small, non-zero values, which depends on the gradients of the mean and variance of that particular layer. These gradients will be much smaller than those that would typically backpropagate to the unit, but if an optimizer such as Adam or RMSProp is used to correct for the gradient norm, then the unit may still be able to take nontrivial steps, which have a chance at propelling it back into the activated regime. We include the full derivation in Appendix A.2, and we demonstrate the effect this can have on both a theoretical model and empirical neural network training in Figure 10 in Appendix C.4. This property is also naturally inherited by layer normalization, which can be viewed as the composition of RMSNorm with a centering transform. While layer normalization can help the network to recover from saturated nonlinearities, it introduces a new source of potential saturation which must be carefully considered, which is something we will do in the next section.

3.2 Parameter norm and effective learning rate decay

While the *output* of a scale-invariant function is insensitive to scalar multiplication of the parameters, its *gradient* magnitude scales inversely with the parameter norm. This results in the opposite behaviour of what we would expect in an unnormalized network: in networks with layer normalization, growth in the norm of the parameters corresponds to a decline in the network's sensitivity to changes in these parameters. In a sense this is preferable, as the glacially slow but stable regime of vanishing gradients is easier to recover from than the unstable exploding gradient regime. However, if the parameter

Algorithm 1 NaP: Normalize-and-Project

Input: network $\mathcal{N}$, input x
for nonlinearity ϕ_l in network **do**
  if ϕ_l not already normalized **then**
    $\phi_l \leftarrow \phi_l \circ \text{LayerNorm}$
  end if
end for
compute $\theta' = \texttt{update}(\theta)$
for parameter W_l in network **do**
  compute $W_l \leftarrow \text{WeightProject}(W_l)$
end for

$\text{WeightProject}(W_l, \rho_l)$:
if W_l is a weight parameter **then**
  $W_l \leftarrow \frac{\rho_l W_l}{\|W_l\|}$
else
  $\sigma_l, \mu_l \leftarrow W_l, d \leftarrow \text{len}(\sigma_l)$
  $\sigma_l \leftarrow \sigma_l \sqrt{\frac{d}{\|\sigma_l\|^2 + \|\mu_l\|^2}}$
  $\mu_l \leftarrow \mu_l \sqrt{\frac{d}{\|\sigma_l\|^2 + \|\mu_l\|^2}}$
end if

norm grows indefinitely then the corresponding reduction in the effective learning rate will eventually cause noticeable slowdowns in learning – indeed, it is quite easy to induce this type of situation as we show in Figure 1. To do so, we take a small base convolutional neural network architecture (detailed in Appendix B.4) and train it on random labels of the CIFAR-10 dataset, akin to the classic setting of Zhang et al. [2021a]. We then re-randomize these labels and continue training, repeating this process 500 times. When we apply this process to a network without normalization layers, the Jacobian norm grows to unstable values as the parameter norm increases; in contrast, an equivalent architecture with normalization layers sees a sharp decline in the Jacobian norm as the parameter norm increases. In both cases, the end result is similar: increased parameter norm accompanies reduced performance.

While this particular problem is artificial, it is a real and widely observed underlying phenomenon that the magnitudes of neural network parameters tend to increase over the course of training [Nikishin et al., 2022, Abbas et al., 2023]. In a supervised learning problem, where one is using a fixed training budget, the ELR decay induced by growing parameter norms might be desirable and help to protect against too-large learning rates [Arora et al., 2018, Salimans and Kingma, 2016]. Allowed to continue to extremes, however, ELR decay becomes problematic [Lyle et al., 2024]. Instead, if we re-normalize the parameters after every task (which does not affect the output of our scale-invariant model) so they recover the norm they had at initialization (but do not change their direction), we see in Figure 1 that we can maintain the performance of a random initialization even after hundreds of training iterations.

3.3 Normalize-and-Project

We conclude from the above investigation that normalizing a network's pre-activations and fixing the parameter norm presents a simple but effective defense against loss of plasticity. In this section, we propose a principled approach to combine these two steps which we call NaP. Our goal for NaP is to provide a flexible recipe which can be applied to essentially any architecture, and which improves the stability of training, motivated by but not limited to non-stationary problems. Our approach can be decomposed into two steps: the **insertion of normalization layers** prior to nonlinearities in the network architecture, and the **periodic projection** of the network's weights onto a fixed-norm radius throughout training, along with a corresponding update to the per-layer learning rates into the optimization process. Algorithm 1 provides an overview of NaP. While some design choices, such as the use of scale and offset terms, can be tailored to a particular problem setting, we will aim to provide principled guidance on how to reason about the effects of these choices.

Layer normalization: Introducing layer normalization allows us to benefit from the properties discussed in Section 3.1, though fully benefiting from these properties depend on normalization being applied each time a linear transform precedes a nonlinearity. While it might seem extreme, this proposal is in line with most popular language and vision architectures. For example, Vaswani et al. [2017] apply normalization after every two fully-connected layers, and recent results suggesting that adding normalization to the key and query matrices in attention heads [Henry et al., 2020] can provide further benefits.

Weight projection: As discussed in Section 3.2, we must then take care in controlling the network's effective learning rate. We propose disentangling the parameter norm from the effective learning rate by enforcing a constant norm on the weight parameters of the network, allowing scaling of the layer outputs to depend only on the learnable scale and offset parameters. This approach is similar to that proposed by Kodryan et al. [2022], but importantly takes care to treat the scale and offset parameters

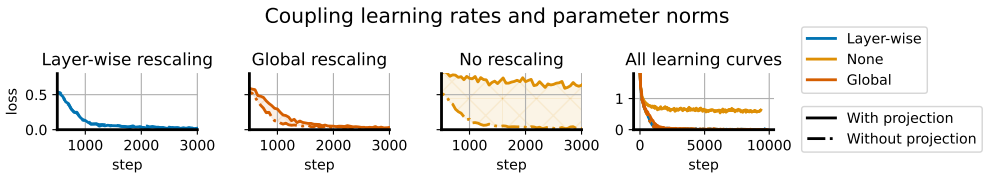


Figure 2: We run a 'coupled networks' experiment as described in the text. All networks exhibit similar learning curves, as seen by the rightmost subplot, however there is small but visible gap between the learning curves obtained by NaP and an unconstrained network with fixed learning rates. Using a global learning rate schedule almost entirely closes this gap, but does not induce a precise equivalence in the dynamics as obtained by layer-wise rescaling (leftmost).

separately from weights. For simplicity, we remove bias terms as these are made redundant by the learnable offset parameters. In order to maintain constant parameter norm, we rescale the parameters of each layer to match their initial norm periodically throughout training – the precise frequency is not important as long as the parameter norm does not meaningfully grow to a point of slowing optimization between projections. For example, we find that in Rainbow agents an interval of 1000 steps and 1 step produce nearly identical empirical results.

We find it is absolutely critical to normalize the weight parameters, as these represent the bulk of trainable parameters in the network. The learnable scale and offset parameters, assuming they are included in the network[3], permit more flexibility and can be dealt with in one of three ways. The strictly correct version in the case of homogeneous activations such as ReLUs is the approach we outline in Appendix D, which projects the concatenated scale-offset vector. This can require some effort to implement due to the dependency between the scale and offset parameters and may not be worthwhile for small training runs – indeed, most of our empirical results did not require this step, though we include it in our analysis of smaller networks in Figure 2. A softer version of this normalization which requires less implementation overhead and which generalizes to non-homogeneous activations is to regularize the scale and offset parameters to their initial values, a strategy which we employ in our continual learning evaluations. Finally, they can be allowed to drift unconstrained from their initialization, a choice we find unproblematic in most shorter, stationary learning problems. For a more detailed discussion on this choice, we refer to Appendix D.

4 Lessons on the effective learning rate

NaP constrains the network's effective learning rate to follow an explicit rather than implicit schedule. In this section, we explore how this property affects network training dynamics, demonstrating how implicit learning rate schedules due to parameter norm growth can be made explicit and be leveraged to improve the performance of NaP in deep RL domains.

4.1 Replicating the dynamics of parameter norm growth in NaP

We begin our study of effective learning rates by illustrating how the implicit learning rate schedule induced by the evolution of the parameter norm can be translated to an explicit schedule in NaP. We study a small CNN described in Appendix B.4 with layer normalization prior to each nonlinearity trained on CIFAR-10 with the usual label set. We train two 'twin' scale-invariant networks with the Adam optimizer in tandem: both networks see the exact same data stream and start from the same initialization, but the per-layer weights of one are projected after every gradient step to have constant norm, while the other is allowed to vary the norms of the weights. We then consider three experimental settings: in the first, we re-scale the per-layer learning rates of the projected network so that the explicit learning rate is equal to the effective learning rate of its twin. In the second, we re-scale the global learning rate based on the ratio of parameter norms between the projected and unprojected network, but do not tune per-layer. In the third, we do no learning rate re-scaling. We see in Figure 2 that the shapes of the learning curves for all networks except for the constant-ELR

[3]We observed in many of our experiments that removing trainable scale and offset parameters often has little effect on network performance. In Rainbow agents, for example, removing the trainable offset parameter even improves performance in several environments.

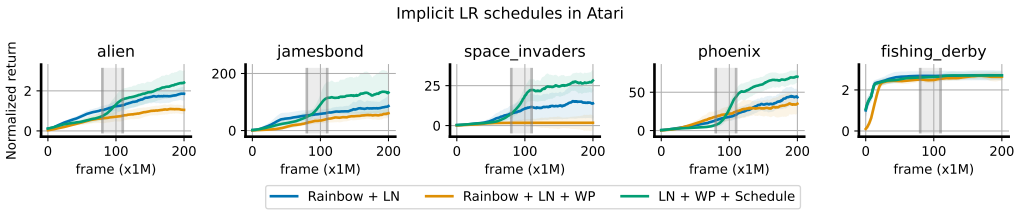


Figure 3: Without an explicit learning rate schedule, a Rainbow trained with NaP may fail to make any performance improvement; while the implicit schedule induced by the parameter norm is clearly important to performance, in several games this is significantly outperformed by a simple linear schedule terminating halfway through training. Intriguingly, we see a characteristic sharp improvement near the end of the decay schedule in several (though not all, e.g. fishing derby) games.

variant are quite similar, with the global learning rate scaling strategy producing a smaller gap than the no-rescaling strategy. By construction, the dynamics of the per-layer rescaling network and its twin are identical. Because global learning rate schedules are standard practice and induce dynamics that are quite close to those obtained by parameter norm growth in Figure 2, we take this approach in the remainder of the paper, leaving layer-specific learning rates and schedules for future work.

4.2 Implicit learning rate schedules in deep RL

When taken to extremes, learning rate decay will eventually prevent the network from making nontrivial learning progress. However, learning rate decay plays an integral role in the training of many modern architectures, and is required to achieve convergence for stochastic training objectives (unless the interpolation applies or Polyak averaging is employed). In this section we will show that, perhaps unsurprisingly, naive application of NaP with a constant effective learning rate can sometimes harm performance in settings where the implicit learning rate schedule induced by parameter norm growth was in fact critical to the optimization process. More surprising is that the domain where this phenomenon is most apparent is one where common wisdom would suggest learning rate decay would be undesirable: deep RL.

RL involves a high degree of nonstationarity. As a result, deep RL algorithms such as DQN and Rainbow often use a constant learning rate schedule. Given that layer normalization has been widely observed to improve performance in value-based RL agents on the arcade learning environment, and that parameter norm tends to increase significantly in these agents, one might at first believe that the performance improvement offered by layer normalization is happening in *spite* of the resulting implicit learning rate decay. A closer look at the literature, however, reveals that several well-known algorithms such as AlphaZero [Schrittwieser et al., 2020], along with many implementations of popular methods such as Proximal Policy Optimization [Schulman et al., 2017], incorporate some form of learning rate decay, suggesting that a constant learning rate is not always desirable. Indeed, Figure 3 shows that constraining the parameter norm to induce a fixed ELR in the Rainbow agent frequently results in *worse* performance compared to unconstrained parameters. This is particularly striking given that many of the benefits supposedly provided by layer normalization, such as better conditioning of the loss landscape [Lyle et al., 2023] and mitigation of overestimation bias [Ball et al., 2023], should be independent of the effective learning rate. Instead, these properties appear to either be irrelevant for optimization or dependent on reductions in the effective learning rate.

We can close this gap by introducing a learning rate schedule (linear decay from the default $6.25 \cdot 10^{-5}$ to 10^{-6}, roughly proportional to the average parameter norm growth across games). We further observe in Appendix C.1 that when we vary the endpoint of the learning rate schedule, we often obtain a corresponding x-axis shift in the learning curves, suggesting that reaching a particular learning rate was necessary to master some aspect of the game. We conclude that, while beneficial, the implicit schedule induced by the parameter norm is not necessarily *optimal* for deep RL agents, and it is possible that a more principled adaptive approach could provide still further improvements.

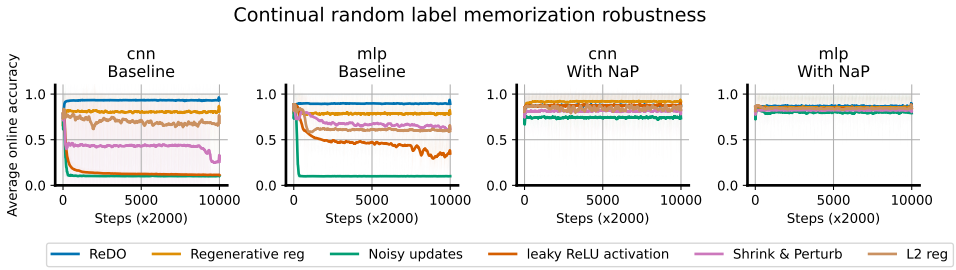


Figure 4: Robustness to nonstationarity: we see that without NaP, there is a wide spread in the effectiveness of various plasticity-preserving methods across two architectures. Once we incorporate NaP, however, the gaps between these methods shrink significantly and almost uniformly improves over the unconstrained baseline.

	CIFAR-10	ImageNet-1k
NaP	**94.64**	77.26
Baseline	**94.65**	77.08
Norm only	94.47	**77.45**

C4	Pile	WikiText	Lambada	SIQA	PIQA
45.7	**47.9**	**45.4**	**56.6**	**44.2**	**68.8**
44.8	47.4	44.2	54.1	43.5	67.3
44.9	47.6	44.3	53.6	43.8	67.1

Table 1: Left: Top-1 prediction accuracy on the test sets of CIFAR-10 and ImageNet-1k. **Right:** per-token accuracy of a 400M transformer model pretrained on the C4 dataset, evaluated on a variety of language benchmarks. See Appendix C.5 for more results with variation measures.

5 Experiments

We now validate the utility of NaP empirically. Our goal in this section is to validate two key properties: first, that NaP does not hurt performance on stationary tasks; second, that NaP can mitigate plasticity loss under a variety of both synthetic and natural nonstationarities.

5.1 Robustness to nonstationarity

We begin with the continual classification problem described in Appendix B.4. We evaluate our approach on a variety of sources of nonstationarity, using two architectures: a small CNN, and a fully-connected MLP (see Appendix B.4. for details). We first evaluate a number of methods designed to maintain plasticity including Regenerative regularization [Kumar et al., 2023], Shrink and Perturb [Ash and Adams, 2020], ReDo [Sokar et al., 2023], along with leaky ReLU units (inspired by the CReLU trick of Abbas et al. [2023]), L2 regularization, and random Gaussian perturbations to the optimizer update, a heuristic form of Langevin Dynamics. We track the average online accuracy over the course of training for 20M steps, equivalent to 200 data relabelings, using a constant learning rate. We find varying degrees of efficacy in these approaches in the base network architectures, with regenerative regularization and ReDO tending to perform the best. When we apply the same suite of methods to networks with NaP, in Figure 4, we observe near-monotonic improvements (with the exception of ReDO, which we conjecture is because the reset method designed for unnormalized networks) in performance and a significant reduction in the gaps between methods, with the performance curves of the different methods nearly indistinguishable in the MLP. Further, we observe constant or increasing slopes in the online accuracy, suggesting that the difference between methods has more to do with their effect on within-task performance than on plasticity loss once the parameter and layer norms have been constrained.

5.2 Stationary supervised benchmarks

Having observed remarkable improvements in synthetic tasks, we now confirm that NaP does not interfere with learning on more widely-studied, natural datasets.

Large-scale image classification. We begin by studying the effect of NaP on two well-established benchmarks: a VGG16-like network [Simonyan and Zisserman, 2014] on CIFAR-10, and a ResNet-50 [He et al., 2016] on the ImageNet-1k dataset. We provide full details in Appendix B.4. In Table 1 we obtain comparable performance in both cases using the same learning rate schedule as the baseline.

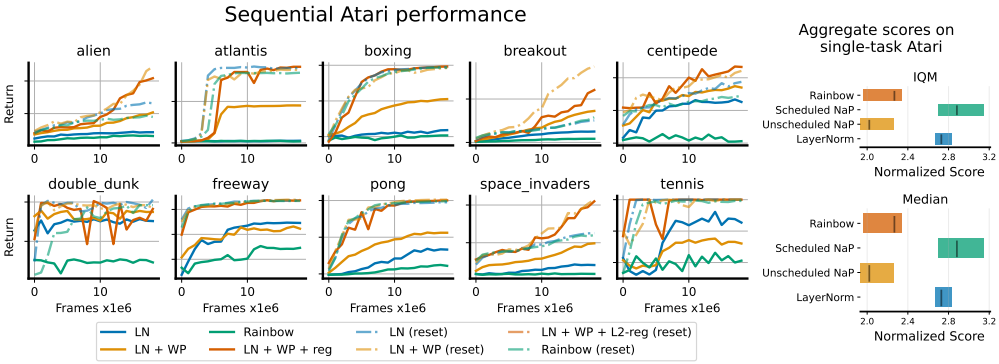


Figure 5: Left: We visualize the learning curves of continual atari agents on sequential ALE training (i.e. 200M frames). Each game is played for 20M frames, and agents pass sequentially from one to another, repeating all ten games twice for a total of 400M training frames. Solid lines indicate performance on the second visit to each game, and dotted lines indicate performance of a randomly initialized network on the game. Even in its second visit to each game, NaP performs comparably the randomly initialized networks, whereas the standard rainbow agent exhibits poor performance on all games in the sequential training regime. **Right:** aggregate effects of normalization on single-task atari, computed via the approach of Agarwal et al. [2021]. Bars indicate 95% confidence intervals over 4 seeds and 57 environments.

Natural language: we now turn our attention to language tasks, training a 400M-parameter transformer architecture (details in Appendix B.3) on the C4 benchmark [Raffel et al., 2020]. Table 1 shows that our approach does not interfere with performance on this task, where we match final performance in terms of training accuracy. When evaluating the pre-trained network on a variety of other datasets, we find that NaP slightly outperforms baselines in terms of performance on a variety of benchmarks, including WikiText-103, Lambada [Paperno et al., 2016], piqa [Bisk et al., 2020], SocialIQA [Sap et al., 2019], and Pile [Gao et al., 2020].

5.3 Deep reinforcement learning

Finally, we evaluate our approach on a setting where maintaining plasticity is critical to performance: RL on the Arcade Learning Environment. We conduct a full sweep over 57 Atari 2600 games comparing the effects of normalization, weight projection, and learning rate schedules on a Rainbow agent [Hessel et al., 2018]. In the RHS of Figure 5 we plot the spread of scores, along with estimates of the Mean and IQM of four agents: standard Rainbow, Rainbow + LayerNorm, Rainbow + NaP without an explicit LR schedule, and Rainbow + NaP with the LR schedule described in Section 4.2. We find that NaP with a linear schedule outperforms the other methods.

We also consider the sequential setting of Abbas et al. [2023]. In this case, we consider an idealized setup where we reset the optimizer state and schedule every time the environment changes, using a cosine schedule with warmup described in Appendix B.2. To evaluate NaP on this regime, we train on each of 10 games for 20M frames, going through this cycle twice. We do not reset parameters of the continual agents between games, but do reset the optimizer. We plot learning curves for the second round of games in the LHS of Figure 5, finding that NaP significantly outperforms a baseline Rainbow agent with and without layer normalization. Indeed, even after 200M steps the networks trained with NaP make similar learning progress to a random initialization.

6 Discussion

This paper has shown that loss of plasticity can be substantially mitigated by constraining the norms of the network's layers and parameters. This finding was made possible by two key insights: first, that in addition to the obvious benefits relating to maintaining constant (pre-)activation norms, layer normalization can also protect against saturated units, and second, that it introduces a correspondence between the parameter norm and the effective learning rate which has significant consequences on performance. We proposed NaP as a means of making the effective learning rate explicit in the optimization process, and leveraged this to gain new insights into the importance of learning rate decay schedules in deep reinforcement learning. In particular, we showed that Rainbow agents trained

on the Arcade Learning Environment in fact under-perform their unconstrained counterparts when a constant effective learning rate is enforced. Provided a suitable decay schedule is used, however, a network's performance and robustness to nonstationarity can both be improved by using NaP. Our analysis suggests a number of exciting directions for future work; in particular, the use of adaptive learning rate schedules in reinforcement learning, and the possibility of tuning not only the global learning rate but also per-layer learning rates in networks which use normalization layers.

Broader Impact

This work concerns basic properties of optimization of neural networks. We do not anticipate any broader societal impacts as a direct consequence of our findings.

References

Zaheer Abbas, Rosie Zhao, Joseph Modayil, Adam White, and Marlos C Machado. Loss of plasticity in continual deep reinforcement learning. *ICML*, 2023.

Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron C Courville, and Marc Bellemare. Deep reinforcement learning at the edge of the statistical precipice. *Advances in neural information processing systems*, 34:29304–29320, 2021.

Atish Agarwala, Fabian Pedregosa, and Jeffrey Pennington. Second-order regression models exhibit progressive sharpening to the edge of stability. *arXiv preprint arXiv:2210.04860*, 2022.

Sanjeev Arora, Zhiyuan Li, and Kaifeng Lyu. Theoretical analysis of auto rate-tuning by batch normalization. In *International Conference on Learning Representations*, 2018.

Devansh Arpit, Yingbo Zhou, Bhargava Kota, and Venu Govindaraju. Normalization propagation: A parametric technique for removing internal covariate shift in deep networks. In Maria Florina Balcan and Kilian Q. Weinberger, editors, *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1168–1176, New York, New York, USA, 20–22 Jun 2016. PMLR.

Jordan Ash and Ryan P Adams. On warm-starting neural network training. *Advances in Neural Information Processing Systems*, 33:3884–3894, 2020.

Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

David Balduzzi, Marcus Frean, Lennox Leary, JP Lewis, Kurt Wan-Duo Ma, and Brian McWilliams. The shattered gradients problem: If resnets are the answer, then what is the question? In *International Conference on Machine Learning*, pages 342–350. PMLR, 2017.

Philip J. Ball, Laura Smith, Ilya Kostrikov, and Sergey Levine. Efficient online reinforcement learning with offline data. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 1577–1594. PMLR, 23–29 Jul 2023.

David GT Barrett and Benoit Dherin. Implicit gradient regularization. *arXiv preprint arXiv:2009.11162*, 2020.

Marc G Bellemare, Yavar Naddaf, Joel Veness, and Michael Bowling. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47: 253–279, 2013.

Tudor Berariu, Wojciech Czarnecki, Soham De, Jorg Bornschein, Samuel Smith, Razvan Pascanu, and Claudia Clopath. A study on the plasticity of neural networks. *arXiv preprint arXiv:2106.00042*, 2021.

Yonatan Bisk, Rowan Zellers, Jianfeng Gao, Yejin Choi, et al. Piqa: Reasoning about physical commonsense in natural language. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 7432–7439, 2020.

A Derivations

A.1 Notation

Analysis of a network's training dynamics depends on characterizing the evolution of (pre-)activations and gradients in the forward and backward passes respectively. We lay out basic notation for fully-connected layers first, and then note additional details which must be considered for convolutional, skip connection, and attention layers. We will write the parameters of a layer as θ_l, and use f to denote a neural network.

Fully-connected layers: We write the forward pass through a network $f : \mathbb{R}^{d_0} \to \mathbb{R}^{d_L}$ as a composition of layer-wise computations $f^l : \mathbb{R}^{d_{l-1}} \to \mathbb{R}^{d_l}$ of the form:

$$a^l = f^l(a^{l-1}) = \phi(\sigma^l f_{\text{LN}}(h^l) + \mu^l), \quad h^l = W^l a^{l-1} \; W^l = \theta_l \tag{2}$$

where ϕ denotes a nonlinearity and f_{LN} is a (possibly absent) normalization operator. We let a^0 denote the network inputs. We will refer to a forward pass through a subset of a network with the notation $f^{l_1:l_2} = f^{l_2} \circ \cdots \circ f^{l_1}$, and use interchangeably $f = f^{1:L}$. We will refer to the full set of network parameters by $\theta = \text{vec}(\{\theta^l\}_{l=1}^L)$.

Convolutional layers: since a convolution can be viewed as a parameterization of a matrix with a particular symmetry, we express these layers identically to the fully-connected layers, with a change in semantics such that W^l is the matrix representation of the convolutional parameters θ_l, where we write the embedding of θ_l into a matrix as $W_{\text{conv}}(\theta_l)$. For simplicity, we ignore the choice of padding.

$$\phi(\sigma^l f_{\text{LN}}(h^l) + \mu^l), \quad h^l = W^l a^{l-1} \text{ and } W^l = W_{\text{conv}}(\theta_l) \tag{3}$$

Skip-connect layers: in some network architectures, a nonlinearity is placed on the outputs of two subnetworks to produce a function of the form

$$f^l = \phi\left(\sum_{l_i \in L} a_{l_i}\right) \quad \text{or} \quad f^l = \phi\left(f_{\text{LN}}\left(\sum_{l_i \in L} a_{l_i}\right)\right) \tag{4}$$

for some index set L. The choice of whether to apply normalization to the sum of the subnetwork outputs or to each output individually depends on the desired signal propagation properties of the network [De and Smith, 2020].

A.2 Derivation of Proposition 1

The result described in proposition 1 follows straightforwardly from the chain rule. For pre-activation RMSNorm we have

$$\frac{d}{dh_i}\phi(f_{\text{RMS}}(h))_j = \phi'(f_{\text{RMS}}(h))_j \frac{d}{dh_i} f_{\text{RMS}}(h)_j \tag{5}$$

$$\frac{d}{dh_i} f_{\text{RMS}}(h)_j = \frac{d}{dh_i} \frac{h_j}{\left(\sum h_k^2\right)^{1/2}} \tag{6}$$

$$= -\frac{1}{2} \frac{h_j}{\left(\sum h_k^2\right)^{3/2}} \frac{d}{dh_i} \sum h_k^2 \tag{7}$$

$$= -\frac{h_j}{\left(\sum h_k^2\right)^{3/2}} h_i \tag{8}$$

$$\frac{d}{dh_i}\phi(f_{\text{RMS}}(h))_j = -\frac{\phi'(f_{\text{RMS}}(h))_j h_j h_i}{\|h\|^3} \tag{9}$$

A.3 Gradients and signal propagation

One perspective which can provide some additional insight into our approach is to consider the following decomposition of the gradient being backpropagated through a layer.

$$\nabla_{W_k}\mathcal{L}(\theta) = \partial_{a_k}\mathcal{L}(\theta) \cdot \partial_{W_k}\phi(f_{\text{LN}}(W_k a_{k-1})) \tag{10}$$

$$= \partial_{a_k}\mathcal{L}(\theta) \cdot D_{\dot{\phi}} \nabla_{h_k} f_{\text{LN}}(h_k)) a_{k-1}^{\top} \tag{11}$$

With this decomposition, we obtain the following interpretation of some common pathologies.

Saturated nonlinearities: a saturated nonlinearity implies that $D_{\dot{\phi}}$, and the gradient is exactly (in the case of ReLU) or very close to (e.g. tanh) zero. As a result, u_t will be zero for the affected coordinates and the corresponding parameters will remain frozen. NaP addresses this problem by using Layer or RMSNorm prior to any nonlinearity in the network.

Saturated normalization layers: the normalization transformation $x \mapsto \frac{x}{\|x\|}$ is vulnerable to saturation as $\|x\|$ grows, in the sense that for a fixed-norm update u we will have

$$\lim_{\|x\|\to\infty} \frac{x+u}{\|x+u\|} - \frac{x}{\|x\|} = 0 \tag{12}$$

and so the term $\nabla_{h_k} f_{\text{LN}}(h_k))$ will vanish. In this case, while the optimizer will be able to update the parameters, these updates will have a diminishing effect on the network's output.

Vanishing and exploding gradients: divergence and disappearance of the activations a_{k-1} and backpropagated gradients $\partial_{a_k}\mathcal{L}(\theta)$ are well-known pathologies which can make networks untrainable. However, even networks which start training from a well-tuned initialization may still encounter exploding gradients due to parameter norm growth over time [Dohare et al., 2021, Wortsman et al., 2023], or vanishing gradients and activations, such as in the case of saturated (i.e. dead/dormant) ReLU units [Sokar et al., 2023].

A.4 Details of NaP

Guiding principles: in general, the goal of NaP is to avoid dramatic distribution shifts in the pre-activation and parameter norms, and to ensure that the network can perform updates to parameters even if a nonlinearity is saturated. With these in mind, there are two key properties that a network designer should aim to maintain:

1. All **parametric** functions entering a nonlinearity should have a normalization layer that at ensures the gradients of all units' parameters are correlated. If there are no parameters between nonlinearities (as is sometimes the case in e.g. resnets) normalization is not essential.
2. Based on our investigations in Appendix C.4, *L2 normalization* of the pre-activations is crucial to obtain the positive benefits of layer normalization, while centering does not have noticeable effects on the network's robustness to unit saturation. As a result, applying at least RMSNorm is crucial prior to nonlinearities, but the choice of whether or not to incorporate centering is up to the designer's discretion.

Batch normalization layers: by default, we put layer normalization prior to batch normalization if an architecture already incorporates batch normalization prior to a nonlinearity. This preserves the property of batch norm that individual units have mean zero across the batch, which may not be the case if layer normalization is applied after. We also always omit offset parameters if layernorm is succeeded by batchnorm, as these offset parameters will be zeroed out by batchnorm.

Skip-connect layers: provided that layer normalization is applied to the outputs of a linear transformation prior to a nonlinearity, NaP is agnostic to whether normalization is applied prior to or after a residual connection's outputs are added to the output of a layer. In particular, if we have a layer of the form $\phi(a_1 + a_2)$ and a_1 and a_2 are the outputs of some subnetwork of the form $\phi_1(f_{\text{LN}}(h_1))$ and $\phi_2(f_{\text{LN}}(h_2))$ where ϕ_1 and ϕ_2 are (possibly trivial) activation functions, then the relevant parameters will already benefit from Proposition 1 and it is not necessary to add an additional normalization layer prior to the activation ϕ.

Attention layers: unlike linear layers, attention layers do not typically include bias terms. To analogize this common practice in NaP, we omit the offset and mean subtraction components of the LayerNorm transform, obtaining

$$\mathrm{MHA}(W_Q X, W_K X) \mapsto \mathrm{MHA}(\sigma_Q f_{\mathrm{RMS}}(W_Q X), \sigma_K f_{\mathrm{RMS}}(W_K X)) \tag{13}$$

where crucially f_{RMS} is not applied along the token axis as this can lead to leakage of information during training on next-token-prediction objectives. A similar problem also prevents us from using normalization directly on the QK product matrix, along with the observation that the intuition of normalizing vectors so that their dot product is equal to the cosine similarity is lost once the dot products have already occurred [Henry et al., 2020]. Empirically, we find that the scale parameters σ_Q and σ_K don't seem to be strictly necessary for expressivity, and that networks can even form selective attention masks for in-context learning without using these parameters to further saturate the softmax.

A.5 Dynamics of NaP

Weight projection non-interference: NaP incorporates a projection onto the ball of constant norm after each update step. A natural question is whether this projection step might simply be the inverse of the update step, leaving the parameters of the network constant. Fortunately, we note that the normalization layers have the effect of projecting gradients onto a subspace which is orthogonal to the current parameter values, i.e.

$$\left(\nabla_{\mathbf{x}} f_{\mathrm{RMS}}(\mathbf{x})\right)(\mathbf{x}) = \mathbf{0}\,. \tag{14}$$

We also note that except for extreme situations such as Neural Collapse [Papyan et al., 2020], real-world gradient updates are almost never colinear with the parameters, meaning that even without normalization layers this problem would be unlikely.

Another concern that arises from the constraints we place on the weights and features is the possibility that these constraints will limit the network's expressivity. Normalization does remove the ability to distinguish colinear inputs of differing norms, meaning that the inputs $\mathbf{x}$ and $\alpha\mathbf{x}$ will map to the same output for all α; however, since many data preprocessing pipelines already normalize inputs, we argue this is not a significant limitation. Indeed, under a more widely-used notion of expressivity, the number of *activation patterns* [Raghu et al., 2017], NaP does not limit expressivity at all. While straightforward, we provide a formal statement and proof of this claim in Appendix A.7.

Layer normalization and parameter growth: In fact, if we incorporate normalization layers into the network we might expect an even more aggressive decay schedule. Recall that in a scale invariant network, we have $\langle \nabla_\theta f(\theta), \theta \rangle = 0$. Thus we know that the gradient at each time step will be orthogonal to the current parameters. In an idealized setting where we use the update rule $\theta_{t+1} \leftarrow \theta_t + \alpha \frac{\nabla_\theta \ell(\theta_t)}{\|\nabla_\theta \ell(\theta_t)\|}$, this would result in the parameter norm growing at a rate $\Theta(t)$, corresponding to an effectively linear learning rate decay.

A.6 Scale-invariance and layer-wise gradient norms

One benefit of NaP is that, because we normalize layer outputs, we limit the extent to which divergence in the norm of one layer's parameters can propagate to the gradients of other layers. For e.g. linear homogeneous activations such as ReLUs, the gradient of some objective function for some input with respect to the parameters of a particular layer contains a sum of matrix products whose norm will depend multilinearly on the norm of each matrix. In particular, in the simplified setting of a deep linear network where $f(\theta, \mathbf{x}) = \prod W^l \mathbf{x}$, we recall Saxe et al. [2013]

$$\nabla_{W^l} f(\theta; \mathbf{x}) = \left[\prod_{k>l} W^k\right]^\top \mathbf{x}^\top \left[\prod_{k<l} W^k\right]^\top \tag{15}$$

In particular, with $\theta' = W^1, \ldots, cW^k, \ldots, W^L$, for $k \neq l$ we would have

$$\implies \nabla_{W^l} f(\theta'; \mathbf{x}) = c \nabla_{W^l} f(\theta; \mathbf{x}) \tag{16}$$

The situation changes little if we add ReLU nonlinearities to the network. In this case, we use the notation $\mathbf{D}_{\phi_l}(\mathbf{x})$ to denote the diagonal matrix indicating whether $a^l[i] > 0$

$$\nabla_{W^l} f(\theta; \mathbf{x}) = \left[\prod_{k>l} \mathbf{D}_{\phi_k}(\mathbf{x}) W^k\right]^\top \mathbf{x}^\top \left[\prod_{k<l} \mathbf{D}_{\phi_k}(\mathbf{x}) W^k\right]^\top \tag{17}$$

$$\implies \nabla_{W^l} f(\theta'; \mathbf{x}) = c \nabla_{W^l} f(\theta; \mathbf{x}) \tag{18}$$

If we incorporate a normalization layer at the end of the network (for simplicity we consider RMSNorm here, but a similar argument applies to standard LayerNorm), the scale-invariance of the resulting output means that the norms of each layer's gradients are independent of the norms of the other layers' parameters, i.e.

$$\nabla_{W^l} f_{\text{RMS}} \circ f(\theta; \mathbf{x}) = \nabla_{W^l} f_{\text{RMS}} \circ f(\theta'; \mathbf{x}) \text{ whenever } \theta' = (W_1, \ldots, cW_k, \ldots, W_L), k \neq l \tag{19}$$

This property is appealing as it means that growth or decay of the norm of a single layer will not interfere with the dynamics of the others. However, it does mean that a layer's effective learning rate will still be sensitive to scaling, which motivates our use of renormalization. It also does not help to avoid saturated units, motivating our use of layer normalization prior to nonlinearities.

A.7 Expressivity of NaP

Finally, we discuss the effect of normalization and weight projection on a notion of expressivity known as the number of activation patterns [Raghu et al., 2017] exhibited by a neural network. This quantity relates to the complexity of the function class a network can compute, giving the following result the corollary that NaP doesn't interfere with this notion of expressivity.

Proposition 2. *Let f be a fully-connected network with ReLU nonlinearities. Let $\tilde{f}$ be the function computed by f after applying NaP. Then the activation pattern of a particular architecture f and parameter θ be $\mathcal{A}_\theta$, we have*

$$\mathcal{A}_f(\theta, \mathbf{x}) = \mathcal{A}_{\tilde{f}}(N(\theta), \mathbf{x}) \,. \tag{20}$$

Further, the decision boundary $\max_{i \in d_{out}} f_i(\mathbf{x})$ is preserved under NaP.

Proof. We apply an inductive argument on each layer. In particular, when ϕ is a ReLU nonlinearity we have

$$\begin{aligned}
\mathcal{A}(\phi(\mathbf{h})) &= \mathcal{A}(\phi(f_{\text{RMS}}(\mathbf{h}))) \\
\phi(f_{\text{RMS}}(\mathbf{h})) &= \frac{\phi(\mathbf{h})}{\|\mathbf{h}\|} \\
\tilde{f}(\mathbf{x}) &= \frac{1}{\Pi_{l=1}^L \|\mathbf{h}_l(\mathbf{x})\|} f(\mathbf{x})
\end{aligned}$$

which trivially results in identical activation patterns in the normalized and unnormalized networks. It is worth noting that one distinguishing factor from a standard ReLU network is that the resulting scaling factor will be different for each $\mathbf{x}$. Thus while the activation patterns will be the same, the two different inputs $\mathbf{x}$ $\mathbf{y}$ might have different scaling factors, which will be a nonlinear function of the input. NaP networks, even with ReLU activations, thus do not have the property of being piecewise linear. □

A.8 Rescaling scale/offset parameters (linear homogeneous networks)

We observe that for any $c > 0$, letting $\phi(x) = \max(x, 0)$ we have:

$$\begin{aligned}
f_{\text{LN}}(\phi(W\sigma\mathbf{x} + \mu)) &= f_{\text{LN}}(c\phi(W\sigma\mathbf{x} + \mu)) && (21) \\
&= f_{\text{LN}}(\phi(cW(\sigma\mathbf{x} + \mu))) \text{ by homogeneity of ReLU} && (22) \\
&= f_{\text{LN}}(\phi(W(c\sigma\mathbf{x} + c\mu))) && (23)
\end{aligned}$$

Then we obtain analogous effective learning rates, letting

$$g_{c\mu} = \nabla_\mu f(c\sigma, c\mu; \mathbf{x}) \quad g_{c\sigma} = \nabla_\sigma f(c\sigma, c\mu; \mathbf{x}) \tag{24}$$

we then have equivalent updates for $\|\sigma^2 + \mu^2\| = 1$ and $\eta_c = c^2$

$$f_{\mathrm{LN}}((c\sigma + \eta_c g_{c\sigma})\mathbf{x} + c\mu + \eta_c g_{c\mu}) = f_{\mathrm{LN}}((c\sigma + c^2 g_{c\sigma})\mathbf{x} + c\mu + c^2 g_{c\mu}) \tag{25}$$

$$= f_{\mathrm{LN}}((c\sigma + c^2 \frac{1}{c} g_\sigma + c\mu + c^2 \frac{1}{c} g_\mu) \tag{26}$$

$$= f_{\mathrm{LN}}(c((\sigma + g_\sigma)\mathbf{x} + \mu + g_\mu) = f_{\mathrm{LN}}((\sigma + g_\sigma)\mathbf{x} + \mu + g_\mu) \tag{27}$$

Finally, we note that the above also holds if we apply a linear transformation W to the output of the scale-offset transform, since

$$f_{\mathrm{LN}}(W(c\sigma\mathbf{x} + c\mu)) = f_{\mathrm{LN}}(cW(\sigma\mathbf{x} + \mu)) = f_{\mathrm{LN}}(W(\sigma\mathbf{x} + \mu)) \tag{28}$$

and so the effective learning rate scales precisely as we had previously for linear layers, but now with respect to the joint norm of the scale and offset parameters μ, σ.

B Experiment details

B.1 Toy experiment details

We conduct a variety of illustrative experiments on toy problem settings and small networks.

Network: in Figures 1 and 2, we use a DQN-style network which consists of two sets of two convolutional layers with 32 and 64 channels respectively. We then apply max pooling and flatten the output, feeding through a 512-unit hidden linear layer before applying a final linear transform to obtain the output logits. The network uses ReLU nonlinearities. When NaP is applied, we add layer normalization prior to each nonlinearity.

B.2 RL details

Single-task atari: We base our RL experiments off of the publicly available implementation of the Rainbow agent [Hessel et al., 2018] in DQN Zoo [Quan and Ostrovski, 2020]. We follow the default hyperparameters detailed in this codebase. In our implementation, we add normalization layers prior to each nonlinearity except for the final softmax. We train for 200M frames on the Atari 57 suite [Bellemare et al., 2013]. We also allow for a learning rate schedule, which we explicitly detail in cases where non-constant learning rates are used.

Sequential ALE: we use the same rainbow implementation as for the single-task results, using a cosine decay learning rate for all variants. We restart the cosine decay schedule at every task change for all agents. Our cosine decay schedule uses an init value of 10^{-8}, a peak value of the default LR for Rainbow (0.000625), 1000 warmup steps after the optimizer is reset, and end-value equal to 10^{-6} as in the single-task settings. We choose cosine decay due to its popularity in supervised learning, and to highlight the versatility of NaP to different LR schedules. We follow the game sequence used by Abbas et al. [2023], training for 20M frames per game.

B.3 Language details

Sequence memorization: we set a dataset size of 1024 and a sequence length of 512. We use a vocabulary size of 256, equivalent to ASCII tokenization. We use the adam optimizer, and train all networks for a minimum of 10 000 steps. We reset the dataset every 1000 optimizer steps, generating a new set of 1024 random strings of length 512. We use as a baseline a transformer architecture [Vaswani et al., 2017, Raffel et al., 2020] consisting of 4 attention blocks, with 8 heads and d_{model} equal to 256. We use a batch size of 128.

In-context learning: our in-context learning experiments use the same overall setup as the sequence memorization experiments, with identical architectures and baseline optimization algorithm. In this

case we train on a dataset consisting of 4096 randomly generated strings, in which the final 100 tokens are a contiguous subsequence of the first 412, selected uniformly at random from indices in [1, 312].

Natural language: we run our natural language experiments on a 400M parameter transformer architecture based on the same backbone as the previous two tasks, this time consisting of 12 blocks with 12 heads and model dimension 1536. We use the standard practice of learning rate warmup followed by cosine decay, setting a peak learning rate of $2 \cdot 10^{-4}$ which is reached after a linear warmup of 1000 steps. We use a batch size of 128 and train for 30,000 steps. We use a weight decay parameter of 0.1 with the adamw optimizer.

B.4 Vision details

CIFAR-10 Memorization: We consider three classes of network architecture: a fully-connected multilayer perceptron (MLP), for which we default to a width of 512 and depth of 4 in our evaluations; a convolutional network with k convolutional layers followed by two fully connected layers, for which we default to depth four, 32 channels, and fully-connected hidden layer width of 256. In all networks, we apply layer normalization before batch normalization if both are used at the same time. By default, we typically do not use batch normalization. We use ReLU activations

Our continual supervised learning domain is constructed from the CIFAR-10 dataset, from which we construct a family of continual classification problems. Each continual classification problem is characterized by a transformation function on the labels. For label transformations, we permute classes (for example, all images with the label 5 will be re-assigned the label 2), and random label assignment, where each input is uniformly at random assigned a new label independent of its class in the underlying classification dataset. Figures in the main paper concern random label assignments, as this is a more challenging task which produces more pronounced effects.

In our figures in the main paper, we run a total of 20M steps and a total of 200 random target resets. All networks are trained using the Adam optimizer. We conducted a sweep over learning rates for the different architectures, settling on 10^{-4} as this provided a reasonable balance between convergence speed and stability in all architectures, to ensure that all networks could at least solve the single-task version of each label and target transformation.

VGGNet and ResNet-50 baselines: we use the standard data augmentation policies for the CIFAR-10 and ImageNet-1k experiments. In our ImageNet-1k experiments, we use the ResNetv2 architecture variant, with a label smoothing parameter of 0.1, weight decay 10^{-4}, and as an optimizer we use SGD with a cosine annealing learning rate schedule, and Nesterov momentum with decay rate 0.9. We use a batch size of 256. Our CIFAR-10 experiments use a VGG-Net architecture. We use the sgd optimizer with a batch size of 32, and set a learning rate schedule which starts at 0.025 and decays by a factor of 0.1 iteratively through training. We use Nesterov momentum with decay rate 0.9. We train for a total of 400 000 steps.

C Additional experimental results

C.1 Arcade learning environment

Our choice of learning rate schedule in the paper is motivated by an attempt to roughly approximate the shape of the implicit schedule obtained by parameter growth on average across games in the suite, with a slightly smaller terminal point than would typically be achieved by the parameter norm alone. We consider learning rate schedules which linearly interpolate between the initial learning rate of 0.000625 and a learning rate of 10^{-6}, which is roughly equivalent to increasing the parameter norm by a factor of 60. We explore the importance of the duration of this decay in Figure 6, where we conduct a sweep over a subset of the full Atari 57 suite (selected by sorting alphabetically and selecting every third environment). We observe that faster decay schedules result in better performance initially, but often plateau at a lower value. Decaying linearly over the entire course of training, in contrast, exhibits slow initial progress but often picks up significantly towards the end of training. We conclude that in many games, reducing the learning rate is necessary for performance improvements in this agent, and the linear decay over the entire training period doesn't give the agent sufficien time to take advantage of the finer-grained updates to its predictions that a lower learning rate affords.

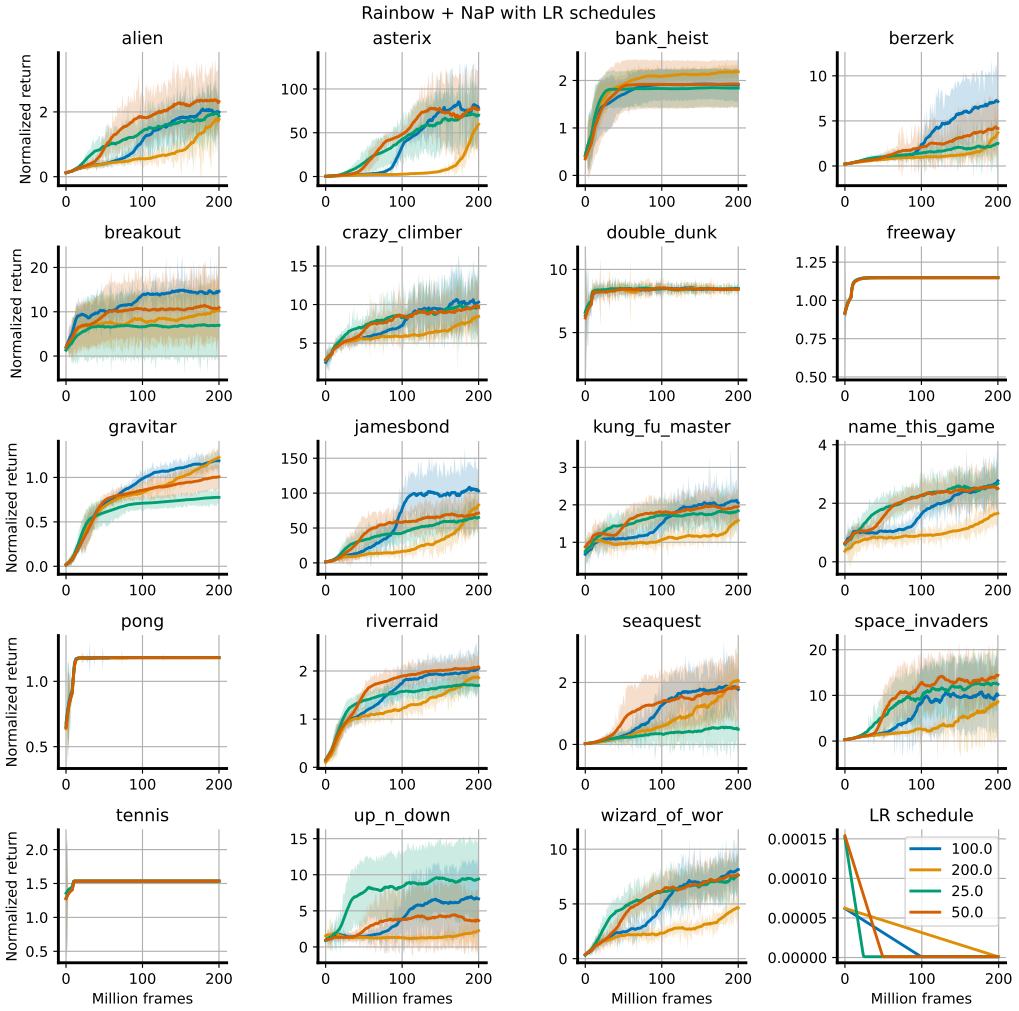


Figure 6: We see that faster LR decays typically accompany fast initial progress followed by plateaus. Terminating the linear schedule halfway through training strikes the best balance of the four settings considered for overall progress.

C.2 Non-stationary MNIST

We include additional network statistics from the experiments shown in Figure 4 in Figure 7.

Linearized units: given a large batch, what fraction of the ReLU units in the penultimate layer are either 0 for all units or nonzero for all units. This gives a slightly more nuanced take on the amount of computation performed in the penultimate layer than the feature rank.

Feature rank: we compute the numerical rank of the penultimate layer features by sampling a batch of data and computing the singular values of the $b \times d$ matrix of $d-$dimensional feature vectors. $\sigma_1, \ldots, \sigma_d$. We then compute the numerical rank as $\sum \mathbb{1}(\sigma_i/\sigma_1 > 0.01)$.

Parameter and gradient norm: these are both computed in the standard way by flattening out the set of parameters / per-parameter gradients and computing the 2-norm of this vector.

C.3 Non-stationary sequence modeling

We find additionally that NaP is capable of improving the robustness of sequence models to nonstationarities, while also not interfering with the formation of in-context learning circuits. We demonstrate

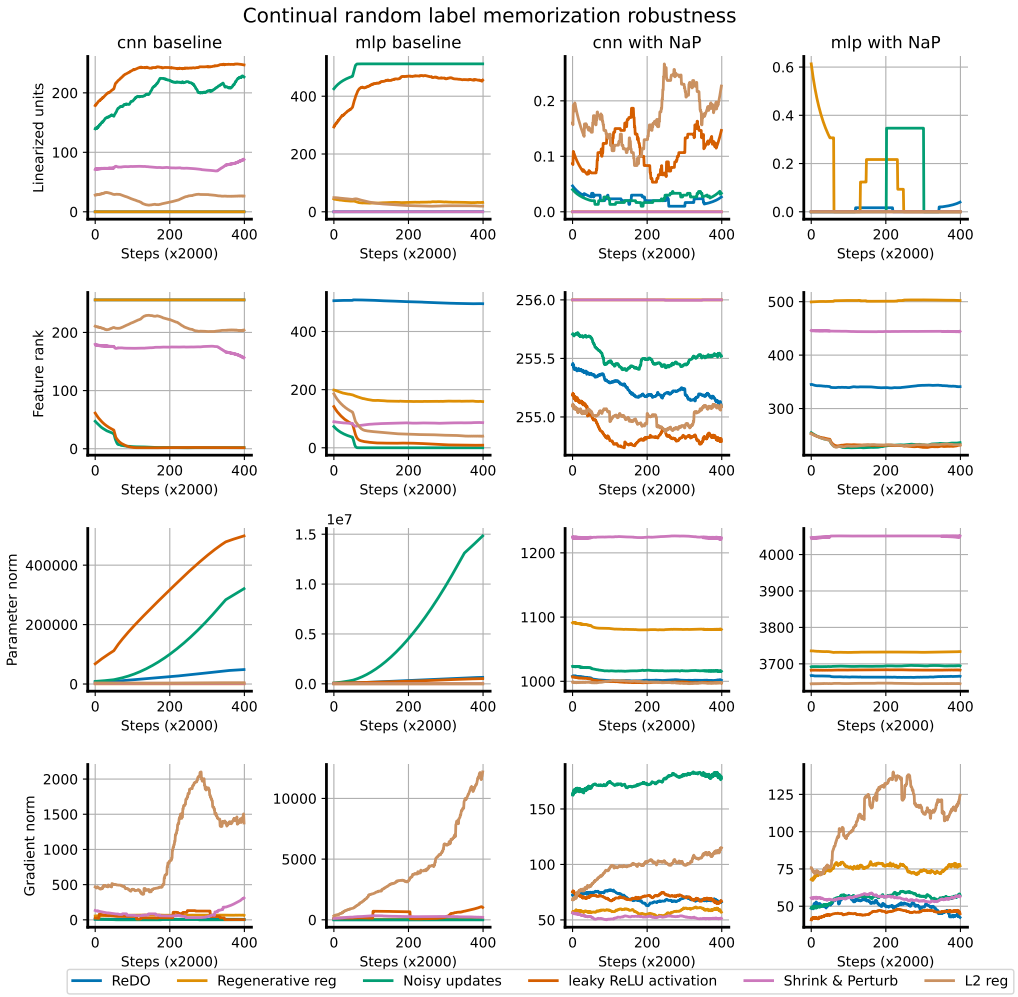


Figure 7: We plot a variety of additional network statistics in the continual CIFAR-10 experiment shown in Figure 4.

the latter point in Figure 8, where we train a small transformer model on a dataset of the form $s_p \oplus s_s$, where s_p is a prefix string of length 100 and s_s is a suffix string of length 50 which is a contiguous subset of the prefix string. We use a fixed dataset, such that in theory the network could memorize all strings, though we use a sufficiently large dataset that this is not possible to achieve within the training budget. We train four protocols using next-token prediction: with and without weight projection, and with and without normalization (the networks are small enough that normalization is not critical for training stability). We evaluate accuracy on the final 48 tokens of s_s as training progresses, and observe that all networks very quickly learn to identify the starting point of the suffix string and copy the relevant subset of the prefix. We observe that normalization accelerates learning of both the retrieval component of the accuracy and the memorization component of the accuracy, and that the weight projection step in fact also accelerates this process.

In Figure 9, we see that normalization and weight projection can also help to improve robustness in a nonstationary random string memorization task, a sequence-modelling analogue of random label memorization in CIFAR-10. We observe that in this case, allowing a learnable scale to evolve unregularized can somewhat slow down learning compared to omitting this parameter, and that weight projection recovers similar dynamics to learning with a large weight decay factor.

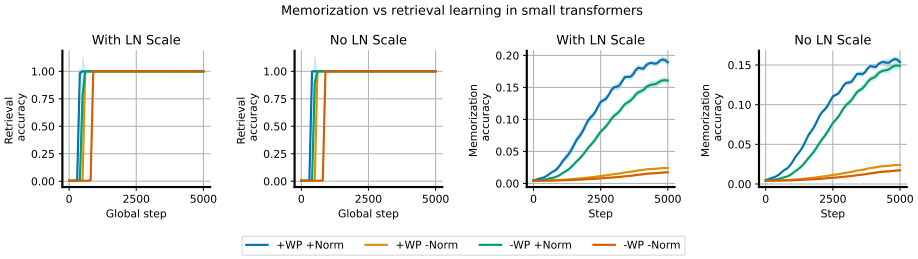


Figure 8: Demonstration that applying normalization and removing the learnable scale parameter does not prevent the network from learning to copy previously-observed subsequences.

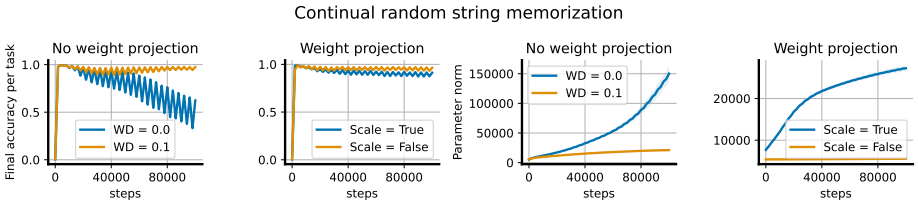


Figure 9: Random string memorization: unregularized networks exhibit plasticity loss when trained to memorize a sequence of random strings, while weight projection and weight decay improve robustness to this nonstationarity.

C.4 ReLU revival experiments

Many previous works have noted that adaptive optimizers are particularly damaging to network plasticity [Dohare et al., 2021, Lyle et al., 2023, Dohare et al., 2023]. The primary mechanism underlying this is due to the sudden distribution shift in gradient moments due to changes in the learning objective – when the gradient norm increases, adaptive optimizers are slow to catch up and can take enormous update steps when this occurs. Layer normalization mitigates this effect due to two facts: first, the gradients of negative preactivations are nonzero, and second, all nonzero gradients are treated essentially the same by adaptive optimizers (up to ϵ). As a result, networks with layer norm can still perform significant updates to parameters feeding into 'dead' units, meaning that these networks have a good chance of turning on again later.

We illustrate this with a simple experimental setting, where we model optimizer updates with isotropic Gaussian-distributed gradient signals and perform a (truncated at zero) random walk. Formally we look at the time evolution of the system:

$$\mathbf{v}_t = \mathbf{v}_{t-1} + \max(\mathbf{v}, \mathbf{0}) \odot \mathbf{z}_t, \quad \mathbf{z}_t \sim \mathcal{N}(0, I) \tag{29}$$

to model the evolution of features under a gradient descent trajectory. To simulate the steps taken by an adaptive optimizer like RMSProp or Adam, where updates to each parameter have fixed norm, we modify this process slightly as follows:

$$\mathbf{v}_t = \mathbf{v}_{t-1} + \text{sign}\left(\max(\mathbf{v}, \mathbf{0}) \odot \mathbf{z}_t\right), \quad \mathbf{z}_t \sim \mathcal{N}(0, I)\ . \tag{30}$$

To simulate layer normalization, we compute the dot product between $\mathbf{z}_t$ and the Jacobian $\nabla_{\mathbf{v}} \frac{\mathbf{v}}{\|\mathbf{v}\|}$ to simulate gradient descent:

$$\mathbf{v}_t = \mathbf{v}_{t-1} + \mathbf{z}_t^\top \nabla_{\mathbf{v}} \max\left(\frac{\mathbf{v}}{\|\mathbf{v}\|}, \mathbf{0}\right), \quad \mathbf{z}_t \sim \mathcal{N}(0, I) \tag{31}$$

and analogously compute the sign of this update to model RMSProp-style optimizers:

$$\mathbf{v}_t = \mathbf{v}_{t-1} + \text{sign}\left(\mathbf{z}_t^\top \nabla_{\mathbf{v}} \max\left(\frac{\mathbf{v}}{\|\mathbf{v}\|}, \mathbf{0}\right)\right), \quad \mathbf{z}_t \sim \mathcal{N}(0, I) \tag{32}$$

In Figure 10 we simulate each of these processes for 1000 steps, and track the number of negative (i.e. 'dead') indices. We observe that layer normalization doesn't avoid dead units in GD, but does reduce

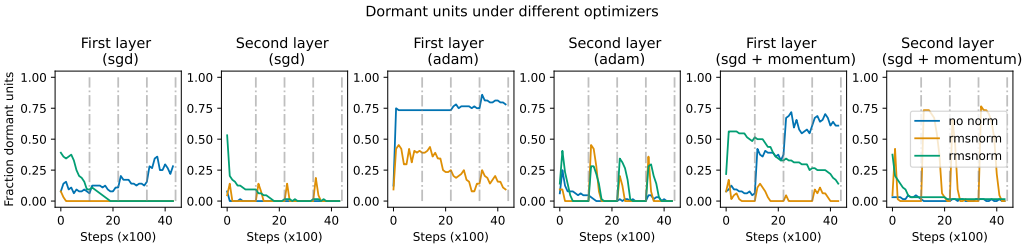


Figure 10: Simple MLP model with dead unit recovery after sudden changes in the classification task. We

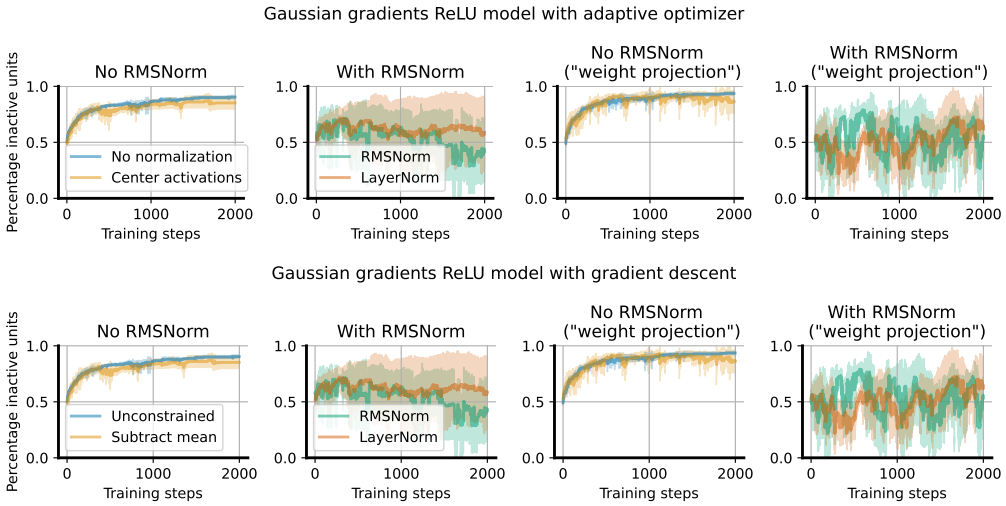


Figure 11: Accumulation of gradients in a random walk model: backpropagated 'gradients' are isotropic random Gaussian vectors and updates are computed by taking the product of these vectors with the layer jacobian. We see that the centering transform actually does relatively little to reduce the risk of dead units, and can in fact produce a 'winner-take-all' effect wherein one large

the rate at which they accumulate. We also observe that layer normalization does avoid monotonic increases in the number of dead units in a model of RMSProp. Intuitively, this makes sense: rather than freezing once they reach a negative value, parameters continue updating once they become negative with equally large steps as they did before, making escape from the dead zone more probable. One visualization of a few trajectories in each setting is shown in Figure 11.

C.5 Stationary supervised benchmarks

We provide the results with standard deviations in Table 2 and Table 3.

	CIFAR-10	**ImageNet-1k**
NaP	**94.64 (0.12)**	77.26 (0.04)
Baseline	**94.65 (0.08)**	77.08 (0.11)
Norm only	94.47 (0.18)	**77.45 (0.08)**

Table 2: Top-1 prediction accuracy on the test sets of CIFAR-10 and ImageNet-1k. Numbers in parentheses are standard deviations.

	C4	Pile	WikiText	Lambada	SIQA	PIQA
NaP	**45.7 (0.0)**	**47.9 (0.1)**	**45.4 (0.1)**	**56.6 (0.4)**	**44.2 (0.2)**	**68.8 (0.7)**
Baseline	44.8 (0.0)	47.4 (0.1)	44.2 (0.0)	54.1 (0.2)	43.5 (0.6)	67.3 (0.2)
Norm only	44.9 (0.0)	47.6 (0.1)	44.3 (0.0)	53.6 (0.3)	43.8 (0.6)	67.1 (0.4)

Table 3: Per-token accuracy of a 400M transformer model pretrained on the C4 dataset, evaluated on a variety of language benchmarks. Numbers in parentheses are standard deviations.

D A note on scale and offset parameters

One loose end from our presentation of NaP is what to do with the learnable scale and offset terms, which are not by default projected and so may drift from their initial values. Most supervised tasks are too short for this drift to present problems, and adding layer-specific regularization or normalization adds additional engineering overhead to an experiment. However, in deep RL or in the synthetic continual tasks we present in Figure 4, this is a real concern. In the case of homogeneous activations such as ReLU, the scale and offset parameters can be viewed identically to the weight and bias terms and normalized accordingly, noting that now all that matters is the relative ratio of the scale and the offset. To account for this, we propose to treat the joint set σ, μ as a single parameter to be normalized. This resolves the issues involved with normalizing a parameter to an initial value of zero, and can be shown not to change the network output (see Appendix A.8 for a derivation of this fact). With non-homogeneous nonlinearities, however, this property will not hold, and we suggest in the general case to use mild weight decay towards a the initial values of 1 and 0 for the scale and offset terms respectively. These two approaches can be summarized in the following two update rules:

$$\mathcal{U}_{\text{norm}}(\sigma, \mu) = \frac{(\sigma, \mu)}{\sqrt{\|\sigma\|^2 + \|\mu\|^2}} \quad \text{and} \quad \mathcal{U}^{\alpha}_{\text{decay}}(\sigma, \mu) = (\alpha\sigma + (1-\alpha)\mathbb{1}, \alpha\mu)\,. \tag{33}$$

Most of our evaluations on single tasks do not use any regularization or projection of the scale and offset parameters, but we do include a regularization-based approach in our evaluations on the sequential ALE in Section 5. In general, if it did not appear that the scale/offset drift was causing problems, we did not introduce additional complexity by adding regularization or normalization. Indeed, in many cases a simpler solution was to omit these parameters entirely; for example we observed that removing offsets was beneficial in several, though not all, games in the Arcade Learning Environment.

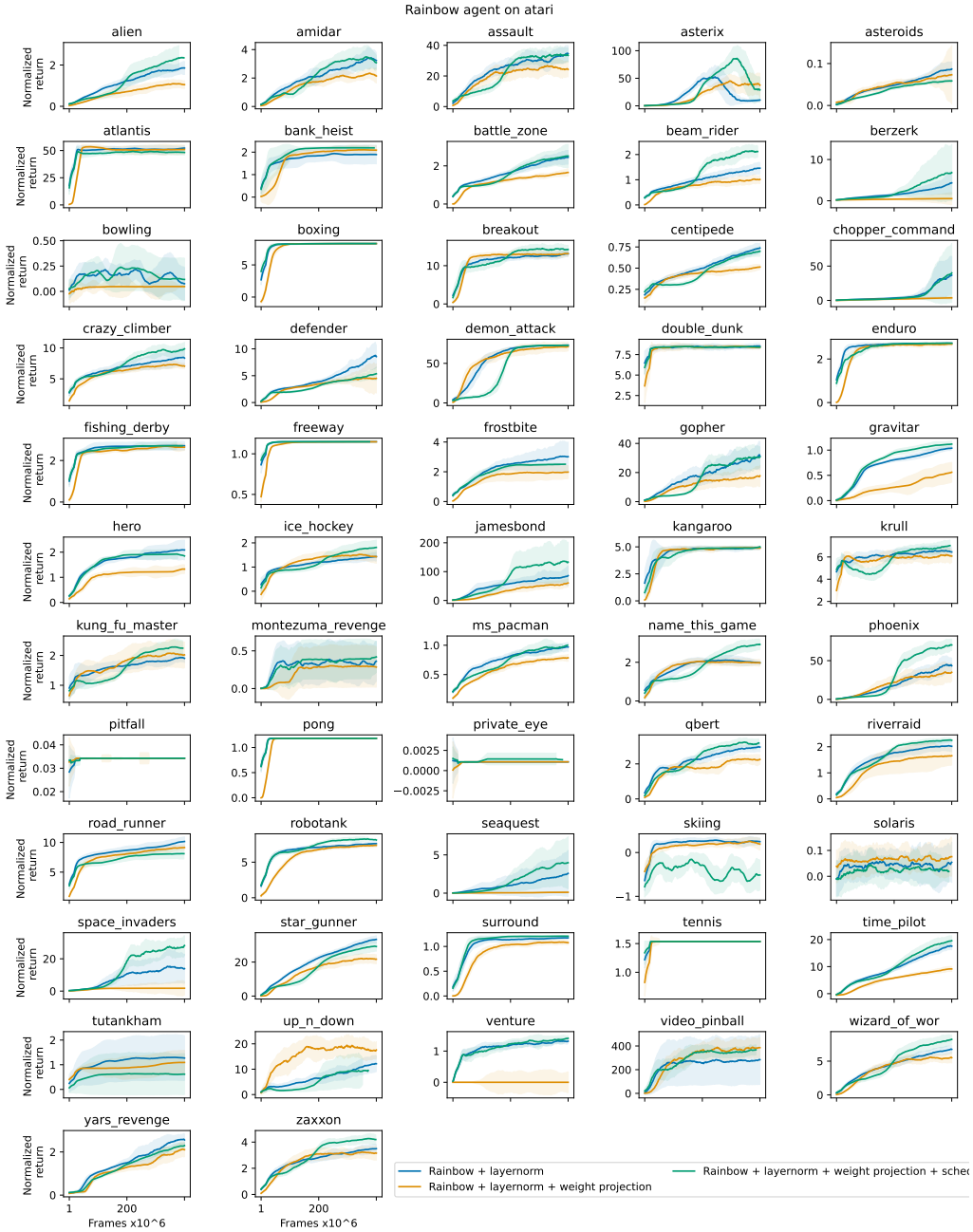


Figure 12: Rainbow agents on atari.

Scaling Off-Policy Reinforcement Learning with Batch and Weight Normalization

Daniel Palenicek [1 2] **Florian Vogt** [3] **Jan Peters** [1 2 4 5]

Abstract

Reinforcement learning has achieved significant milestones, but sample efficiency remains a bottleneck for real-world applications. Recently, CrossQ has demonstrated state-of-the-art sample efficiency with a low update-to-data (UTD) ratio of 1. In this work, we explore CrossQ's scaling behavior with higher UTD ratios. We identify challenges in the training dynamics, which are emphasized by higher UTD ratios. To address these, we integrate weight normalization into the CrossQ framework, a solution that stabilizes training, has been shown to prevent potential loss of plasticity and keeps the effective learning rate constant. Our proposed approach reliably scales with increasing UTD ratios, achieving competitive performance across 25 challenging continuous control tasks on the DeepMind Control Suite and Myosuite benchmarks, notably the complex `dog` and `humanoid` environments. This work eliminates the need for drastic interventions, such as network resets, and offers a simple yet robust pathway for improving sample efficiency and scalability in model-free reinforcement learning.

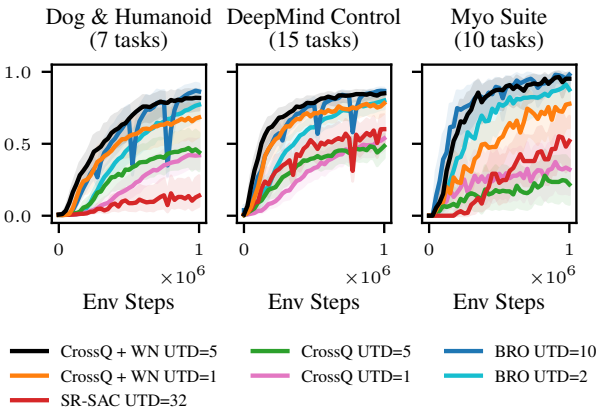


Figure 1. CrossQ + WN UTD=5 is competitive to BRO UTD=10. In comparison, our proposed CrossQ + WN is a simple algorithm that does not require extra exploration policies or full parameter resets. We present results for 25 complex continuous control tasks from the DMC and MyoSuite benchmarking suites. 1.0 marks the maximum score achievable on the respective benchmarks (DMC *return* up to 1000 / Myosuite up to 100% *success rate*).

1. Introduction

Reinforcement Learning (RL) has shown great successes in recent years, achieving breakthroughs in diverse areas. Despite these advancements, a fundamental challenge that remains in RL is enhancing the sample efficiency of algorithms. Indeed, in real-world applications, such as robotics, collecting large amounts of data can be time-consuming, costly, and sometimes impractical due to physical constraints or safety concerns. Thus, addressing this is crucial to make RL methods more accessible and scalable.

Different approaches have been explored to address the problem of low sample efficiency in RL. Model-based RL, on the one hand, attempts to increase sample efficiency by learning dynamic models that reduce the need for collecting real data, a process often expensive and time-consuming (Sutton, 1990; Janner et al., 2019; Feinberg et al., 2018; Heess et al., 2015). Model-free RL approaches, on the other hand, have explored increasing the number of gradient updates on the available data, referred to as the update-to-data (UTD) ratio (Nikishin et al., 2022; D'Oro et al., 2022), modifying network architectures (Bhatt et al., 2024), or both (Chen et al., 2021; Hiraoka et al., 2021; Hussing et al., 2024; Nauman et al., 2024).

In this work, we build upon CrossQ (Bhatt et al., 2024), a recent model-free RL algorithm that recently showed state-of-the-art sample efficiency on the `MuJoCo` (Todorov et al., 2012) continuous control benchmarking tasks. Notably, the authors achieved this by carefully utilizing Batch Normal-

[1]Intelligent Autonomous Systems Group, Technical University of Darmstadt, Germany [2]hessian.AI, Darmstadt, Germany [3]Department of Computer Science, University of Freiburg, Germany [4]German Research Center for Artificial Intelligence (DFKI) [5]Robotics Institute Germany (RIG). Correspondence to: Daniel Palenicek <daniel.palenicek@tu-darmstadt.de>.

Preprint. Under review.

ization (BN, Ioffe (2015)) within the actor-critic architecture. A technique previously thought not to work in RL, as famously reported by Hiraoka et al. (2021) and others. The insight that Bhatt et al. (2024) offered is that one needs to carefully consider the different state-action distributions within the Bellman equation and handle them correctly to succeed. This novelty allowed CrossQ at a low UTD of 1 to outperform the then state-of-the-art algorithms that scaled their UTD ratios up to 20. Even though higher UTD ratios are more computationally expensive, they allow for larger policy improvements using the same amount of data.

This naturally raises the question: *How can we extend the sample efficiency benefits of CrossQ and* BN *to the high* UTD *training regime?* Which we address in this manuscript.

Contributions. In this work, we show that the vanilla CrossQ algorithm is brittle to tune on DeepMind Control (DMC) and Myosuite environments and can fail to scale reliably with increased compute. To address these limitations, we propose the addition of weight normalization (WN), which we show to be a simple yet effective enhancement that stabilizes CrossQ. We motivate the combined use of WN and BN on insights from the continual learning and loss of plasticity literature and connections to the effective learning rate. Our experiments show that incorporating WN not only improves the stability of CrossQ but also allows us to scale its UTD, thereby significantly enhancing sample efficiency.

2. Preliminaries

This section briefly outlines the required background knowledge for this paper.

Reinforcement learning. A Markov Decision Process (MDP) (Puterman, 2014) is a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, r, \mu_0, \gamma \rangle$, with state space $\mathcal{S} \subseteq \mathbb{R}^n$, action space $\mathcal{A} \subseteq \mathbb{R}^m$, transition probability $\mathcal{P} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, the reward function $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$, initial state distribution μ_0 and discount factor γ. We define the RL problem according to Sutton & Barto (2018). A policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ is a behavior plan, which maps a state $\boldsymbol{s}$ to a distribution over actions $\boldsymbol{a}$. The discounted cumulative return is defined as

$$\mathcal{R}(\boldsymbol{s}, \boldsymbol{a}) = \textstyle\sum_{t=0}^{\infty} \gamma^t r(\boldsymbol{s}_t, \boldsymbol{a}_t),$$

where $\boldsymbol{s}_0 = \boldsymbol{s}$ and $\boldsymbol{a}_0 = \boldsymbol{a}$. Further, $\boldsymbol{s}_{t+1} \sim \mathcal{P}(\,\cdot\,|\boldsymbol{s}_t, \boldsymbol{a}_t)$ and $\boldsymbol{a}_t \sim \pi(\,\cdot\,|\boldsymbol{s}_t)$. The Q-function of a policy π is the expected discounted return $Q^\pi(\boldsymbol{s}, \boldsymbol{a}) = \mathbb{E}_{\pi,\mathcal{P}}[\mathcal{R}(\boldsymbol{s}, \boldsymbol{a})]$.

The goal of an RL agent is to find an optimal policy π^* that maximizes the expected return from the initial state distribution

$$\pi^* = \arg\max_\pi \mathbb{E}_{\boldsymbol{s}\sim\mu_0}\left[Q^\pi(\boldsymbol{s}, \boldsymbol{a})\right].$$

Soft Actor-Critic (SAC, Haarnoja et al. (2018)) addresses this optimization problem by jointly learning neural network representations for the Q-function and the policy. The policy network is optimized to maximize the Q-values, while the Q-function is optimized to minimize the Bellmann error

$$\mathcal{L} = \mathop{\mathbb{E}}_{\mathcal{D}}\left[\frac{1}{2}\left(Q_\theta(\boldsymbol{s}_t, \boldsymbol{a}_t) - \left(r(\boldsymbol{s}_t, \boldsymbol{a}_t) + \gamma \mathop{\mathbb{E}}_{\mathcal{P}}[V(\boldsymbol{s}_{t+1})]\right)\right)^2\right],$$

where the value function is computed by taking an expectation over the learned Q function

$$V(\boldsymbol{s}_{t+1}) = \mathbb{E}_{\mathcal{P},\pi_\theta}\left[Q_{\bar{\theta}}(\boldsymbol{s}_{t+1}, \boldsymbol{a}_{t+1})\right]. \tag{1}$$

To stabilize the Q-function learning, Haarnoja et al. (2018) found it necessary to use a target Q-network in the computation of the value function instead of the regular Q-network. The target Q-network is structurally equal to the regular Q-network, and its parameters $\bar{\theta}$ are obtained via Polyak Averaging over the learned parameters θ. While this scheme ensures stability during training by explicitly delaying value function updates, it also arguably slows down online learning (Plappert et al., 2018; Kim et al., 2019; Morales, 2020).

Instead of relying on target networks, CrossQ (Bhatt et al., 2024) addresses training stability issues by introducing Batch Normalization (BN, Ioffe (2015)) in its Q-function and achieves substantial improvements in sample and computational efficiency over SAC. A central challenge when using BN in Q networks is distribution mismatch: during training, the Q-function is optimized with samples $\boldsymbol{s}_t, \boldsymbol{a}_t$ from the replay buffer. However, when the Q-function is evaluated to compute the target values (Equation (1)), it receives actions sampled from the current policy $\boldsymbol{a}_{t+1} \sim \pi_\theta(\,\cdot\,|\boldsymbol{s}_{t+1})$. Those samples have no guarantee of lying within the training distribution of the Q-function. BN is known to struggle with out-of-distribution samples, as such, training can become unstable if the distribution mismatch is not correctly accounted for (Bhatt et al., 2024). To deal with this issue, CrossQ removes the separate target Q-function and evaluates both Q values during the critic update in a single forward pass, which causes the BN layers to compute shared statistics over the samples from the replay buffer and the current policy. This scheme effectively tackles distribution mismatch problems, ensuring that all inputs and intermediate activations are effectively forced to lie within the training distribution.

Normalization techniques in RL. Normalization techniques are widely recognized for improving the training of neural networks, as they generally accelerate training and improve generalization (Huang et al., 2023). There are many ways of introducing different types of normalizations into the RL framework. Most commonly, authors have used Layer Normalization (LN) within the network architectures to stabilize training (Hiraoka et al., 2021; Lyle et al., 2024). Recently, CrossQ has been the first algorithm to successfully

use BN layers in RL (Bhatt et al., 2024). The addition of BN leads to substantial gains in sample efficiency. In contrast to LN, however, one needs to carefully consider the different state-action distributions within the critic loss when integrating BN. In a different line of work, Hussing et al. (2024) proposed the integration of unit ball normalization and projected the output features of the penultimate layer onto the unit ball in order to reduce Q-function overestimation.

Increasing update-to-data ratios. Although scaling up the UTD ratio is an intuitive approach to increase the sample efficiency, in practice, it comes with several challenges. Nikishin et al. (2022) demonstrated that overfitting on early training data can inhibit the agent from learning anything later in the training. The authors dub this phenomenon the primacy bias. To address the primacy bias, they suggest to periodically reset the network parameters while retraining the replay buffer. Many works that followed have adapted this intervention (D'Oro et al., 2022; Nauman et al., 2024). While often effective, regularly resetting is a very drastic intervention and by design induces regular drops in performance. Since the agent has to start learning from scratch repeatedly, it is also not very computing efficient. Finally, the exact reasons why parameter resets work well in practice are not yet well understood (Li et al., 2023). Instead of resetting there have also been other types of regularization that allowed practitioners to train stably with high UTD ratios. Janner et al. (2019) generate additional modeled data, by virtually increasing the UTD. In REDQ, Chen et al. (2021) leverage ensembles of Q-functions, while Hiraoka et al. (2021) use dropout and LN to effectively scale to higher UTD ratios.

3. CrossQ fails to scale up stably

Bhatt et al. (2024) demonstrated CrossQ's state-of-the-art sample efficiency on the `MuJoCo` task suite (Todorov et al., 2012), while at the same time also being very computationally efficient. However, on the more extensive DMC and Myosuite task suites, we find that CrossQ requires tuning. We further find that it works on some, but not all, environments stably and reliably.

Mixed performance of CrossQ. Figure 2 shows CrossQ training performance on a subset of DMC tasks. Namely, the `dog-stand`, `dog-trot` and `humanoid-walk`, selected for their varying difficulty levels to demonstrate a wide range of behaviors, including both successful learning and challenges encountered during training. The figure compares a SAC baseline with standard hyperparameters against tuned CrossQ agents with UTD ratios of 1 and 5, where the hyperparameters were identified through a grid search over learning rates and network sizes, as detailed in Table 1. The first row of the figure shows the IQM training

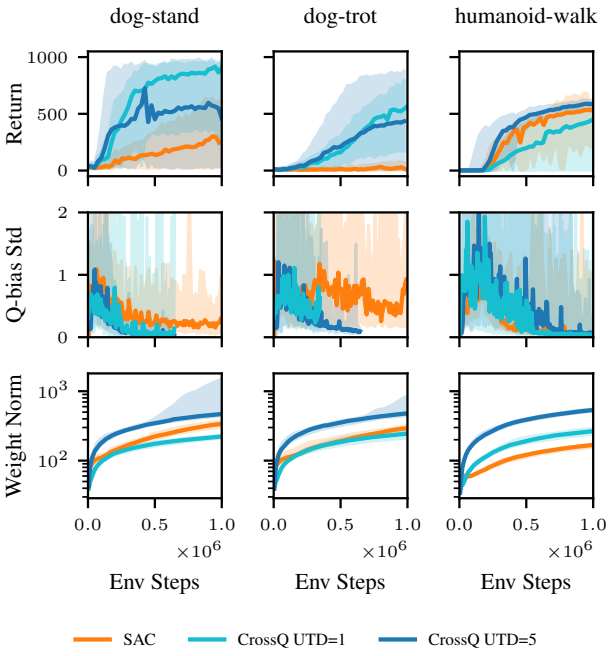


Figure 2. Q-bias and weight norms. CrossQ critic weight norms increase significantly with increasing UTD ratios.

performance and 95% confidence intervals for each agent across 10 seeds. Here, we identify three different training behaviors. On `dog-stand` CrossQ trains stably at UTD= 1, but increasing the UTD to 5 introduces instabilities and decreases performance. On `dog-trot`, both UTD ratios perform very similarly. Finally, on `humanoid-walk`, UTD=5 outperforms UTD=1, although it merely manages to catch up to the SAC baseline in this case. Overall, for all CrossQ runs we notice very large confidence intervals.

Q-function bias analysis. The second row in Figure 2 shows the standard deviation of the normalized Q-function bias. This bias measures how much the Q-function is overestimating or underestimating the true expected return of the current policy.

To compute the normalized Q-function bias, we follow the protocol of Chen et al. (2021). We gather 5 trajectories in the environment using the current policy and use each trajectory's first 350 state-action pairs to calculate the bias. For this, we compare the cumulative discounted rewards for each state-action pair with its Q-value predicted by the Q-function. This bias is then normalized using the cumulative discounted rewards. The mean over these normalized Q-function biases measures the expected bias. Even if the mean is high, as long as the bias is consistent, the selected actions of the policy do not change and, therefore, it is not a problem (Van Hasselt et al., 2016). If the standard deviation

is high, the change in bias is high, which might hinder learning. Thus, following the work of Chen et al. (2021); Bhatt et al. (2024), we focus on the standard deviation.

We find large fluctuations for the Q-function bias standard deviation with all three agents across environments. And even on `dog-stand`, where CrossQ UTD=5 does not learn reliably, it maintains a small Q-function bias standard deviation. From these mixed results, we conclude that we cannot directly link the standard deviation of the Q-function bias to the learning performance. While this does not mean that for larger UTD ratios, the Q-bias would not explode. Rather, it means that, in our case, the Q-bias is not at fault, and the root cause for unstable training lies elsewhere.

Growing network parameter norms. The third row of Figure 2 displays the sum over the L2 norms of the dense layers in the critic network. This includes three dense layers, each with a hidden dimension of 512.

All three baselines exhibit growing network weights over the course of training. We find that the effect is particularly pronounced for CrossQ with increasing UTD ratios. We further find that in the second training phase, the spread of network weight norms increases for the `dog` runs with large confidence intervals. This is visualized by the large spread of the shaded areas, which show the 95% inter percentile ranges for the weight norms.

Growing network weights have been linked to a loss of plasticity, a phenomenon where networks become increasingly resistant to parameter update, which can lead to premature convergence (Elsayed et al., 2024). Additionally, the growing magnitudes pose a challenge for optimization, connected to the issue of growing activations, which has recently been analyzed by Hussing et al. (2024). Further, growing network weights decrease the effective learning rate when the networks contain normalization layers (Van Hasselt et al., 2019; Lyle et al., 2024).

In summary, the scaling results for vanilla CrossQ are mixed. While increasing UTD ratios is known to yield increased sample efficiency, if careful regularization is used (Janner et al., 2019; Chen et al., 2021; Nikishin et al., 2022), CrossQ alone with BN cannot benefit from it. We notice that with increasing UTD ratios, CrossQ's weight layer norms grow significantly faster and overall larger. This observation motivates us to further study the weight norms in CrossQ to increase UTD ratios.

4. Combining batch normalization and weight normalization for scaling up

Inspired by the combined insights of Van Hasselt et al. (2019) and Lyle et al. (2024), we propose to integrate CrossQ with Weight Normalization (WN) as a means of counteracting the rapid growth of weight norms we observe with increasing update-to-data (UTD) ratios.

Our approach is based on the following reasoning: Due to the use of BN in CrossQ, the critic network exhibits scale invariance, as previously noted by Van Laarhoven (2017).

Theorem 4.1 (Van Laarhoven (2017))**.** *Let $f(\boldsymbol{X}; \boldsymbol{w}, b, \gamma, \beta)$ be a function, with inputs $\boldsymbol{X}$ and parameters $\boldsymbol{w}$ and $\boldsymbol{b}$ and γ and β batch normalization parameters. When f is normalized with batch normalization, f becomes scale-invariant with respect to its parameters, i.e.,*

$$f(\boldsymbol{X}; c\boldsymbol{w}, cb, \gamma, \beta) = f(\boldsymbol{X}; \boldsymbol{w}, b, \gamma, \beta),$$

with scaling factor $c > 0$.

Proof. Appendix A

This property allows us to introduce WN as a mechanism to regulate the growth of weight norms in CrossQ without affecting the critic's outputs. Further, it can be shown, that for such a scale invariant function, the gradient scales inversely proportionally to the scaling factor $c > 0$.

Theorem 4.2 (Van Laarhoven (2017))**.** *Let $f(\boldsymbol{X}; \boldsymbol{w}, b, \gamma, \beta)$ be a scale-invariant function. Then, the gradients of f scale inversely proportional to the scaling factor $c \in \mathbb{R}$ of its parameters $\boldsymbol{w}$,*

$$\nabla f(\boldsymbol{X}; c\boldsymbol{w}, cb, \gamma, \beta) = \nabla f(\boldsymbol{X}; \boldsymbol{w}, b, \gamma, \beta)/c.$$

Proof. Appendix B

Recently, Lyle et al. (2024) demonstrated that the combination of LN and WN can help mitigate loss of plasticity. Since the gradient scale is inversely proportional to c, keeping norms constant helps to maintain a stable effective learning rate (ELR,Van Hasselt et al. (2019)), further enhancing training stability.

We conjecture that maintaining a stable ELR could also be beneficial when increasing the UTD ratios in continuous control RL. As the UTD ratio increases, the networks are updated more frequently with each environment interaction. Empirically, we find that the network norms tend to grow quicker with increased UTD ratios (Figure 2), which in turn decreases the ELR even quicker and could be the case for instabilities and low training performance. As such, we empirically investigate the effectiveness of combining CrossQ with WN with increasing UTD ratios.

Implementation details. We apply WN to the first two linear layers, ensuring that their weights remain unit norm after each gradient step by projecting them onto the unit ball, similar to Lyle et al. (2024). Since this constraint effectively stabilizes the ELR in these layers, we found it beneficial to slightly reduce the learning rate from $1e{-}3$ to $1e{-}4$. While

we could employ a learning rate schedule (Lyle et al., 2024) we did not investigate this here. Additionally, we impose weight decay on all parameters that remain unbounded—specifically the final dense output layer. In practice, we use AdamW (Loshchilov, 2017) with a decay of 0 (which falls back to vanilla Adam (Kingma, 2014)) for the normalized intermediate dense layers and $1e{-}2$ otherwise. We chose a maximum UTD of 5 for our experiments due to our computational budget and good strong in sample efficiency.

Target networks. CrossQ famously removed the target networks from the actor-critic framework and showed that using BN training remains stable even without them (Bhatt et al., 2024). While we find this to be true in many cases, we find that especially in DMC, the re-integration of target networks can help stabilize training overall (see Section 5.4). However, not surprisingly, we find that the integration of target networks with BN requires careful consideration of the different state-action distributions between the $\boldsymbol{s}, \boldsymbol{a}$ and $\boldsymbol{s}', \boldsymbol{a}' \sim \pi(\boldsymbol{s}')$ exactly as proposed by Bhatt et al. (2024). To satisfy this, we keep the joined forward pass through both the critic network as well as the target critic network. We evaluate both networks in *training mode*, i.e., they calculate the joined state-action batch statistics on the current batches. As is common, we use Polyak-averaging with a $\tau = 0.005$ from the critic network to the target network.

5. Experiments

To evaluate the effectiveness of our proposed CrossQ + WN method, we conduct a comprehensive set of experiments on the DeepMind Control Suite (Tassa et al., 2018) and MyoSuite (Caggiano et al., 2022) benchmarks. Our primary goal is to investigate the scalability of CrossQ + WN with increasing UTD ratios and to assess the stabilizing effects of combining CrossQ with WN. We compare our approach to several baselines, including the recent BRO (Nauman et al., 2024), CrossQ (Bhatt et al., 2024) and SR-SAC (D'Oro et al., 2022) a version of SAC (Haarnoja et al., 2018) with high UTD ratios and network resets.

5.1. Experimental setup

Our implementation is based on the SAC implementation of `jaxrl` codebase (Kostrikov, 2021). We implement CrossQ following the author's original codebase and add the architectural modifications introduced by (Bhatt et al., 2024), incorporating batch normalization in the actor and critic networks. We extend this approach by introducing WN to regulate the growth of weight norms and prevent loss of plasticity and add target networks. We perform a grid search to focus on learning rate selection and layer width.

We evaluate 25 diverse continuous control tasks, 15 from DMC and 10 from MyoSuite. These tasks vary significantly in complexity, requiring different levels of fine motor control and policy adaptation with high dimensional state spaces up to $\mathbb{R}^{223}$.

Each experiment is run for 1 million environment steps and across 10 random seeds to ensure statistical robustness. We evaluate agents every $25,000$ environment steps for 5 trajectories. As proposed by Agarwal et al. (2021), we report the interquartile mean (IQM) and 95% stratified bootstrap confidence intervals (CIs) of the return, if not otherwise stated.

For the BRO baseline results, for computational reasons, we take the official evaluation data the authors provide. The official BRO codebase is also based on `jaxrl`, and the authors followed the same evaluation protocol, making it a fair comparison.

5.2. Weight normalization allows CrossQ to scale effectively

We provide empirical evidence for our hypothesis that controlling the weight norm and, thereby, the ELR can stabilize training. We show that through the addition of WN, CrossQ + WN shows stable training and can stably scale with increasing UTD ratios.

Figure 3 shows per environment results of our experiments encompassing all 25 tasks evaluated across 10 seeds each. Based on that, Figure 1 shows aggregated performance over all environments from Figure 3 per task suite, with a separate aggregation for the most complex `dog` and `humanoid` environments.

These results show that CrossQ + WN UTD=5 is competitive to the BRO baseline on both DMC and Myosuite, especially on the more complex `dog` and `humanoid` tasks. Notably, CrossQ + WN UTD=5 uses only half the UTD of BRO and does not require any parameter resets and no additional exploration policy. Further, it uses $\sim 90\%$ fewer network parameters—BRO reports $\sim 5M$, while our proposed CrossQ + WN uses only $\sim 600k$ (these numbers vary slightly per environment, depending on the state and action dimensionalities).

In contrast, vanilla CrossQ UTD=1 exhibits much slower learning on most tasks and, in some environments, fails to learn performant policies. Moreover, the instability of vanilla CrossQ at UTD=5 is particularly notable, as it does not reliably converge across environments.

These findings highlight the necessity of incorporating additional normalization techniques to sustain effective training at higher UTD ratios. This leads us to conclude that CrossQ benefits from the addition of WN, which results in stable training and scales well with higher UTD ratios. The resulting algorithm can match or outperform state-of-the-art

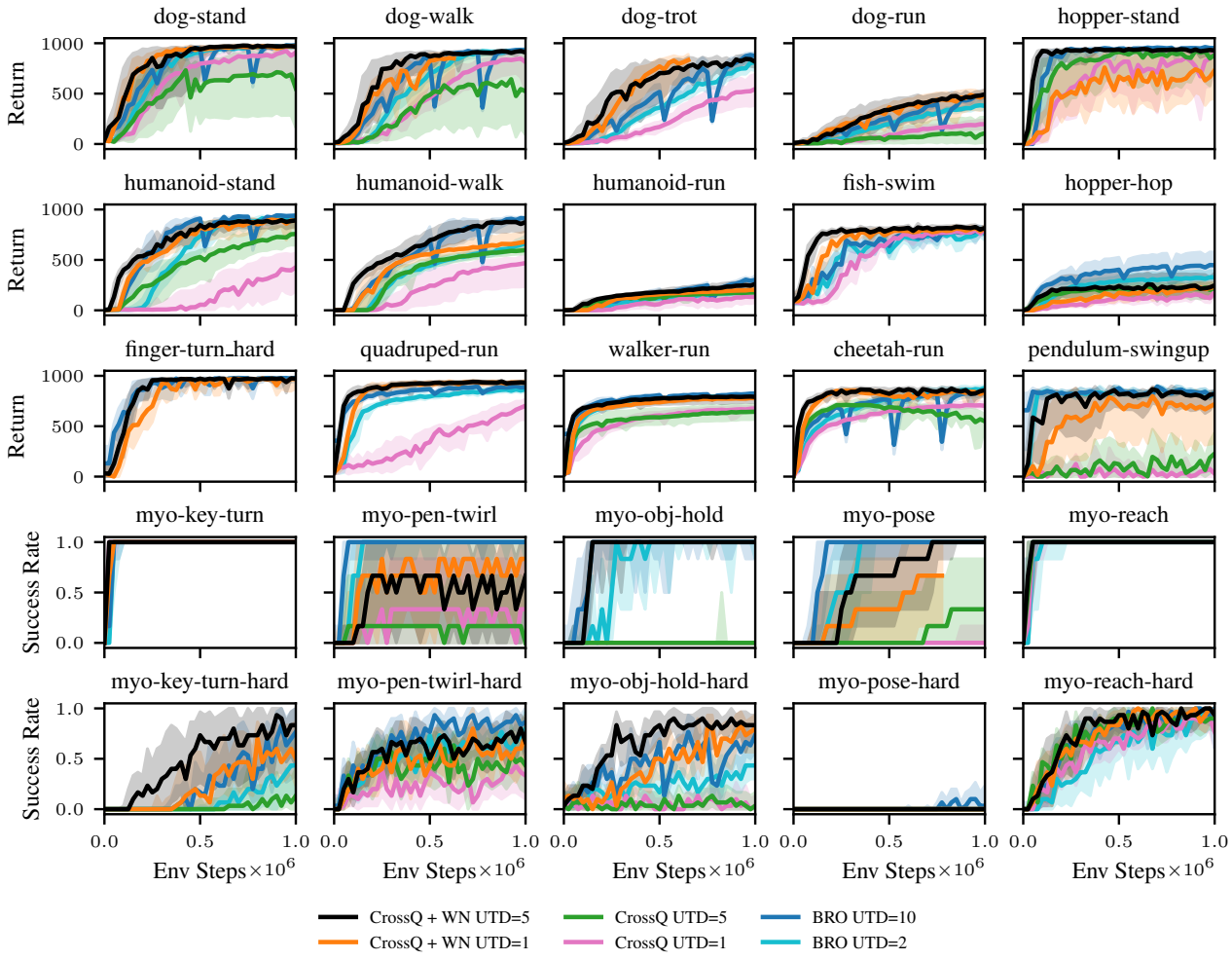


Figure 3. CrossQ WN + UTD=5 *against baselines.* We compare our proposed CrossQ + WN UTD=5 against two baselines, BRO (Nauman et al., 2024) and SR-SAC UTD=32. Results are reported on all 15 DMC and 10 Myosuite tasks. We plot the IQM and 95% CIs over 10 random random seeds. Our proposed approach proves competitive to BRO and outperforms the CrossQ baseline. We want to note that our approach achieves this performance without requiring any parameter resetting or additional exploration policies.

baselines on the continuous control DMC and Myosuite benchmarks while being much simpler algorithmically.

5.3. Stable scaling of CrossQ + WN with UTD ratios

To visualize the stable scaling behavior of CrossQ + WN we ablate across three different UTD ratios $\in \{1, 2, 5\}$. Figure 4 shows training curves aggregated over all 15 DMC tasks. As expected, CrossQ + WN shows reliable scaling behavior, with the learning curves ordered in increasing order accordance to their respective UTD ratio.

5.4. Hyperparameter ablation studies

We also ablate the different hyperparameters of CrossQ + WN UTD=5, by changing each one at a time. Figure 5 shows aggregated results of the final performances of each ablation. We will briefly discuss each ablation individually.

Removing weight normalization. Not performing weight normalization results in the biggest drop in performance across all our ablations. This loss is most drastic on the Myosuite tasks and often results in no meaningful learning. Showing that, as hypothesized, the inclusion of WN into the CrossQ framework yields great improvements in terms of sample efficiency and training stability, especially for larger UTD ratios.

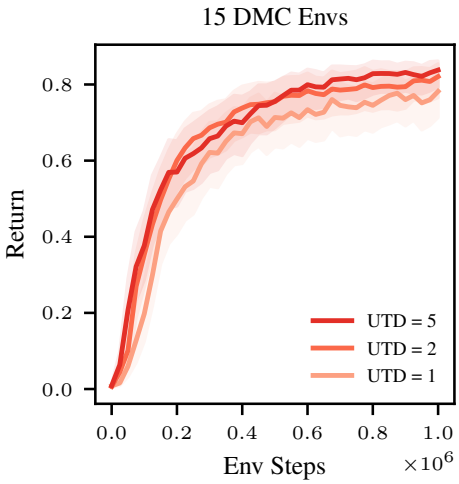


Figure 4. CrossQ WN UTD *scaling behavior.* We plot the IQM return and 95% confidence intervals for different UTD ratios $\in \{1, 2, 5\}$. The results are aggregated over 15 DMC environments and 10 random seeds each according to Agarwal et al. (2021). The sample efficiency scales reliably with increasing UTD ratios.

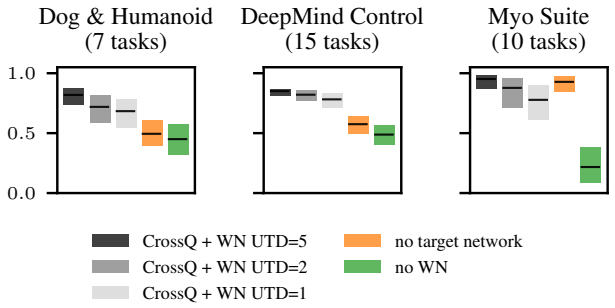


Figure 5. Hyperparameter ablations. We ablate CrossQ + WN with respect to the WN, target networks and different UTD.

Update-to-data ratio. As expected, decreasing the UTD ratio decreases the performance of CrossQ + WN, as demonstrated in the previous section. Aggregated over all DMC environments, this effect is the smallest, since this aggregation includes easier environments as well. Looking at the harder `dog` and `humanoid` tasks, as well as Myosuite, the effect is more pronounced. However, lower UTDs are already reasonably competitive in overall performance.

Target networks. Ablating the target networks shows that on Myosuite, there is no significant difference between using a target network and or no target network. Results on DMC differ significantly. There, removing target networks leads to a significant drop in performance, nearly as large as removing weight normalization. This finding is interesting, as it suggests that CrossQ + WN without target networks is not inherently unstable. But there are situations where the inclusion of target networks is required. Further investigating the role and necessity of target networks in RL is an interesting direction for future research.

6. Related work

RL has demonstrated remarkable success across various domains, yet sample efficiency remains a significant challenge, especially in real-world applications where data collection is expensive or impractical. Various approaches have been explored to address this issue, including model-based RL, UTD ratio scaling, and architectural modifications.

Model-based RL methods enhance sample efficiency by constructing predictive models of the environment to reduce reliance on real data collection (Sutton, 1990; Janner et al., 2019; Feinberg et al., 2018; Heess et al., 2015). However, such methods introduce additional complexity, computational overhead, and potential biases due to model inaccuracies.

Update-to-data ratio scaling. Model-free RL methods, including those utilizing higher UTD ratios, focus on increasing the number of gradient updates per collected sample to maximize learning from available data. High UTD training introduces several challenges, such as overfitting to early training data, a phenomenon known as primacy bias (Nikishin et al., 2022). This can be counteracted by periodically resetting the network parameters (Nikishin et al., 2022; D'Oro et al., 2022). However, network resets introduce abrupt performance drops. Alternative approaches use techniques such as Q-function ensembles (Chen et al., 2021; Hiraoka et al., 2021) and architectural changes (Nauman et al., 2024).

Normalization techniques in RL. Normalization techniques have long been recognized for their impact on neural network training. LN (Ba et al., 2016) and other architectural modifications have been used to stabilize learning in RL (Hiraoka et al., 2021; Nauman et al., 2024). Yet BN has only recently been successfully applied in this context (Bhatt et al., 2024), challenging previous findings, where BN in critics caused training to fail (Hiraoka et al., 2021). WN has been shown to keep ELRs stable and prevent loss of plasticity (Lyle et al., 2024), when combined with LN, making it a promising candidate for integration into existing RL frameworks.

7. Limitations & future work

In this work, we only consider continuous state-action benchmarking tasks. While our proposed CrossQ + WN performs competitively on these tasks, its performance on discrete state-action spaces or vision-based tasks remains unexplored. We plan to investigate this in future work.

8. Conclusion

In this work, we have addressed the instability and scalability limitations of CrossQ in RL by integrating WN. Our empirical results demonstrate that WN effectively stabilizes training and allows CrossQ to scale reliably with higher UTD ratios. The proposed CrossQ + WN approach achieves competitive or superior performance compared to state-of-the-art baselines across a diverse set of 25 complex continuous control tasks from the DMC and Myosuite benchmarks. These tasks include complex and high-dimensional humanoid and dog environments. This extension preserves simplicity while enhancing robustness and scalability by eliminating the need for drastic interventions such as network resets.

Acknowledgments

We wish to thank Joe Watson for helpful discussions and advice during the project. We would also like to thank Tim Schneider, Cristiana de Farias, João Carvalho and Theo Gruner for proofreading and constructive criticism on the manuscript. This research was funded by the research cluster "Third Wave of AI", funded by the excellence program of the Hessian Ministry of Higher Education, Science, Research and the Arts, hessian.AI.

References

Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A. C., and Bellemare, M. Deep reinforcement learning at the edge of the statistical precipice. *Advances in neural information processing systems*, 2021.

Ba, J. L., Kiros, J. R., and Hinton, G. E. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

Bhatt, A., Palenicek, D., Belousov, B., Argus, M., Amiranashvili, A., Brox, T., and Peters, J. CrossQ: Batch normalization in deep reinforcement learning for greater sample efficiency and simplicity. In *International conference on learning representations*, 2024.

Caggiano, V., Wang, H., Durandau, G., Sartori, M., and Kumar, V. Myosuite–a contact-rich simulation suite for musculoskeletal motor control. *arXiv preprint arXiv:2205.13600*, 2022.

Chen, X., Wang, C., Zhou, Z., and Ross, K. Randomized ensembled double Q-learning: Learning fast without a model. In *International conference on learning representations*, 2021.

D'Oro, P., Schwarzer, M., Nikishin, E., Bacon, P.-L., Bellemare, M. G., and Courville, A. Sample-efficient reinforcement learning by breaking the replay ratio barrier. In *International conference on learning representations*, 2022.

Elsayed, M., Lan, Q., Lyle, C., and Mahmood, A. R. Weight clipping for deep continual and reinforcement learning. In *Reinforcement Learning Conference*, 2024.

Feinberg, V., Wan, A., Stoica, I., Jordan, M. I., Gonzalez, J. E., and Levine, S. Model-based value estimation for efficient model-free reinforcement learning. In *International Conference on Machine Learning*, 2018.

Haarnoja, T., Zhou, A., Hartikainen, K., Tucker, G., Ha, S., Tan, J., Kumar, V., Zhu, H., Gupta, A., Abbeel, P., and Levine, S. Soft actor-critic algorithms and applications. *arXiv preprint arXiv:1812.05905*, 2018.

Heess, N., Wayne, G., Silver, D., Lillicrap, T., Erez, T., and Tassa, Y. Learning continuous control policies by stochastic value gradients. In *Advances in neural information processing systems*, 2015.

Hiraoka, T., Imagawa, T., Hashimoto, T., Onishi, T., and Tsuruoka, Y. Dropout q-functions for doubly efficient reinforcement learning. In *International conference on learning representations*, 2021.

Huang, L., Qin, J., Zhou, Y., Zhu, F., Liu, L., and Shao, L. Normalization techniques in training dnns: Methodology, analysis and application. *IEEE transactions on pattern analysis and machine intelligence*, 2023.

Hussing, M., Voelcker, C., Gilitschenski, I., Farahmand, A.-m., and Eaton, E. Dissecting deep rl with high update ratios: Combatting value overestimation and divergence. *arXiv preprint arXiv:2403.05996*, 2024.

Ioffe, S. Batch normalization: Accelerating deep network training by reducing internal covariate shift. *arXiv preprint arXiv:1502.03167*, 2015.

Janner, M., Fu, J., Zhang, M., and Levine, S. When to trust your model: Model-based policy optimization. In *Advances in neural information processing systems*, 2019.

Kim, S., Asadi, K., Littman, M., and Konidaris, G. Deepmellow: Removing the need for a target network in deep q-learning. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence,*

A. Proof Scale Invariance

Proof of Theorem 4.1.

$$\begin{aligned} f(\boldsymbol{X}; c\boldsymbol{w}, cb, \gamma, \beta) &= \frac{g(c\boldsymbol{X}\boldsymbol{w} + cb) - \mu(g(c\boldsymbol{X}\boldsymbol{w} + cb))}{\sigma(g(c\boldsymbol{X}\boldsymbol{w} + cb))}\gamma + \beta \\ &= \frac{cg(\boldsymbol{X}\boldsymbol{w} + b) - c\mu(g(\boldsymbol{X}\boldsymbol{w} + b))}{|c|\sigma(g(\boldsymbol{X}\boldsymbol{w} + b))}\gamma + \beta \\ &= \frac{g(\boldsymbol{X}\boldsymbol{w} + b) - \mu(g(\boldsymbol{X}\boldsymbol{w} + b))}{\sigma(g(\boldsymbol{X}\boldsymbol{w} + b))}\gamma + \beta = f(\boldsymbol{X}; \boldsymbol{w}, b, \gamma, \beta) \end{aligned}$$

B. Proof Inverse Proportional Gradients

To show that the gradients scale inversely proportional to the parameter norm, we can first write

$$\begin{aligned} f(\boldsymbol{X}; c\boldsymbol{w}, cb, \gamma, \beta) &= \frac{g(c\boldsymbol{X}\boldsymbol{w} + cb) - \mu(g(c\boldsymbol{X}\boldsymbol{w} + cb))}{\sigma(g(c\boldsymbol{X}\boldsymbol{w} + cb))}\gamma + \beta \\ &= \frac{g(c\boldsymbol{X}\boldsymbol{w} + cb)}{\sigma(g(c\boldsymbol{X}\boldsymbol{w} + cb))}\gamma - \frac{\mu(g(c\boldsymbol{X}\boldsymbol{w} + cb))}{\sigma(g(c\boldsymbol{X}\boldsymbol{w} + cb))}\gamma + \beta. \end{aligned}$$

As the gradient of the weights is not backpropagated through the mean and standard deviation, we have

$$\nabla_w f(\boldsymbol{X}; c\boldsymbol{w}, cb, \gamma, \beta) = \frac{g'(c\boldsymbol{X}\boldsymbol{w} + cb)X}{|c|\sigma(g(\boldsymbol{X}\boldsymbol{w} + b))}\gamma.$$

The gradient of the bias can be computed analogously

$$\nabla_b f(\boldsymbol{X}; c\boldsymbol{w}, cb, \gamma, \beta) = \frac{g'(c\boldsymbol{X}\boldsymbol{w} + cb)}{|c|\sigma(g(\boldsymbol{X}\boldsymbol{w} + b))}\gamma.$$

C. Hyperparameters

Table 1 gives an overview of the hyperparameters that were used for each algorithm that was considered in this work.

Table 1. Hyperparameters

Hyperparameter	**CrossQ**	**CrossQ + WN**	**SAC**	**SR-SAC**	**BRO**
Critic learning rate	0.0001	0.0001	0.0003	0.0003	0.0003
Critic hidden dim	512	512	256	256	512
Actor learning rate	0.0001	0.0001	0.0003	0.0003	0.0003
Actor hidden dim	256	256	256	256	256
Initial temperature	1.0	1.0	1.0	1.0	1.0
Temperature learning rate	0.0001	0.0001	0.0003	0.0003	0.0003
Target entropy	$\|A\|$	$\|A\|$	$\|A\|$	$\|A\|$	$\|A\|$
Target network momentum	0.005	0.005	0.005	0.005	0.005
UTD	1,5	1,5	1	32	10
Number of critics	2	2	2	2	1
Action repeat	2	2	2	2	1
Discount	0.99 (DMC) 0.95 (Myo)	0.99 (DMC) 0.95 (Myo)	0.99 (DMC) 0.95 (Myo)	0.99 (DMC) 0.95 (Myo)	0.99 (DMC) 0.99 (Myo)
Optimizer	Adam	AdamW	Adam	Adam	AdamW
Optimizer momentum (β_1, β_2)	(0.9, 0.999)	(0.9, 0.999)	(0.9, 0.999)	(0.9, 0.999)	(0.9, 0.999)
Policy delay	3	3	1	1	1
Warmup transitions	5000	5000	10000	10000	10000
AdamW weight decay critic	0.0	0.01	0.0	0.0	0.0001
AdamW weight decay actor	0.0	0.01	0.0	0.0	0.0001
AdamW weight decay temperature	0.0	0.0	0.0	0.0	0.0
Batch Normalization momentum	0.99	0.99	N/A	N/A	N/A
Reset Interval of networks	N/A	N/A	N/A	every 80k steps	*at 15k, 50k, 250k, 500k and 750k steps*
Batch Size	256	256	256	256	128

D. Individual training curves for ablation results

Here, we provide detailed individual training curves for each ablation experiment conducted in our study. Figure 6 shows experiments with no WN, no target network, CrossQ with WN and UTD=1, and CrossQ with WN and UTD=5.

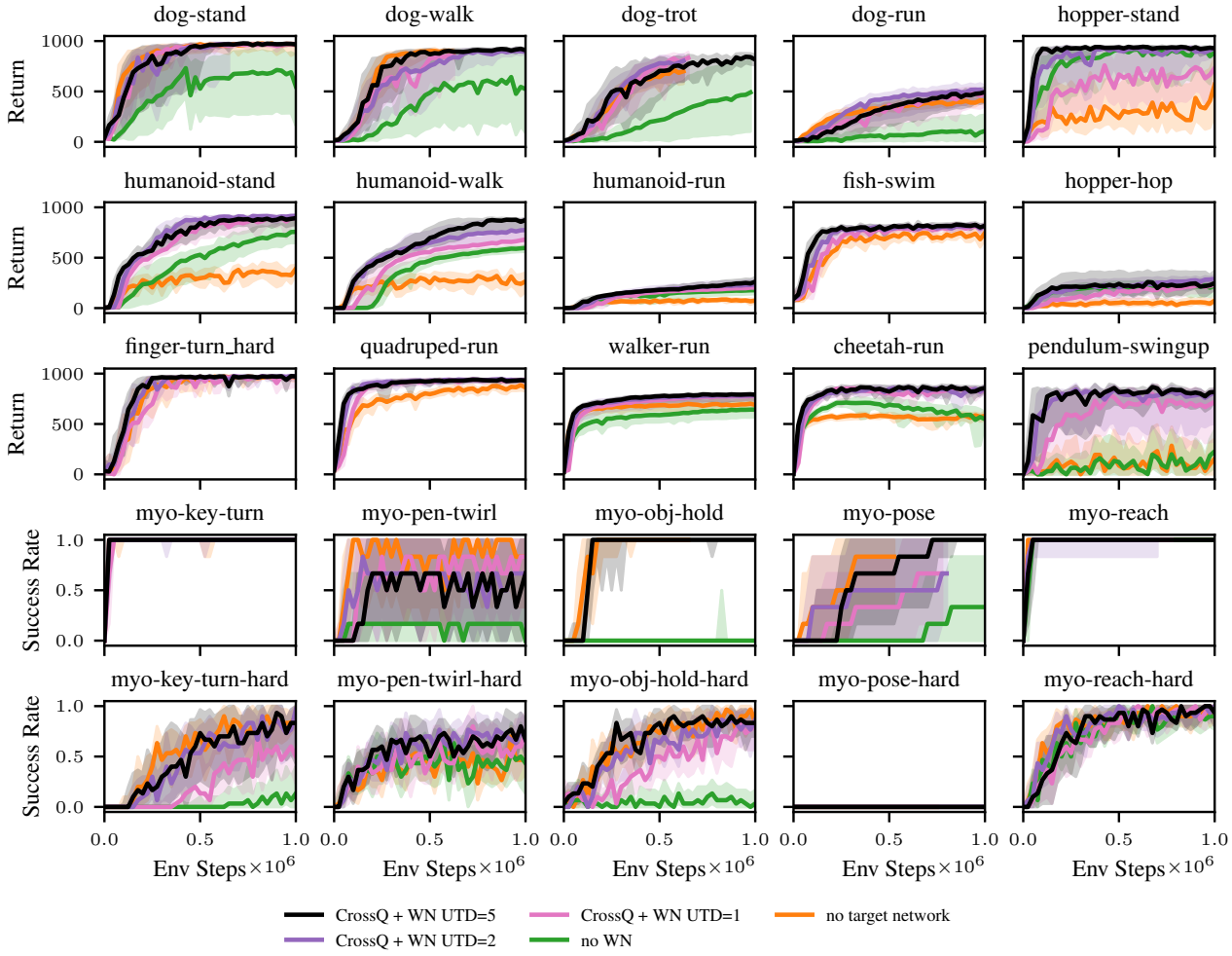


Figure 6. Individual training curves for ablations

On the importance of initialization and momentum in deep learning

Ilya Sutskever[1] ILYASU@GOOGLE.COM
James Martens JMARTENS@CS.TORONTO.EDU
George Dahl GDAHL@CS.TORONTO.EDU
Geoffrey Hinton HINTON@CS.TORONTO.EDU

Abstract

Deep and recurrent neural networks (DNNs and RNNs respectively) are powerful models that were considered to be almost impossible to train using stochastic gradient descent with momentum. In this paper, we show that when stochastic gradient descent with momentum uses a well-designed random initialization and a particular type of slowly increasing schedule for the momentum parameter, it can train both DNNs and RNNs (on datasets with long-term dependencies) to levels of performance that were previously achievable only with Hessian-Free optimization. We find that both the initialization and the momentum are crucial since poorly initialized networks cannot be trained with momentum and well-initialized networks perform markedly worse when the momentum is absent or poorly tuned.

Our success training these models suggests that previous attempts to train deep and recurrent neural networks from random initializations have likely failed due to poor initialization schemes. Furthermore, carefully tuned momentum methods suffice for dealing with the curvature issues in deep and recurrent network training objectives without the need for sophisticated second-order methods.

1. Introduction

Deep and recurrent neural networks (DNNs and RNNs, respectively) are powerful models that achieve high performance on difficult pattern recognition problems in vision, and speech (Krizhevsky et al., 2012; Hinton et al., 2012; Dahl et al., 2012; Graves, 2012).

Although their representational power is appealing, the difficulty of training DNNs has prevented their widepread use until fairly recently. DNNs became the subject of renewed attention following the work of Hinton et al. (2006) who introduced the idea of greedy layerwise pre-training. This approach has since branched into a family of methods (Bengio et al., 2007), all of which train the layers of the DNN in a sequence using an auxiliary objective and then "fine-tune" the entire network with standard optimization methods such as stochastic gradient descent (SGD). More recently, Martens (2010) attracted considerable attention by showing that a type of truncated-Newton method called Hessian-free Optimization (HF) is capable of training DNNs from certain random initializations without the use of pre-training, and can achieve lower errors for the various auto-encoding tasks considered by Hinton & Salakhutdinov (2006).

Recurrent neural networks (RNNs), the temporal analogue of DNNs, are highly expressive sequence models that can model complex sequence relationships. They can be viewed as very deep neural networks that have a "layer" for each time-step with parameter sharing across the layers and, for this reason, they are considered to be even harder to train than DNNs. Recently, Martens & Sutskever (2011) showed that the HF method of Martens (2010) could effectively train RNNs on artificial problems that exhibit very long-range dependencies (Hochreiter & Schmidhuber, 1997). Without resorting to special types of memory units, these problems were considered to be impossibly difficult for first-order optimization methods due to the well known vanishing gradient problem (Bengio et al., 1994). Sutskever et al. (2011) and later Mikolov et al. (2012) then applied HF to train RNNs to perform character-level language modeling and achieved excellent results.

Recently, several results have appeared to challenge the commonly held belief that simpler first-order methods are incapable of learning deep models from random initializations. The work of Glorot & Bengio (2010), Mohamed et al. (2012), and Krizhevsky et al. (2012) reported little difficulty training neural networks with depths up to 8 from certain well-chosen

Proceedings of the 30^{th} International Conference on Machine Learning, Atlanta, Georgia, USA, 2013. JMLR: W&CP volume 28.

[1]Work was done while the author was at the University of Toronto.

random initializations. Notably, Chapelle & Erhan (2011) used the random initialization of Glorot & Bengio (2010) and SGD to train the 11-layer autoencoder of Hinton & Salakhutdinov (2006), and were able to surpass the results reported by Hinton & Salakhutdinov (2006). While these results still fall short of those reported in Martens (2010) for the same tasks, they indicate that learning deep networks is not nearly as hard as was previously believed.

The first contribution of this paper is a much more thorough investigation of the difficulty of training deep and temporal networks than has been previously done. In particular, we study the effectiveness of SGD when combined with well-chosen initialization schemes and various forms of momentum-based acceleration. We show that while a definite performance gap seems to exist between plain SGD and HF on certain deep and temporal learning problems, this gap can be eliminated or nearly eliminated (depending on the problem) by careful use of classical momentum methods or Nesterov's accelerated gradient. In particular, we show how certain carefully designed schedules for the constant of momentum μ, which are inspired by various theoretical convergence-rate theorems (Nesterov, 1983; 2003), produce results that even surpass those reported by Martens (2010) on certain deep-autencoder training tasks. For the long-term dependency RNN tasks examined in Martens & Sutskever (2011), which first appeared in Hochreiter & Schmidhuber (1997), we obtain results that fall just short of those reported in that work, where a considerably more complex approach was used.

Our results are particularly surprising given that momentum and its use within neural network optimization has been studied extensively before, such as in the work of Orr (1996), and it was never found to have such an important role in deep learning. One explanation is that previous theoretical analyses and practical benchmarking focused on local convergence in the stochastic setting, which is more of an estimation problem than an optimization one (Bottou & LeCun, 2004). In deep learning problems this final phase of learning is not nearly as long or important as the initial "transient phase" (Darken & Moody, 1993), where a better argument can be made for the beneficial effects of momentum.

In addition to the inappropriate focus on purely local convergence rates, we believe that the use of poorly designed standard random initializations, such as those in Hinton & Salakhutdinov (2006), and suboptimal meta-parameter schedules (for the momentum constant in particular) has hampered the discovery of the true effectiveness of first-order momentum methods in deep learning. We carefully avoid both of these pitfalls in our experiments and provide a simple to understand and easy to use framework for deep learning that is surprisingly effective and can be naturally combined with techniques such as those in Raiko et al. (2011).

We will also discuss the links between classical momentum and Nesterov's accelerated gradient method (which has been the subject of much recent study in convex optimization theory), arguing that the latter can be viewed as a simple modification of the former which increases stability, and can sometimes provide a distinct improvement in performance we demonstrated in our experiments. We perform a theoretical analysis which makes clear the precise difference in local behavior of these two algorithms. Additionally, we show how HF employs what can be viewed as a type of "momentum" through its use of special initializations to conjugate gradient that are computed from the update at the previous time-step. We use this property to develop a more momentum-like version of HF which combines some of the advantages of both methods to further improve on the results of Martens (2010).

2. Momentum and Nesterov's Accelerated Gradient

The momentum method (Polyak, 1964), which we refer to as classical momentum (CM), is a technique for accelerating gradient descent that accumulates a velocity vector in directions of persistent reduction in the objective across iterations. Given an objective function $f(\theta)$ to be minimized, classical momentum is given by:

$$v_{t+1} = \mu v_t - \varepsilon \nabla f(\theta_t) \quad (1)$$

$$\theta_{t+1} = \theta_t + v_{t+1} \quad (2)$$

where $\varepsilon > 0$ is the learning rate, $\mu \in [0, 1]$ is the momentum coefficient, and $\nabla f(\theta_t)$ is the gradient at θ_t.

Since directions d of low-curvature have, by definition, slower local change in their rate of reduction (i.e., $d^\top \nabla f$), they will tend to persist across iterations and be amplified by CM. Second-order methods also amplify steps in low-curvature directions, but instead of accumulating changes they reweight the update along each eigen-direction of the curvature matrix by the inverse of the associated curvature. And just as second-order methods enjoy improved local convergence rates, Polyak (1964) showed that CM can considerably accelerate convergence to a local minimum, requiring $\sqrt{R}$-times fewer iterations than steepest descent to reach the same level of accuracy, where R is the condition number of the curvature at the minimum and μ is set to $(\sqrt{R} - 1)/(\sqrt{R} + 1)$.

Nesterov's Accelerated Gradient (abbrv. NAG; Nesterov, 1983) has been the subject of much recent attention by the convex optimization community (e.g., Cotter et al., 2011; Lan, 2010). Like momentum, NAG is a first-order optimization method with better convergence rate guarantee than gradient descent in

certain situations. In particular, for general smooth (non-strongly) convex functions and a deterministic gradient, NAG achieves a global convergence rate of $O(1/T^2)$ (versus the $O(1/T)$ of gradient descent), with constant proportional to the Lipschitz coefficient of the derivative and the squared Euclidean distance to the solution. While NAG is not typically thought of as a type of momentum, it indeed turns out to be closely related to classical momentum, differing only in the precise update of the velocity vector v, the significance of which we will discuss in the next sub-section. Specifically, as shown in the appendix, the NAG update may be rewritten as:

$$v_{t+1} = \mu v_t - \varepsilon \nabla f(\theta_t + \mu v_t) \quad (3)$$
$$\theta_{t+1} = \theta_t + v_{t+1} \quad (4)$$

While the classical convergence theories for both methods rely on noiseless gradient estimates (i.e., not stochastic), with some care in practice they are both applicable to the stochastic setting. However, the theory predicts that any advantages in terms of asymptotic local rate of convergence will be lost (Orr, 1996; Wiegerinck et al., 1999), a result also confirmed in experiments (LeCun et al., 1998). For these reasons, interest in momentum methods diminished after they had received substantial attention in the 90's. And because of this apparent incompatibility with stochastic optimization, some authors even discourage using momentum or downplay its potential advantages (LeCun et al., 1998).

However, while local convergence is all that matters in terms of asymptotic convergence rates (and on certain very simple/shallow neural network optimization problems it may even dominate the total learning time), in practice, the "transient phase" of convergence (Darken & Moody, 1993), which occurs before fine local convergence sets in, seems to matter a lot more for optimizing deep neural networks. In this transient phase of learning, directions of reduction in the objective tend to persist across many successive gradient estimates and are not completely swamped by noise.

Although the transient phase of learning is most noticeable in training deep learning models, it is still noticeable in convex objectives. The convergence rate of stochastic gradient descent on smooth convex functions is given by $O(L/T + \sigma/\sqrt{T})$, where σ is the variance in the gradient estimate and L is the Lipshits coefficient of ∇f. In contrast, the convergence rate of an accelerated gradient method of Lan (2010) (which is related to but different from NAG, in that it combines Nesterov style momentum with dual averaging) is $O(L/T^2 + \sigma/\sqrt{T})$. Thus, for convex objectives, momentum-based methods will outperform SGD in the early or transient stages of the optimization where L/T is the dominant term. However, the two methods will be equally effective during the final stages of the optimization where $\sigma/\sqrt{T}$ is the dominant term (i.e., when the optimization problem resembles an estimation one).

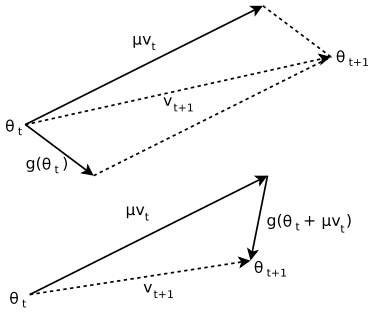


Figure 1. **(Top)** Classical Momentum **(Bottom)** Nesterov Accelerated Gradient

2.1. The Relationship between CM and NAG

From Eqs. 1-4 we see that both CM and NAG compute the new velocity by applying a gradient-based correction to the previous velocity vector (which is decayed), and then add the velocity to θ_t. But while CM computes the gradient update from the current position θ_t, NAG first performs a partial update to θ_t, computing $\theta_t + \mu v_t$, which is similar to θ_{t+1}, but missing the as yet unknown correction. This benign-looking difference seems to allow NAG to change v in a quicker and more responsive way, letting it behave more stably than CM in many situations, especially for higher values of μ.

Indeed, consider the situation where the addition of μv_t results in an immediate undesirable increase in the objective f. The gradient correction to the velocity v_t is computed at position $\theta_t + \mu v_t$ and if μv_t is indeed a poor update, then $\nabla f(\theta_t + \mu v_t)$ will point back towards θ_t more strongly than $\nabla f(\theta_t)$ does, thus providing a larger and more timely correction to v_t than CM. See fig. 1 for a diagram which illustrates this phenomenon geometrically. While each iteration of NAG may only be slightly more effective than CM at correcting a large and inappropriate velocity, this difference in effectiveness may compound as the algorithms iterate. To demonstrate this compounding, we applied both NAG and CM to a two-dimensional oblong quadratic objective, both with the same momentum and learning rate constants (see fig. 2 in the appendix). While the optimization path taken by CM exhibits large oscillations along the high-curvature vertical direction, NAG is able to avoid these oscillations almost entirely, confirming the intuition that it is much more effective than CM at decelerating over the course of multiple iterations, thus making NAG more tolerant of large values of μ compared to CM.

In order to make these intuitions more rigorous and

help quantify precisely the way in which CM and NAG differ, we analyzed the behavior of each method when applied to a positive definite quadratic objective $q(x) = x^\top Ax/2 + b^\top x$. We can think of CM and NAG as operating independently over the different eigendirections of A. NAG operates along any one of these directions equivalently to CM, except with an effective value of μ that is given by $\mu(1 - \lambda\varepsilon)$, where λ is the associated eigenvalue/curvature.

The first step of this argument is to reparameterize $q(x)$ in terms of the coefficients of x under the basis of eigenvectors of A. Note that since $A = U^\top DU$ for a diagonal D and orthonormal U (as A is symmetric), we can reparameterize $q(x)$ by the matrix transform U and optimize $y = Ux$ using the objective $p(y) \equiv q(x) = q(U^\top y) = y^\top U \left(U^\top DU\right) U^\top y/2 + b^\top U^\top y = y^\top Dy/2 + c^\top y$, where $c = Ub$. We can further rewrite p as $p(y) = \sum_{i=1}^n [p]_i([y]_i)$, where $[p]_i(t) = \lambda_i t^2/2 + [c]_i t$ and $\lambda_i > 0$ are the diagonal entries of D (and thus the eigenvalues of A) and correspond to the curvature along the associated eigenvector directions. As shown in the appendix (Proposition 6.1), both CM and NAG, being first-order methods, are "invariant" to these kinds of reparameterizations by orthonormal transformations such as U. Thus when analyzing the behavior of either algorithm applied to $q(x)$, we can instead apply them to $p(y)$, and transform the resulting sequence of iterates back to the default parameterization (via multiplication by $U^{-1} = U^\top$).

Theorem 2.1. *Let $p(y) = \sum_{i=1}^n [p]_i([y]_i)$ such that $[p]_i(t) = \lambda_i t^2/2 + c_i t$. Let ε be arbitrary and fixed. Denote by $CM_{\mathrm{x}}(\mu, p, y, v)$ and $CM_{\mathrm{v}}(\mu, p, y, v)$ the parameter vector and the velocity vector respectively, obtained by applying one step of CM (i.e., Eq. 1 and then Eq. 2) to the function p at point y, with velocity v, momentum coefficient μ, and learning rate ε. Define NAG_{x} and NAG_{v} analogously. Then the following holds for $\mathrm{z} \in \{\mathrm{x}, \mathrm{v}\}$:*

$$CM_{\mathrm{z}}(\mu, p, y, v) = \begin{bmatrix} CM_{\mathrm{z}}(\mu, [p]_1, [y]_1, [v]_1) \\ \vdots \\ CM_{\mathrm{z}}(\mu, [p]_n, [y]_n, [v]_n) \end{bmatrix}$$

$$NAG_{\mathrm{z}}(\mu, p, y, v) = \begin{bmatrix} CM_{\mathrm{z}}(\mu(1 - \lambda_1\varepsilon), [p]_1, [y]_1, [v]_1) \\ \vdots \\ CM_{\mathrm{z}}(\mu(1 - \lambda_n\varepsilon), [p]_n, [y]_n, [v]_n) \end{bmatrix}$$

Proof. See the appendix. □

The theorem has several implications. First, CM and NAG become equivalent when ε is small (when $\varepsilon\lambda \ll 1$ for every eigenvalue λ of A), so NAG and CM are distinct only when ε is reasonably large. When ε is relatively large, NAG uses smaller effective momentum for the high-curvature eigen-directions, which prevents oscillations (or divergence) and thus allows the use of a larger μ than is possible with CM for a given ε.

3. Deep Autoencoders

The aim of our experiments is three-fold. First, to investigate the attainable performance of stochastic momentum methods on deep autoencoders starting from well-designed random initializations; second, to explore the importance and effect of the schedule for the momentum parameter μ assuming an optimal fixed choice of the learning rate ε; and third, to compare the performance of NAG versus CM.

For our experiments with feed-forward nets, we focused on training the three deep autoencoder problems described in Hinton & Salakhutdinov (2006) (see sec. A.2 for details). The task of the neural network autoencoder is to reconstruct its own input subject to the constraint that one of its hidden layers is of low-dimension. This "bottleneck" layer acts as a low-dimensional code for the original input, similar to other dimensionality reduction techniques like Principle Component Analysis (PCA). These autoencoders are some of the deepest neural networks with published results, ranging between 7 and 11 layers, and have become a standard benchmarking problem (e.g., Martens, 2010; Glorot & Bengio, 2010; Chapelle & Erhan, 2011; Raiko et al., 2011). See the appendix for more details.

Because the focus of this study is on optimization, we only report training errors in our experiments. Test error depends strongly on the amount of overfitting in these problems, which in turn depends on the type and amount of regularization used during training. While regularization is an issue of vital importance when designing systems of practical utility, it is outside the scope of our discussion. And while it could be objected that the gains achieved using better optimization methods are only due to more exact fitting of the training set in a manner that does not generalize, this is simply not the case in these problems, where undertrained solutions are known to perform poorly on both the training and test sets (underfitting).

The networks we trained used the standard sigmoid nonlinearity and were initialized using the "sparse initialization" technique (SI) of Martens (2010) that is described in sec. 3.1. Each trial consists of 750,000 parameter updates on minibatches of size 200. No regularization is used. The schedule for μ was given by the following formula:

$$\mu_t = \min(1 - 2^{-1-\log_2(\lfloor t/250 \rfloor + 1)}, \mu_{\max}) \qquad (5)$$

where $\mu_{\max}$ was chosen from $\{0.999, 0.995, 0.99, 0.9, 0\}$. This schedule was motivated by Nesterov (1983) who advocates using what amounts to $\mu_t = 1 - 3/(t+5)$ after some manipulation

task	$0_{(\text{SGD})}$	0.9N	0.99N	0.995N	0.999N	0.9M	0.99M	0.995M	0.999M	SGD_C	$\text{HF}^\dagger$	HF^*
Curves	0.48	0.16	0.096	0.091	**0.074**	0.15	0.10	0.10	0.10	0.16	0.058	0.11
Mnist	2.1	1.0	**0.73**	0.75	0.80	1.0	0.77	0.84	0.90	0.9	0.69	1.40
Faces	36.4	14.2	8.5	7.8	**7.7**	15.3	8.7	8.3	9.3	NA	7.5	12.0

Table 1. The table reports the squared errors on the problems for each combination of $\mu_{\max}$ and a momentum type (NAG, CM). When $\mu_{\max}$ is 0 the choice of NAG vs CM is of no consequence so the training errors are presented in a single column. For each choice of $\mu_{\max}$, the highest-performing learning rate is used. The column SGD_C lists the results of Chapelle & Erhan (2011) who used 1.7M SGD steps and tanh networks. The column $HF^\dagger$ lists the results of HF without L2 regularization, as described in sec. 5; and the column HF^* lists the results of Martens (2010).

problem	before	after
Curves	0.096	0.074
Mnist	1.20	0.73
Faces	10.83	7.7

Table 2. The effect of low-momentum finetuning for NAG. The table shows the training squared errors before and after the momentum coefficient is reduced. During the primary ("transient") phase of learning we used the optimal momentum and learning rates.

(see appendix), and by Nesterov (2003) who advocates a constant μ_t that depends on (essentially) the condition number. The constant μ_t achieves exponential convergence on strongly convex functions, while the $1-3/(t+5)$ schedule is appropriate when the function is not strongly convex. The schedule of Eq. 5 blends these proposals. For each choice of $\mu_{\max}$, we report the learning rate that achieved the best training error. Given the schedule for μ, the learning rate ε was chosen from $\{0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001\}$ in order to achieve the lowest final error training error after our fixed number of updates.

Table 1 summarizes the results of these experiments. It shows that NAG achieves the lowest published results on this set of problems, including those of Martens (2010). It also shows that larger values of $\mu_{\max}$ tend to achieve better performance and that NAG usually outperforms CM, especially when $\mu_{\max}$ is 0.995 and 0.999. Most surprising and importantly, the results demonstrate that NAG can achieve results that are comparable with some of the best HF results for training deep autoencoders. Note that the previously published results on HF used L2 regularization, so they cannot be directly compared. However, the table also includes experiments we performed with an improved version of HF (see sec. 2.1) where weight decay was removed towards the end of training.

We found it beneficial to reduce μ to 0.9 (unless μ is 0, in which case it is unchanged) during the final 1000 parameter updates of the optimization without reducing the learning rate, as shown in Table 2. It appears that reducing the momentum coefficient allows for finer convergence to take place whereas otherwise the overly aggressive nature of CM or NAG would prevent this. This phase shift between optimization that favors fast accelerated motion along the error surface (the "transient phase") followed by more careful optimization-as-estimation phase seems consistent with the picture presented by Darken & Moody (1993). However, while asymptotically it is the second phase which must eventually dominate computation time, in practice it seems that for deeper networks in particular, the first phase dominates overall computation time as long as the second phase is cut off before the remaining potential gains become either insignificant or entirely dominated by overfitting (or both).

It may be tempting then to use lower values of μ from the outset, or to reduce it immediately when progress in reducing the error appears to slow down. However, in our experiments we found that doing this was detrimental in terms of the final errors we could achieve, and that despite appearing to not make much progress, or even becoming significantly non-monotonic, the optimizers were doing something apparently useful over these extended periods of time at higher values of μ.

A speculative explanation as to why we see this behavior is as follows. While a large value of μ allows the momentum methods to make useful progress along slowly-changing directions of low-curvature, this may not immediately result in a significant reduction in error, due to the failure of these methods to converge in the more turbulent high-curvature directions (which is especially hard when μ is large). Nevertheless, this progress in low-curvature directions takes the optimizers to new regions of the parameter space that are characterized by closer proximity to the optimum (in the case of a convex objective), or just higher-quality local minimia (in the case of non-convex optimization). Thus, while it is important to adopt a more careful scheme that allows fine convergence to take place along the high-curvature directions, this must be done with care. Reducing μ and moving to this fine convergence regime too early may make it difficult for the optimization to make significant progress along the low-curvature directions, since without the benefit of momentum-based acceleration, first-order methods are notoriously bad at this (which is what motivated the use of second-order methods like HF for deep learning).

SI scale multiplier	0.25	0.5	1	2	4
error	16	16	0.074	0.083	0.35

Table 3. The table reports the training squared error that is attained by changing the scale of the initialization.

3.1. Random Initializations

The results in the previous section were obtained with standard logistic sigmoid neural networks that were initialized with the sparse initialization technique (SI) described in Martens (2010). In this scheme, each random unit is connected to 15 randomly chosen units in the previous layer, whose weights are drawn from a unit Gaussian, and the biases are set to zero. The intuitive justification is that the total amount of input to each unit will not depend on the size of the previous layer and hence they will not as easily saturate. Meanwhile, because the inputs to each unit are not all randomly weighted blends of the outputs of many 100s or 1000s of units in the previous layer, they will tend to be qualitatively more "diverse" in their response to inputs. When using tanh units, we transform the weights to simulate sigmoid units by setting the biases to 0.5 and rescaling the weights by 0.25.

We investigated the performance of the optimization as a function of the scale constant used in SI (which defaults to 1 for sigmoid units). We found that SI works reasonably well if it is rescaled by a factor of 2, but leads to noticeable (but not severe) slow down when scaled by a factor of 3. When we used the factor 1/2 or 5 we did not achieve sensible results.

4. Recurrent Neural Networks

Echo-State Networks (ESNs) is a family of RNNs with an unusually simple training method: their hidden-to-output connections are learned from data, but their recurrent connections are fixed to a random draw from a specific distribution and are not learned. Despite their simplicity, ESNs with many hidden units (or with units with explicit temporal integration, like the LSTM) have achieved high performance on tasks with long range dependencies (**?**). In this section, we investigate the effectiveness of momentum-based methods with ESN-inspired initialization at training RNNs with conventional size and standard (i.e., non-integrating) neurons. We find that momentum-accelerated SGD can successfully train such RNNs on various artificial datasets exhibiting considerable long-range temporal dependencies. This is unexpected because RNNs were believed to be almost impossible to successfully train on such datasets with first-order methods, due to various difficulties such as vanishing/exploding gradients (Bengio et al., 1994). While we found that the use of momentum significantly improved performance and robustness, we obtained nontrivial results even with standard SGD, provided that the learning rate was set low enough.

connection type	sparsity	scale
in-to-hid (add,mul)	dense	$0.001 \cdot N(0, 1)$
in-to-hid (mem)	dense	$0.1 \cdot N(0, 1)$
hid-to-hid	15 fan-in	spectral radius of 1.1
hid-to-out	dense	$0.1 \cdot N(0, 1)$
hid-bias	dense	0
out-bias	dense	average of outputs

Table 4. The RNN initialization used in the experiments. The scale of the vis-hid connections is problem dependent.

Each task involved optimizing the parameters of a randomly initialized RNN with 100 standard tanh hidden units (the same model used by Martens & Sutskever (2011)). The tasks were designed by Hochreiter & Schmidhuber (1997), and are referred to as training "problems". See sec. A.3 of the appendix for details.

4.1. ESN-based Initialization

As argued by Jaeger & Haas (2004), the spectral radius of the hidden-to-hidden matrix has a profound effect on the dynamics of the RNN's hidden state (with a tanh nonlinearity). When it is smaller than 1, the dynamics will have a tendency to quickly "forget" whatever input signal they may have been exposed to. When it is much larger than 1, the dynamics become oscillatory and chaotic, allowing it to generate responses that are varied for different input histories. While this allows information to be retained over many time steps, it can also lead to severe exploding gradients that make gradient-based learning much more difficult. However, when the spectral radius is only slightly greater than 1, the dynamics remain oscillatory and chaotic while the gradient are no longer exploding (and if they do explode, then only "slightly so"), so learning may be possible with a spectral radius of this order. This suggests that a spectral radius of around 1.1 may be effective.

To achieve robust results, we also found it is essential to carefully set the initial scale of the input-to-hidden connections. When training RNNs to solve those tasks that possess many random and irrelevant distractor inputs, we found that having the scale of these connections set too high at the start led to relevant information in the hidden state being too easily "overwritten" by the many irrelevant signals, which ultimately led the optimizer to converge towards an extremely poor local minimum where useful information was never relayed over long distances. Conversely, we found that if this scale was set too low, it led to significantly slower learning. Having experimented with multiple scales we found that a Gaussian draw with a standard deviation of 0.001 achieved a good balance between these concerns. However, unlike the value of 1.1 for the spectral radius of the dynamics matrix, which worked well on all tasks, we found that good choices for initial scale of the input-to-hidden weights depended a lot on the particular characteristics of the particular task (such

as its dimensionality or the input variance). Indeed, for tasks that do not have many irrelevant inputs, a larger scale of the input-to-hidden weights (namely, 0.1) worked better, because the aforementioned disadvantage of large input-to-hidden weights does not apply. See table 4 for a summary of the initializations used in the experiments. Finally, we found centering (mean subtraction) of both the inputs and the outputs to be important to reliably solve all of the training problems. See the appendix for more details.

4.2. Experimental Results

We conducted experiments to determine the efficacy of our initializations, the effect of momentum, and to compare NAG with CM. Every learning trial used the aforementioned initialization, 50,000 parameter updates and on minibatches of 100 sequences, and the following schedule for the momentum coefficient μ: $\mu = 0.9$ for the first 1000 parameter, after which $\mu = \mu_0$, where μ_0 can take the following values $\{0, 0.9, 0.98, 0.995\}$. For each μ_0, we use the empirically best learning rate chosen from $\{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$.

The results are presented in Table 5, which are the average loss over 4 different random seeds. Instead of reporting the loss being minimized (which is the squared error or cross entropy), we use a more interpretable zero-one loss, as is standard practice with these problems. For the bit memorization, we report the fraction of timesteps that are computed incorrectly. And for the addition and the multiplication problems, we report the fraction of cases where the RNN the error in the final output prediction exceeded 0.04.

Our results show that despite the considerable long-range dependencies present in training data for these problems, RNNs can be successfully and robustly trained to solve them, through the use of the initialization discussed in sec. 4.1, momentum of the NAG type, a large μ_0, and a particularly small learning rate (as compared with feedforward networks). Our results also suggest that with larger values of μ_0 achieve better results with NAG but not with CM, possibly due to NAG's tolerance of larger μ_0's (as discussed in sec. 2).

Although we were able to achieve surprisingly good training performance on these problems using a sufficiently strong momentum, the results of Martens & Sutskever (2011) appear to be moderately better and more robust. They achieved lower error rates and their initialization was chosen with less care, although the initializations are in many ways similar to ours. Notably, Martens & Sutskever (2011) were able to solve these problems without centering, while we had to use centering to solve the multiplication problem (the other problems are already centered). This suggests that the initialization proposed here, together with the method of Martens & Sutskever (2011), could achieve even better performance. But the main achievement of these results is a demonstration of the ability of momentum methods to cope with long-range temporal dependency training tasks to a level which seems sufficient for most practical purposes. Moreover, our approach seems to be more tolerant of smaller minibatches, and is considerably simpler than the particular version of HF proposed in Martens & Sutskever (2011), which used a specialized update damping technique whose benefits seemed mostly limited to training RNNs to solve these kinds of extreme temporal dependency problems.

5. Momentum and HF

Truncated Newton methods, that include the HF method of Martens (2010) as a particular example, work by optimizing a local quadratic model of the objective via the linear conjugate gradient algorithm (CG), which is a first-order method. While HF, like all truncated-Newton methods, takes steps computed using partially converged calls to CG, it is naturally accelerated along at least some directions of lower curvature compared to the gradient. It can even be shown (Martens & Sutskever, 2012) that CG will tend to favor convergence to the exact solution to the quadratic sub-problem first along higher curvature directions (with a bias towards those which are more clustered together in their curvature-scalars/eigenvalues).

While CG accumulates information as it iterates which allows it to be optimal in a much stronger sense than any other first-order method (like NAG), once it is terminated, this information is lost. Thus, standard truncated Newton methods can be thought of as persisting information which accelerates convergence (of the current quadratic) only over the number of iterations CG performs. By contrast, momentum methods persist information that can inform new updates across an arbitrary number of iterations.

One key difference between standard truncated Newton methods and HF is the use of "hot-started" calls to CG, which use as their initial solution the one found at the previous call to CG. While this solution was computed using old gradient and curvature information from a previous point in parameter space and possibly a different set of training data, it may be well-converged along certain eigen-directions of the new quadratic, despite being very poorly converged along others (perhaps worse than the default initial solution of $\vec{0}$). However, to the extent to which the new local quadratic model resembles the old one, and in particular in the more difficult to optimize directions of low-curvature (which will arguably be more likely to persist across nearby locations in parameter space), the previous solution will be a preferable starting point to 0, and may even allow for gradually increasing levels of convergence along certain directions which persist

problem	biases	0	0.9N	0.98N	0.995N	0.9M	0.98M	0.995M
add $T = 80$	0.82	0.39	0.02	0.21	**0.00025**	0.43	0.62	0.036
mul $T = 80$	0.84	0.48	0.36	0.22	**0.0013**	0.029	0.025	0.37
mem-5 $T = 200$	2.5	1.27	1.02	0.96	**0.63**	1.12	1.09	0.92
mem-20 $T = 80$	8.0	5.37	2.77	0.0144	**0.00005**	1.75	0.0017	0.053

Table 5. Each column reports the errors (zero-one losses; sec. 4.2) on different problems for each combination of μ_0 and momentum type (NAG, CM), averaged over 4 different random seeds. The "biases" column lists the error attainable by learning the output biases and ignoring the hidden state. This is the error of an RNN that failed to "establish communication" between its inputs and targets. For each μ_0, we used the fixed learning rate that gave the best performance.

in the local quadratic models across many updates.

The connection between HF and momentum methods can be made more concrete by noticing that a single step of CG is effectively a gradient update taken from the current point, plus the previous update reapplied, just as with NAG, and that if CG terminated after just 1 step, HF becomes equivalent to NAG, except that it uses a special formula based on the curvature matrix for the learning rate instead of a fixed constant. The most effective implementations of HF even employ a "decay" constant (Martens & Sutskever, 2012) which acts analogously to the momentum constant μ. Thus, in this sense, the CG initializations used by HF allow us to view it as a hybrid of NAG and an exact second-order method, with the number of CG iterations used to compute each update effectively acting as a dial between the two extremes.

Inspired by the surprising success of momentum-based methods for deep learning problems, we experimented with making HF behave even more like NAG than it already does. The resulting approach performed surprisingly well (see Table 1). For a more detailed account of these experiments, see sec. A.6 of the appendix.

If viewed on the basis of each CG step (instead of each update to parameters θ), HF can be thought of as a peculiar type of first-order method which approximates the objective as a series of quadratics only so that it can make use of the powerful first-order CG method. So apart from any potential benefit to global convergence from its tendency to prefer certain directions of movement in parameter space over others, perhaps the main theoretical benefit to using HF over a first-order method like NAG is its use of CG, which, while itself a first-order method, is well known to have strongly optimal convergence properties for quadratics, and can take advantage of clustered eigenvalues to accelerate convergence (see Martens & Sutskever (2012) for a detailed account of this well-known phenomenon). However, it is known that in the worst case that CG, when run in batch mode, will converge asymptotically no faster than NAG (also run in batch mode) for certain specially designed quadratics with very evenly distributed eigenvalues/curvatures. Thus it is worth asking whether the quadratics which arise during the optimization of neural networks by HF are such that CG has a distinct advantage in optimizing them over NAG, or if they are closer to the aforementioned worst-case examples. To examine this question we took a quadratic generated during the middle of a typical run of HF on the curves dataset and compared the convergence rate of CG, initialized from zero, to NAG (also initialized from zero). Figure 5 in the appendix presents the results of this experiment. While this experiment indicates some potential advantages to HF, the closeness of the performance of NAG and HF suggests that these results might be explained by the solutions leaving the area of trust in the quadratics before any extra speed kicks in, or more subtly, that the faithfulness of approximation goes down just enough as CG iterates to offset the benefit of the acceleration it provides.

6. Discussion

Martens (2010) and Martens & Sutskever (2011) demonstrated the effectiveness of the HF method as a tool for performing optimizations for which previous attempts to apply simpler first-order methods had failed. While some recent work (Chapelle & Erhan, 2011; Glorot & Bengio, 2010) suggested that first-order methods can actually achieve some success on these kinds of problems when used in conjunction with good initializations, their results still fell short of those reported for HF. In this paper we have completed this picture and demonstrated conclusively that a large part of the remaining performance gap that is not addressed by using a well-designed random initialization is in fact addressed by careful use of momentum-based acceleration (possibly of the Nesterov type). We showed that careful attention must be paid to the momentum constant μ, as predicted by the theory for local and convex optimization.

Momentum-accelerated SGD, despite being a first-order approach, is capable of accelerating directions of low-curvature just like an approximate Newton method such as HF. Our experiments support the idea that this is important, as we observed that the use of stronger momentum (as determined by μ) had a dramatic effect on optimization performance, particularly for the RNNs. Moreover, we showed that HF can be viewed as a first-order method, and as a generalization of NAG in particular, and that it already derives some of its benefits through a momentum-like mechanism.

ARTICLE IN PRESS

Neural Networks ▮ (▮▮▮▮) ▮▮▮–▮▮▮

Contents lists available at SciVerse ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

2012 Special Issue

Autonomous reinforcement learning with experience replay

Paweł Wawrzyński [a,*], Ajay Kumar Tanwani [b]

[a] Warsaw University of Technology, Institute of Control and Computation Engineering, Poland

[b] École Polytechnique Fédérale De Lausanne, Switzerland

ARTICLE INFO

Keywords:
Reinforcement learning
Autonomous learning
Step-size estimation
Actor–critic

ABSTRACT

This paper considers the issues of efficiency and autonomy that are required to make reinforcement learning suitable for real-life control tasks. A real-time reinforcement learning algorithm is presented that repeatedly adjusts the control policy with the use of previously collected samples, and autonomously estimates the appropriate step-sizes for the learning updates. The algorithm is based on the actor–critic with experience replay whose step-sizes are determined on-line by an enhanced fixed point algorithm for on-line neural network training. An experimental study with simulated octopus arm and half-cheetah demonstrates the feasibility of the proposed algorithm to solve difficult learning control problems in an autonomous way within reasonably short time.

1. Introduction

Reinforcement learning (RL) is a potentially useful methodology for control optimization. It may be the only feasible approach to many problems. However, its widespread use is limited by the total time many RL algorithms require to solve a given problem. This time encompasses duration of a single run in collecting appropriate number of samples multiplied by the number of runs necessary to tune problem dependent parameters. Selection of appropriate parameter setting requires learning to start over again, thereby, making the previously collected data useless. The cost of collecting data from control process and the elapsed time to handcraft the parameters makes learning infeasible for many real world tasks. Ideally, the learner should interact with the environment in a 'plug-and-play' manner and optimize a control policy in a single trial with minimum number of data samples. The main focus of this research is to make reinforcement learning suitable for real control applications and minimize the associated implementation costs.

A broad class of RL algorithms that encompass Actor–Critics (Bhatnagar, Sutton, Ghavamzadeh, & Lee, 2009; Konda & Tsitsiklis, 2003; Peters, Vijayakumar, & Schaal, 2005) and policy gradient methods (Sutton, McAllester, Singh, & Mansour, 2000; Williams, 1992) optimizes a control policy on the basis of stochastic gradient estimates. In these algorithms, the control policy is parameterized by a vector that is optimized incrementally on the basis of the data coming from the control process. The policy vector adjustment is local in the sense that the algorithm moves a small step in the estimate of the improvement direction; its length is defined by the step-size of the learning process. The appropriate length of the step-size depends on the stage of the learning process and setting it wrong can make the process unstable or prohibitively slow. In this paper, we present a real-time reinforcement learning algorithm that is considerably efficient and autonomous due to: (1) *experience replay*—it stores the agent–environment interaction samples in a database and repeatedly draws previous samples—simultaneous to the interaction—to estimate the improvement direction of policy vector and speed up learning (Wawrzyński, 2009), and (2) *step-size estimation*—it autonomously determines the appropriate step-size to optimize the learning updates and eliminate the overhead of its manual tuning (Wawrzyński, 2010).

Optimization by means of improvement direction estimates is a tool commonly used in RL but it is not unique to this field. It is the main principle of a wide class of stochastic approximation algorithms (Kushner & Yin, 1997). In all these methods the step-size plays a key role in controlling their speed and stability. Despite significance of the problem of step-size estimation, its generally applicable solution has not been found.

1.1. Related work

A lot of effort is made in the reinforcement learning community to speed up the learning methods by using the previously collected samples. The concept of reusing samples evolved from recomputing previous experience by means of dynamic programming (Sutton, 1990). Application of the *importance sampling* technique (Rubinstein, 1981) made it possible to optimize a parametric control policy with the use of experience gained with other policies

* Corresponding author.
E-mail addresses: pwawrzynski@gmail.com, p.wawrzynski@elka.pw.edu.pl (P. Wawrzyński), ajay.tanwani@epfl.ch (A.K. Tanwani).

0893-6080/$ – see front matter © 2012 Elsevier Ltd. All rights reserved.
doi:10.1016/j.neunet.2012.11.007

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

(Hachiya, Peters, & Sugiyama, 2011; Peshkin & Mukherjee, 2001). Kober and Peters (2011) proposed an algorithm called PoWER—Policy Learning by Weighting Exploration with the Returns—that combines the previous experience by per-decision importance weighting technique. The approach to sample reuse developed here is experience replay (Adam, Busoniu, & Babuska, 2012; Cichosz, 1999) applied to actor–critics with the use of the randomized truncated estimators (Wawrzyński, 2009).

Real world applications of reinforcement learning require a number of parameters to be defined and empirically tuned, e.g., reward function, policy structure, step-size, exploration noise, discount factor, and others specific to the learning algorithm in use. Despite several promising approaches to provide leverage for these overheads such as the use of apprenticeship learning to find reward function (Abbeel & Ng, 2005), dynamic evolution of policy parameters to get its appropriate size and representation (Kormushev, Ugurlu, Calinon, Tsagarakis, & Caldwell, 2011), the parameter that remains difficult to determine autonomously in reinforcement learning is the step-size of the learning process. The work (Sutton, 1992a, 1992b) established *delta-bar-delta* algorithm, introduced earlier by Jacobs (1988) as a generally applicable approach to step-size estimation in RL. The method requires the initial step-size to be specified and collapses if this value is too large, which delimits its autonomous use. Consequently, the solution to the problem of step-size determination in reinforcement learning is still a subject of active research (George & Powell, 2006; Noda, 2009; Schraudolph, Yu, & Aberdeen, 2006).

Many approaches to step-size estimation have been designed as tools for online neural network training. Algorithms of this type include *delta-delta* (Silva & Almeida, 1990), aforementioned *delta-bar-delta*, and stochastic meta-descent (Schraudolph & Giannakopoulos, 2000). Among others, step-sizes can be estimated to ensure Lyapunov-stability of the process (Behera, Kumar, & Patnaik, 2006; Kathirvalavakumar & Subavathi, 2009).

Most methods of autonomous or adaptive step-size estimation are designed as recursions that are updated from one sample to another (George & Powell, 2006). This design is often based on some problem-dependent parameters and may be sensitive to wrong initialization or nonstationarity. In this work we build upon the fixed-point method of step-size estimation (Wawrzyński, 2010) which has a different design to remedy the aforementioned problems. By computing gradient estimates always at two points it captures global characteristics of the process, thereby, obtaining robustness and independence from any parameters. In Wawrzyński and Papis (2011) this approach was specialized to neural-network on-line training, e.g., each weight of the network was assigned a separate step-size. Here the original algorithm is enhanced and adopted to provide autonomy to RL with experience replay.

1.2. Contributions

The purpose of this paper is to present a reinforcement learning framework that is fast and autonomous enough for on-line optimization of control of systems with complex dynamics and multidimensional, possibly continuous, state and input spaces. The contribution of this paper may be summarized in the following points:

- An enhanced fixed point method of step-size estimation (Wawrzyński, 2010) is presented.
- The above method of step-size estimation is combined with experience replay for actor–critic class of RL algorithms (Wawrzyński, 2009).
- The proposed approach is implemented on two simulated robot-control problems, both with complex dynamics and rich state and action spaces, namely *octopus arm* and *half-cheetah*.

The rest of the paper is organized as follows: Section 2 formulates the reinforcement learning problem followed by the description of actor–critic algorithm with experience replay in Section 3. In Section 4, we describe the fixed point method of step-size adaptation in a generic stochastic descent procedure. The resulting actor–critic reinforcement learning algorithm with experience replay and step-size estimation is proposed in Section 5. Experimental study is presented in Section 6 on challenging learning control problems. The last section concludes the paper with an outlook to future work.

2. Problem formulation

We consider the standard reinforcement learning setup under the Markov Decision Process (MDP) framework (Sutton & Barto, 1998). The problem concerns an agent that observes the state of its environment, s_t, in discrete time, $t = 1, 2, 3, \ldots$, performs an action, a_t, which moves the environment to the next state, s_{t+1}, and gives the agent a reward, $r_t \in \Re$. The environment is in general stochastic which means that the consecutive state, s_{t+1}, is a result of sampling from the transition distribution conditioned on the preceding state, s_t, and the action, a_t. Mathematically,

$$s_{t+1} \sim P_s(\cdot|s_t, a_t).$$

The reward may depend deterministically on the current action and the next state, $r_t = r(a_t, s_{t+1})$. A particular MDP is a tuple $\langle \mathcal{S}, \mathcal{A}, P_s, r\rangle$ where $\mathcal{S}$ and $\mathcal{A}$ are the state and action spaces, respectively; $\{P_s(\cdot|s, a) : s \in \mathcal{S}, a \in \mathcal{A}\}$ is a set of state transition distributions and r is the reward function. The transition distributions, P_s, and the reward function, r, are initially unknown to the agent. The goal of learning is to determine a stochastic control policy π that assigns actions to states such that in each state the agent may expect the highest rewards in the future. Note that the control policy maps each state into a distribution of actions rather than a single action. Performing different actions in each state, the agent is able to differentiate good actions from inferior ones in each state. The control policy π is parameterized by the policy vector, θ, e.g., a set of weights in a neural network. It can be represented as

$$a \sim \pi(\cdot; s, \theta).$$

Let us denote by π_θ the policy of the form π with the fixed parameter θ. The quality of this policy is measured by the *value-function* V^{π_θ} defined as the expected value of the discounted sum of future rewards the agent may gain in the future, starting from the given state, s, at a certain time, t, and the policy π_θ:

$$V^{\pi_\theta}(s) = E\left(\sum_{i\geq 0} \gamma^i r_{t+i} \middle| s_t = s,\ \text{policy} = \pi_\theta\right). \tag{1}$$

The parameter $\gamma \in [0, 1)$ is the discount factor that defines the weight of distant rewards in relation to those obtained sooner. The quality of a policy may also be evaluated using the *action-value function* Q^{π_θ}, defined as

$$Q^{\pi_\theta}(s, a) = E\left(\sum_{i\geq 0} \gamma^i r_{t+i} \middle| s_t = s,\ a_t = a,\ \text{policy} = \pi_\theta\right). \tag{2}$$

It is the expected value of the same sum, but here it also depends on the first action to be performed, that is the action performed in the given state. This function is crucial for the policy optimization as it tells good actions from inferior ones in the same state.

We seek a learning algorithm that can optimize the control policy (i) without parameters determined experimentally, (ii) within short time of agent–environment interaction, but not necessarily after small amount of computation.

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

Algorithm 1. The classic actor–critic.

0: Set $y_\theta = 0, y_\upsilon = 0, t = 1$. Initialize θ and υ
1: Choose the action: $a_t \sim \pi(\cdot\,; s_t, \theta)$
2: Execute a_t, evaluate s_{t+1} and r_t
3: Calculate the temporal difference $d_t(\upsilon)$
4: $\quad d_t(\upsilon) = r_t + \gamma \bar{V}(s_{t+1}; \upsilon) - \bar{V}(s_t; \upsilon)$
5: Update the actor:
6: $\quad y_\theta := (\gamma\lambda) y_\theta + \beta_t^\theta \nabla_\theta \ln \pi(a_t; s_t, \theta)$
7: $\quad \theta := \theta + y_\theta d_t$
8: Update the critic:
9: $\quad y_\upsilon := (\gamma\lambda) y_\upsilon + \beta_t^\theta \nabla_\upsilon \bar{V}(s_t; \upsilon)$
10: $\quad \upsilon := \upsilon + y_\upsilon d_t$
11: Assign $t := t + 1$ and repeat from Point 1

3. Actor–critic with experience replay

This section presents actor–critic algorithms (Barto, Sutton, & Anderson, 1983; Bhatnagar et al., 2009; Kimura & Kobayashi, 1998; Konda & Tsitsiklis, 2003) and the way of its speeding-up by experience replay (Adam et al., 2012; Cichosz, 1999; Wawrzyński, 2009). Particular attention is paid to the 'classic' actor–critic in the form presented by Kimura and Kobayashi (1998) because this algorithm is conceptually the simplest among its family to which the ideas presented in further sections are directly applicable.

3.1. Classic Actor–Critic

The classic actor–critic algorithm employs two data structures. The first one is an actor, $\pi(a; s, \theta)$, that represents a stochastic control policy to generate random actions on the basis of the given state and the policy vector $\theta \in \Re^{n_\theta}$. The second one is a critic, $\bar{V}(s; \upsilon)$, that represents an approximator of the value function (1) parameterized by the critic vector $\upsilon \in \Re^{n_\upsilon}$. The algorithm is based on a special estimator of the action-value function $Q^{\pi_\theta}(s_t, a_t)$ (Eq. (2)). This estimator has the form:

$$R_t^\lambda = r_t + \gamma \bar{V}(s_{t+1}; \upsilon) + \sum_{i \geq 1} (\gamma\lambda)^i \left(r_{t+i} + \gamma \bar{V}(s_{t+i+1}; \upsilon) - \bar{V}(s_{t+i}; \upsilon) \right). \tag{3}$$

The parameter $\lambda \in [0, 1]$ defines how much R_t^λ is influenced by the value-function approximator, $\bar{V}$, and the true rewards, r. For $\lambda = 1$ the estimator is based on true rewards only and thus unbiased, but its variance may be very large. For $\lambda = 0$ variance of the estimator is the smallest, but it is biased by the inaccuracy of the value-function approximator. Moderate values of λ balance the bias and the variance of the estimator.

The classic actor–critic is sketched in Algorithm 1. It can be verified that after a visit in state s_t, for $t = 1, 2, \ldots$, the algorithm modifies the policy vector θ by the product $\beta_t^\theta \widehat{\phi}_t$ where β_t^θ is a step-size (small positive number) and $\widehat{\phi}_t$ is the policy vector improvement direction estimate given by

$$\widehat{\phi}_t = \left(R_t^\lambda - \bar{V}(s_t; \upsilon) \right) \nabla_\theta \ln \pi(a_t; s_t, \theta), \tag{4}$$

where ∇_θ means gradient with respect to θ. (For $\lambda = 0$, the modification takes place in step t, for $\lambda > 0$ it is also partially distributed over further steps). Consequently, if the action a_t turns out to bring rewards, R_t^λ, larger than $\bar{V}(s_t; \upsilon)$, i.e., the rewards expected in state s_t, then the probability of action a_t in state s_t is being increased. If, conversely, the action turns out to bring rewards smaller than expected, then its probability is being decreased.

The purpose of the critic, $\bar{V}(s; \upsilon)$ is to approximate the expected value of R_t^λ. Therefore, a visit in state s_t induces a modification of the critic vector υ by the product $\beta_t^\upsilon \widehat{\psi}_t$ where β_t^υ is a step-size and

Algorithm 2. Actor–Critic with Experience Replay. Estimators mentioned in Steps 6 and 7 are based on the samples in a database.

1: $t := 1$, Initialize θ and υ
2: Choose and execute an action, $a_t \sim \pi(\cdot\,; s_t, \theta)$
3: Assign $\pi_t = \pi(a_t; s_t, \theta)$
4: Repeat $\nu(t)$ times, begin
5: $\quad$ Draw $i \in \{t - N + 1, t - N + 2, \ldots, t\}$
6: $\quad$ Adjust θ along an estimator of $\phi(s_i, \theta, \upsilon)$:
7: $\quad\quad \theta := \theta + \beta_t^\theta \widehat{\phi}_i^r(\theta, \upsilon)$
8: $\quad$ Adjust υ along an estimator of $\psi(s_i, \theta, \upsilon)$:
9: $\quad\quad \upsilon := \upsilon + \beta_t^\upsilon \widehat{\psi}_i^r(\theta, \upsilon)$
10: End
11: Register the tuple $\langle s_t, a_t, \pi_t, r_t, s_{t+1} \rangle$ in the database
12: Make sure only N most recent tuples remain in the database.
13: Assign $t := t + 1$ and repeat from Line 2

$\widehat{\psi}_t$ is the critic vector improvement direction estimate:

$$\widehat{\psi}_t = \left(R_t^\lambda - \bar{V}(s_t; \upsilon) \right) \nabla_\upsilon \bar{V}(s_t; \upsilon). \tag{5}$$

3.2. Classic actor–critic with experience replay

The idea of experience replay is to repeatedly recall previous pieces of experience and apply them to update the current policy as a sequential RL algorithm (like the classic actor–critic) would if they have just happened. This idea was applied to actor–critics in Wawrzyński (2009). Namely, let us denote by

$$\phi(s, \theta, \upsilon) = E\left(\widehat{\phi}_t \middle| s_t = s, \theta, \upsilon \right) \tag{6}$$

a direction in which θ is on average modified to increase the rewards expected in state s. Also, let us denote by

$$\psi(s, \theta, \upsilon) = E\left(\widehat{\psi}_t \middle| s_t = s, \theta, \upsilon \right) \tag{7}$$

a direction in which υ is on average modified to increase the accuracy of the value function approximation in state s.

After each instant t, the classic actor–critic estimates $\phi(s_t, \theta, \upsilon)$, i.e., the direction of policy improvement, and adjusts the policy vector θ along the estimate. Before the next instant t, the modified algorithm repeatedly chooses one of the recently visited states at random, s_i, estimates $\phi(s_i, \theta, \upsilon)$, and modifies the policy vector along the estimate. Essentially both algorithms achieve the same goal, but the modified one improves the current actor and critic with the use of the whole gathered experience rather than only the present event, like the classic method. Due to more exhaustive exploitation of information experience replay leads to faster learning at the cost of additional computation.

In Wawrzyński (2009), the *randomized-truncated estimators* are introduced for estimating $\phi(s_i, \theta, \upsilon)$ and $\psi(s_i, \theta, \upsilon)$. To define them, let b be the upper limit of the randomized truncated estimators with $b > 1$, θ_{i+j} be the policy vector applied to generate a_{i+j}, $\alpha \in [0, 1)$ and K be drawn independently from $Geom(\rho)$ the geometric distribution[1] with parameter $\rho \in [0, 1)$. The randomized-truncated estimators, $\widehat{\phi}_i^r(\theta, \upsilon)$ and $\widehat{\psi}_i^r(\theta, \upsilon)$ have a generic form that in the case of classic actor–critic is reduced to

$$\widehat{\phi}_i^r(\theta, \upsilon) = \nabla_\theta \ln \pi(a_t; s_t, \theta) \times \sum_{k=0}^{K} \alpha^k d_{i+k}(\upsilon) \min \left\{ \prod_{j=0}^{k} \frac{\pi(a_{i+j}; s_{i+j}, \theta)}{\pi_{i+j}}, b \right\}, \tag{8}$$

[1] That is, random variable K of values in $\{0, 1, 2, \ldots\}$ has distribution $Geom(\rho)$, iff $P(K = m) = (1 - \rho)\rho^m$ for nonnegative integer m.

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

$$\widehat{\psi}_i^r(\theta, \upsilon) = \nabla_\upsilon \bar{V}(s_t; \upsilon) \times \sum_{k=0}^{K} \alpha^k d_{i+k}(\upsilon) \min\left\{ \prod_{j=0}^{k} \frac{\pi(a_{i+j}; s_{i+j}, \theta)}{\pi_{i+j}}, b \right\} \tag{9}$$

where

$$d_i(\upsilon) = r_i + \gamma \bar{V}(s_{i+1}; \upsilon) - \bar{V}(s_i; \upsilon),$$

is the temporal difference and π_{i+j} is the probability of the action taken at instant $i+j$ (or it is the density if $\mathcal{A}$ is continuous). Remembering the definition of R_t^λ (Eq. (3)) makes it easy to notice the similarity between $\widehat{\phi}_t$ (Eq. (4)) and $\widehat{\phi}_i^r$ (Eq. (8)), and between $\widehat{\psi}_t$ (Eq. (5)) and $\widehat{\psi}_i^r$ (Eq. (9)). Though, there are two important differences. First, the infinite sum in Eq. (3) is replaced by the finite one with appropriately designed random limit in Eqs. (8) and (9). Second, truncated density ratios are introduced in order to compensate for the difference between the current policy and the ones that generated the actions $a_t, a_{t-1}, \ldots$ present in the database.

The actor–critic with experience replay is presented in Algorithm 2. The number of computation steps after t-th control step is defined by the function $\nu(t)$. Since computation is carried out on-line, this function is bounded from above. It should also be limited for small t to prevent overtraining with too few samples in the database. (In Section 5 this function is discussed in more detail.)

3.3. Generalized actor–critic with experience replay

Experience replay may be applied to the classic actor–critic to increase its learning speed, but also to a wide class of sequential actor–critic algorithms. As shown in Wawrzyński (2009) the sufficient condition is that each step in the environment prompts the original algorithm to do modifications (possibly distributed in time) of the policy vector and the critic vector with estimates, respectively, $\widehat{\phi}_t$ and $\widehat{\psi}_t$ of the generic form

$$G_t(\theta, \upsilon) \sum_{k \geq 0} (\alpha\rho)^k z_{t,k}(\theta, \upsilon)$$

where G_t is a vector defined by s_t and a_t, $\alpha \in [0, 1)$, $\rho \in [0, 1)$, and $z_{t,k} \in \Re$ is defined by s_{t+k}, a_{t+k}, r_{t+k}, s_{t+k+1}, and possibly a_{t+k+1}.

4. Fixed point method of step-size estimation

The actor–critic algorithm with experience replay repeatedly samples data and uses it to compute a certain vector with which it updates parameters θ and υ. This pattern of operation places it in a wide class of optimization algorithms based on stochastic gradient descent (Kushner & Yin, 1997). Each algorithm of that type defines a sequence

$$\theta_{t+1} = \theta_t - \beta_t g(\theta_t; \xi_t), \quad t = 1, 2, \ldots \tag{10}$$

in which $\theta \in \Re^{n_\theta}$ is a parameter to be optimized, β_t, $t = 1, 2, \ldots$ is a sequence of step-sizes, ξ_t, $t = 1, 2, \ldots$ is a sequence of data samples and g is a function such that for random ξ the expected value of $-g(\theta, \xi)$ defines the direction towards the local minimum of a certain quality index, $J : \Re^{n_\theta} \mapsto \Re$. Typically this direction is the opposite of the gradient

$$Eg(\theta, \xi) = \nabla J(\theta).$$

For example, suppose a feedforward neural network is trained to approximate a function from given samples. Then θ is the vector of neural weights, J is the quality index, ξ represents an input–output pair, and g is the gradient computed by means of e.g., error backpropagation. In order to train the network, the input–output pairs may be sampled repeatedly, and its weights may be adjusted according to Eq. (10), which is exactly how the well established back-propagation algorithm works (Rumelhart, Hinton, & Williams, 1988).

For the sequence (10) to have convergence guarantees, a set of requirements needs to be fulfilled (Kushner & Yin, 1997). Among others, the sequence of step-sizes has to be vanishing at an appropriate pace:

$$\begin{cases} \sum_{t \geq 1} \beta_t = +\infty, \\ \sum_{t \geq 1} \beta_t^2 < +\infty. \end{cases} \tag{11}$$

Unfortunately the practical value of the above conditions is limited. Even if they are satisfied, the step-sizes may still be too large and hence make the convergence process unstable, or too small and hence yield a very slow convergence.

The fixed point method of step-size estimation addresses this issue by autonomously setting the step-sizes (Wawrzyński, 2010). The method is based on the following principles:

1. The process (10) is divided into parts such that within each part the step-size remains constant. The length, n, of a part starting at moment t is kept large enough to enable estimation of the expected value of $g(\theta_t, \xi)$ with sufficient accuracy.
2. The sums $G_{t,n}$ and $G^*_{t,n}$ are computed for each part beginning at time t and ending at time $t+n$

$$G_{t,n} = \sum_{i=0}^{n-1} g(\theta_{t+i}, \xi_{t+i}), \qquad G^*_{t,n} = \sum_{i=0}^{n-1} g(\theta_t, \xi_{t+i}). \tag{12}$$

   where the vector $G_{t,n}$ is proportional to the displacement of the point θ between instant t and $t+n$, and the vector $G^*_{t,n}$ is proportional to the displacement of the point θ if all the gradient estimators g are calculated using θ_t.
3. At the end of each part the discrepancy between $G^*_{t,n}$ and $G_{t,n}$ is registered. On average, it increases with the ratio of the step-size used and the optimal one giving fastest convergence. Therefore, if the discrepancy is too small, the step-size is being increased. If it is too large, it is decreased. If it is radically too large, then instability of the process is detected, the step-size is radically decreased, and θ moves back to the point where the part began.

This last principle is illustrated in Fig. 1. For a very small β, the vector θ_{t+i} moves through a flat slope of the function J. This results in small differences between $g(\theta_t, \xi_{t+i})$ and $g(\theta_{t+i}, \xi_{t+i})$, thereby, making the discrepancy between $G_{t,n}$ and $G^*_{t,n}$ small as well. The speed of the local minimum search can be increased with larger values of β. However, if β is too large, θ_{t+i} jumps over the minimum of the function to be optimized. This creates large differences between $g(\theta_t, \xi_{t+i})$ and $g(\theta_{t+i}, \xi_{t+i})$ manifested by large discrepancy between $G_{t,n}$ and $G^*_{t,n}$.

The papers (Wawrzyński, 2010; Wawrzyński & Papis, 2011) present a statistical analysis to formalize the above intuitive principles in a simple, unidimensional model. Since each $g(\theta_t, \xi_{t+i})$ in (12) is an unbiased estimate of $\nabla J(\theta_t)$, the value $\frac{1}{n} G^*_{t,n}$ is also an unbiased estimator of $\nabla J(\theta_t)$ and its quality increases with n. The model considers n that ensures signal-to-noise ratio of 1 in $\frac{1}{n} G^*_{t,n}$, namely

$$n = \frac{E_{\theta_t} \|g(\theta_t, \xi)\|^2}{\|\nabla J(\theta_t)\|^2}. \tag{13}$$

For n of that size, and constant $\alpha_0 \cong 0.15$ the difference

$$\alpha_0 \|G^*_{t,n}\|^2 - \|G^*_{t,n} - G_{t,n}\|^2 \tag{14}$$

is on average equal to 0 if the step-size is optimal for fast convergence of (10). If the step-size is smaller than optimal, the

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

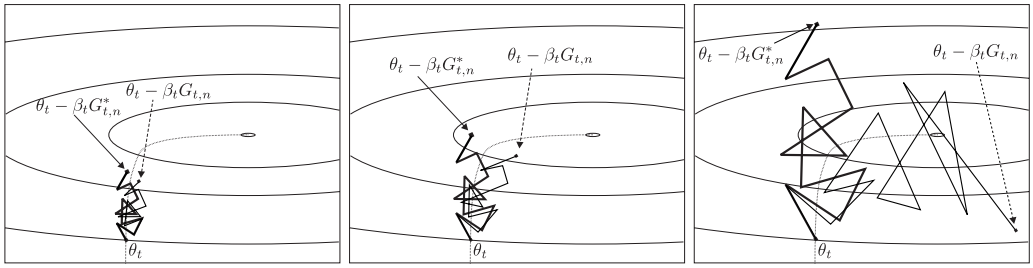


Fig. 1. The sketches depict contour lines of three different functions J with equal $\nabla J(\theta_t)$, β, and $g(\theta_t, \xi_{t+i})$ for $i \geq 0$. In n steps starting from t, the parameter in the J domain evolves from θ_t to $\theta_{t+n} = \theta_t - \beta_t G_{t,n}$. *Left:* Small curvature of J in relation to β, causes small difference of $-\beta_t G_{t,n}$ and $-\beta_t G^*_{t,n}$. *Middle:* Moderate curvature of J causes moderate difference between $-\beta_t G_{t,n}$ and $-\beta_t G^*_{t,n}$. *Right:* Very large curvature of J relatively to β causes instability of the optimization process which is manifested by divergence of $-\beta_t G_{t,n}$ and $-\beta_t G^*_{t,n}$ that grows with n.

first term of (14) is larger than the latter, and for too large step-size, the relation is opposite. Therefore, the difference (14) on average indicates the direction in which the step-size should be modified, i.e., increased or decreased. In other words, the step-size is stabilized at the best level when on average the discrepancy $\|G^*_{t,n} - G_{t,n}\|^2$ is stabilized at the value of $\alpha_0 \|G^*_{t,n}\|^2$. Also, if the term (14) with larger α_0 is applied to adjust the step-size, then the step-size is being stabilized at a higher level than one that is best for the purpose of static optimization (10). This last property will become useful in the next section.

In order to apply the same principle to multidimensional θ, we will consider only the direction of gradient estimate, that is we will project $G_{t,n}$ on $G^*_{t,n}$ and consider the discrepancy between $G^*_{t,n}$ and the projection. The indicator of step-size adjustment (14) will thus take the form

$$\alpha_0 \|G^*_{t,n}\|^2 - \|G^*_{t,n} - proj(G_{t,n}, G^*_{t,n})\|^2, \tag{15}$$

for

$$proj(G, G^*) = G^* G^T G^* / (\|G^*\|^2 + \epsilon) \tag{16}$$

and ϵ being a small number that prevents division by zero.

The discussion above leads to Algorithm 3 for autonomously setting the step-sizes. It divides the learning process into parts or *estimation periods* in which the step-size β remains fixed. In order to compute n according to (13) the fraction (L/M) estimates

$$E_{\theta_t} \|g(\theta_t, \xi)\|^2$$

such that L adds $\|g(\theta_t, \xi_{t+i})\|^2$ and M counts them. The fraction (L'/M') estimates $\|\nabla J(\theta_t)\|^2$ on the basis of the following property. Let ξ and ξ' be sampled independently from the same distribution. Then we have

$$\begin{aligned} E_{\theta_t} g(\theta_t, \xi)^T g(\theta_t, \xi') &= E_{\theta_t} g(\theta_t, \xi)^T E_{\theta_t} g(\theta_t, \xi') \\ &= \|\nabla J(\theta_t)\|^2. \end{aligned}$$

The estimator of $\|\nabla J(\theta_t)\|^2$ can thus take the form of the statistics

$$\frac{\sum_{i=1}^{n} \sum_{i'=0}^{i-1} g(\theta_t, \xi_{t+i})^T g(\theta_t, \xi_{t+i'})}{\sum_{i=1}^{n} i} = \frac{\sum_{i=1}^{n} g(\theta_t, \xi_{t+i})^T G^*_{t,i}}{\sum_{i=1}^{n} i}, \tag{17}$$

whose numerator and denominator are denoted by L' and M', respectively.

We now explain the algorithm in a step-wise manner for better understanding. In Line 3, it receives a new training example. In Lines 6–8, it initializes variables of the fixed point algorithm. In Lines 9–12, it updates estimators L/M and L'/M'. In Lines 13–16 it updates the values $G_{t,n}$, θ_{t+n}, $G^*_{t,n}$, and

$$h = \max_{i=1,\ldots,n} \|G^*_{t,i}\|^2,$$

respectively.

In Line 19, the algorithm checks whether the learning process is stable. Its instability is manifested by $\|G_{t,n} - G^*_{t,n}\|^2$ being larger than $h = \max_{1 \leq i \leq n} \|G^*_{t,i}\|^2$. This condition is additionally tightened in the k_0 initial periods. If instability is detected, the estimation period is reinitialized with halved step-size.

In Lines 21–22, the algorithm checks if the present estimation period has come to its end. It happens when

$$n \geq \frac{L/M}{L'/M'},$$

that is when the estimator of $Eg(\theta_t, \xi)$ has appropriate quality (13). The period is also finished when

$$n \geq \frac{3}{2} n',$$

where n' is the duration of the previous period. This prevents the algorithm from getting stuck in local minima ($Eg(\theta, \xi) = 0$ in a minimum and appropriate duration of an estimation period becomes infinite).

In the end of an estimation period, the step-size is modified (Line 25) and a new estimation period is started. Namely, the step-size is incremented proportionally to the statistics (15). The increment is scaled with the variable $\bar{h}$, that is determined in Line 26 as the maximal value of previous $\|G^*_{t,n}\|^2/(1 - \alpha_1)$. Line 24 prevents very small values of the scaling factor $\bar{h}$, for instance in the first period.

Every time a new estimation period begins, the estimators (L/M) and (L'/M') become obsolete and the amount of information they carry is decreased. In Lines 27–28, the algorithm decreases proportionally the weight of previous samples in comparison to forthcoming ones.

The algorithm updates the step-size after each estimation period to optimize the process (10). One such period is defined by the number of steps sufficient to estimate the direction of movement of θ with unity information-noise ratio. While various processes may differ substantially, the algorithm divides them to similar sequences of periods. Coefficients of the algorithm are chosen to make these sequences converge fast for all underlying processes (10) at its all stages.

In comparison to its version presented in Wawrzyński (2010), the algorithm is improved in two directions. First, it estimates $\|Eg(\theta_t, \xi)\|^2$ better by means of the statistics (17). Second, after each of first k_0 estimation periods it moves the process back to its origin; at this time only data is collected to estimate $\|Eg(\theta_0, \xi)\|^2$ and $E\|g(\theta_0, \xi)\|^2$. The goal of these modifications is to make the algorithm exploit data better and be robust to too large initial step-size.

5. Actor–critic with experience replay and the fixed point method of step-size estimation

This section discuses combination of the actor–critic algorithm with experience replay (ac&er) and the fixed point method of

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

Algorithm 3. The fixed point method of step-size estimation. The coefficients applied: $\alpha_1 = 0.9$, $\alpha_2 = 0.5$, $k_0 = 10$, initial $\beta = 1$.

FIXED_POINT

FP_INITIALIZE(θ_0)
Repeat
  Get ξ_{t+n}
  FP_LOOP$\big(g(\theta_t, \xi_{t+n}), g(\theta_{t+n}, \xi_{t+n})\big)$
Until some-stopping-condition

FP_INITIALIZE(θ_0)

Assign $t = n := 0$, $k := 1$, $L = L' = M = M' := 0$
Assign $G_{0,0} = G^*_{0,0} := 0$, $h = \bar{h} := 0$, $n' = +\infty$
Initialize θ_0, and β

FP_LOOP$\big(g(\theta_t, \xi_{t+n}), g(\theta_{t+n}, \xi_{t+n})\big)$

$L := L + \|g(\theta_t, \xi_{t+n})\|^2$
$M := M + 1$
$L' := \max\{L' + g(\theta_t, \xi_{t+n})^T G^*_{t,n}, 0\}$
$M' := M' + n$
$G_{t,n+1} := G_{t,n} + g(\theta_{t+n}, \xi_{t+n})$
$\theta_{t+n+1} := \theta_{t+n} - \beta g(\theta_{t+n}, \xi_{t+n})$
$G^*_{t,n+1} := G^*_{t,n} + g(\theta_t, \xi_{t+n})$
$h := \max\{h, \|G^*_{t,n+1}\|^2\}$
$n := n + 1$
newperiod := *false*
If $\|G_{t,n} - G^*_{t,n}\|^2 > h \vee (k < k_0 \wedge \|G_{t,n} - G^*_{t,n}\|^2 > 0.3h)$ then
  $\beta := \beta/2$, *newperiod* := *true*
If $\neg$*newperiod* $\wedge$ $n \geq 3$ $\wedge$ $L' > 0$
    $\wedge$ $n \geq \min\{(L/M)/(L'/M'), \frac{3}{2}n'\}$ then begin
  If $k \geq k_0$ then begin
    $\bar{h} := \max\{\bar{h}, \alpha_0\|G^*_{t,n}\|^2, \|proj(G_{t,n}, G^*_{t,n}) - G^*_{t,n}\|^2\}$
    $\beta := \beta \exp((\alpha_0\|G^*_{t,n}\|^2 - \|proj(G_{t,n}, G^*_{t,n}) - G^*_{t,n}\|^2)/\bar{h})$
    $\bar{h} := \max\{\alpha_1\bar{h}, (L/M)n/(1-\alpha_1)\}$
    $L := \alpha_1 L$, $M := \alpha_1 M$
    $L' := \alpha_2 L'$, $M' := \alpha_2 M'$
    $n' := n$
  end
  $t := t + n$
  $k := k + 1$
  newperiod := *true*
end
If *newperiod* then
  $n := 0$, $G_{t,0} = G^*_{t,0} := 0$, $h := h/2$

step-size adaptation. The original ac&er repeatedly selects data samples and computes the improvement direction estimators $\widehat{\phi}^r_i$ and $\widehat{\psi}^r_i$ to adjust the parameters of the actor and critic respectively. The combined algorithm optimizes the actor and critic parameters with separate modules that realize the fixed point method of step-size estimation (separate objects, as we would say in terms of object-oriented programming). Below, the generic terms of Section 4 are translated to the components of the ac&er algorithm. For the actor step-size optimization:

- ξ translates into the sequence of data that is required to compute $\widehat{\phi}^r_i$.
- the vector that undergoes optimization is the actor parameter, θ,
- its fixed value will be denoted by θ^*,
- $g(\theta, \xi)$ represents $-\widehat{\phi}^r_i(\theta, \upsilon)$; the minus results from the fact, that originally the fixed point method was meant for minimization while here the direction of optimization is opposite,
- $g(\theta^*, \xi)$ represents $-\widehat{\phi}^r_i(\theta^*, \upsilon)$.

For the critic step-size estimation we have:

- as previously, ξ translates into the sequence of data that is required to compute $\widehat{\psi}^r_i$.
- the vector being optimized is now the critic parameter, υ,
- its fixed value will be denoted by υ^*,
- $g(\theta, \xi)$ represents $-\widehat{\psi}^r_i(\theta, \upsilon)$; notice that now θ in the first term corresponds to υ in the second while θ in the second is only one of the entities that define this random vector,
- $g(\theta^*, \xi)$ represents $-\widehat{\psi}^r_i(\theta, \upsilon^*)$.

The above explanation suffices for straightforward combination of ac&er and the fixed point method of Section 4. This combination may be further enhanced by addressing the issues of overtraining, wrong fixed point initialization, and mutual adjustments of the actor and critic.

Overtraining. In general, a model is overtrained when it fits the data very well, but because the data is not sufficiently representative, the model does not describe the actual system that has generated the data. In the case of ac&er, the actor and critic may become overtrained when they are optimized on the basis of a short history of experiences. For instance, suppose the experience contains only several state–action pairs among which there are pairs with the same state, and the ac&er has enough resources for full optimization of the policy on the basis of these pairs. Then, it would readily make the better action in the same state infinitely more probable than the worse one. Consequently, the resulting policy would not try the worse actions at all which could lead to inability to explore certain area of the action space.

A way to overcome the above difficulty is to introduce regularization. Let

$$l : \mathcal{S} \times \mathfrak{R}^{n_\theta} \mapsto \mathfrak{R}_+$$

be a loss function that is continuous and assigns, for each state, zero to a bounded set of policy vectors that ensure exploration in a state, and an increasing positive score ($\in \mathfrak{R}_+$) to the policy vectors with growing distance from this set. Then, the policy vector adjustment takes the form

$$\theta := \theta + \beta^\theta \left(\widehat{\phi}^r_i(\theta, \upsilon) - \nabla l_\theta(s_i, \theta)\right).$$

As a result, even if the unrepresentative data pushes the policy vector outside the feasible exploration region, the loss function pulls it back.

Wrong fixed point initialization. In the initial period of the fixed point method, the algorithm only collects data and estimates $\|Eg(\theta, \xi)\|^2$ and $E\|g(\theta, \xi)\|^2$. It is assumed then that the g vectors are independently sampled from the appropriate distribution. If there are very few samples in the database, then this assumption is obviously violated and the estimator of $E\|g(\theta, \xi)\|^2$ becomes positively biased. As a result, the length of the estimation periods gets smaller. In order to stabilize the discrepancy between G and G^* for too short estimation periods, the step-sizes are being increased. This may lead to oscillations in the early stages of learning process.

A simple way to overcome this problem is to delay learning until a certain number of samples are collected in the database. To this end, we use the ν function to regulate the intensity of replaying experience, of the form

$$\nu(t) = \begin{cases} 0 & \iff t \leq \mathit{min\_capacity} \\ \mathit{comp\_steps} & \iff t > \mathit{min\_capacity} \end{cases} \tag{18}$$

for *min_capacity* > *comp_steps* > 0, i.e., $\nu(t)$ is a piecewise constant function that triggers the intensity of replaying experience given by *comp_steps* only after a certain time has elapsed denoted

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

Algorithm 4. Actor–Critic with experience replay and step-size estimation based on the fixed point method.

1: $t := 1$, Initialize θ and υ
2: `Actor.FP_INITIALIZE`(θ_0)
3: `Critic.FP_INITIALIZE`(υ_0)
4: Draw and execute an action, $a_t \sim \pi(\cdot; s_t, \theta)$
5: Assign $\pi_t = \pi(a_t; s_t, \theta)$
6: Repeat $\nu(t)$ times, begin
7: Draw $i \in \{t - N, t - N + 2, \ldots, t - 1\}$
8: Adjust θ along an estimator of $\phi(s_i, \theta, \upsilon)$:
9: `Actor.FP_LOOP`$\big(-\widehat{\phi}_i^r(\theta^*, \upsilon) + \nabla_{\theta^*} l(s_i, \theta^*),$
$-\widehat{\phi}_i^r(\theta, \upsilon) + \nabla_\theta l(s_i, \theta)\big)$
10: Adjust υ along an estimator of $\psi(s_i, \theta, \upsilon)$:
11: `Critic.FP_LOOP`$\big(-\widehat{\psi}_i^r(\theta, \upsilon^*), -\widehat{\psi}_i^r(\theta, \upsilon)\big)$
12: End
13: Register the tuple $\langle s_t, a_t, \pi_t, r_t, s_{t+1}\rangle$ in the database
14: Make sure only N most recent tuples remain in the database
15: Assign $t := t + 1$ and repeat from Line 4

by *min_capacity*. While *comp_steps* is defined by available computation power, *min_capacity* should be set to exclude the use of too large computational power to optimize the actor and critic vectors on the basis of too few data samples.

Mutual adjustments of the actor and critic. The fixed point method of step-size estimation is originally designed as a tool for stochastic optimization. Actor–critic with experience replay may be understood as an algorithm of that type in which the actor parameter, θ, is optimized to adapt it to the current critic, and the critic parameter, υ, is optimized to adapt it to the current actor. In both the cases, the ultimate goal of adaptation is constantly changing or running away. Consequently, the step-sizes should be estimated taking into account larger distance to the goal than currently perceived. The discrepancy between G and G^* in the fixed point method should be thus stabilized at a relatively higher level in ac&er as compared to stationary stochastic optimization. It is also inherent in Actor–Critics that the critic should adapt to the actor faster than *vice versa* (see e.g., Bhatnagar et al., 2009; Konda & Tsitsiklis, 2003).

The above remarks translate into the following guideline for the use of the fixed point method in ac&er: the discrepancies between G and G^*, regulated by the α_0 parameter, should be stabilized at relatively higher level for the critic than for the actor, and in both cases should be larger than optimal for stationary stochastic optimization.

The combined algorithm is presented in Algorithm 4. It defines the agent–environment interaction and the improvement directions for the actor and the critic. Their step-sizes and parameters are adjusted with the two fixed-point modules.

6. Experimental study

In the previous section, a reinforcement learning algorithm is introduced that combines experience replay with autonomous step-size estimation. In this section, it is confronted with two simulated control problems in multidimensional continuous spaces. The problems are selected to have general characteristics and the level of complexity corresponding to real-life robotic applications. The tasks analyzed are: (1) point reaching movement of octopus arm, and (2) cyclic running motion of half-cheetah.

6.1. Octopus arm

Octopus is well-known for exhibiting a high level of flexibility in controlling arm movements. The highly developed limbs of octopus make it capable of bending, stretching, shortening and twisting its arm at any point and in any direction with virtually unlimited degrees of freedom. Artificial octopus arm has found its niche in continuum robotics to replace conventional serial manipulators with smooth, continuous and flexible links (McMahan et al., 2006). However, their use is severely limited by existing motion planning and control algorithms. Classical approaches of controlling redundant manipulators by applying functional constraints to the additional degrees of freedom with null-space projection do not scale to continuum robots due to added computational complexity (Chiaverini, Oriolo, & Walker, 2008). While the theoretical foundation of continuum kinematics is in its nascent stages, learning based approaches provide an interesting alternative to control such robots. To this end, Engel, Szabó, and Volkinshtein (2005) used the Gaussian process temporal difference based reinforcement learning algorithm to successfully demonstrate various learning tasks of a simulated octopus arm model. Because the original state and action space of an octopus arm has huge number of dimensions, it often serves as a test bed for solutions to the problem of dimensionality in RL (Koutnik, Gomez, & Schmidhuber, 2010; Woolley & Stanley, 2010). Here the proposed reinforcement learning algorithm is tested for solving the task of reaching a given goal by an octopus arm; the state and action spaces are reduced but still rich enough to represent current position and dynamics of the arm.

6.1.1. Control problem

We are interested in learning a reactive control policy for the octopus arm to reach an arbitrary goal point starting from some random position. The octopus arm is represented by a planar simulator model composed of 10 quadrilateral compartments with each segment containing a longitudinal or a transverse muscle, as described by Yekutieli et al. (2005). The mass of each compartment is distributed in its four corners as point masses which are connected by linear damped springs representing muscles. Each compartment follows the muscular hydrostat principle of maintaining a constant volume (area in this case) assuming the muscles to be incompressible. The activation pattern in muscles acts as input to the model and the resulting position of point masses is obtained on the basis of the net force acting on the arm. The forces modeled in the simulation are of four types: (1) internal force generated by muscles, (2) vertical force due to gravity and buoyancy of arm in seawater, (3) external force produced by water drag, and (4) constraint force induced by change in pressure to keep the area of each compartment constant. Moreover, the base is non-rotating in the model, i.e., the first spring is fixed to the ground. The planar dynamic model of the octopus arm provides a trade-off between its fidelity and real-time processing overheads. The model is simulated with the use of software made available for 2006 RL competition (Octopus-sources, 2006) originally written in Java and rewritten for the experiments presented here to C#. The simulator worked in its standard setting with 10 compartments, fixed base, each muscle 1 unit long, and action duration 0.1 s.

We now define the state and action spaces along with the associated reward function to formulate the control problem of octopus arm as MDP for reinforcement learning.

State space. In this study, we define three moving frames in polar coordinates that are used to localize the octopus arm movement towards its goal. The first frame (r_1, ϕ_1) is located at the center of the point masses of all the quadrilateral compartments, the second one (r_2, ϕ_2) is located at the center of the point masses of last 5 quadrilateral compartments, and the third one (r_3, ϕ_3) is located at the center of mass of the last compartment. The first two frames are movable with respect to the arm as they may or may not be located inside the model depending upon its configuration (see Fig. 2).

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

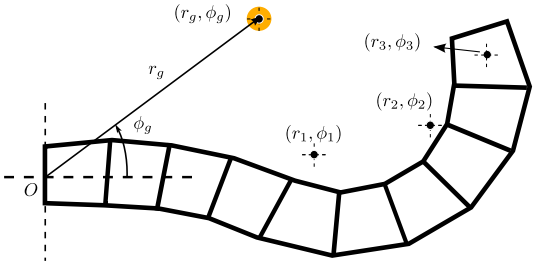


Fig. 2. Simulated octopus arm model with geometric description.

Table 1
State variables of octopus arm.

$s_0 = (r_g - 10)/3$	$s_4 = \dot{r_1}/0.01$	$s_8 = (r_3 - r_g)/3$
$s_1 = \phi_g$	$s_5 = \dot{\phi_1}/0.01$	$s_9 = \phi_3 - \phi_g$
$s_2 = (r_1 - r_g)/3$	$s_6 = (r_2 - r_g)/3$	$s_{10} = \text{sign}(r_3 - r_g)$
$s_3 = \phi_1 - \phi_g$	$s_7 = \phi_2 - \phi_g$	$s_{11} = \text{sign}(\phi_3 - \phi_g)$

Based on the three frames and the goal frame (r_g, ϕ_g), the state space of our model consists of 12 state variables given in Table 1. Each state variable is normalized to roughly cover the interval $[-1, 1]$. Majority of the variables define the position of three frames with reference to the goal frame. Additionally, variables s_4 and s_5 express velocity of the center of mass, and discrete variables s_{10} and s_{11} describe whether the last compartment is 'in front of' or 'behind' the target. While these variables are redundant in defining arm configuration, they are quite useful inputs to the control policy.

Action space. The action space consists of a set of 6 actions each of which pre-defines the activation level in the arm's muscles, as used in Engel et al. (2005). The action space is shown in Fig. 3.

Episodes and reward function. The learning task is carried out in episodes. An episode terminates with success when the last compartment touches the goal (which is transparent for other compartments). If this goal is not reached within 500 steps, the episode terminates. A new target position is sampled in the beginning of every episode within the area described by a circular region:

$$r_g \in (7.5, 9.5)$$
$$\phi_g \in (-\pi/4, \pi/4).$$

The position of the arm is initialized such that it is set in its default position, an action is chosen randomly from the set {Action 1, Action 2}, and it is applied to the arm for a number of steps chosen randomly from the set $\{1, \ldots, 100\}$.

The reward function is defined such that the controller is penalized with a negative score of -1 for all the learning steps in which the goal has not been reached. Moreover, the arm is rewarded for moving towards the goal in proportion to its speed. The reward function is given by:

$$r_t = \begin{cases} d_t - d_{t+1} - 1 & \text{if the target is not reached at } t+1 \\ d_t & \text{if the target is reached at } t+1. \end{cases} \quad (19)$$

The above term d_t denotes Euclidean distance between the tip of the arm and the goal position measured at instant t. The goal of the octopus arm is to maximize the reward function by reaching the goal position as quickly as possible. The total reward to be obtained within an episode is therefore equal to the difference between the initial distance of the last compartment to the goal and the number of steps in which the goal is reached.

6.1.2. Actor and critic structure

The actor and critic are based on feedforward neural networks, namely 2-layer perceptrons with sigmoidal neurons in their hidden layers and linear neurons in their output layers. The input of both the networks is the scaled state of the octopus arm. All neurons in the networks have a constant input (bias). The initial weights in the hidden layers are drawn randomly from the normal distribution $N(0, 1)$ and weights of the output layers are initially set to zero.

The actor is the combination of a neural network and a generator of discrete numbers. The network has N_A neurons in its hidden layer and 6 neurons in the output layer corresponding to the size of the discrete action set. Given state s and the observed neural network outputs

$$\mu_i(s, \theta), \quad i = 1 \ldots 6,$$

the likelihood function for choosing a discrete action, Action i, is given by:

$$P(\text{Action } i|s) = \frac{\exp(\mu_i(s, \theta)/10)}{\sum_{j=1}^{6} \exp(\mu_j(s, \theta)/10)}. \quad (20)$$

The exploration takes place as each action has a non-zero probability. This is ensured by the penalty function of the form

$$l(s, \theta) = \sum_{i=1}^{6} \max\{0, |\mu_i(s, \theta)| - 20\}^2.$$

As a result, the policy is not penalized for the network's outputs in the range $[-20, 20]$, which means that the best action is not more than $e^4 \approx 54$ times more probable than the worst one.

The critic is a 2-layer perceptron with N_C neurons in its hidden layer and one neuron in the output layer.

6.1.3. Experiments and results

The experiments of the learning control problem of an octopus arm with the proposed algorithm are now presented. The parameter setting of the actor–critic reinforcement learning algorithm is as follows: $N_A = 50, N_C = 100, \gamma = 0.98, b = 2, \alpha = \gamma = 0.98$, and $\rho = \lambda = 0.8$. The experience database contained $M = 10^4$ events. The ν function took the form (18) with *min_capacity* $= 1000$ and *comp_steps* $= 100$. The fixed point algorithm parameters were: $\alpha_1 = 0.9, \alpha_2 = 0.5, \alpha_0 = 0.2$ for the actor and $\alpha_0 = 0.3$ for the critic.

Fig. 4 shows different stages of the octopus arm in reaching the desired goal. Figs. 5–9 present the result of experiments performed with the above parameter setting. The actor–critic with experience replay and various constant step-sizes is compared against those estimated by the fixed point method. Fig. 5 presents learning curves (average rewards vs. episode number), Fig. 6 presents efficiency (ratio of runs in which the goal was reached), Figs. 7 and 8 present average duration of episodes, and Fig. 9 presents the step-sizes estimated by the fixed point method.

What is not demonstrated in the figures is that the learning process is unstable for constant step-sizes greater or equal to 0.1 for both the actor and the critic. For $\beta^\theta = \beta^\nu = 10^{-2}$, the algorithm does not converge and produces poor performance. Smaller step-sizes lead to better performance but for $\beta^\theta \leq 10^{-4}$ or $\beta^\nu \leq 10^{-4}$ the learning process becomes very slow. The fixed point method of adjusting step-sizes enables at least as good performance of the learning algorithm, at any stage of its operation, as given by any constant step-size. The training is completed (no further improvement is observed) after 240 episodes which is equivalent to 80 min of Octopus time.

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

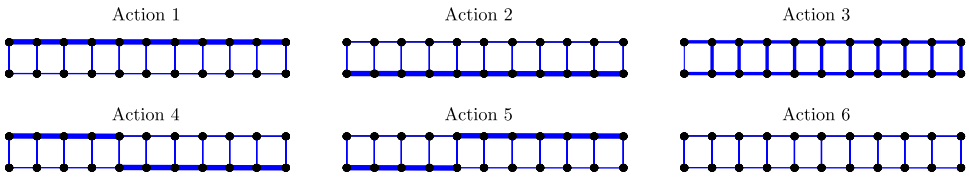


Fig. 3. Action set. The activated muscles are indicated by thick blue lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

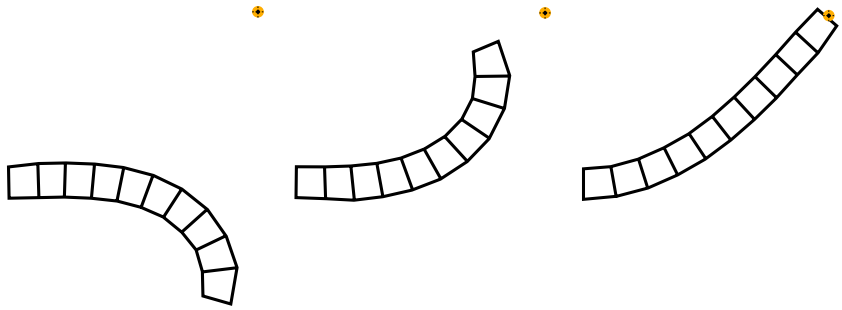


Fig. 4. Simulated goal reaching movement of octopus arm. *Left:* Arm position in the start of episode. *Middle:* Arm position trying to reach the goal. *Right:* Arm position when reached the goal. After learning, the arm reaches the goal in about 100 steps with maximum efficiency.

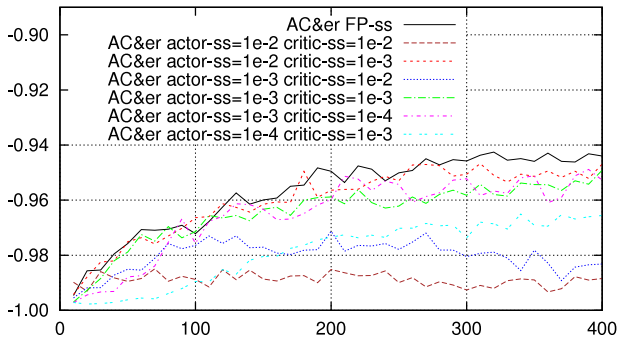


Fig. 5. *Octopus arm:* Average rewards vs. episode no. for actor–critic with experience replay and constant step-sizes against those estimated by the fixed point method. Each curve averages 10 runs.

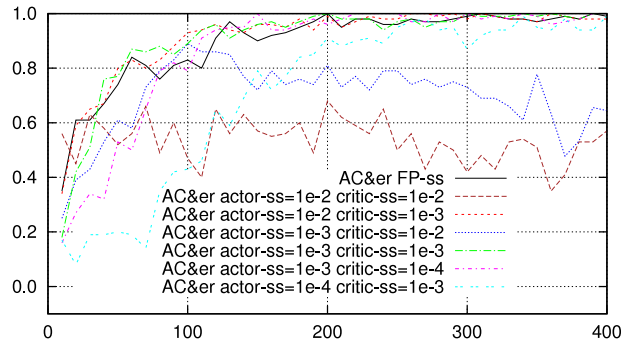


Fig. 6. *Octopus arm:* Efficiency vs. episode no. for actor–critic with experience replay, step-sizes estimated by the fixed point method, and constant step-sizes. Each curve averages 10 runs.

Fig. 9 shows the evolution of step-sizes driven by the fixed point method. They are averaged over runs and episodes according to the formula

$$\bar{\beta} = \exp\left(\frac{1}{N}\sum_{i}^{N} \ln \beta_i\right), \tag{21}$$

where N is the number of registered step-sizes. This way of averaging is better here than the normal one because it does not underweight very small step-sizes. It is seen that the actor step-size starts from a large initial value and then gradually decreases. The critic step-size initially decreases to a rather small value and then, after about 100 episodes, almost remains the same. This suggests that slowing down of the actor learning does not necessarily corresponds to slowing down of the critic learning.

6.2. Half-cheetah

Running in biological systems has largely inspired the development of legged robotics. Starting from the seminal spring-loaded hopping robots to the state-of-the-art quadrupeds and bipeds with compliant legs, a number of prototypes have been proposed to mimic versatility, speed and robustness in natural running. The dramatic progress is hindered by the lack of ability in robots to acquire these skills by online learning. In this section, we apply our proposed algorithm on a planar model of a large cat, called half-cheetah, to learn the complex skill of running.

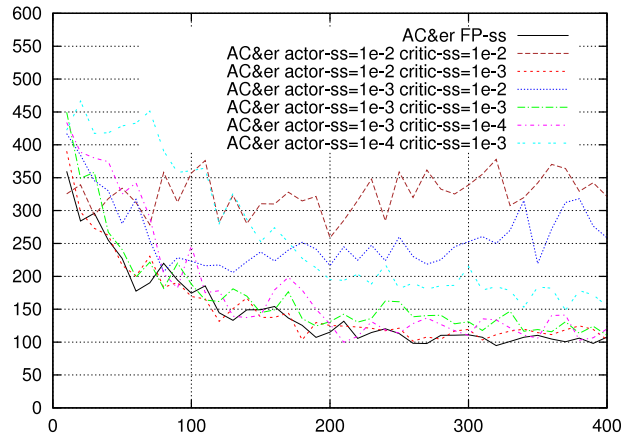


Fig. 7. *Octopus arm:* Duration in steps vs. episode no. for actor–critic with experience replay, step-sizes estimated by the fixed point method, and constant step-sizes. Each curve averages 10 runs.

6.2.1. Control problem

Half-Cheetah is a 6 degrees of freedom planar robot introduced in Wawrzyński (2009). It is composed of nine links, eight joints and two paws (see Fig. 10). The angles of the fourth and the fifth joint are fixed while others are controllable. The torque applied at each

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

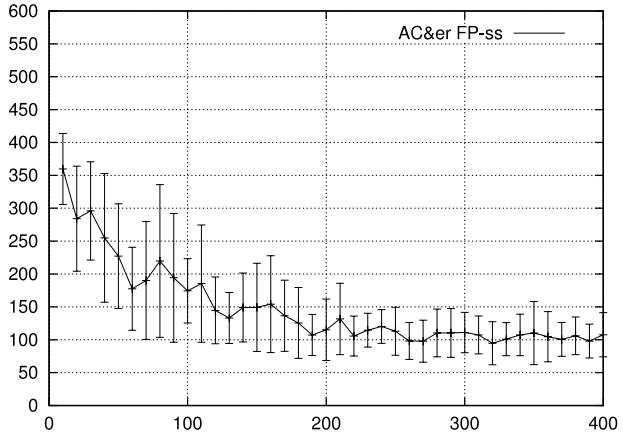


Fig. 8. *Octopus arm:* Duration in steps vs. episode no. for actor–critic with experience replay and step-sizes estimated by the fixed point method. The one-sigma limits are calculated to show run-to-run variability of trial averages .

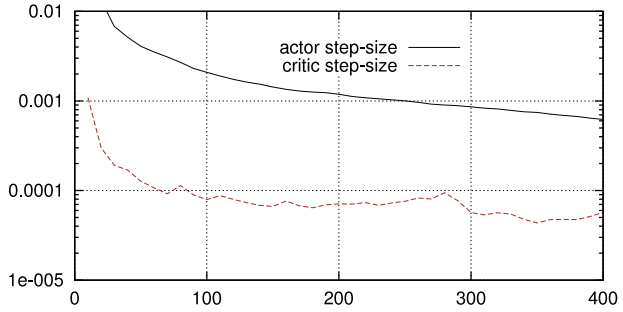


Fig. 9. *Octopus arm:* The fixed point method of step-size estimation vs. episode no. for actor–critic with experience replay. Each curve averages 10 runs.

joint acts as input to the model and the next position of the robot is obtained as output by integrating its dynamic equations of motion.

The control problem is to learn a reactive policy under the MDP framework to make half-cheetah run as fast as possible. State of half-cheetah is defined by 31 variables. The action space is continuous, contrary to that of octopus arm, with 6 dimensions each corresponding to one actuated joint independently. Learning is divided into episodes with an average duration of 250 steps, for 0.02 s duration of each step. The robot is mainly rewarded for its speed of moving forward. Other components are minor penalties for violating torque limits, joint limits, not moving the trunk in idle position and touching the ground with other body parts than paws. In this study, we use the simulator of half-cheetah, as reported in Wawrzyński (2009), with same experimental settings including its dynamics, state and action definition, and reward function.

6.2.2. Actor and critic structure

The actor, π, is composed of two parts: a neural network and a normal distribution. The input of the network is the state of half-cheetah. The network has a hidden layer with N_A sigmoidal neurons and six linear output neurons corresponding to the dimensionality of the action space. The output, $\mu(s;\theta)$, becomes a mean value of the normal distribution with covariance matrix equal to $I\sigma^2$. The distribution generates actions. Because each component of the action is projected on the interval $[-30, 30]$, the penalty function keeps each network output in this interval for a given state. It is of the form

$$l(s,\theta) = \sum_{i=1}^{6} 10^{-3} \max\{0, |\mu_i(s,\theta)| - 30\}^2.$$

The critic is a neural network of the same structure as the actor network with N_C neurons in its hidden layer. The initial weights in the hidden layers are drawn randomly from the normal distribution $N(0, 1)$ and weights of the output layers are initially set to zero.

6.2.3. Experiments and results

The experiments to make half-cheetah run are configured with following parameters: $N_A = 160, N_C = 160, \sigma = 5, \gamma = 0.99$, $b = 2, \alpha = \gamma = 0.99, \rho = \lambda = 0.9$. The experience database contained $M = 10^5$ events. The ν function took the form (18) with *min_capacity* $= 100$ and *comp_steps* $= 30$. The fixed point algorithm parameters were kept same as for octopus arm with: $\alpha_1 = 0.9, \alpha_2 = 0.5, \alpha_0 = 0.2$ for the actor and $\alpha_0 = 0.3$ for the critic.

Fig. 10 shows various stages of the learned running gait of half-cheetah. The cat robot starts from a standing position and first learns to move forward with the added noise in the control system. The awkward walk transforms into a trot gait which at the end of training becomes a smooth nimble run. The use of experience replay speeds up the learning process of running in proportion to the intensity of replaying computations.

In this study, we are interested in analyzing the effect of the proposed fixed-point method of step-size estimation on the learned policy. To this end, Figs. 11 and 12 compare the learning curve of the adaptive step-size estimation algorithm with the ones obtained by setting various constant step-sizes. Among the learning curves with constant step-sizes, the learning algorithm produces optimum results for $\beta^\theta = \beta^\upsilon = 10^{-5}$. For larger values, the policy evolves with a reasonable speed in the initial period but the ultimate performance is worse. Smaller step-sizes slow down the learning speed significantly. The fixed point method of estimating the step-sizes makes the learning algorithm perform better at every stage of the learning process than with any manually selected step-sizes. The algorithm required 3500 episodes (about 5 h of half-cheetah time) to learn to receive average reward of 6, which corresponds to a really nimble run.

Fig. 13 presents evolution of step-sizes driven by the fixed point method. They are averaged over runs and episodes according to formula (21). It is seen that the actor and critic step-sizes start descending from the same value and then gradually converge to about 5.10^{-6} and 10^{-6} respectively.

6.3. Discussion

The purpose of this experimental study is to evaluate the feasibility of autonomously determining the step-sizes for actor–critic class of reinforcement learning algorithms with experience replay. The results are promising: the augmented algorithm has been used to find control policies for two challenging problems similar to those encountered in robotics. The algorithm gave at least as good performance at any stage of the learning process as Actor–Critic with experience replay and constant step-sizes.

In all the experiments, the step-sizes of the fixed point method were initialized blindly, with a value of 1.0 (Algorithm 3, Line 8). As experiments with constant step-sizes showed, the value of 1.0 was very large and surely caused instability of the learning, which was identified by the algorithm (Line 19), and the step-sizes were halved (Line 20) until they ensured normal learning. Therefore, despite their large initial value, improper for the problems at hand, the step-sizes are always smaller than 1.0 in Figs. 9 and 13. Further, the step-sizes were autonomously adjusted with time (Algorithm 3, Line 25), which usually (but not always) meant their slow decrease.

The fixed point method of estimating step-sizes gave at least as good performance as the manually defined constant step-sizes used in the experiments. We argue that constant step-sizes generally do not ensure good performance as even their

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

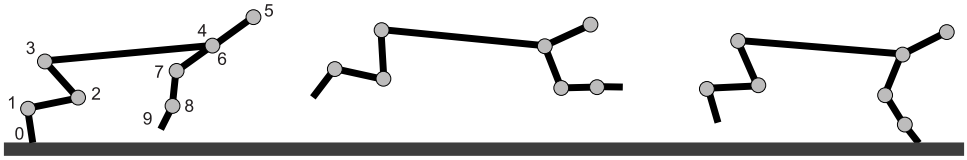


Fig. 10. Simulated run of Half-Cheetah. *Left:* Initial stance. *Middle:* Middle stance with feet in the air. *Right:* Landing stance. After learning, half-cheetah runs with a speed of about 6.5 m/s.

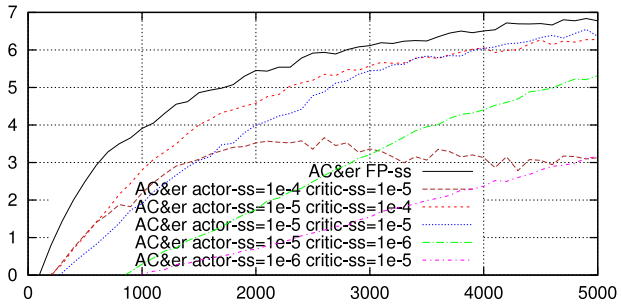


Fig. 11. *Half-Cheetah:* Average rewards vs. episode no. for actor–critic with experience replay and constant step-sizes against those estimated by the fixed point method. Each curve averages 10 runs .

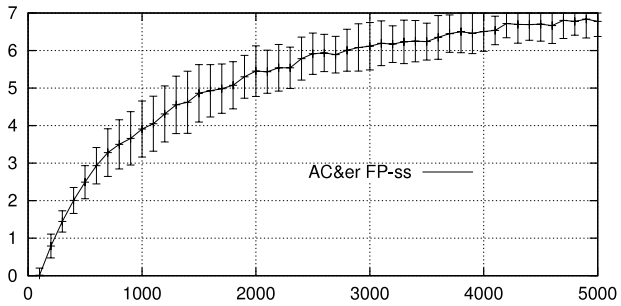


Fig. 12. *Half-Cheetah:* Average rewards vs. episode no. for actor–critic with experience replay and step-sizes estimated by the fixed point method. The one-sigma limits are calculated to show run-to-run variability of trial averages .

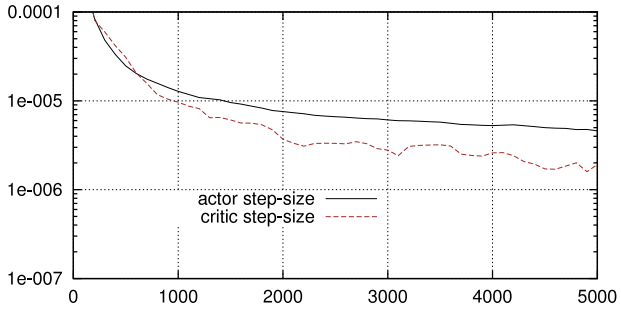


Fig. 13. *Half-Cheetah:* The fixed point method of step-size estimation vs. episode no. for actor–critic with experience replay. Each curve averages 10 runs.

handcrafted values yield slow learning in the early stages and fluctuations in the final stages. The fixed point method estimates large step-sizes initially and gradually decreases them during the course of learning. However, it has been observed in our supplementary experiments that the critic step-sizes that were twice larger than estimated by the fixed-point algorithm led to even better performance. Step-size estimation for the critic is an especially difficult problem as the critic learns mainly on the basis of values that it produces alone, thereby, going beyond the model of stochastic optimization for which the fixed-point algorithm along with most other methods of step-size estimation is designed. Still, the results presented here suggest the correctness of the principles that govern the fixed-point method.

Step-sizes are not the only input parameters of the learning algorithm. The rest of the parameters, however, either have the same values for both tasks or can be assigned by the experimenters based on their experience. The first category encompasses the parameters of the fixed-point method and parameter b of the randomized-truncated estimators (Eqs. (8) and (9)). The second category includes the parameters whose values are in principle problem-dependent, but the learning is not very sensitive to them and not much experience of an experimenter is required to set their proper values. Those are: (1) the sizes of the neural networks, N_A and N_C—the more difficult the control problem gets, the larger is the size of required networks, (2) the discount factor, γ, and the λ parameter—the more farsighted the control policy is required to be, the larger are the values of γ and λ, (3) the size, M, of the database with experience—it should be sufficiently large for this experience to be representative, (4) the intensity of experience replay, *comp_steps*, results from the available computer power.

In comparison to the parameters discussed above, the step-sizes have different nature: (1) the learning is quite sensitive to them, (2) their proper values differ orders of magnitude from one task to another, (3) for the same task they may differ substantially between stages of the learning. Their on-line estimation is therefore crucial for autonomy of learning.

7. Conclusions and future work

In this paper, an actor–critic reinforcement learning algorithm was introduced that combines experience replay (Wawrzyński, 2009) with the fixed point method of step-size estimation, improved upon its version presented in Wawrzyński (2010). Potentially problematic issues associated with this combination were addressed, namely overtraining, wrong fixed point initialization, and mutual adjustment of the actor and critic. The experimental study on octopus arm and half-cheetah revealed that the algorithm learned control policies in multidimensional continuous space within a short time. It is also autonomous in the sense that it achieved its goal essentially within a single run, without auxiliary experiments aiming at manual tuning of the problematic step-size parameters.

As the proposed approach has been designed for robotic applications, we plan its evaluation on a real humanoid walking robot. While the fixed point method of step-size estimation is justified by a simplified model and empirical performance, there is room for its further development and analysis of its convergence. This method has been applied here for optimization of the actor and critic step-sizes separately, while the learning of actor and of critic are coupled processes. The question whether the principles underlying the fixed point method can be formally extended to such coupling is another interesting subject of further research.

References

Abbeel, P., & Ng, A. Y. (2005). Exploration and apprenticeship learning in reinforcement learning. In *Proc. of the 22nd ICML* (pp. 1–8). ACM.

Adam, S., Busoniu, L., & Babuska, R. (2012). Experience replay for real-time reinforcement learning control. *IEEE Transactions on Systems, Man, and Cybernetics, Part C*, *42*(2), 201–212.

Barto, A. G., Sutton, R. S., & Anderson, C. W. (1983). Neuronlike adaptive elements that can learn difficult learning control problems. *IEEE Transactions on Systems, Man, and Cybernetics*, *13*, 834–846.

Please cite this article in press as: Wawrzyński, P., & Tanwani, A. K. Autonomous reinforcement learning with experience replay. Neural Networks (2012), doi:10.1016/j.neunet.2012.11.007

Prioritized Level Replay

Minqi Jiang [1 2] **Edward Grefenstette** [1 2] **Tim Rocktäschel** [1 2]

Abstract

Environments with procedurally generated content serve as important benchmarks for testing systematic generalization in deep reinforcement learning. In this setting, each *level* is an algorithmically created environment instance with a unique configuration of its factors of variation. Training on a prespecified subset of levels allows for testing generalization to unseen levels. What can be learned from a level depends on the current policy, yet prior work defaults to uniform sampling of training levels independently of the policy. We introduce *Prioritized Level Replay* (PLR), a general framework for selectively sampling the next training level by prioritizing those with higher estimated learning potential when revisited in the future. We show TD-errors effectively estimate a level's future learning potential and, when used to guide the sampling procedure, induce an emergent curriculum of increasingly difficult levels. By adapting the sampling of training levels, PLR significantly improves sample efficiency and generalization on Procgen Benchmark—matching the previous state-of-the-art in test return—and readily combines with other methods. Combined with the previous leading method, PLR raises the state-of-the-art to over 76% improvement in test return relative to standard RL baselines.

1. Introduction

Deep reinforcement learning (RL) easily overfits to training experiences, making generalization a key open challenge to widespread deployment of these methods. Procedural content generation (PCG) environments have emerged as a promising class of problems with which to probe and address this core weakness (Risi & Togelius, 2020; Chevalier-Boisvert et al., 2018; Cobbe et al., 2020a; Juliani et al., 2019; Zhong et al., 2020; Küttler et al., 2020). Unlike singleton environments, like the Arcade Learning Environment games (Bellemare et al., 2013), PCG environments take on algorithmically created configurations at the start of each training episode, potentially varying the layout, asset appearances, or even game dynamics. Each environment instance generated this way is called a *level*. In mapping an identifier, such as a random seed, to each level, PCG environments allow us to measure a policy's generalization to held-out test levels. In this paper, we assume only a blackbox generation process that returns a level given an identifier. We avoid the strong assumption of control over the generation procedure itself, explored by several prior works (see Section 6). Further, we assume levels follow common latent dynamics, so that in aggregate, experiences collected in individual levels reveal general rules governing the entire set of levels.

Despite its humble origin in games, the PCG abstraction proves far-reaching: Many control problems, such as teaching a robotic arm to stack blocks in a specific formation, easily conform to PCG. Here, each level may consist of a unique combination of initial block positions, arm state, and target formation. In a vastly different domain, Hanabi (Bard et al., 2020) also conforms to PCG, where levels map to initial deck orderings. These examples illustrate the generality of the PCG abstraction: Most challenging RL problems entail generalizing across instances (or levels) differing along some underlying factors of variation and thereby can be aptly framed as PCG. This highlights the importance of effective deep RL methods for PCG environments.

Many techniques have been proposed to improve generalization in the PCG setting (see Section 6), requiring changes to the model architecture, learning algorithm, observation space, or environment structure. Notably, these approaches default to uniform sampling of training levels. We instead hypothesize that the variation across levels implies that at each point of training, each level likely holds different potential for an agent to learn about the structure shared across levels to improve generalization. Inspired by this insight and selective sampling methods from active learning, we investigate whether sampling the next training level weighed by its learning potential can improve generalization.

We introduce Prioritized Level Replay (PLR), illustrated in Figure 1, a method for sampling training levels that exploits

[1]Facebook AI Research, London, United Kingdom [2]University College London, London, United Kingdom. Correspondence to: Minqi Jiang <msj@fb.com>.

Proceedings of the 38^{th} International Conference on Machine Learning, PMLR 139, 2021.

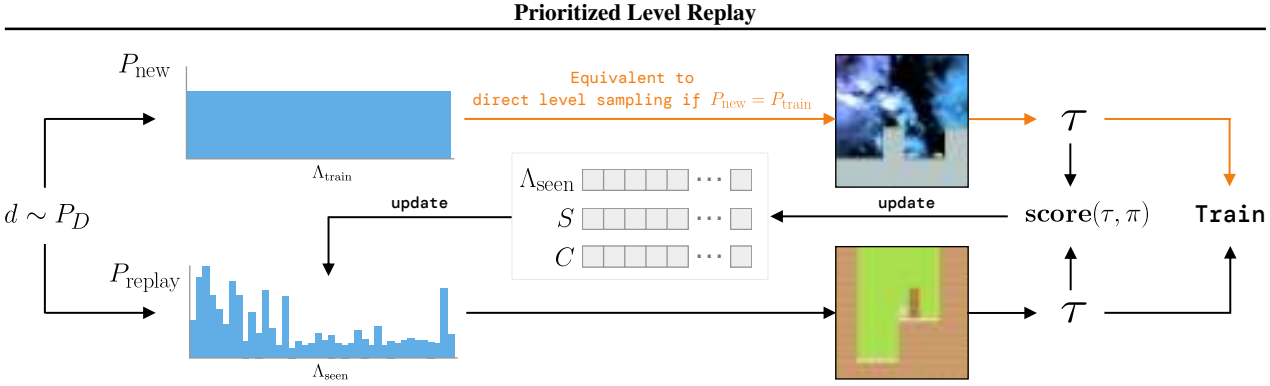


Figure 1. Overview of Prioritized Level Replay. The next level is either sampled from a distribution with support over unseen levels (top), which could be the environment's (perhaps implicit) full training-level distribution, or alternatively, sampled from the replay distribution, which prioritizes levels based on future learning potential (bottom). In either case, a trajectory τ is sampled from the next level and used to update the replay distribution. This update depends on the lists of previously seen levels Λ_{seen}, their latest estimated learning potentials S, and last sampled timestamps C.

the differences in learning potential among levels to improve both sample efficiency and generalization. PLR selectively samples the next training level based on an estimated learning potential of replaying each level anew. During training, our method updates scores estimating each level's learning potential as a function of the agent's policy and temporal-difference (TD) errors collected along the last trajectory sampled on that level. Our method then samples the next training level from a distribution derived from a normalization procedure over these level scores. PLR makes no assumptions about how the policy is updated, so it can be used in tandem with any RL algorithm and combined with many other methods such as data augmentation. Our method also does not assume any external, predefined ordering of levels by difficulty or other criteria, but instead derives level scores dynamically during training based on how the policy interacts with the environment. The only requirements are as follows—satisfied by almost any problem that can be framed as PCG, including RL environments implemented as seeded simulators: (i) Some notion of "level" exists, such that levels follow common latent dynamics; (ii) such levels can be sampled from the environment in an identifiable way; and (iii) given a level identifier, the environment can be set to that level to collect new experiences from it.

While previous works in off-policy RL devised effective methods to directly reuse *past* experiences for training (Schaul et al., 2016; Andrychowicz et al., 2017), PLR uses past experiences to inform the collection of *future* experiences by estimating how much replaying each level anew will benefit learning. Our method can thus be seen as a forward-view variation of prioritized experience replay, and an online counterpart to this off-policy method.

This paper makes the following contributions[1]: (i) We introduce a computationally-efficient algorithm for sampling levels during training based on an estimate of the future learning potential of collecting new experiences from each level; (ii) we show our method significantly improves generalization on 10 of 16 environments in Procgen Benchmark and two challenging MiniGrid environments; (iii) we demonstrate our method combines with the previous leading method to set a new state-of-the-art on Procgen Benchmark; and (iv) we show our method induces an implicit curriculum over training levels in sparse reward settings.

2. Background

In this paper, we refer to a *PCG environment* as any computational process that, given a level identifier (e.g. an index or a random seed), generates a *level*, defined as an environment instance exhibiting a unique configuration of its underlying factors of variation, such as layout, asset appearances, or specific environment dynamics (Risi & Togelius, 2020). For example, MiniGrid's MultiRoom environment instantiates mazes with varying numbers of rooms based on the seed (Chevalier-Boisvert et al., 2018). We refer to sampling a new trajectory generated from the agent's latest policy acting on a given level l as *replaying* that level l.

The level diversity of PCG environments makes them useful testbeds for studying the robustness and generalization ability of RL agents, measured by agent performance on unseen test levels. The standard test evaluation protocol for PCG environments consists of training the agent on a finite number of training levels Λ_{train}, and evaluating performance on unseen test levels Λ_{test}, drawn from the set of all levels. Training levels are sampled from an arbitrary distribution $P_{\text{train}}(l|\Lambda_{\text{train}})$. We call this training process *direct level sampling*. A common variation of this protocol sets Λ_{train} to the set of all levels, though in practice, the agent will still only effectively see a finite set of levels after training for a finite

[1]Our code is available at https://github.com/facebookresearch/level-replay.

number of steps. In the case of a finite training set, typically $P_{\text{train}}(l|\Lambda_{\text{train}}) = \textbf{Uniform}(l; \Lambda_{\text{train}})$. See Appendix C for the pseudocode outlining this procedure.

PCG environments naturally lend themselves to curriculum learning. Prior works have shown that directly altering levels to match their difficulty to the agent's abilities can improve generalization (Justesen et al., 2018; Dennis et al., 2020; Chevalier-Boisvert et al., 2018; Zhong et al., 2020). These findings further suggest the levels most useful for improving an agent's policy vary throughout the course of training. In this work, we consider how to automatically discover a curriculum that improves generalization for a general black-box PCG environment—crucially, without assuming any knowledge or control of how levels are generated (beyond providing the random seed or other indicial level identifier).

3. Prioritized Level Replay

In this section, we present *Prioritized Level Replay* (PLR), an algorithm for selectively sampling the next training level given the current policy, by prioritizing levels with higher estimated learning potential when replayed (that is, revisited). PLR is a drop-in replacement for the experience-collection process used in a wide range of RL algorithms. Algorithm 1 shows how it is straightforward to incorporate PLR into a generic policy-gradient training loop. For clarity, we focus on training on batches of complete trajectories (see Appendix C for pseudocode of PLR with T-step rollouts).

Our method, illustrated in Figure 1 and fully specified in Algorithm 2, induces a dynamic, nonparametric sampling distribution $P_{\text{replay}}(l|\Lambda_{\text{seen}})$ over previously visited training levels Λ_{seen} that prioritizes visited levels with higher learning potential based on properties of the agent's past trajectories. We refer to $P_{\text{replay}}(l|\Lambda_{\text{seen}})$ as the *replay distribution*. Throughout training, our method updates this replay distribution according to a heuristic score, assigning greater weight to visited levels with higher *future* learning potential. Using dynamic arrays S and C of equal length to Λ_{seen}, PLR tracks level scores $S_i \in S$ for each visited training level l_i based on the latest episode trajectory on l_i, as well as the episode count $C_i \in C$ at which each level $l_i \in \Lambda_{\text{seen}}$ was last sampled. Our method updates P_{replay} after each terminated episode by computing a mixture of two distributions, P_S, based on the level scores, and P_C, based on how long ago each level was last sampled:

$$P_{\text{replay}} = (1-\rho)\cdot P_S + \rho \cdot P_C, \tag{1}$$

where the staleness coefficient $\rho \in [0, 1]$ is a hyperparameter. We discuss how we compute level scores S_i, parameterizing the scoring distribution P_S, and the staleness distribution P_C, in Sections 3.1 and 3.2, respectively.

PLR chooses the next level at the start of every training episode by first sampling a replay-decision from a Bernoulli (or similar) distribution $P_D(d)$ to determine whether to replay a level sampled from the replay distribution $P_{\text{replay}}(l|\Lambda_{\text{seen}})$ or to experience a new, unseen level from Λ_{train}, according to some distribution $P_{\text{new}}(l|\Lambda_{\text{train}} \setminus \Lambda_{\text{seen}})$. In practice, for the case of a finite number of training levels, we implement P_{new} as a uniform distribution over the remaining unseen levels. For the case of a countably infinite number of training levels, we simulate P_{new} by sampling levels from P_{train} until encountering an unseen level. In our experiments based on a finite number of training levels, we opt to naturally anneal $P_D(d=1)$ as $|\Lambda_{\text{seen}}|/|\Lambda_{\text{train}}|$, so replay occurs more often as more training levels are visited.

The following sections describes how Prioritized Level Replay updates the replay distribution $P_{\text{replay}}(l|\Lambda_{\text{seen}})$, namely through level scoring and staleness-aware prioritization.

Algorithm 1 Policy-gradient training loop with PLR

Input: Training levels Λ_{train}, policy π_θ,
policy update function $\mathcal{U}(\mathcal{B}, \theta) \to \theta'$, and batch size N_b.
Initialize level scores S and level timestamps C
Initialize global episode counter $c \leftarrow 0$
Initialize the ordered set of visited levels $\Lambda_{\text{seen}} = \varnothing$
Initialize experience buffer $\mathcal{B} = \varnothing$
while training **do**
  $\mathcal{B} \leftarrow \varnothing$
  while collecting experiences **do**
    $\mathcal{B} \leftarrow \mathcal{B} \cup$ **collect_experiences**$(\Lambda_{\text{train}}, \Lambda_{\text{seen}}, \pi_\theta, S, C, c)$
    Using Algorithm 2
  end while
  $\theta \leftarrow \mathcal{U}(\mathcal{B}, \theta)$ *Update policy using collected experiences*
end while

Algorithm 2 Experience collection with PLR

Input: Training levels Λ_{train}, visited levels Λ_{seen}, policy π,
global level scores S, global level timestamps C, and
global episode counter c.
Output: A sampled trajectory τ
$c \leftarrow c + 1$
Sample replay decision $d \sim P_D(d)$
if $d = 0$ **and** $|\Lambda_{\text{train}} \setminus \Lambda_{\text{seen}}| > 0$ **then**
  Define new index $i \leftarrow |S| + 1$
  Sample $l_i \sim P_{\text{new}}(l|\Lambda_{\text{train}}, \Lambda_{\text{seen}})$ *Sample an unseen level*
  Add l_i to Λ_{seen}
  Add initial value $S_i = 0$ to S and $C_i = 0$ to C
else
  Sample $l_i \sim (1-\rho)\cdot P_S(l|\Lambda_{\text{seen}}, S) + \rho \cdot P_C(l|\Lambda_{\text{seen}}, C, c)$
  Sample a level for replay
end if
Sample $\tau \sim P_\pi(\tau|l_i)$
Update score $S_i \leftarrow \textbf{score}(\tau, \pi)$ and timestamp $C_i \leftarrow c$

3.1. Scoring Levels for Learning Potential

After collecting each complete episode trajectory τ on level l_i using policy π, our method assigns l_i a score $S_i = \textbf{score}(\tau, \pi)$ measuring the learning potential of replaying l_i in the future. We employ a function of the TD-error at timestep t, $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, as a proxy

for this learning potential. The expectation of the TD-error over next states is equivalent to the advantage estimate, and therefore higher-magnitude TD-errors imply greater discrepancy between expected and actual returns, making δ_t a useful measure of the learning potential in revisiting a particular state transition. To prioritize the learning potential of future experiences resulting from replaying a level, we use a scoring function based on the *average magnitude* of the Generalized Advantage Estimate (GAE; Schulman et al., 2016) over each of T time steps in the latest trajectory τ from that level:

$$S_i = \textbf{score}(\tau, \pi) = \frac{1}{T}\sum_{t=0}^{T}\left|\sum_{k=t}^{T}(\gamma\lambda)^{k-t}\delta_k\right|. \quad (2)$$

While the GAE at time t is most commonly expressed as the discounted sum of all 1-step TD-errors starting at t as in Equation 2, it is equivalent to an exponentially-discounted sum of all k-step TD-errors from t, with discount factor λ. By considering all k-step TD-errors, the GAE mitigates the bias introduced by the bootstrap term in 1-step TD-errors. The discount factor λ then controls the trade-off between bias and variance. Our scoring function considers the absolute value of the GAE, as we assume the learning potential grows with the magnitude of the TD-error irrespective of its sign. This also avoids opposite signed errors canceling out.

Another useful interpretation of Equation 2 comes from observing that the GAE magnitude at t is equivalent to the L1 value loss $|\hat{V}_t - V_t|$ under a policy-gradient algorithm that uses GAE for its own advantage estimates (and therefore value targets $\hat{V}_t$), as done in state-of-the-art implementations of PPO (Schulman et al., 2017) used in our experiments. Unless otherwise indicated, PLR refers to the instantiation of our algorithm with L1 value loss as the scoring function.

We further formulate the *Value Correction Hypothesis* to motivate our approach: In sparse reward settings, prioritizing the sampling of training levels with greatest average absolute value loss leads to a curriculum that improves both sample efficiency and generalization. We reason that on threshold levels (i.e. those at the limit of the agent's current abilities) the agent will see non-stationary returns (or value targets)—and therefore incur relatively high value errors—until it learns to solve them consistently. In contrast, levels beyond the agent's current abilities tend to result in stationary value targets signaling failure and therefore low value errors, until the agent learns useful, transferable behaviors from threshold levels. Prioritizing levels by value loss then naturally guides the agent along the expanding threshold of its ability—without the need for any externally provided measure of difficulty. We believe that learning behaviors systematically aligned with the inherent complexities of the environment in this way may lead to better generalization, and will seek to verify this empirically in Section 5.2.

While we provide principled motivations for our specific choice of scoring function, we emphasize that in general, the scoring function can be any approximation of learning potential based on trajectory values. Note that candidate scoring functions should asymptotically decrease with frequency of level visitation to avoid mode collapse of P_{replay} to a limited set of levels and possible overfitting. In Section 4, we compare our choice of the GAE magnitude, or equivalently, the L1 value loss, to alternative TD-error-based and uncertainty-based scoring approaches, listed in Table 1.

Given level scores, we use normalized outputs of a prioritization function h evaluated over these scores and tuned using a temperature parameter β to define the score-prioritized distribution $P_S(\Lambda_{\text{train}})$ over the training levels, under which

$$P_S(l_i|\Lambda_{\text{seen}}, S) = \frac{h(S_i)^{1/\beta}}{\sum_j h(S_j)^{1/\beta}}. \quad (3)$$

The function h defines how differences in level scores translate into differences in prioritization. The temperature parameter β allows us to tune how much $h(S)$ ultimately determines the resulting distribution. We make the design choice of using rank prioritization, for which $h(S_i) = 1/\text{rank}(S_i)$, where $\text{rank}(S_i)$ is the rank of level score S_i among all scores sorted in descending order. We also experimented with proportional prioritization ($h(S_i) = S_i$) as well as greedy prioritization (the level with the highest score receives probability 1), both of which tend to perform worse.

3.2. Staleness-Aware Prioritization

As the scores used to parameterize P_S are a function of the state of the policy at the time the associated level was last played, they come to reflect a gradually more off-policy measure the longer they remain without an update through replay. We mitigate this drift towards "off-policy-ness" by explicitly mixing the sampling distribution with a staleness-prioritized distribution P_C:

$$P_C(l_i|\Lambda_{\text{seen}}, C, c) = \frac{c - C_i}{\sum_{C_j \in C} c - C_j} \quad (4)$$

which assigns probability mass to each level l_i in proportion to the level's *staleness* $c - C_i$. Here, c is the count of total episodes sampled so far in training and C_i (referred to as the level's timestamp) is the episode count at which l_i was last sampled. By pushing support to levels with staler scores, P_C ensures no score drifts too far off-policy.

Plugging Equations 3 and 4 into Equation 1 gives us a replay distribution that is calculated as

$$P_{\text{replay}}(l_i) = (1-\rho)\cdot P_S(l_i|\Lambda_{\text{seen}}, S) + \rho \cdot P_C(l_i|\Lambda_{\text{seen}}, C, c).$$

Thus, a level has a greater chance of being sampled when its score is high or it has not been sampled for a long time.

4. Experimental Setting

We evaluate PLR on several PCG environments with various combinations of scoring functions and prioritization schemes, and compare to the most common direct level sampling baseline of $P_{\text{train}}(l|\Lambda_{\text{train}}) = \textbf{Uniform}(l; \Lambda_{\text{train}})$. We train and test on all 16 environments in the Procgen Benchmark on easy and hard difficulties, but focus discussion on the easy results, which allow direct comparison to several prior studies. We compare to UCB-DrAC (Raileanu et al., 2021), the state-of-the-art image augmentation method on this benchmark, and mixreg (Wang et al., 2020a), a recently introduced data augmentation method. We also compare to TSCL Window (Matiisen et al., 2020), which resembles PLR with an alternative scoring function using the slope of recent returns and no staleness sampling. For fair comparison, we also evaluate a custom TSCL Window variant that mixes in the staleness distribution P_C weighted by $\rho > 0$. Further, to demonstrate the ease of combining PLR with other methods, we evaluate UCB-DrAC using PLR for sampling training levels. Finally, we test the Value Correction Hypothesis on two challenging MiniGrid environments.

We measure episodic *test returns* per game throughout training, as well as the performance of the final policy over 100 unseen test levels of each game relative to PPO with uniform sampling. We also evaluate the mean normalized episodic test return and mean generalization gap, averaged over all games (10 runs each). We normalize returns according to Cobbe et al. (2019) and compute the generalization gap as train returns minus test returns. Thus, a larger gap indicates more overfitting, making it an apt measure of generalization. We assess statistical significance at $p = 0.05$, using the Welch t-test.

In line with the standard baseline for these environments, all experiments use PPO with GAE for training. For Procgen, we use the same ResBlock architecture as Cobbe et al. (2020a) and train for 25M total steps on 200 levels on the easy setting as in the original baselines. For MiniGrid, we use a 3-layer CNN architecture based on Igl et al. (2019), and provide approximately 1000 levels of each difficulty per environment during training. Detailed descriptions of the environments, architectures, and hyperparameters used in our experiments (and how they were set or obtained) can be found in Appendix A. See Table 1 for the full set of scoring functions investigated in our experiments.

Additionally, we extend PLR to support training on an unbounded number of levels by tracking a rolling, finite buffer of the top levels so far encountered by learning potential. Appendix B.3 reports the results of this extended PLR algorithm when training on the full level distribution of the MiniGrid environments featured in our main experiments.

Table 1. Scoring functions investigated in this work.

Scoring function	$\textbf{score}(\tau, \pi)$
Policy entropy	$\frac{1}{T}\sum_{t=0}^{T}\sum_{a}\pi(a,s_t)\log\pi(a,s_t)$
Policy min-margin	$\frac{1}{T}\sum_{t=0}^{T}(\max_a \pi(a,s_t) - \max_{a\neq\max_a\pi(a,s_t)}\pi(a,s_t))$
Policy least-confidence	$\frac{1}{T}\sum_{t=0}^{T}(1-\max_a\pi(a,s_t))$
1-step TD error	$\frac{1}{T}\sum_{t=0}^{T}\lvert\delta_t\rvert$
GAE	$\frac{1}{T}\sum_{t=0}^{T}\sum_{k=t}^{T}(\gamma\lambda)^{k-t}\delta_k$
GAE magnitude (L1 value loss)	$\frac{1}{T}\sum_{t=0}^{T}\left\lvert\sum_{k=t}^{T}(\gamma\lambda)^{k-t}\delta_k\right\rvert$

5. Results and Discussion

Our main findings are that (i) PLR significantly improves both sample efficiency and generalization, attaining the highest normalized mean test and train returns and mean reduction in generalization gap on Procgen out of all individual methods evaluated, while matching UCB-DrAC in test improvement relative to PPO; (ii) alternative scoring functions lead to inconsistent improvements across environments; (iii) PLR combined with UCB-DrAC sets a new state-of-the-art on Procgen; and (iv) PLR induces an implicit curriculum over training levels, which substantially aids training in two challenging MiniGrid environments.

5.1. Procgen Benchmark

Our results, summarized in Figure 2, show PLR with rank prioritization ($\beta = 0.1$, $\rho = 0.1$) leads to the largest statistically significant gains in mean normalized test and train returns and reduction in generalization gap compared to uniform sampling, outperforming all other methods besides UCB-DrAC + PLR. PLR combined with UCB-DrAC sees the most drastic improvements in these metrics. As reported in Table 2, UCB-DrAC + PLR yields a 76% improvement in mean test return relative to PPO with uniform sampling, and a 28% improvement relative to the previous state-of-the-art set by UCB-DrAC. While PLR with rank prioritization leads to statistically significant gains in test return on 10 of 16 environments and proportional prioritization, on 11 of 16 games, we prefer rank prioritization: While we find the two comparable in mean normalized returns, Figure 3 shows rank prioritization results in higher mean *unnormalized* test returns and a significantly lower mean generalization gap, averaged over all environments.

Further, Figure 3 shows that gains only occur when P_{replay} considers *both* level scores and staleness ($0 < \rho < 1$), highlighting the importance of staleness-based sampling in keeping scores from drifting off-policy. Lastly, we also

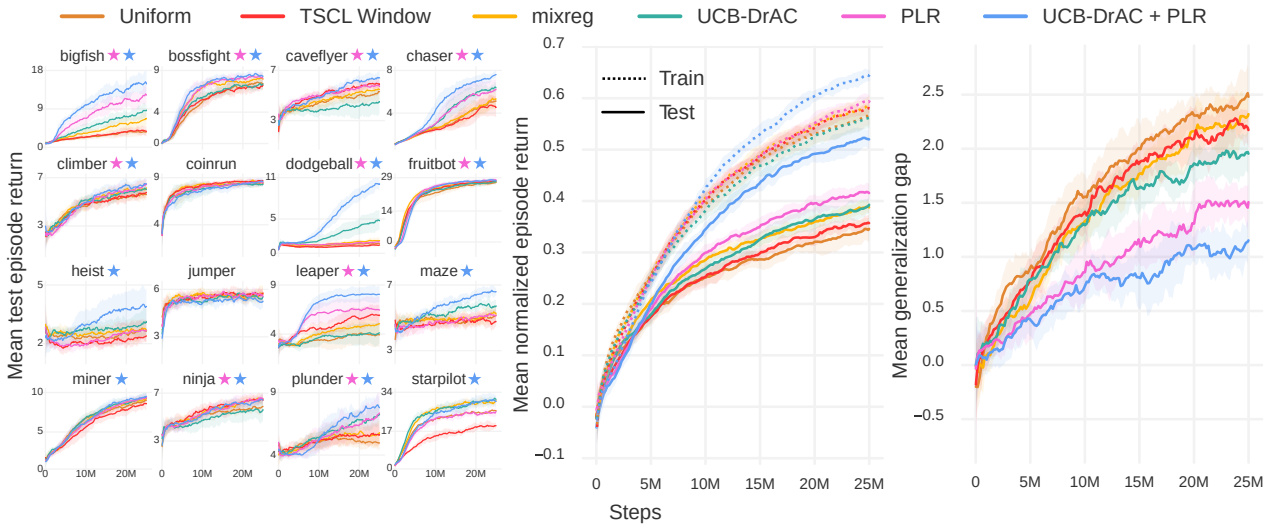


Figure 2. Left: Mean episodic test returns (10 runs) of each method. Each colored ★ indicates statistically significant ($p < 0.05$) gains in final test performance or sample complexity along the curve, relative to uniform sampling, for the PLR-based method of the same color. Center: Mean normalized train and test returns averaged across all games. Right: Mean generalization gaps averaged across all games.

benchmarked PLR on the hard setting against the same set of methods, where it again leads with 35% greater test returns relative to uniform sampling and 83% greater test returns when combined with UCB-DrAC. Figures 10–18 and Table 4 in Appendix B report additional details on these results.

The alternative scoring metrics based on TD-error and classifier uncertainty perform inconsistently across games. While certain games, such as BigFish, see improved sample-efficiency and generalization under various scoring functions, others, such as Ninja, see no improvement or worse, degraded performance. See Figure 3 for an example of this inconsistent effect across games. We find the best-performing variant of TSCL Window does not incorporate staleness information ($\rho = 0$) and similarly leads to inconsistent outcomes across games at test time, notably significantly worsening performance on StarPilot, as seen in Figure 2, and increasing the generalization gap on some environments as revealed in Figure 15 of Appendix B.

5.2. MiniGrid

We provide empirical support for the Value Correction Hypothesis (defined in Section 3) on two challenging MiniGrid environments, whose levels fall into discrete difficulties (e.g. by number of rooms to be traversed). In both, PLR with rank prioritization significantly improves sample efficiency and generalization over uniform sampling, demonstrating our method also works well in discrete state spaces. We find a staleness coefficient of $\rho = 0.3$ leads to the best test performance on MiniGrid. The top row of Figure 4 summarizes these results.

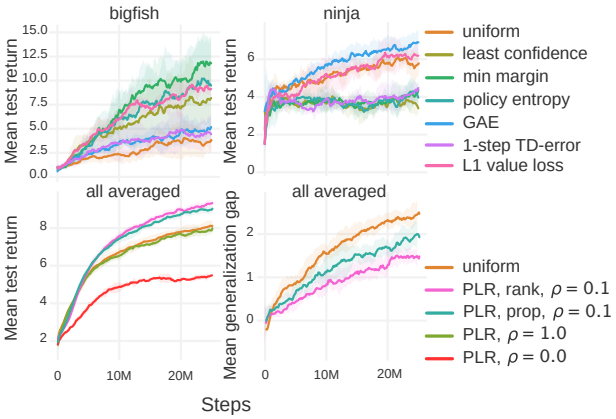


Figure 3. Top: Two example Procgen environments, between which all scoring functions except L1 value loss show inconsistent improvements to test performance (rank prioritization, $\beta = 0.1$, $\rho = 0.3$). This inconsistency holds across settings in our grid search. Bottom: Mean unnormalized episodic test returns (left) and mean generalization gap (right) for various PLR settings.

To test our hypothesis, we bin each level into its corresponding difficulty, expressed as ascending, discrete values (note that PLR does not have access to this privileged information). In the bottom row of Figure 4, we see how the expected difficulty of levels sampled using PLR changes during training for each environment. We observe that as P_{replay} is updated, levels become sampled according to an implicit curriculum over the training levels that prioritizes progressively harder levels. Of particular note, PLR seems to struggle to discover a useful curriculum for around the first 4,000 updates on ObstructedMazeGamut-Medium, at which point it discovers

Table 2. Test returns of policies trained using each method with its best hyperparameters. Following Raileanu et al. (2021), the reported mean and standard deviations per environment are computed by evaluating the final policy's average return on 100 test episodes, aggregated across multiple training runs (10 runs for Procgen Benchmark and 3 for MiniGrid, each initialized with a different training seed). Normalized test returns per run are computed by dividing the average test return per run for each environment by the corresponding average test return of the uniform-sampling baseline over all runs. We then report the means and standard deviations of normalized test returns aggregated across runs. We report the normalized return statistics for Procgen and MiniGrid environments separately. Bolded methods are not significantly different from the method with highest mean, unless all are, in which case none are bolded.

Environment	Uniform	TSCL	mixreg	UCB-DrAC	PLR	UCB-DrAC + PLR
BigFish	3.7 ± 1.2	4.3 ± 1.3	6.9 ± 1.6	8.7 ± 1.1	10.9 ± 2.8	$\mathbf{14.3 \pm 2.1}$
BossFight	7.7 ± 0.4	7.4 ± 0.8	8.1 ± 0.7	7.7 ± 0.7	$\mathbf{8.9 \pm 0.4}$	$\mathbf{8.8 \pm 0.8}$
CaveFlyer	5.4 ± 0.8	$\mathbf{6.3 \pm 0.6}$	6.0 ± 0.6	4.6 ± 0.9	$\mathbf{6.3 \pm 0.5}$	$\mathbf{6.8 \pm 0.7}$
Chaser	5.2 ± 0.7	4.9 ± 1.0	5.7 ± 1.1	6.8 ± 0.9	6.9 ± 1.2	$\mathbf{8.0 \pm 0.6}$
Climber	5.9 ± 0.6	6.0 ± 0.8	$\mathbf{6.6 \pm 0.7}$	$\mathbf{6.4 \pm 0.9}$	$\mathbf{6.3 \pm 0.8}$	$\mathbf{6.8 \pm 0.7}$
CoinRun	8.6 ± 0.4	$\mathbf{9.2 \pm 0.2}$	8.6 ± 0.3	8.6 ± 0.4	8.8 ± 0.5	$\mathbf{9.0 \pm 0.4}$
Dodgeball	1.7 ± 0.2	1.2 ± 0.4	1.8 ± 0.4	5.1 ± 1.6	1.8 ± 0.5	$\mathbf{10.3 \pm 1.4}$
FruitBot	27.3 ± 0.9	27.1 ± 1.6	27.7 ± 0.8	27.0 ± 1.3	28.0 ± 1.3	27.6 ± 1.5
Heist	2.8 ± 0.9	2.5 ± 0.6	2.7 ± 0.4	3.2 ± 0.7	2.9 ± 0.5	$\mathbf{4.9 \pm 1.3}$
Jumper	5.7 ± 0.4	6.1 ± 0.6	6.1 ± 0.3	5.6 ± 0.5	5.8 ± 0.5	5.9 ± 0.3
Leaper	4.2 ± 1.3	6.4 ± 1.2	5.2 ± 1.1	4.4 ± 1.4	6.8 ± 1.2	$\mathbf{8.7 \pm 1.0}$
Maze	5.5 ± 0.4	5.0 ± 0.3	5.4 ± 0.5	6.2 ± 0.5	5.5 ± 0.8	$\mathbf{7.2 \pm 0.8}$
Miner	8.7 ± 0.7	8.9 ± 0.6	9.5 ± 0.4	$\mathbf{10.1 \pm 0.6}$	$\mathbf{9.6 \pm 0.6}$	$\mathbf{10.0 \pm 0.5}$
Ninja	6.0 ± 0.4	$\mathbf{6.8 \pm 0.5}$	$\mathbf{6.9 \pm 0.5}$	5.8 ± 0.8	$\mathbf{7.2 \pm 0.4}$	$\mathbf{7.0 \pm 0.5}$
Plunder	5.1 ± 0.6	5.9 ± 1.1	5.7 ± 0.5	$\mathbf{7.8 \pm 0.9}$	$\mathbf{8.7 \pm 2.2}$	$\mathbf{7.7 \pm 0.9}$
StarPilot	26.8 ± 1.5	19.8 ± 3.4	$\mathbf{32.7 \pm 1.5}$	$\mathbf{31.7 \pm 2.4}$	27.9 ± 4.4	29.6 ± 2.2
Normalized test returns (%)	100.0 ± 4.5	103.0 ± 3.6	113.8 ± 2.8	129.8 ± 8.2	128.3 ± 5.8	$\mathbf{176.4 \pm 6.1}$
MultiRoom-N4-Random	0.80 ± 0.04	–	–	–	$\mathbf{0.81 \pm 0.01}$	–
ObstructedMazeGamut-Easy	0.53 ± 0.04	–	–	–	$\mathbf{0.85 \pm 0.04}$	–
ObstructedMazeGamut-Med	0.65 ± 0.01	–	–	–	$\mathbf{0.73 \pm 0.07}$	–
Normalized test returns (%)	100.0 ± 2.5	–	–	–	$\mathbf{124.3 \pm 4.7}$	–

a curriculum that gradually assigns more weight to harder levels. This curriculum enables PLR with access to only $6{,}000$ training levels to attain even higher mean test returns than the uniform-sampling baseline with access to the full set of training levels, of which there are roughly 4 billion (so our training levels constitute 0.00015% of the total number).

We further tested an extended version of PLR that trains on the full level distribution on these two environments by tracking a buffer of levels with the highest estimated learning potential. We find it outperforms uniform sampling with access to the full level distribution. These additional results are presented in Appendix B.3.

6. Related Work

Several methods for improving generalization in deep RL adapt techniques from supervised learning, including stochastic regularization (Igl et al., 2019; Cobbe et al., 2020a), data augmentation (Kostrikov et al., 2020; Raileanu et al., 2021; Wang et al., 2020a), and feature distillation (Igl et al., 2020; Cobbe et al., 2020b). In contrast, PLR modifies only how the next training level is sampled, thereby easily combining with any model or RL algorithm.

The selective-sampling performed by PLR makes it a form of active learning (Cohn et al., 1994; Settles, 2009). Our work also echoes ideas from Graves et al. (2017), who train a multi-armed bandit to choose the next task in multi-task supervised learning, so to maximize gradient-based progress signals. Sharma et al. (2018) extend these ideas to multi-task RL, but add the additional requirement of knowing a maximum target return for each task a priori. Zhang et al. (2020b) use an ensemble of value functions for selective goal sampling in the off-policy continuous control setting, requiring prior knowledge of the environment structure to generate candidate goals. Unlike PLR, these methods assume the ability to sample tasks or levels based on their structural properties, an assumption that does not typically hold for PCG simulators. Instead, our method automatically uncovers similarly difficult levels, giving rise to a curriculum without prior knowledge of the environment.

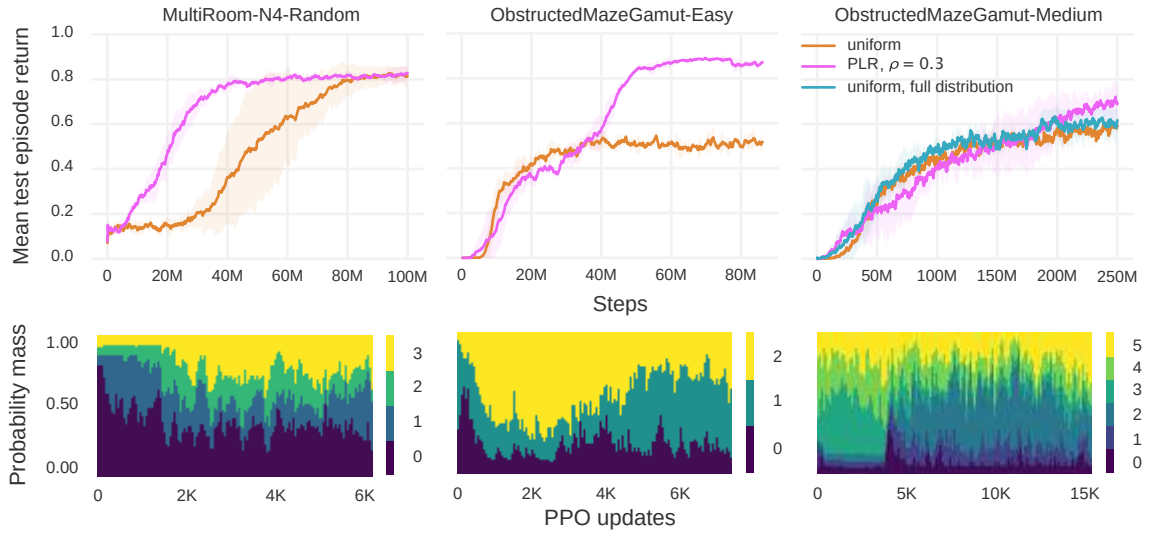


Figure 4. Top: Mean episodic test returns of PLR and the uniform-sampling baseline on MultiRoom-N4-Random (4 runs), ObstructedMazeGamut-Easy (3 runs), and ObstructedMazeGamut-Medium (3 runs). Bottom: The probability mass assigned to levels of varying difficulty over the course of training in a single, randomly selected run for the respective environment.

A recent theme in the PCG setting explores adaptively generating levels to facilitate learning (Sukhbaatar et al., 2017; Wang et al., 2019; 2020b; Khalifa et al., 2020; Akkaya et al., 2019; Campero et al., 2020; Dennis et al., 2020). Unlike these approaches, our method does not assume control over level generation, requiring only the ability to replay previously visited levels. These methods also require parameterizing level generation with additional learning modules. In contrast, our approach does not require such extensions of the environment, for example including teacher-specific action spaces in the case of Campero et al. (2020). Most similar to our method, Matiisen et al. (2020) proposes a teacher-student curriculum learning (TSCL) algorithm that samples training levels by considering the change in episodic returns per level, though they neither design nor test the method for generalization. As shown in Section 5.1, it provides inconsistent benefits at test time. Further, unlike TSCL, PLR does not assume access to all levels at the start of training, and as we show in Appendix B.3, PLR can be extended to improve sample efficiency and generalization by training on an unbounded number of training levels.

Like our method, Schaul et al. (2016) and Kapturowski et al. (2019) use TD-errors to estimate learning potential. While these methods make use of TD-errors to prioritize learning from *past* experiences, our method uses such estimates to prioritize revisiting levels for generating entirely new *future* experiences for learning.

Generalization requires sufficient exploration of environment states and dynamics. Thus, recent exploration strategies (e.g. Raileanu & Rocktäschel, 2020; Campero et al., 2020; Zhang et al., 2020a; Zha et al., 2021) shown to benefit simple PCG settings are complementary to the aims of this work. However, as these studies focus on PCG environments with low-dimensional state spaces, whether such methods can be successfully applied to more complex PCG environments like Procgen Benchmark remains to be seen. If so, they may potentially combine with PLR to yield additive improvements. We believe the interplay between such exploration methods and PLR to be a promising direction for future research.

7. Conclusion and Future Work

We introduced Prioritized Level Replay (PLR), an algorithm for selectively sampling the next training level in PCG environments based on the estimated learning potential of revisiting each level for the current policy. We showed that our method remarkably improves both the sample efficiency and generalization of deep RL agents in PCG environments, including the majority of environments in Procgen Benchmark and two challenging MiniGrid environments. We further combined PLR with the prior leading method to set a new state-of-the-art on Procgen Benchmark. Further, on MiniGrid environments, we showed PLR induces an emergent curriculum of increasingly more difficult levels.

The flexibility of the PCG abstraction makes PLR applicable to many problems of practical importance, for example, robotic object manipulation tasks, where domain randomized environment instances map to the notion of levels. We believe PLR may even be applicable to singleton environments, given a procedure for generating variations of the underlying MDP as a function of a level identifier, for ex-

ample, by varying the starting positions of entities. Another natural extension of PLR is to adapt the method to operate in the goal-conditioned setting, by incorporating goals into the level parameterization.

Despite the wide applicability of PCG and consequently PLR, not all problem domains can be effectively represented in seed-based simulation. Many real world problems require transfer into domains too complex to be adequately captured by simulation, such as car driving, where realizing a completely faithful simulation would entail solving the very same control problem of interest, creating a chicken-and-egg dilemma. Further, environment resets are not universally available, such as in the continual learning setting, where the agent interacts with the environment without explicit episode boundaries—arguably, a more realistic interaction model for a learning agent deployed in the wild.

Still, pre-training in simulation with resets can nevertheless benefit such settings, where the target domain is rife with open-ended complexity and where resets are unavailable, especially as training through real-world interactions can be slow, expensive, and precarious. In fact, in practice, we almost exclusively train deep RL policies in simulation for these reasons. As PLR provides a simple method to more fully exploit the simulator for improved test-time performance, we believe PLR can also be adapted to improve learning in these settings.

We further note that while we empirically demonstrated that the L1 value loss acts as a highly effective scoring function, there likely exist even more potent choices. Directly learning such functions may reveal even better alternatives. Lastly, combining PLR with various exploration strategies may further improve test performance in hard exploration environments. We look forward to investigating each of these promising directions in future work, prioritized accordingly, by learning potential.

Acknowledgements

We thank Roberta Raileanu, Heinrich Küttler, and Jakob Foerster for useful discussions and feedback on this work, and our anonymous reviewers, for their recommendations on improving this paper.

References

Akkaya, I., Andrychowicz, M., Chociej, M., Litwin, M., McGrew, B., Petron, A., Paino, A., Plappert, M., Powell, G., Ribas, R., Schneider, J., Tezak, N., Tworek, J., Welinder, P., Weng, L., Yuan, Q., Zaremba, W., and Zhang, L. Solving rubik's cube with a robot hand. *CoRR*, abs/1910.07113, 2019. URL http://arxiv.org/abs/1910.07113.

Andrychowicz, M., Wolski, F., Ray, A., Schneider, J., Fong, R., Welinder, P., McGrew, B., Tobin, J., Abbeel, O. P., and Zaremba, W. Hindsight experience replay. In *Advances in neural information processing systems*, pp. 5048–5058, 2017.

Bard, N., Foerster, J. N., Chandar, S., Burch, N., Lanctot, M., Song, H. F., Parisotto, E., Dumoulin, V., Moitra, S., Hughes, E., Dunning, I., Mourad, S., Larochelle, H., Bellemare, M. G., and Bowling, M. The hanabi challenge: A new frontier for AI research. *Artif. Intell.*, 280:103216, 2020. doi: 10.1016/j.artint.2019.103216. URL https://doi.org/10.1016/j.artint.2019.103216.

Bellemare, M. G., Naddaf, Y., Veness, J., and Bowling, M. The arcade learning environment: An evaluation platform for general agents. *Journal of Artificial Intelligence Research*, 47:253–279, Jun 2013. ISSN 1076-9757. doi: 10.1613/jair.3912. URL http://dx.doi.org/10.1613/jair.3912.

Campero, A., Raileanu, R., Küttler, H., Tenenbaum, J. B., Rocktäschel, T., and Grefenstette, E. Learning with AMIGo: Adversarially Motivated Intrinsic Goals. *CoRR*, abs/2006.12122, 2020. URL https://arxiv.org/abs/2006.12122.

Chevalier-Boisvert, M., Bahdanau, D., Lahlou, S., Willems, L., Saharia, C., Nguyen, T. H., and Bengio, Y. BabyAI: First steps towards grounded language learning with a human in the loop. *CoRR*, abs/1810.08272, 2018. URL http://arxiv.org/abs/1810.08272.

Chevalier-Boisvert, M., Willems, L., and Pal, S. Minimalistic gridworld environment for OpenAI Gym. https://github.com/maximecb/gym-minigrid, 2018.

Cobbe, K., Klimov, O., Hesse, C., Kim, T., and Schulman, J. Quantifying generalization in reinforcement learning. In *International Conference on Machine Learning*, pp. 1282–1289. PMLR, 2019.

Cobbe, K., Hesse, C., Hilton, J., and Schulman, J. Leveraging Procedural Generation to Benchmark Reinforcement Learning. In *International Conference on Machine Learning*, pp. 2048–2056. PMLR, November 2020a. URL http://proceedings.mlr.press/v119/cobbe20a.html. ISSN: 2640-3498.

Cobbe, K., Hilton, J., Klimov, O., and Schulman, J. Phasic Policy Gradient. *CoRR*, abs/2009.04416, 2020b. URL https://arxiv.org/abs/2009.04416.

Cohn, D., Atlas, L., and Ladner, R. Improving generalization with active learning. *Machine learning*, 15(2): 201–221, 1994.

A. Experiment Details and Hyperparameters

A.1. Procgen Benchmark

Procgen Benchmark consists of 16 PCG environments of varying styles, exhibiting a diversity of gameplay similar to that of the ALE benchmark. Game levels are determined by a random seed and can vary in navigational layout, visual appearance, and starting positions of entities. All Procgen environments share the same discrete 15-dimensional action space and produce $64 \times 64 \times 3$ RGB observations. (Cobbe et al., 2020a) provides a comprehensive description of each of the 16 environments. State-of-the-art RL algorithms, like PPO, lead to significant generalization gaps between test and train performance in all games, making Procgen a useful benchmark for assessing generalization performance.

We follow the standard protocol for testing generalization performance on Procgen: We train an agent for each game on a finite number of levels, N_{train}, and sample test levels from the full distribution of levels. Normalized test returns are computed as $(R - R_{\text{min}})/(R_{\text{max}} - R_{\text{min}})$, where R is the unnormalized return and each game's minimum return, R_{min}, and maximum return, R_{max}, are provided in Cobbe et al. (2020a), which uses this same normalization.

To make the most efficient use of our computational resources, we perform hyperparameter sweeps on the easy setting. This also makes our results directly comparable to most prior works benchmarked on Procgen, which have likewise focused on the easy setting. In Procgen easy, our experiments use the recommended settings of $N_{\text{train}} = 200$ and 25M steps of training, as well as the same ResNet policy architecture and PPO hyperparameters shared across all games as in Cobbe et al. (2020a) and Raileanu et al. (2021). We find 25M steps to be sufficient for uncovering differences in generalization performance among our methods and standard baselines. Moreover, under this setup, we find Procgen training runs require much less wall-clock time than training runs on the two MiniGrid environments of interest over an equivalent number of steps needed to uncover differences in generalization performance. Therefore we survey the empirical differences across various settings of PLR on Procgen easy rather than MiniGrid.

To find the best hyperparameters for PLR, we evaluate each combination of the scoring function choices in Table 1 with both rank and proportional prioritization, performing a coarse grid search for each pair over different settings of the temperature parameter β in $\{0.1, 0.5, 1.0, 1.4, 2.0\}$ and the staleness coefficient ρ in $\{0.1, 0.3, 1.0\}$. For each setting, we run 4 trials across all 16 of games of the Procgen Benchmark, evaluating based on mean unnormalized test return across games. In our TD-error-based scoring functions, we set γ and λ equal to the same respective values used by the GAE in PPO during training. We found PLR offered the most pronounced gains at $\beta = 0.1$ and $\rho = 0.1$ on Procgen, but these benefits also held for higher values ($\beta = 0.5$ and $\rho = 0.3$), though to a lesser degree.

For UCB-DrAC, we make use of the best-reported hyperparameters on the easy setting of Procgen in Raileanu et al. (2021), listed in Table 3.

We found the default setting of mixreg's $\alpha = 0.2$, as used by Wang et al. (2020a) in the hard setting, performs poorly on the easy setting. Instead, we conducted a grid search over α in $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.8, 0.2, 0.5, 0.8, 1\}$.

Since the TSCL Window algorithm was not previously evaluated on Procgen Benchmark, we perform a grid search over different settings for both Boltzmann and ϵ-greedy variants of the algorithm to determine the best hyperparameter settings for Procgen easy. We searched over window size K in $\{10, 100, 1000, 10000\}$, bandit learning rate α in $\{0.01, 0.1, 0.5, 1.0\}$, random exploration probability ϵ in $\{0.0, 0.01, 0.1, 0.5\}$ for the ϵ-greedy variant, and temperature τ in $\{0.1, 0.5, 1.0\}$ for the Boltzmann variant. Additionally, for a fairer comparison to PLR we further evaluated a variant of TSCL Window that, like PLR, incorporates the staleness distribution, by additionally searching over values of the staleness coefficient ρ in $\{0.0, 0.1, 0.3, 0.5\}$, though we ultimately found that TSCL Window performed best without staleness sampling ($\rho = 0$).

See Table 3 for a comprehensive overview of the hyperparameters used for PPO, UCB-DrAC, mixreg, and TSCL Window, shared across all Procgen environments to generate our reported results on Procgen easy.

The evaluation protocol on the hard setting entails training on 500 levels over 200M steps (Cobbe et al., 2020a), making it more compute-intensive than the easy setting. To save on computational resources, we make use of the same hyperparameters found in the easy setting for each method on Procgen hard, with one exception: As our PPO implementation does not use multi-GPU training, we were unable to quadruple our GPU actors as done in Cobbe et al. (2020a) and Wang et al. (2020a). Instead, we resorted to doubling the number of environments in our single actor to 128, resulting in mini-batch sizes half as large as used in these two prior works. As such, our baseline results on hard are not directly comparable to theirs. We found setting mixreg's $\alpha = 0.2$ as done in Wang et al. (2020a) led to poor performance under this reduced batch size. We conducted an additional grid search, finding $\alpha = 0.01$ to perform best, as on Procgen easy.

A.2. MiniGrid

The MiniGrid suite (Chevalier-Boisvert et al., 2018) features a series of highly structured environments of increasing difficulty. Each environment features a task in a grid world

Table 3. Hyperparameters used for training on Procgen Benchmark and MiniGrid environments.

Parameter	Procgen	MiniGrid
PPO		
γ	0.999	0.999
λ_{GAE}	0.95	0.95
PPO rollout length	256	256
PPO epochs	3	4
PPO minibatches per epoch	8	8
PPO clip range	0.2	0.2
PPO number of workers	64	64
Adam learning rate	5e-4	7e-4
Adam ϵ	1e-5	1e-5
return normalization	yes	yes
entropy bonus coefficient	0.01	0.01
value loss coefficient	0.5	0.5
PLR		
Prioritization	rank	rank
Temperature, β, 0.1	0.1	0.1
Staleness coefficient, ρ	0.1	0.3
UCB-DrAC		
Window size, K	10	-
Regularization coefficient, α_r	0.1	-
UCB exploration coefficient, c	0.1	-
mixreg		
Beta shape, α	0.01	-
TSCL Window		
Bandit exploration strategy	ϵ-greedy	-
Window size, K	10	-
Bandit learning rate, α	1.0	-
Exploration probability, ϵ	0.5	-

setting, and as in Procgen, environment levels are determined by a seed. Harder levels require the agent to perform longer action sequences over a combinatorially-rich set of game entities, on increasingly larger grids. The clear ordering of difficulty over subsets of MiniGrid environments allows us to track the relative difficulty of levels sampled by PLR over the course of training.

MiniGrid environments share a discrete 7-dimensional action space and produce a 3-channel integer state encoding of the 7×7 grid immediately including and in front of the agent. However, following the training setup in Igl et al. (2019), we modify the environment to produce an $N \times M \times 3$ encoding of the full grid, where N and M vary according to the maximum grid dimensions of each environment. Full observability makes generalization harder, requiring the agent to generalize across different level layouts in their entirety.

We evaluate PLR with rank prioritization on two MiniGrid environments whose levels are uniformly distributed across several difficulty settings. Training on levels of varying difficulties helps agents make use of the easier levels as stepping stones to learn useful behaviors that help the agent make progress on harder levels. However, under the uniform-sampling baseline, learning may be inefficient, as the training process does not selectively train the agent on levels of increasing difficulty, leading to wasted training steps when a difficult level is sampled early in training. On the contrary, if PLR scores levels according to the time-averaged L1 value loss of recently experienced level trajectories, the average difficulty of the sampled levels should adapt to the agent's current abilities, following the reasoning outlined in the Value Correction Hypothesis, stated in Section 3.

As in Igl et al. (2019), we parameterize the agent policy as a 3-layer CNN with 16, 32, and 32 channels, with a final hidden layer of size 64. All kernels are 2×2 and use a stride of 1. For the ObstructedMazeGamut environments, we increase the number of channels of the final CNN layer to 64. We follow the same high-level generalization evaluation protocol used for Procgen, training the agent on a fixed set of 4000 levels for MultiRoom-N4-Random, 3000 levels for ObstructedMazeGamut-Easy, and 6000 levels for ObstructedMazeGamut-Medium, and testing on the full level distribution. We chose these values for $|\Lambda_{\text{train}}|$ to ensure roughly 1000 training levels of each difficulty setting of each environment. We model our PPO parameters on those used by Igl et al. (2019) in their MiniGrid experiments. We performed a grid search to find that PLR with rank prioritization, $\beta = 0.1$, and $\rho = 0.3$ learned most quickly on the MultiRoom environment, and used this setting for all our MiniGrid experiments. Table 3 summarizes these hyperparameter choices.

The remainder of this section provides more details about the various MiniGrid environments used in this work.

MultiRoom-N4-Random This environment requires the agent to navigate through 1, 2, 3, or 4 rooms respectively to reach a goal object, resulting in a natural ordering of levels over four levels of difficulty. The agent always starts at a random position in the furthest room from the goal object, facing a random direction. The goal object is also initialized to a random position within its room. See Figure 5 for screenshots of example levels.

ObstructedMazeGamut-Easy This environment consists of levels uniformly distributed across the first three difficulty settings of the ObstructedMaze environment, in which the agent must locate and pick up the key in order to unlock the door to pick up a goal object in a second room. The agent and goal object are always initialized in random positions in different rooms separated by the locked door.

Figure 5. Example levels of each of the four difficulty levels of MultiRoom-N4-Random, in order of increasing difficulty from left to right. The agent (red triangle) must reach the goal (green square).

The second difficulty setting further requires the agent to first uncover the key from under a box before picking up the key. The third difficulty level further requires the agent to first move a ball blocking the door before entering the door. See Figure 6 for screenshots of example levels.

Figure 6. Example levels of each of the three difficulty levels of ObstructedMazeGamut-Easy, in order of increasing difficulty from left to right. The agent must find the key, which may be hidden under a box, to unlock a door, which may be blocked by an obstacle, to reach the goal object (blue circle).

ObstructedMazeGamut-Hard This environment consists of levels uniformly distributed across the first six difficulty levels of the ObstructedMaze environment. Harder levels corresponding to the fourth, fifth, and sixth difficulty settings include two additional rooms with no goal object to distract the agent. Each instance of these harder levels also contain two pairs of keys of different colors, each opening a door of the same color. The agent always starts one room away from the randomly positioned goal object. Each of the two keys is visible in the fourth difficulty setting and doors are unobstructed. The fifth difficulty setting hides the keys under boxes, and the sixth again places obstacles that must be removed before entering two of the doors, one of which is always the door to the goal-containing room. See Figure 7 for example screenshots.

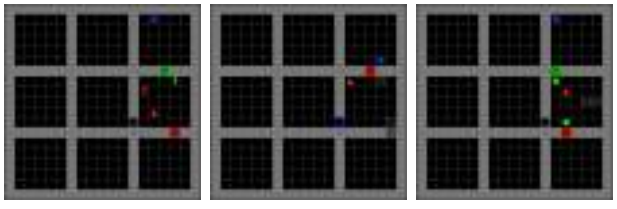


Figure 7. Example levels in increasing difficulty from left to right of each additional difficulty setting introduced by ObstructedMazeGamut-Hard in addition to those in ObstructedMazeGamut-Easy.

B. Additional Experimental Results

B.1. Extended Results on Procgen Benchmark

We present an overview of the improvements in test performance of each method across all 16 Procgen Benchmark games over 10 runs in Figure 14. For each game, Figure 15 further shows how the generalization gap changes over the course of training under each method tested. We show in Figures 16 and 17, the mean test episodic returns on the Procgen Benchmark (easy) for PLR with rank and proportional prioritization, respectively. In both of these plots, we can see that using only staleness ($\rho = 1$) or only L1 value loss scores ($\rho = 0$) is considerably worse than direct level sampling. Thus, we only observe gains compared to the baseline when both level scores and staleness are used for the sampling distribution. Comparing Figures 16 with 17 we find that PLR with proportional instead of rank prioritization provides statistically significant gains over uniform level sampling on an additional game (CoinRun), but rank prioritization leads to slightly larger mean improvements on several games.

Figure 18 shows that when PLR improves generalization performance, it also either matches or improves training sample efficiency, suggesting that when beneficial to test performance, the representations learned via the auto-curriculum induced by PLR prove similarly useful on training levels. However we see that our method reduces training sample efficiency on two games on which our method does not improve generalization performance. Since our method does not discover useful auto-curricula for these games, it is likely that uniformly sampling levels at training time allows the agent to better memorize useful behaviors on each of the training levels compared to the selective sampling performed by our method.

Finally, we also benchmarked PLR and UCB-DrAC + PLR against uniform sampling, TSCL Window, mixreg, and UCB-DrAC on Procgen hard across 5 runs per environment. Due to the high computational cost of the evaluation protocol for Procgen hard, which entails 200M training steps, we directly use the best hyperparameters found in the easy setting for each method. The results in Figure 10 show the two PLR-based methods significantly outperform all other methods in terms of normalized mean train and test episodic return, as well as reduction in mean generalization gap, attaining even greater margins of improvement than in the easy setting. As summarized by Table 4, the gains of PLR and UCB + PLR in mean normalized test return relative to uniform sampling in the hard setting are comparable to those in the easy setting. We provide plots of episodic test return over training for each individual environment in Figure 12.

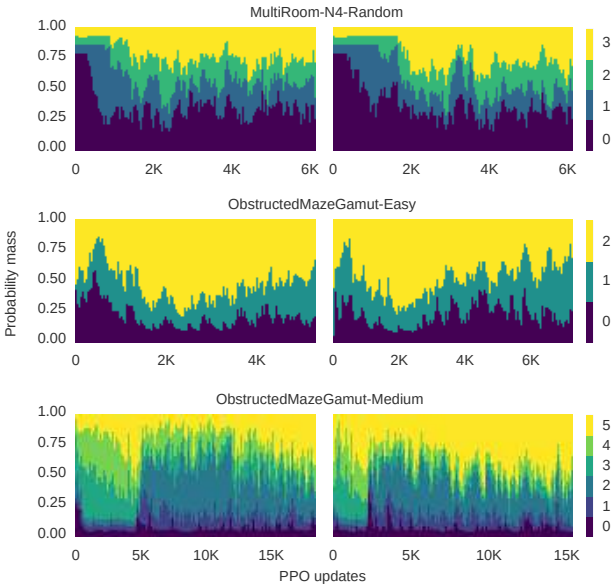


Figure 8. PLR consistently induces emergent curricula from easier to harder levels during training. Left and right correspond to two additional training runs independent of that in Figure 4.

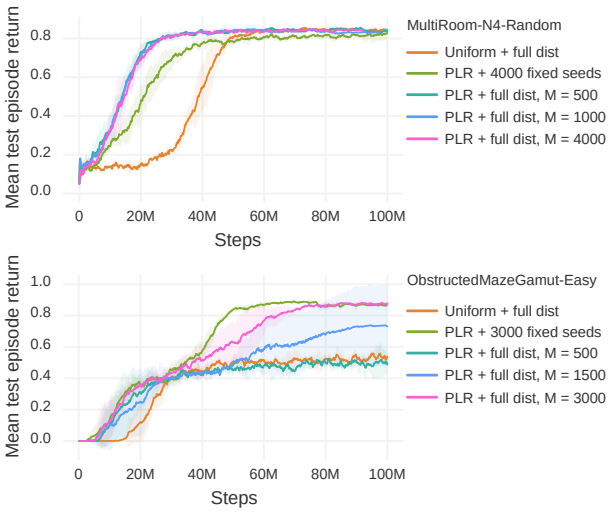


Figure 9. Mean test episodic returns on MultiRoom-N4-Random (top) and ObstructedMazeGamut-Easy (bottom) with access to the full level distribution at training. Plots are averaged over 3 runs. We set P_D to a Bernoulli parameterized as $p = 0.5$ for MultiRoom-N4-Random and $p = 0.95$ for ObstructedMazeGamut-Easy (found via grid search). As with all MiniGrid experiments using PLR, we use rank prioritization, $\beta = 0.1$, and $\rho = 0.3$.

B.2. Extended Results on Minigrid

To demonstrate that PLR consistently induces an emergent curriculum, we present plots showing the change in probability mass over different difficulty bins for additional training runs in Figure 8. Like in Figure 4, we see the probability mass assigned by P_{replay} gradually shifts from easier to harder levels over the course of training.

B.3. Training on the Full Level Distribution

While assessing generalization performance calls for using a fixed set of training levels, ideally our method can also make use of the full level distribution if given access to it. We take advantage of an unbounded number of training levels by modifying the list structures for storing scores and timestamps (see Algorithm 1 and 2) to track the top M levels by learning potential in our finite level buffer. When the lists are full, we set the next level for replacement to be

$$l_{\text{min}} = \arg\min_{l} P_{\text{replay}}(l).$$

When the outcome of the Bernoulli P_D entails sampling a new level l, the score and timestamps of l replace those of l_{min} only if the score of l_{min} is lower than that of l. In this way, PLR keeps a running buffer throughout training of the top M levels appraised to have the highest learning potential for replaying anew.

Figure 9 shows that with access to the full level distribution at training, PLR improves sample efficiency and generalization performance in both environments compared to uniform sampling on the full distribution. In MultiRoom-N4-Random, the value M makes little difference to test performance, and training with PLR on the full level distribution leads to a policy outperforming one trained with PLR on a fixed set of training levels. However, on ObstructedMazeGamut-Easy, a smaller M leads to worse test performance. Nevertheless, for all but $M = 500$, including the case of a fixed set of 3000 training levels, PLR leads to better mean test performance than uniform sampling on the full level distribution.

C. Algorithms

In this section, we provide detailed pseudocode for how PLR can be used for experience collection when using T-step rollouts. Algorithm 3 presents the extension of the generic policy-gradient training loop presented in Algorithm 1 to the case of T-step rollouts, and Algorithm 4 presents an implementation of experience collection in this setting (extending Algorithm 2). Note that when using T-step rollouts in the training loop, rollouts may start and end between episode boundaries. To compute level scores on full trajectories segmented across rollouts, we compute scores of partial episodes according to Equation 2, and record these partial scores alongside the partial episode step count in a separate buffer $\tilde{S}$. The function **score** then technically, optionally takes the additional input $\tilde{S}$ (through an abuse of notation) as an additional argument to stitch together this partial information into scores of full episodic trajectories.

Table 4. Comparison of test scores of PPO with PLR against PPO with uniform-sampling on the hard setting of Procgen Benchmark. Following (Raileanu et al., 2021), reported figures represent the mean and standard deviation of average test scores over 100 episodes aggregated across 5 runs, each initialized with a unique training seed. For each run, a normalized average return is computed by dividing the average test return for each game by the corresponding average test return of the uniform-sampling baseline over all 500 test episodes of that game, followed by averaging these normalized returns over all 16 games. The final row reports the mean and standard deviation of the normalized returns aggregated across runs. Bolded methods are not significantly different from the method with highest mean, unless all are, in which case none are bolded.

Environment	Uniform	TSCL	mixreg	UCB-DrAC	PLR	UCB-DrAC + PLR
BigFish	9.7 ± 1.8	$\mathbf{11.9 \pm 2.5}$	$\mathbf{12.0 \pm 2.5}$	10.9 ± 1.6	$\mathbf{15.3 \pm 3.6}$	$\mathbf{15.5 \pm 2.8}$
BossFight	$\mathbf{9.6 \pm 0.2}$	8.4 ± 0.7	$\mathbf{9.3 \pm 0.9}$	9.0 ± 0.2	$\mathbf{9.7 \pm 0.4}$	$\mathbf{9.5 \pm 1.1}$
CaveFlyer	3.5 ± 0.8	6.3 ± 0.6	4.0 ± 1.0	2.6 ± 0.8	6.4 ± 0.6	$\mathbf{8.0 \pm 0.9}$
Chaser	5.9 ± 0.5	6.2 ± 1.0	$\mathbf{6.5 \pm 0.8}$	$\mathbf{7.0 \pm 0.6}$	$\mathbf{6.8 \pm 2.2}$	$\mathbf{7.6 \pm 0.2}$
Climber	5.3 ± 1.1	5.2 ± 0.7	5.7 ± 0.7	6.1 ± 1.0	7.4 ± 0.6	7.6 ± 1.8
CoinRun	4.5 ± 0.4	5.8 ± 0.8	$\mathbf{6.2 \pm 1.0}$	5.2 ± 1.0	$\mathbf{6.8 \pm 0.6}$	$\mathbf{7.1 \pm 0.5}$
Dodgeball	3.9 ± 0.6	1.9 ± 0.9	4.7 ± 1.0	9.9 ± 1.2	7.4 ± 1.3	$\mathbf{12.4 \pm 0.7}$
FruitBot	$\mathbf{11.9 \pm 4.2}$	13.1 ± 2.3	$\mathbf{14.7 \pm 2.2}$	$\mathbf{15.6 \pm 3.7}$	$\mathbf{16.7 \pm 1.0}$	$\mathbf{12.9 \pm 5.1}$
Heist	1.5 ± 0.4	0.9 ± 0.3	1.2 ± 0.4	1.1 ± 0.3	1.3 ± 0.4	2.6 ± 2.2
Jumper	3.2 ± 0.3	3.2 ± 0.3	3.3 ± 0.4	2.9 ± 0.9	3.5 ± 0.5	3.3 ± 0.8
Leaper	7.1 ± 0.3	$\mathbf{7.5 \pm 0.5}$	$\mathbf{7.5 \pm 0.5}$	3.8 ± 1.6	$\mathbf{7.4 \pm 0.2}$	$\mathbf{8.2 \pm 0.7}$
Maze	3.6 ± 0.7	3.8 ± 0.6	3.9 ± 0.5	4.4 ± 0.2	4.0 ± 0.4	$\mathbf{6.2 \pm 0.4}$
Miner	12.8 ± 1.4	11.7 ± 0.9	13.3 ± 1.6	$\mathbf{16.1 \pm 0.6}$	11.3 ± 0.7	$\mathbf{15.3 \pm 0.8}$
Ninja	5.2 ± 0.1	5.9 ± 0.8	5.0 ± 1.0	5.2 ± 1.0	$\mathbf{6.1 \pm 0.6}$	$\mathbf{6.9 \pm 0.3}$
Plunder	3.2 ± 0.1	5.4 ± 1.1	3.7 ± 0.4	7.8 ± 1.1	8.6 ± 2.7	$\mathbf{17.5 \pm 1.3}$
StarPilot	5.5 ± 0.6	2.1 ± 0.4	6.9 ± 0.6	$\mathbf{11.2 \pm 1.7}$	5.4 ± 0.8	$\mathbf{12.3 \pm 1.5}$
Normalized test returns (%)	100.0 ± 2.0	103.9 ± 3.5	110.6 ± 3.9	126.6 ± 3.0	135.0 ± 6.1	$\mathbf{182.9 \pm 8.2}$

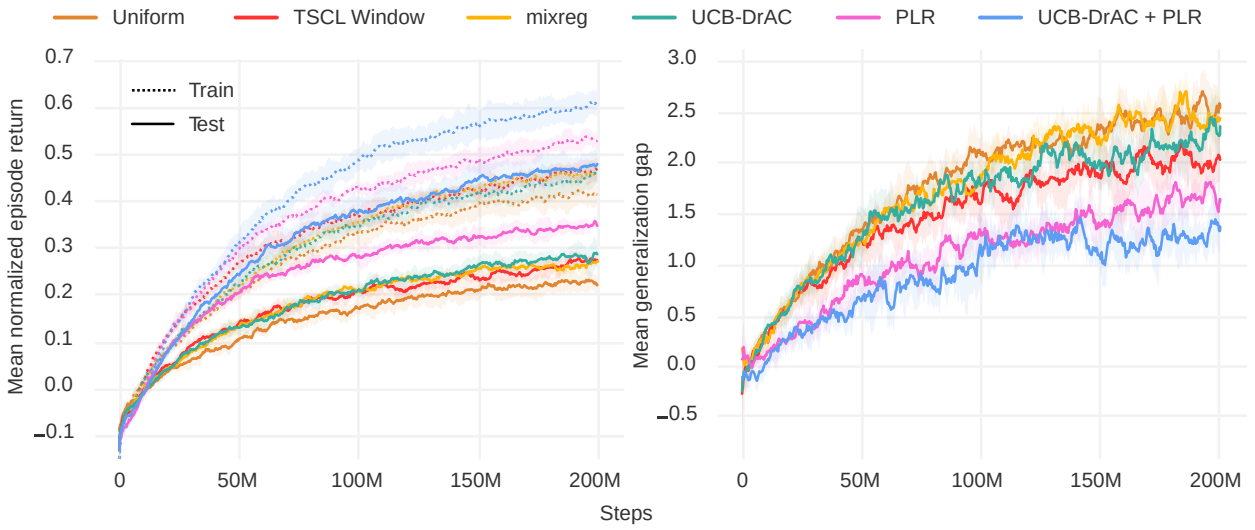


Figure 10. Left: Mean normalized train and test episode returns on Procgen Benchmark (hard). Right: Corresponding generalization gaps during training. All curves are averaged across all environments over 5 runs. The shaded area indicates one standard deviation around the mean. PLR-based methods statistically significantly outperform all others in both train and test returns. Only the PLR-based methods statistically significantly reduce the generalization gap ($p < 0.05$).

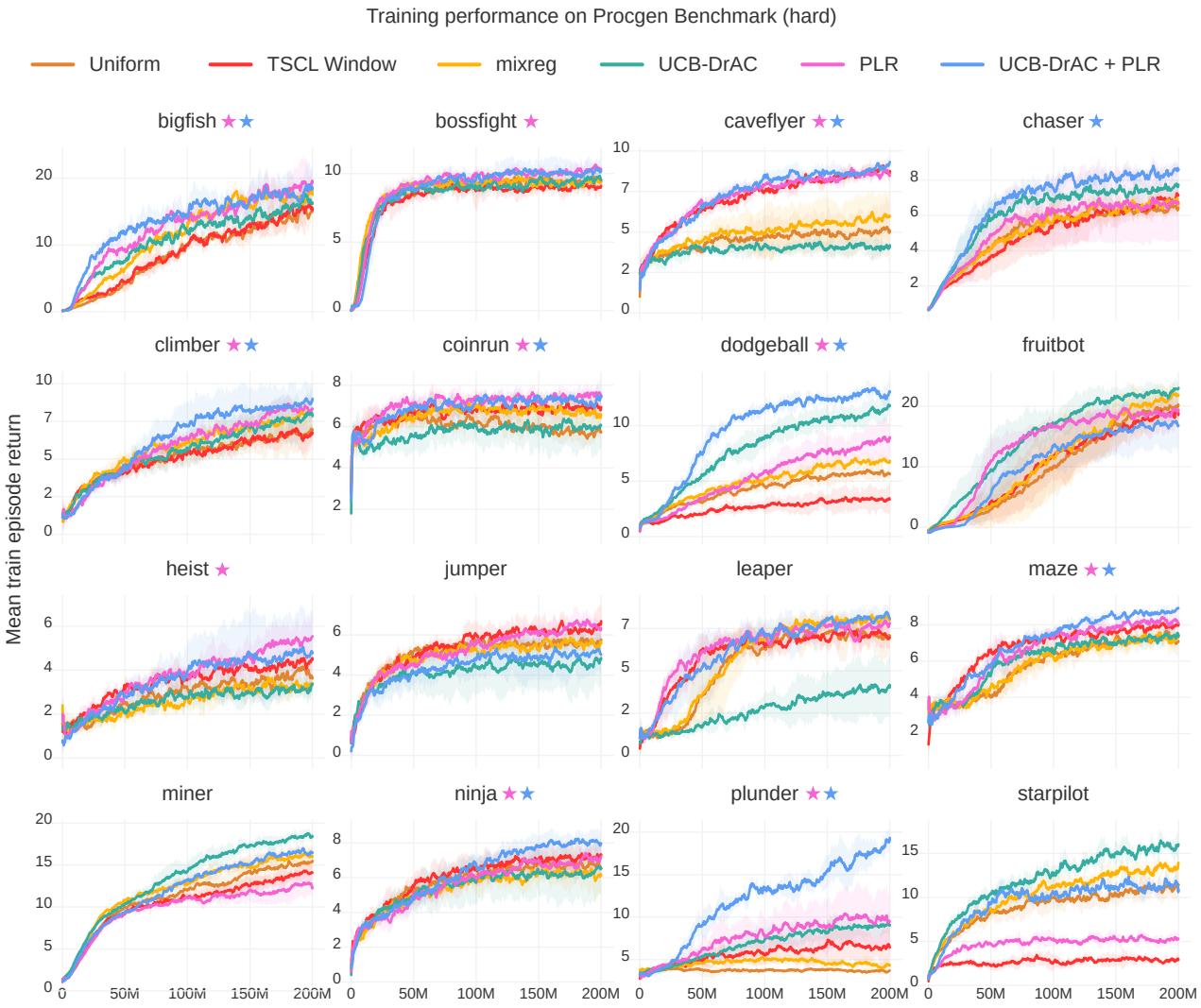


Figure 11. Mean train episode returns (5 runs) on Procgen Benchmark (hard), using the best hyperparameters found on the easy setting. The shaded area indicates one standard deviation around the mean. A ★ indicates statistically significant improvement over the uniform-sampling baseline by the PLR-based method of the matching color ($p < 0.05$). Note that while PLR reduces training performance on StarPilot, it performs comparably to the uniform-sampling baseline at test time, indicating less overfitting to training levels.

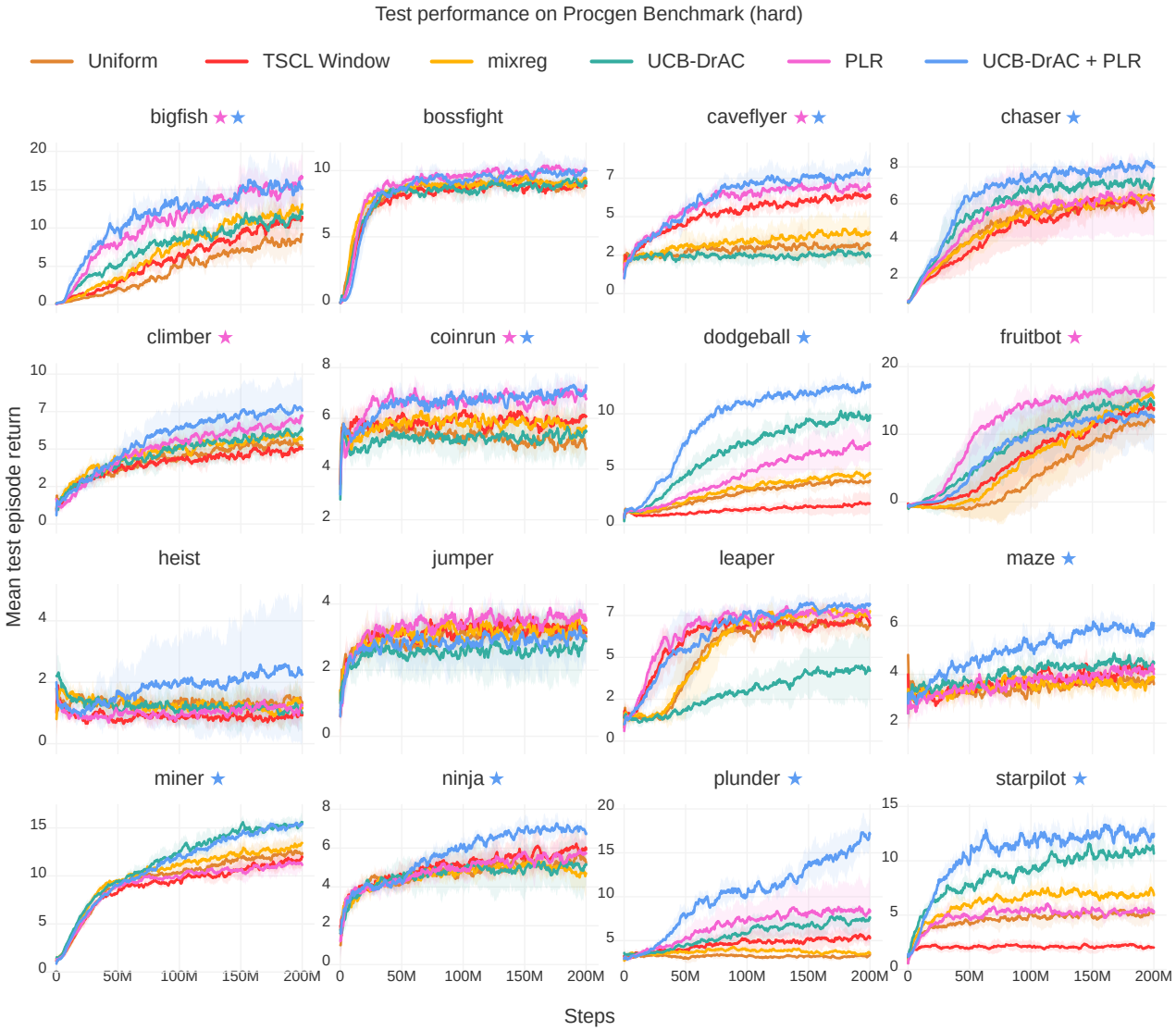


Figure 12. Mean test episode returns (5 runs) on Procgen Benchmark (hard), using best hyperparameters found on the easy setting. The shaded area indicates one standard deviation around the mean. A ★ indicates statistically significant improvement over the uniform-sampling baseline by the PLR-based method of the matching color ($p < 0.05$).

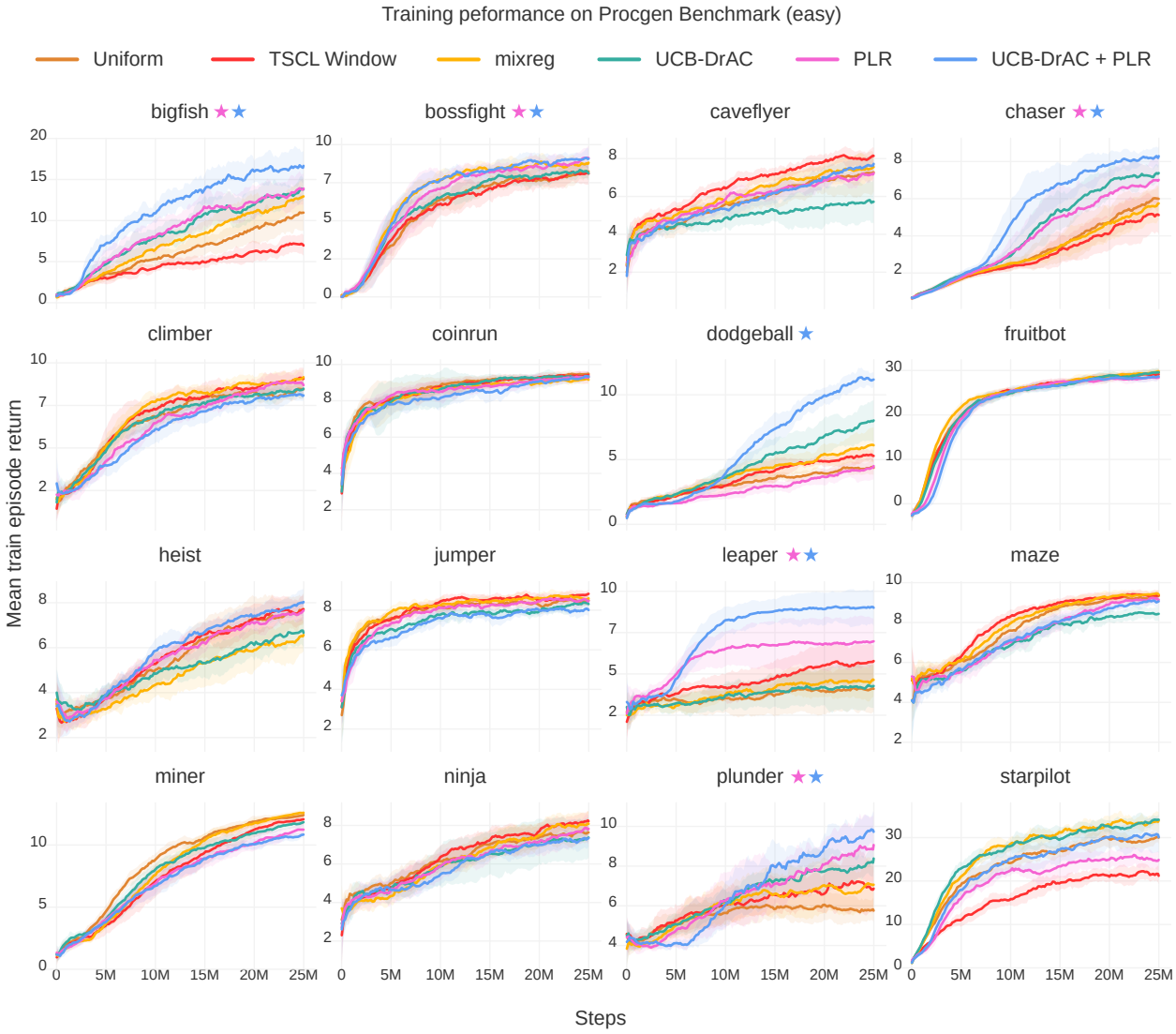


Figure 13. Mean train episode returns (5 runs) on Procgen Benchmark (easy). The shaded area indicates one standard deviation around the mean. A ★ indicates statistically significant improvement over the uniform-sampling baseline by the PLR-based method of the matching color ($p < 0.05$). PLR tends to improve or match training sample efficiency. Note that while PLR reduces training performance on StarPilot, it performs comparably to the uniform-sampling baseline at test time, indicating less overfitting to training levels.

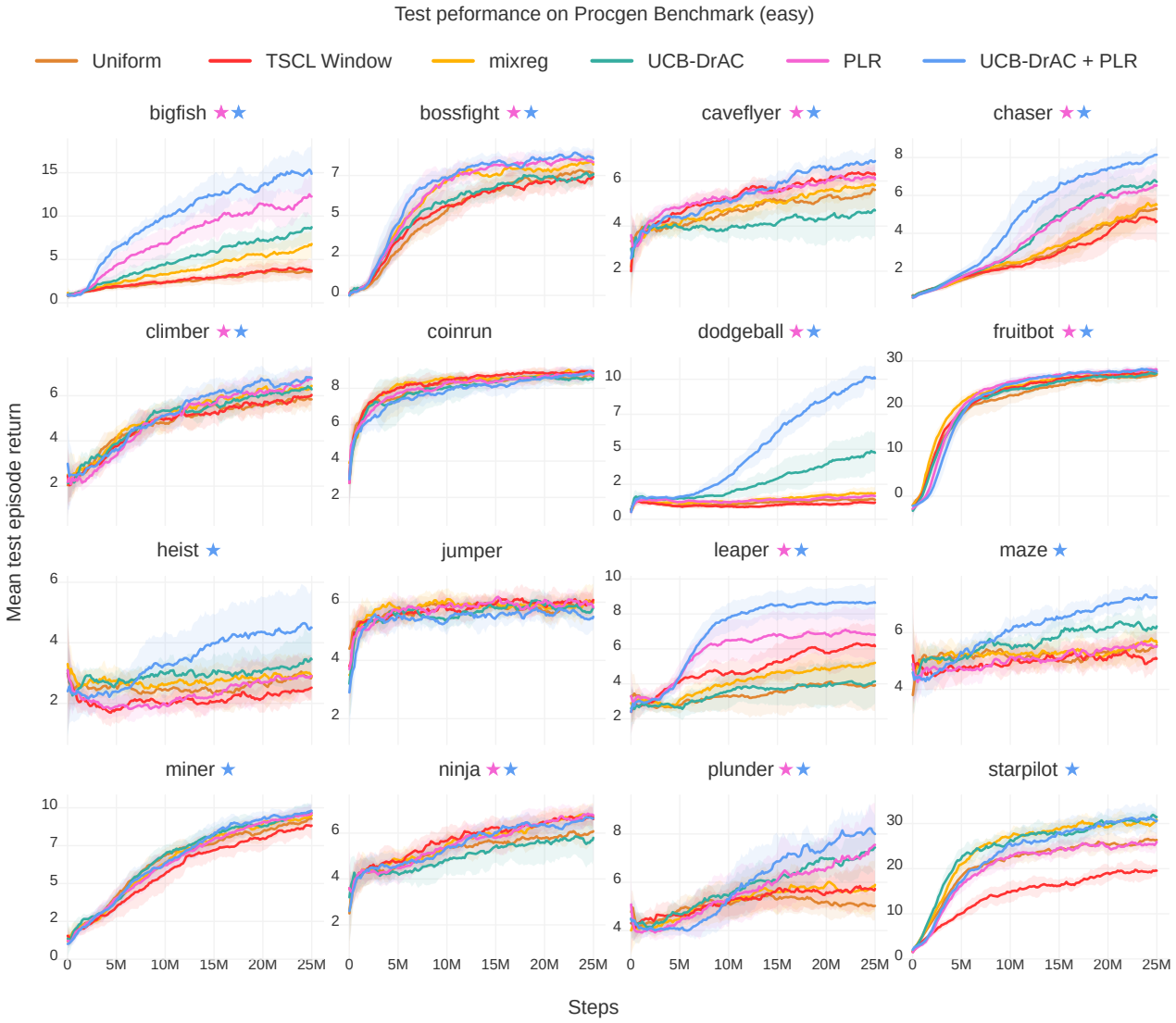


Figure 14. Mean test episode return (10 runs) on each Procgen Benchmark game (easy). The shaded area indicates one standard deviation around the mean. PLR-based methods consistently match or outperform uniform sampling with statistically significance ($p < 0.05$), indicated by a ★ of the corresponding color. We see that TSCL results in inconsistent outcomes across games, notably drastically lower test returns on StarPilot.

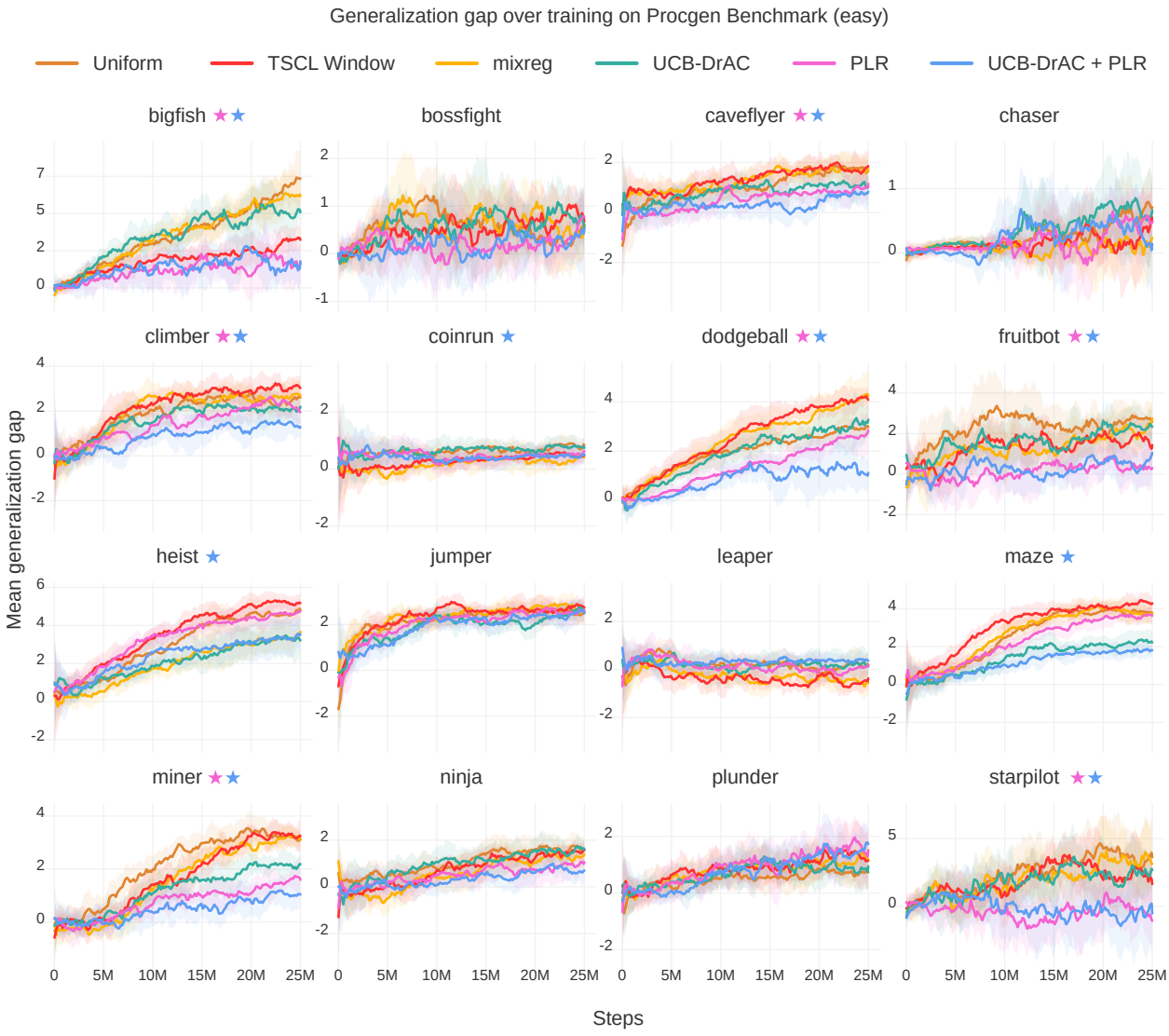


Figure 15. Mean generalization gaps throughout training (10 runs) on each Procgen Benchmark game (easy). The shaded area indicates one standard deviation around the mean. A ★ indicates the method of matching color results in a statistically significant ($p < 0.05$) reduction in generalization gap compared to the uniform-sampling baseline. By itself, PLR significantly reduces the generalization gap on 7 games, and UCB-DrAC, on 5 games. This number jumps to 10 of 16 games when these two methods are combined. TSCL only significantly reduces generalization gap on 2 of 16 games relative to uniform sampling, while increasing it on others, most notably on Dodgeball.

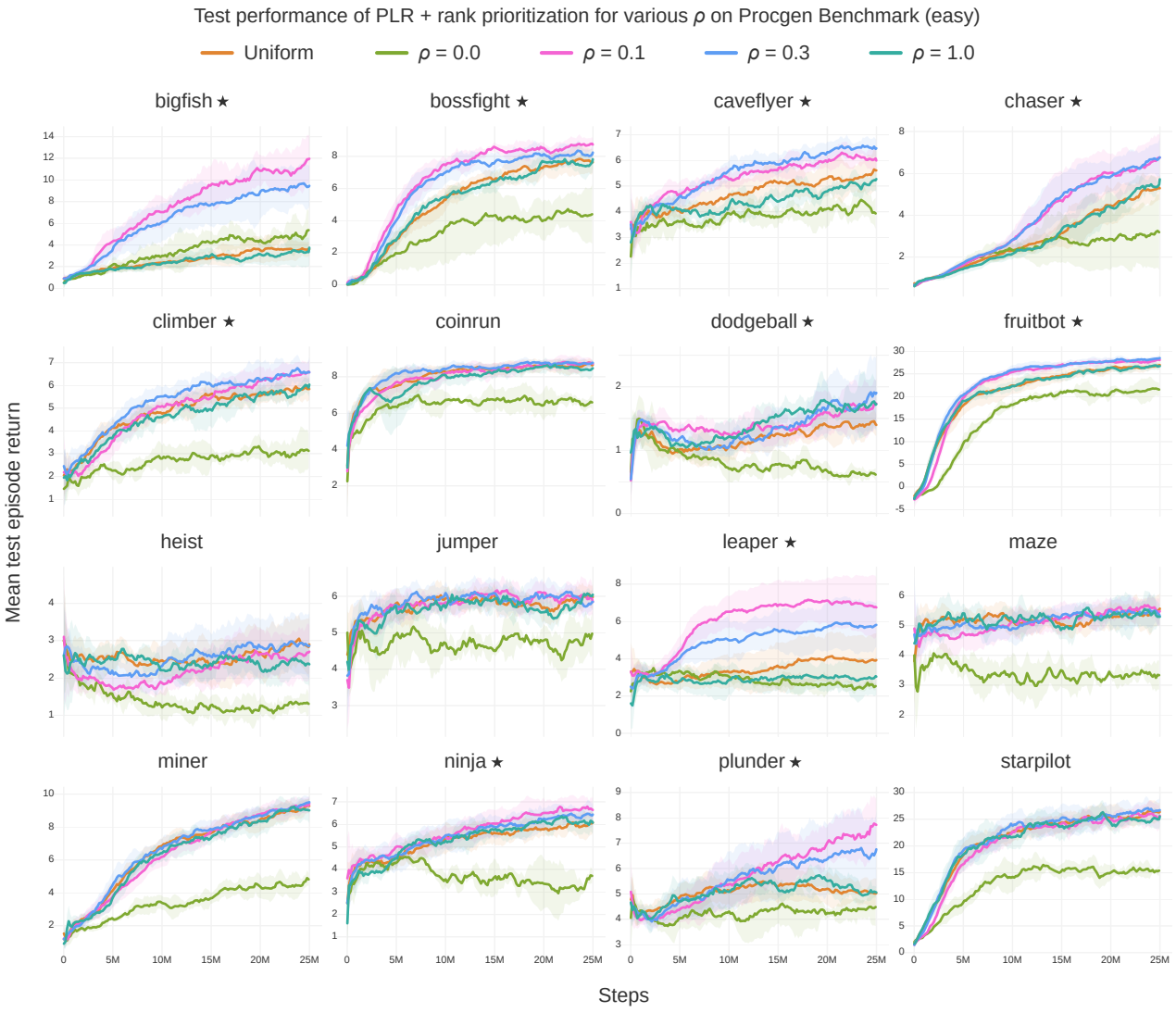


Figure 16. Mean test episode returns (10 runs) on the Procgen Benchmark (easy) for PLR with rank prioritization and $\beta = 0.1$ across a range of staleness coefficient values, ρ. The replay distribution must consider both the L1 value-loss and staleness values to realize improvements to generalization and sample efficiency. The shaded area indicates one standard deviation around the mean. A ★ next to the game name indicates that $\rho = 0.1$ exhibits statistically significantly better final test returns or sample efficiency along the test curve ($p < 0.05$), which we observe in 10 of 16 games.

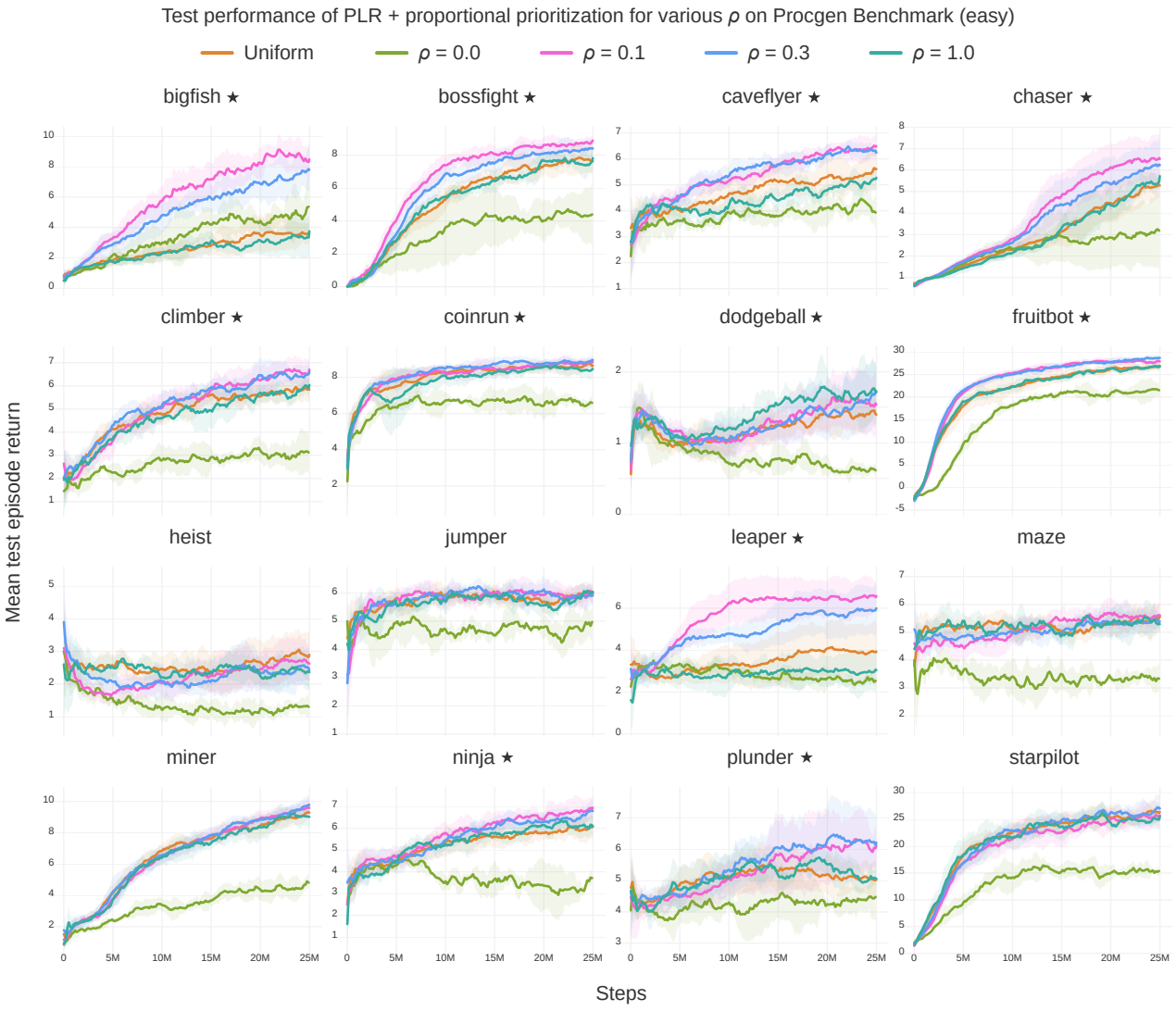


Figure 17. Mean test episode returns (10 runs) on the Procgen Benchmark (easy) for PLR with proportional prioritization and $\beta = 0.1$ across a range of values of ρ. As in the case of rank prioritization, the replay distribution must consider both the L1 value loss score and staleness values in order to realize performance improvements. The shaded area indicates one standard deviation around the mean. A ★ next to the game name indicates the condition $\rho = 0.1$ exhibits statistically significantly better final test returns or sample efficiency along the test curve ($p < 0.05$), which we observe in 11 of 16 games.

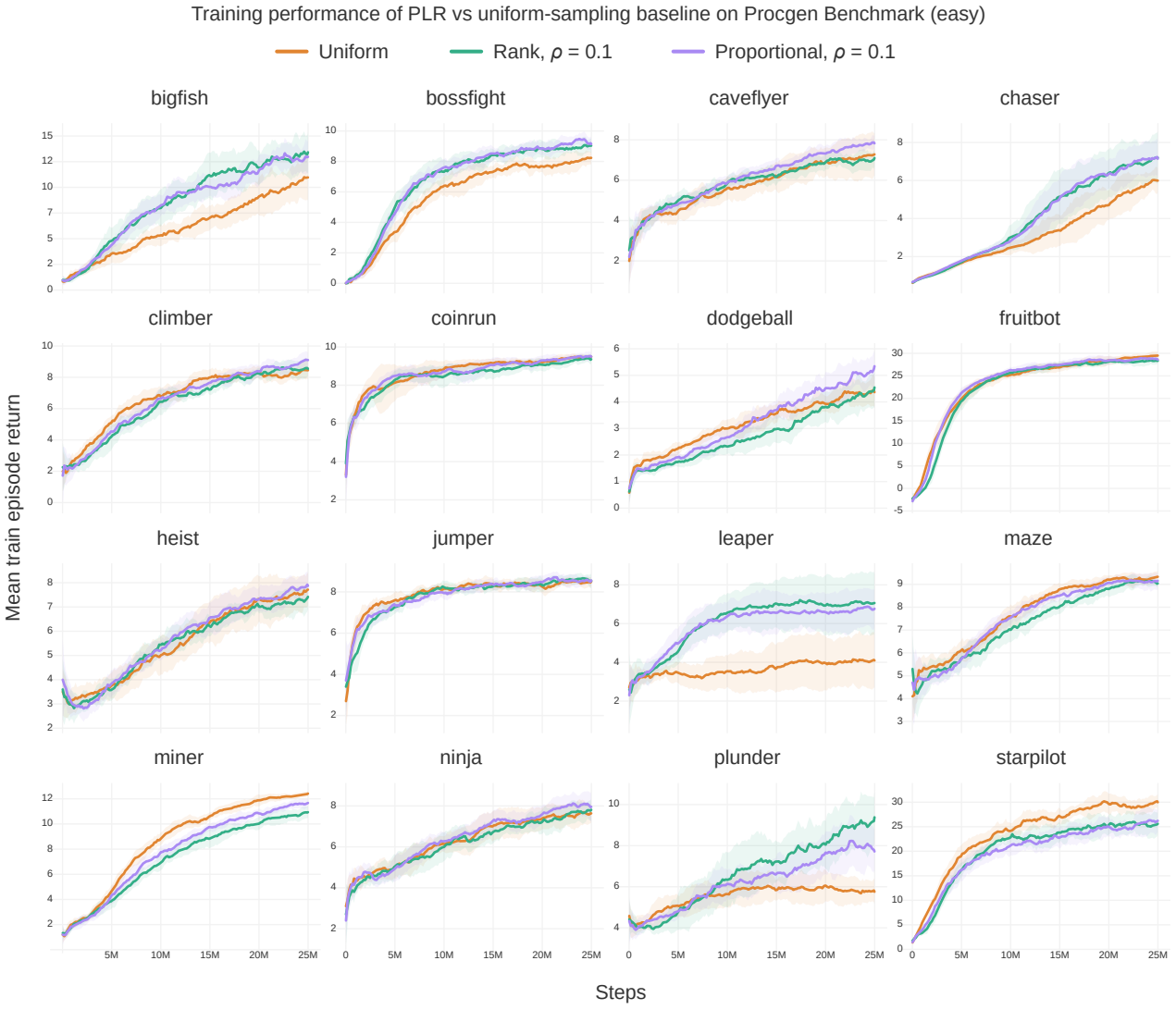


Figure 18. Mean training episode returns (10 runs) on the Procgen Benchmark for (easy) PLR with $\beta = 0.1$, $\rho = 0.1$, and each of rank and proportional prioritization. On some games, PLR improves both training sample efficiency and generalization performance (e.g. BigFish and Chaser), while on others, only generalization performance (e.g. CaveFlyer with rank prioritization). The shaded area indicates one standard deviation around the mean.

Algorithm 3 Generic T-step policy-gradient training loop with prioritized level replay

Input: Training levels Λ_{train} of an environment, policy π_θ, rollout length T, number of updates N_u, batch size N_b, policy update function $\mathcal{U}(\mathcal{B}, \theta) \rightarrow \theta'$.

Initialize level scores S, partial level scores $\tilde{S}$, and level timestamps C
Initialize global episode count $c \leftarrow 0$
Initialize set of visited levels $\Lambda_{\text{seen}} = \varnothing$
Initialize experience buffer $\mathcal{B} = \varnothing$
Initialize N_b parallel environment instances E, each set to a random level in $\in \Lambda_{\text{train}}$
for $u = 1$ **to** N_u **do**
  $\mathcal{B} \leftarrow \varnothing$
  for $k = 1$ **to** N_b **do**
    $\mathcal{B} \leftarrow \mathcal{B} \cup$ **collect_experiences**$(k, E, \Lambda_{\text{train}}, \Lambda_{\text{seen}}, \pi_\theta, T, S, \tilde{S}, C, c)$ *Using Algorithm 4*
  end for
  $\theta \leftarrow \mathcal{U}(\mathcal{B}, \theta)$
end for

Algorithm 4 Collect T-step rollouts with prioritized level replay

Input: Actor index k, batch environments E, training levels Λ_{train}, visited levels Λ_{seen}, current level l, policy π_θ, rollout length T, scoring function $\mathbf{score}$, level scores S, partial scores $\tilde{S}$, staleness values C, and global episode count c.
Output: Experience buffer $\mathcal{B}$

Initialize $\mathcal{B} = \varnothing$, and set current level $l_i = E_k$
Observe current state s_0, termination flag d_0
if d_0 **then**
  Define new index $i \leftarrow |S| + 1$
  Choose current level $l_i \leftarrow \mathbf{sampleNextLevel}(\Lambda_{\text{train}}, S, C, c)$ and $E_k \leftarrow l_i$
  Update level timestamp $C_i \leftarrow c$
  Observe initial state s_0
end if
Choose $a_0 \sim \pi_\theta(\cdot|s_0)$
t = 1
Initialize episodic trajectory buffer $\tau = \varnothing$
while $t < T$ **do**
  Observe (s_t, r_t, d_t)
  $\mathcal{B} \leftarrow \mathcal{B} \cup (s_{t-1}, a_{t-1}, s_t, r_t, d_t, \log \pi_\theta(a))$
  $\tau \leftarrow \tau \cup (s_{t-1}, a_{t-1}, s_t, r_t, d_t, \log \pi_\theta(a))$
  if d_t **then**
    Update level score $S_i \leftarrow \mathbf{score}(\tau, \pi_\theta, \tilde{S}_i)$ and partial score $\tilde{S}_i \leftarrow 0$
    $\tau \leftarrow \varnothing$
    Define new index $i \leftarrow |S| + 1$
    Update current level $l_i \leftarrow$ **sampleNextLevel**$(\Lambda_{\text{train}}, S, C, c)$ and $E_k \leftarrow l_i$
    Update level timestamp $C_i \leftarrow c$
  end if
  Choose $a_{t+1} \sim \pi_\theta(\cdot|s_t)$
  $t \leftarrow t + 1$
end while
if not d_t **then**
  $\tilde{S}_i \leftarrow (\textbf{score}(\tau, \pi_\theta), |\tau|)$ *Track partial time-averaged score and $|\tau|$*
end if

function sampleNextLevel$(\Lambda_{\text{train}}, S, C, c)$
  $c \leftarrow c + 1$
  Sample replay decision $d \sim P_D(d)$
  if $d = 0$ **and** $|\Lambda_{\text{train}} \setminus \Lambda_{\text{seen}}| > 0$ **then**
    Define new index $i \leftarrow |S| + 1$
    Sample $l_i \sim P_{\text{new}}(l|\Lambda_{\text{train}}, \Lambda_{\text{seen}})$ *Sample an unseen level, if any*
    Add l_i to Λ_{seen}, add initial value $S_i = 0$ to S and $C_i = 0$ to C
  else
    Sample $l_i \sim (1 - \rho) \cdot P_S(l|\Lambda_{\text{seen}}, S) + \rho \cdot P_C(l|\Lambda_{\text{seen}}, C, c)$ *Sample a level for replay*
  end if
  return l_i
end function

ADAM: A METHOD FOR STOCHASTIC OPTIMIZATION

Diederik P. Kingma*
University of Amsterdam, OpenAI
dpkingma@openai.com

Jimmy Lei Ba*
University of Toronto
jimmy@psi.utoronto.ca

ABSTRACT

We introduce *Adam*, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients. The hyper-parameters have intuitive interpretations and typically require little tuning. Some connections to related algorithms, on which *Adam* was inspired, are discussed. We also analyze the theoretical convergence properties of the algorithm and provide a regret bound on the convergence rate that is comparable to the best known results under the online convex optimization framework. Empirical results demonstrate that Adam works well in practice and compares favorably to other stochastic optimization methods. Finally, we discuss *AdaMax*, a variant of *Adam* based on the infinity norm.

1 INTRODUCTION

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems in these fields can be cast as the optimization of some scalar parameterized objective function requiring maximization or minimization with respect to its parameters. If the function is differentiable w.r.t. its parameters, gradient descent is a relatively efficient optimization method, since the computation of first-order partial derivatives w.r.t. all the parameters is of the same computational complexity as just evaluating the function. Often, objective functions are stochastic. For example, many objective functions are composed of a sum of subfunctions evaluated at different subsamples of data; in this case optimization can be made more efficient by taking gradient steps w.r.t. individual subfunctions, i.e. stochastic gradient descent (SGD) or ascent. SGD proved itself as an efficient and effective optimization method that was central in many machine learning success stories, such as recent advances in deep learning (Deng et al., 2013; Krizhevsky et al., 2012; Hinton & Salakhutdinov, 2006; Hinton et al., 2012a; Graves et al., 2013). Objectives may also have other sources of noise than data subsampling, such as dropout (Hinton et al., 2012b) regularization. For all such noisy objectives, efficient stochastic optimization techniques are required. The focus of this paper is on the optimization of stochastic objectives with high-dimensional parameters spaces. In these cases, higher-order optimization methods are ill-suited, and discussion in this paper will be restricted to first-order methods.

We propose *Adam*, a method for efficient stochastic optimization that only requires first-order gradients with little memory requirement. The method computes individual adaptive learning rates for different parameters from estimates of first and second moments of the gradients; the name *Adam* is derived from adaptive moment estimation. Our method is designed to combine the advantages of two recently popular methods: AdaGrad (Duchi et al., 2011), which works well with sparse gradients, and RMSProp (Tieleman & Hinton, 2012), which works well in on-line and non-stationary settings; important connections to these and other stochastic optimization methods are clarified in section 5. Some of Adam's advantages are that the magnitudes of parameter updates are invariant to rescaling of the gradient, its stepsizes are approximately bounded by the stepsize hyperparameter, it does not require a stationary objective, it works with sparse gradients, and it naturally performs a form of step size annealing.

*Equal contribution. Author ordering determined by coin flip over a Google Hangout.

Algorithm 1: *Adam*, our proposed algorithm for stochastic optimization. See section 2 for details, and for a slightly more efficient (but less clear) order of computation. g_t^2 indicates the elementwise square $g_t \odot g_t$. Good default settings for the tested machine learning problems are $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. All operations on vectors are element-wise. With β_1^t and β_2^t we denote β_1 and β_2 to the power t.

Require: α: Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates for the moment estimates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0: Initial parameter vector
  $m_0 \leftarrow 0$ (Initialize 1st moment vector)
  $v_0 \leftarrow 0$ (Initialize 2nd moment vector)
  $t \leftarrow 0$ (Initialize timestep)
  while θ_t not converged **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
    $v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (Update biased second raw moment estimate)
    $\widehat{m}_t \leftarrow m_t / (1 - \beta_1^t)$ (Compute bias-corrected first moment estimate)
    $\widehat{v}_t \leftarrow v_t / (1 - \beta_2^t)$ (Compute bias-corrected second raw moment estimate)
    $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \widehat{m}_t / (\sqrt{\widehat{v}_t} + \epsilon)$ (Update parameters)
  end while
  return θ_t (Resulting parameters)

In section 2 we describe the algorithm and the properties of its update rule. Section 3 explains our initialization bias correction technique, and section 4 provides a theoretical analysis of Adam's convergence in online convex programming. Empirically, our method consistently outperforms other methods for a variety of models and datasets, as shown in section 6. Overall, we show that Adam is a versatile algorithm that scales to large-scale high-dimensional machine learning problems.

2 ALGORITHM

See algorithm 1 for pseudo-code of our proposed algorithm *Adam*. Let $f(\theta)$ be a noisy objective function: a stochastic scalar function that is differentiable w.r.t. parameters θ. We are interested in minimizing the expected value of this function, $\mathbb{E}[f(\theta)]$ w.r.t. its parameters θ. With $f_1(\theta), ..., f_T(\theta)$ we denote the realisations of the stochastic function at subsequent timesteps $1, ..., T$. The stochasticity might come from the evaluation at random subsamples (minibatches) of datapoints, or arise from inherent function noise. With $g_t = \nabla_\theta f_t(\theta)$ we denote the gradient, i.e. the vector of partial derivatives of f_t, w.r.t θ evaluated at timestep t.

The algorithm updates exponential moving averages of the gradient (m_t) and the squared gradient (v_t) where the hyper-parameters $\beta_1, \beta_2 \in [0, 1)$ control the exponential decay rates of these moving averages. The moving averages themselves are estimates of the 1st moment (the mean) and the 2nd raw moment (the uncentered variance) of the gradient. However, these moving averages are initialized as (vectors of) 0's, leading to moment estimates that are biased towards zero, especially during the initial timesteps, and especially when the decay rates are small (i.e. the βs are close to 1). The good news is that this initialization bias can be easily counteracted, resulting in bias-corrected estimates $\widehat{m}_t$ and $\widehat{v}_t$. See section 3 for more details.

Note that the efficiency of algorithm 1 can, at the expense of clarity, be improved upon by changing the order of computation, e.g. by replacing the last three lines in the loop with the following lines: $\alpha_t = \alpha \cdot \sqrt{1 - \beta_2^t}/(1 - \beta_1^t)$ and $\theta_t \leftarrow \theta_{t-1} - \alpha_t \cdot m_t / (\sqrt{v_t} + \hat{\epsilon})$.

2.1 ADAM'S UPDATE RULE

An important property of Adam's update rule is its careful choice of stepsizes. Assuming $\epsilon = 0$, the effective step taken in parameter space at timestep t is $\Delta_t = \alpha \cdot \widehat{m}_t / \sqrt{\widehat{v}_t}$. The effective stepsize has two upper bounds: $|\Delta_t| \leq \alpha \cdot (1 - \beta_1)/\sqrt{1 - \beta_2}$ in the case $(1 - \beta_1) > \sqrt{1 - \beta_2}$, and $|\Delta_t| \leq \alpha$

otherwise. The first case only happens in the most severe case of sparsity: when a gradient has been zero at all timesteps except at the current timestep. For less sparse cases, the effective stepsize will be smaller. When $(1-\beta_1) = \sqrt{1-\beta_2}$ we have that $|\widehat{m}_t/\sqrt{\widehat{v}_t}| < 1$ therefore $|\Delta_t| < \alpha$. In more common scenarios, we will have that $\widehat{m}_t/\sqrt{\widehat{v}_t} \approx \pm 1$ since $|\mathbb{E}[g]/\sqrt{\mathbb{E}[g^2]}| \leq 1$. The effective magnitude of the steps taken in parameter space at each timestep are approximately bounded by the stepsize setting α, i.e., $|\Delta_t| \lessapprox \alpha$. This can be understood as establishing a *trust region* around the current parameter value, beyond which the current gradient estimate does not provide sufficient information. This typically makes it relatively easy to know the right scale of α in advance. For many machine learning models, for instance, we often know in advance that good optima are with high probability within some set region in parameter space; it is not uncommon, for example, to have a prior distribution over the parameters. Since α sets (an upper bound of) the magnitude of steps in parameter space, we can often deduce the right order of magnitude of α such that optima can be reached from θ_0 within some number of iterations. With a slight abuse of terminology, we will call the ratio $\widehat{m}_t/\sqrt{\widehat{v}_t}$ the *signal-to-noise* ratio (SNR). With a smaller SNR the effective stepsize Δ_t will be closer to zero. This is a desirable property, since a smaller SNR means that there is greater uncertainty about whether the direction of $\widehat{m}_t$ corresponds to the direction of the true gradient. For example, the SNR value typically becomes closer to 0 towards an optimum, leading to smaller effective steps in parameter space: a form of automatic annealing. The effective stepsize Δ_t is also invariant to the scale of the gradients; rescaling the gradients g with factor c will scale $\widehat{m}_t$ with a factor c and $\widehat{v}_t$ with a factor c^2, which cancel out: $(c \cdot \widehat{m}_t)/(\sqrt{c^2 \cdot \widehat{v}_t}) = \widehat{m}_t/\sqrt{\widehat{v}_t}$.

3 INITIALIZATION BIAS CORRECTION

As explained in section 2, Adam utilizes initialization bias correction terms. We will here derive the term for the second moment estimate; the derivation for the first moment estimate is completely analogous. Let g be the gradient of the stochastic objective f, and we wish to estimate its second raw moment (uncentered variance) using an exponential moving average of the squared gradient, with decay rate β_2. Let $g_1, ..., g_T$ be the gradients at subsequent timesteps, each a draw from an underlying gradient distribution $g_t \sim p(g_t)$. Let us initialize the exponential moving average as $v_0 = 0$ (a vector of zeros). First note that the update at timestep t of the exponential moving average $v_t = \beta_2 \cdot v_{t-1} + (1-\beta_2) \cdot g_t^2$ (where g_t^2 indicates the elementwise square $g_t \odot g_t$) can be written as a function of the gradients at all previous timesteps:

$$v_t = (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} \cdot g_i^2 \tag{1}$$

We wish to know how $\mathbb{E}[v_t]$, the expected value of the exponential moving average at timestep t, relates to the true second moment $\mathbb{E}[g_t^2]$, so we can correct for the discrepancy between the two. Taking expectations of the left-hand and right-hand sides of eq. (1):

$$\mathbb{E}[v_t] = \mathbb{E}\left[(1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} \cdot g_i^2\right] \tag{2}$$

$$= \mathbb{E}[g_t^2] \cdot (1-\beta_2) \sum_{i=1}^{t} \beta_2^{t-i} + \zeta \tag{3}$$

$$= \mathbb{E}[g_t^2] \cdot (1-\beta_2^t) + \zeta \tag{4}$$

where $\zeta = 0$ if the true second moment $\mathbb{E}[g_i^2]$ is stationary; otherwise ζ can be kept small since the exponential decay rate β_1 can (and should) be chosen such that the exponential moving average assigns small weights to gradients too far in the past. What is left is the term $(1-\beta_2^t)$ which is caused by initializing the running average with zeros. In algorithm 1 we therefore divide by this term to correct the initialization bias.

In case of sparse gradients, for a reliable estimate of the second moment one needs to average over many gradients by chosing a small value of β_2; however it is exactly this case of small β_2 where a lack of initialisation bias correction would lead to initial steps that are much larger.

4 CONVERGENCE ANALYSIS

We analyze the convergence of Adam using the online learning framework proposed in (Zinkevich, 2003). Given an arbitrary, unknown sequence of convex cost functions $f_1(\theta)$, $f_2(\theta)$,..., $f_T(\theta)$. At each time t, our goal is to predict the parameter θ_t and evaluate it on a previously unknown cost function f_t. Since the nature of the sequence is unknown in advance, we evaluate our algorithm using the regret, that is the sum of all the previous difference between the online prediction $f_t(\theta_t)$ and the best fixed point parameter $f_t(\theta^*)$ from a feasible set $\mathcal{X}$ for all the previous steps. Concretely, the regret is defined as:

$$R(T) = \sum_{t=1}^{T} [f_t(\theta_t) - f_t(\theta^*)] \tag{5}$$

where $\theta^* = \arg\min_{\theta \in \mathcal{X}} \sum_{t=1}^{T} f_t(\theta)$. We show Adam has $O(\sqrt{T})$ regret bound and a proof is given in the appendix. Our result is comparable to the best known bound for this general convex online learning problem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the i^{th} element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the i^{th} dimension of the gradients over all iterations till t, $g_{1:t,i} = [g_{1,i}, g_{2,i}, \cdots, g_{t,i}]$. Also, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ, that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 4.1. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in R^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, ..., T\}$, and β_1, $\beta_2 \in [0, 1)$ satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. Let $\alpha_t = \frac{\alpha}{\sqrt{t}}$ and $\beta_{1,t} = \beta_1 \lambda^{t-1}$, $\lambda \in (0, 1)$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Our Theorem 4.1 implies when the data features are sparse and bounded gradients, the summation term can be much smaller than its upper bound $\sum_{i=1}^{d} \|g_{1:T,i}\|_2 << dG_\infty\sqrt{T}$ and $\sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} << dG_\infty\sqrt{T}$, in particular if the class of function and data features are in the form of section 1.2 in (Duchi et al., 2011). Their results for the expected value $\mathbb{E}[\sum_{i=1}^{d} \|g_{1:T,i}\|_2]$ also apply to Adam. In particular, the adaptive method, such as Adam and Adagrad, can achieve $O(\log d\sqrt{T})$, an improvement over $O(\sqrt{dT})$ for the non-adaptive method. Decaying $\beta_{1,t}$ towards zero is important in our theoretical analysis and also matches previous empirical findings, e.g. (Sutskever et al., 2013) suggests reducing the momentum coefficient in the end of training can improve convergence.

Finally, we can show the average regret of Adam converges,

Corollary 4.2. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in R^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, ..., T\}$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$\frac{R(T)}{T} = O(\frac{1}{\sqrt{T}})$$

This result can be obtained by using Theorem 4.1 and $\sum_{i=1}^{d} \|g_{1:T,i}\|_2 \leq dG_\infty\sqrt{T}$. Thus, $\lim_{T\to\infty} \frac{R(T)}{T} = 0$.

5 RELATED WORK

Optimization methods bearing a direct relation to Adam are RMSProp (Tieleman & Hinton, 2012; Graves, 2013) and AdaGrad (Duchi et al., 2011); these relationships are discussed below. Other stochastic optimization methods include vSGD (Schaul et al., 2012), AdaDelta (Zeiler, 2012) and the natural Newton method from Roux & Fitzgibbon (2010), all setting stepsizes by estimating curvature

from first-order information. The Sum-of-Functions Optimizer (SFO) (Sohl-Dickstein et al., 2014) is a quasi-Newton method based on minibatches, but (unlike Adam) has memory requirements linear in the number of minibatch partitions of a dataset, which is often infeasible on memory-constrained systems such as a GPU. Like natural gradient descent (NGD) (Amari, 1998), Adam employs a preconditioner that adapts to the geometry of the data, since $\widehat{v}_t$ is an approximation to the diagonal of the Fisher information matrix (Pascanu & Bengio, 2013); however, Adam's preconditioner (like AdaGrad's) is more conservative in its adaption than vanilla NGD by preconditioning with the square root of the inverse of the diagonal Fisher information matrix approximation.

RMSProp: An optimization method closely related to Adam is RMSProp (Tieleman & Hinton, 2012). A version with momentum has sometimes been used (Graves, 2013). There are a few important differences between RMSProp with momentum and Adam: RMSProp with momentum generates its parameter updates using a momentum on the rescaled gradient, whereas Adam updates are directly estimated using a running average of first and second moment of the gradient. RMSProp also lacks a bias-correction term; this matters most in case of a value of β_2 close to 1 (required in case of sparse gradients), since in that case not correcting the bias leads to very large stepsizes and often divergence, as we also empirically demonstrate in section 6.4.

AdaGrad: An algorithm that works well for sparse gradients is AdaGrad (Duchi et al., 2011). Its basic version updates parameters as $\theta_{t+1} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^{t} g_t^2}$. Note that if we choose β_2 to be infinitesimally close to 1 from below, then $\lim_{\beta_2 \to 1} \widehat{v}_t = t^{-1} \cdot \sum_{i=1}^{t} g_t^2$. AdaGrad corresponds to a version of Adam with $\beta_1 = 0$, infinitesimal $(1 - \beta_2)$ and a replacement of α by an annealed version $\alpha_t = \alpha \cdot t^{-1/2}$, namely $\theta_t - \alpha \cdot t^{-1/2} \cdot \widehat{m}_t / \sqrt{\lim_{\beta_2 \to 1} \widehat{v}_t} = \theta_t - \alpha \cdot t^{-1/2} \cdot g_t / \sqrt{t^{-1} \cdot \sum_{i=1}^{t} g_t^2} = \theta_t - \alpha \cdot g_t / \sqrt{\sum_{i=1}^{t} g_t^2}$. Note that this direct correspondence between Adam and Adagrad does not hold when removing the bias-correction terms; without bias correction, like in RMSProp, a β_2 infinitesimally close to 1 would lead to infinitely large bias, and infinitely large parameter updates.

6 EXPERIMENTS

To empirically evaluate the proposed method, we investigated different popular machine learning models, including logistic regression, multilayer fully connected neural networks and deep convolutional neural networks. Using large models and datasets, we demonstrate Adam can efficiently solve practical deep learning problems.

We use the same parameter initialization when comparing different optimization algorithms. The hyper-parameters, such as learning rate and momentum, are searched over a dense grid and the results are reported using the best hyper-parameter setting.

6.1 EXPERIMENT: LOGISTIC REGRESSION

We evaluate our proposed method on L2-regularized multi-class logistic regression using the MNIST dataset. Logistic regression has a well-studied convex objective, making it suitable for comparison of different optimizers without worrying about local minimum issues. The stepsize α in our logistic regression experiments is adjusted by $1/\sqrt{t}$ decay, namely $\alpha_t = \frac{\alpha}{\sqrt{t}}$ that matches with our theoratical prediction from section 4. The logistic regression classifies the class label directly on the 784 dimension image vectors. We compare Adam to accelerated SGD with Nesterov momentum and Adagrad using minibatch size of 128. According to Figure 1, we found that the Adam yields similar convergence as SGD with momentum and both converge faster than Adagrad.

As discussed in (Duchi et al., 2011), Adagrad can efficiently deal with sparse features and gradients as one of its main theoretical results whereas SGD is low at learning rare features. Adam with $1/\sqrt{t}$ decay on its stepsize should theoratically match the performance of Adagrad. We examine the sparse feature problem using IMDB movie review dataset from (Maas et al., 2011). We pre-process the IMDB movie reviews into bag-of-words (BoW) feature vectors including the first 10,000 most frequent words. The 10,000 dimension BoW feature vector for each review is highly sparse. As suggested in (Wang & Manning, 2013), 50% dropout noise can be applied to the BoW features during

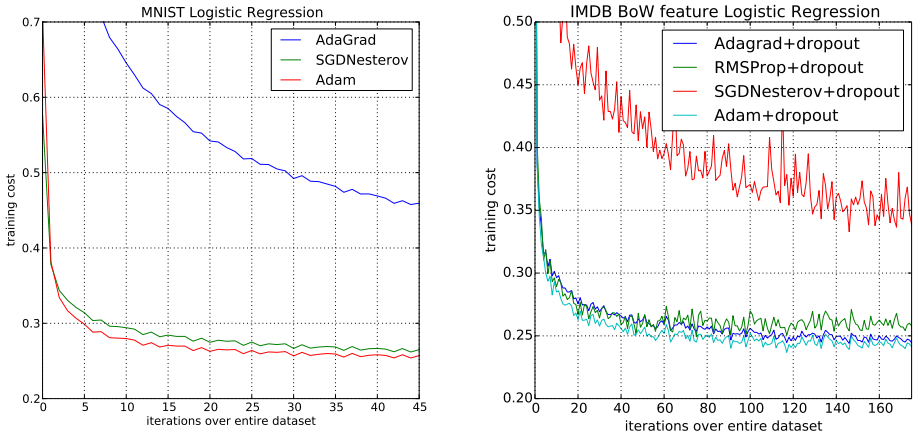


Figure 1: Logistic regression training negative log likelihood on MNIST images and IMDB movie reviews with 10,000 bag-of-words (BoW) feature vectors.

training to prevent over-fitting. In figure 1, Adagrad outperforms SGD with Nesterov momentum by a large margin both with and without dropout noise. Adam converges as fast as Adagrad. The empirical performance of Adam is consistent with our theoretical findings in sections 2 and 4. Similar to Adagrad, Adam can take advantage of sparse features and obtain faster convergence rate than normal SGD with momentum.

6.2 Experiment: Multi-layer Neural Networks

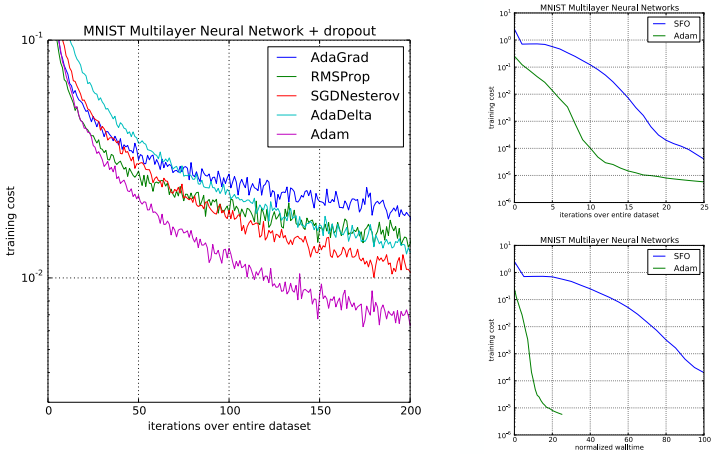
Multi-layer neural network are powerful models with non-convex objective functions. Although our convergence analysis does not apply to non-convex problems, we empirically found that Adam often outperforms other methods in such cases. In our experiments, we made model choices that are consistent with previous publications in the area; a neural network model with two fully connected hidden layers with 1000 hidden units each and ReLU activation are used for this experiment with minibatch size of 128.

First, we study different optimizers using the standard deterministic cross-entropy objective function with L_2 weight decay on the parameters to prevent over-fitting. The sum-of-functions (SFO) method (Sohl-Dickstein et al., 2014) is a recently proposed quasi-Newton method that works with minibatches of data and has shown good performance on optimization of multi-layer neural networks. We used their implementation and compared with Adam to train such models. Figure 2 shows that Adam makes faster progress in terms of both the number of iterations and wall-clock time. Due to the cost of updating curvature information, SFO is 5-10x slower per iteration compared to Adam, and has a memory requirement that is linear in the number minibatches.

Stochastic regularization methods, such as dropout, are an effective way to prevent over-fitting and often used in practice due to their simplicity. SFO assumes deterministic subfunctions, and indeed failed to converge on cost functions with stochastic regularization. We compare the effectiveness of Adam to other stochastic first order methods on multi-layer neural networks trained with dropout noise. Figure 2 shows our results; Adam shows better convergence than other methods.

6.3 Experiment: Convolutional Neural Networks

Convolutional neural networks (CNNs) with several layers of convolution, pooling and non-linear units have shown considerable success in computer vision tasks. Unlike most fully connected neural nets, weight sharing in CNNs results in vastly different gradients in different layers. A smaller learning rate for the convolution layers is often used in practice when applying SGD. We show the effectiveness of Adam in deep CNNs. Our CNN architecture has three alternating stages of 5x5 convolution filters and 3x3 max pooling with stride of 2 that are followed by a fully connected layer of 1000 rectified linear hidden units (ReLU's). The input image are pre-processed by whitening, and



(a) (b)

Figure 2: Training of multilayer neural networks on MNIST images. (a) Neural networks using dropout stochastic regularization. (b) Neural networks with deterministic cost function. We compare with the sum-of-functions (SFO) optimizer (Sohl-Dickstein et al., 2014)

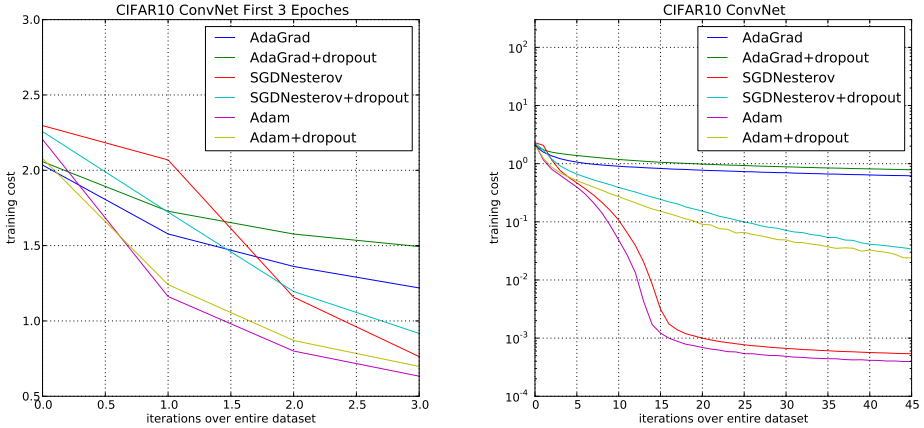
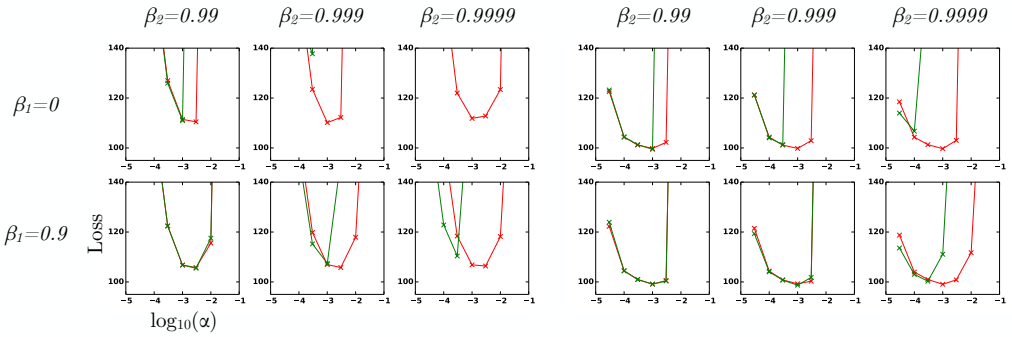


Figure 3: Convolutional neural networks training cost. (left) Training cost for the first three epochs. (right) Training cost over 45 epochs. CIFAR-10 with c64-c64-c128-1000 architecture.

dropout noise is applied to the input layer and fully connected layer. The minibatch size is also set to 128 similar to previous experiments.

Interestingly, although both Adam and Adagrad make rapid progress lowering the cost in the initial stage of the training, shown in Figure 3 (left), Adam and SGD eventually converge considerably faster than Adagrad for CNNs shown in Figure 3 (right). We notice the second moment estimate $\widehat{v}_t$ vanishes to zeros after a few epochs and is dominated by the ϵ in algorithm 1. The second moment estimate is therefore a poor approximation to the geometry of the cost function in CNNs comparing to fully connected network from Section 6.2. Whereas, reducing the minibatch variance through the first moment is more important in CNNs and contributes to the speed-up. As a result, Adagrad converges much slower than others in this particular experiment. Though Adam shows marginal improvement over SGD with momentum, it adapts learning rate scale for different layers instead of hand picking manually as in SGD.



(a) after 10 epochs (b) after 100 epochs

Figure 4: Effect of bias-correction terms (red line) versus no bias correction terms (green line) after 10 epochs (left) and 100 epochs (right) on the loss (y-axes) when learning a Variational Auto-Encoder (VAE) (Kingma & Welling, 2013), for different settings of stepsize α (x-axes) and hyper-parameters β_1 and β_2.

6.4 Experiment: bias-correction term

We also empirically evaluate the effect of the bias correction terms explained in sections 2 and 3. Discussed in section 5, removal of the bias correction terms results in a version of RMSProp (Tieleman & Hinton, 2012) with momentum. We vary the β_1 and β_2 when training a variational auto-encoder (VAE) with the same architecture as in (Kingma & Welling, 2013) with a single hidden layer with 500 hidden units with softplus nonlinearities and a 50-dimensional spherical Gaussian latent variable. We iterated over a broad range of hyper-parameter choices, i.e. $\beta_1 \in [0, 0.9]$ and $\beta_2 \in [0.99, 0.999, 0.9999]$, and $\log_{10}(\alpha) \in [-5, ..., -1]$. Values of β_2 close to 1, required for robustness to sparse gradients, results in larger initialization bias; therefore we expect the bias correction term is important in such cases of slow decay, preventing an adverse effect on optimization.

In Figure 4, values β_2 close to 1 indeed lead to instabilities in training when no bias correction term was present, especially at first few epochs of the training. The best results were achieved with small values of $(1-\beta_2)$ and bias correction; this was more apparent towards the end of optimization when gradients tends to become sparser as hidden units specialize to specific patterns. In summary, Adam performed equal or better than RMSProp, regardless of hyper-parameter setting.

7 Extensions

7.1 AdaMax

In Adam, the update rule for individual weights is to scale their gradients inversely proportional to a (scaled) L^2 norm of their individual current and past gradients. We can generalize the L^2 norm based update rule to a L^p norm based update rule. Such variants become numerically unstable for large p. However, in the special case where we let $p \to \infty$, a surprisingly simple and stable algorithm emerges; see algorithm 2. We'll now derive the algorithm. Let, in case of the L^p norm, the stepsize at time t be inversely proportional to $v_t^{1/p}$, where:

$$v_t = \beta_2^p v_{t-1} + (1-\beta_2^p)|g_t|^p \tag{6}$$

$$= (1-\beta_2^p)\sum_{i=1}^{t} \beta_2^{p(t-i)} \cdot |g_i|^p \tag{7}$$

Algorithm 2: *AdaMax*, a variant of Adam based on the infinity norm. See section 7.1 for details. Good default settings for the tested machine learning problems are $\alpha = 0.002$, $\beta_1 = 0.9$ and $\beta_2 = 0.999$. With β_1^t we denote β_1 to the power t. Here, $(\alpha/(1-\beta_1^t))$ is the learning rate with the bias-correction term for the first moment. All operations on vectors are element-wise.

Require: α: Stepsize
Require: $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates
Require: $f(\theta)$: Stochastic objective function with parameters θ
Require: θ_0: Initial parameter vector
  $m_0 \leftarrow 0$ (Initialize 1$^{\text{st}}$ moment vector)
  $u_0 \leftarrow 0$ (Initialize the exponentially weighted infinity norm)
  $t \leftarrow 0$ (Initialize timestep)
  while θ_t not converged **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_\theta f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)
    $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)
    $u_t \leftarrow \max(\beta_2 \cdot u_{t-1}, |g_t|)$ (Update the exponentially weighted infinity norm)
    $\theta_t \leftarrow \theta_{t-1} - (\alpha/(1-\beta_1^t)) \cdot m_t/u_t$ (Update parameters)
  end while
  return θ_t (Resulting parameters)

Note that the decay term is here equivalently parameterised as β_2^p instead of β_2. Now let $p \to \infty$, and define $u_t = \lim_{p\to\infty}(v_t)^{1/p}$, then:

$$u_t = \lim_{p\to\infty}(v_t)^{1/p} = \lim_{p\to\infty}\left((1-\beta_2^p)\sum_{i=1}^{t}\beta_2^{p(t-i)}\cdot|g_i|^p\right)^{1/p} \tag{8}$$

$$= \lim_{p\to\infty}(1-\beta_2^p)^{1/p}\left(\sum_{i=1}^{t}\beta_2^{p(t-i)}\cdot|g_i|^p\right)^{1/p} \tag{9}$$

$$= \lim_{p\to\infty}\left(\sum_{i=1}^{t}\left(\beta_2^{(t-i)}\cdot|g_i|\right)^p\right)^{1/p} \tag{10}$$

$$= \max\left(\beta_2^{t-1}|g_1|, \beta_2^{t-2}|g_2|, \ldots, \beta_2|g_{t-1}|, |g_t|\right) \tag{11}$$

Which corresponds to the remarkably simple recursive formula:

$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|) \tag{12}$$

with initial value $u_0 = 0$. Note that, conveniently enough, we don't need to correct for initialization bias in this case. Also note that the magnitude of parameter updates has a simpler bound with AdaMax than Adam, namely: $|\Delta_t| \leq \alpha$.

7.2 Temporal averaging

Since the last iterate is noisy due to stochastic approximation, better generalization performance is often achieved by averaging. Previously in Moulines & Bach (2011), Polyak-Ruppert averaging (Polyak & Juditsky, 1992; Ruppert, 1988) has been shown to improve the convergence of standard SGD, where $\bar{\theta}_t = \frac{1}{t}\sum_{k=1}^{n}\theta_k$. Alternatively, an exponential moving average over the parameters can be used, giving higher weight to more recent parameter values. This can be trivially implemented by adding one line to the inner loop of algorithms 1 and 2: $\bar{\theta}_t \leftarrow \beta_2 \cdot \bar{\theta}_{t-1} + (1-\beta_2)\theta_t$, with $\bar{\theta}_0 = 0$. Initalization bias can again be corrected by the estimator $\widehat{\theta}_t = \bar{\theta}_t/(1-\beta_2^t)$.

8 Conclusion

We have introduced a simple and computationally efficient algorithm for gradient-based optimization of stochastic objective functions. Our method is aimed towards machine learning problems with

large datasets and/or high-dimensional parameter spaces. The method combines the advantages of two recently popular optimization methods: the ability of AdaGrad to deal with sparse gradients, and the ability of RMSProp to deal with non-stationary objectives. The method is straightforward to implement and requires little memory. The experiments confirm the analysis on the rate of convergence in convex problems. Overall, we found Adam to be robust and well-suited to a wide range of non-convex optimization problems in the field machine learning.

9 ACKNOWLEDGMENTS

This paper would probably not have existed without the support of Google Deepmind. We would like to give special thanks to Ivo Danihelka, and Tom Schaul for coining the name Adam. Thanks to Kai Fan from Duke University for spotting an error in the original AdaMax derivation. Experiments in this work were partly carried out on the Dutch national e-infrastructure with the support of SURF Foundation. Diederik Kingma is supported by the Google European Doctorate Fellowship in Deep Learning.

REFERENCES

Amari, Shun-Ichi. Natural gradient works efficiently in learning. *Neural computation*, 10(2):251–276, 1998.

Deng, Li, Li, Jinyu, Huang, Jui-Ting, Yao, Kaisheng, Yu, Dong, Seide, Frank, Seltzer, Michael, Zweig, Geoff, He, Xiaodong, Williams, Jason, et al. Recent advances in deep learning for speech research at microsoft. *ICASSP 2013*, 2013.

Duchi, John, Hazan, Elad, and Singer, Yoram. Adaptive subgradient methods for online learning and stochastic optimization. *The Journal of Machine Learning Research*, 12:2121–2159, 2011.

Graves, Alex. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.

Graves, Alex, Mohamed, Abdel-rahman, and Hinton, Geoffrey. Speech recognition with deep recurrent neural networks. In *Acoustics, Speech and Signal Processing (ICASSP), 2013 IEEE International Conference on*, pp. 6645–6649. IEEE, 2013.

Hinton, G.E. and Salakhutdinov, R.R. Reducing the dimensionality of data with neural networks. *Science*, 313 (5786):504–507, 2006.

Hinton, Geoffrey, Deng, Li, Yu, Dong, Dahl, George E, Mohamed, Abdel-rahman, Jaitly, Navdeep, Senior, Andrew, Vanhoucke, Vincent, Nguyen, Patrick, Sainath, Tara N, et al. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *Signal Processing Magazine, IEEE*, 29(6):82–97, 2012a.

Hinton, Geoffrey E, Srivastava, Nitish, Krizhevsky, Alex, Sutskever, Ilya, and Salakhutdinov, Ruslan R. Improving neural networks by preventing co-adaptation of feature detectors. *arXiv preprint arXiv:1207.0580*, 2012b.

Kingma, Diederik P and Welling, Max. Auto-Encoding Variational Bayes. In *The 2nd International Conference on Learning Representations (ICLR)*, 2013.

Krizhevsky, Alex, Sutskever, Ilya, and Hinton, Geoffrey E. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*, pp. 1097–1105, 2012.

Maas, Andrew L, Daly, Raymond E, Pham, Peter T, Huang, Dan, Ng, Andrew Y, and Potts, Christopher. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies-Volume 1*, pp. 142–150. Association for Computational Linguistics, 2011.

Moulines, Eric and Bach, Francis R. Non-asymptotic analysis of stochastic approximation algorithms for machine learning. In *Advances in Neural Information Processing Systems*, pp. 451–459, 2011.

Pascanu, Razvan and Bengio, Yoshua. Revisiting natural gradient for deep networks. *arXiv preprint arXiv:1301.3584*, 2013.

Polyak, Boris T and Juditsky, Anatoli B. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4):838–855, 1992.

10 APPENDIX

10.1 CONVERGENCE PROOF

Definition 10.1. *A function $f : R^d \to R$ is convex if for all $x, y \in R^d$, for all $\lambda \in [0, 1]$,*

$$\lambda f(x) + (1 - \lambda) f(y) \geq f(\lambda x + (1 - \lambda) y)$$

Also, notice that a convex function can be lower bounded by a hyperplane at its tangent.

Lemma 10.2. *If a function $f : R^d \to R$ is convex, then for all $x, y \in R^d$,*

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

The above lemma can be used to upper bound the regret and our proof for the main theorem is constructed by substituting the hyperplane with the Adam update rules.

The following two lemmas are used to support our main theorem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the i^{th} element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the i^{th} dimension of the gradients over all iterations till t, $g_{1:t,i} = [g_{1,i}, g_{2,i}, \cdots, g_{t,i}]$

Lemma 10.3. *Let $g_t = \nabla f_t(\theta_t)$ and $g_{1:t}$ be defined as above and bounded, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$. Then,*

$$\sum_{t=1}^{T} \sqrt{\frac{g_{t,i}^2}{t}} \leq 2 G_\infty \|g_{1:T,i}\|_2$$

Proof. We will prove the inequality using induction over T.

The base case for $T = 1$, we have $\sqrt{g_{1,i}^2} \leq 2 G_\infty \|g_{1,i}\|_2$.

For the inductive step,

$$\begin{aligned}
\sum_{t=1}^{T} \sqrt{\frac{g_{t,i}^2}{t}} &= \sum_{t=1}^{T-1} \sqrt{\frac{g_{t,i}^2}{t}} + \sqrt{\frac{g_{T,i}^2}{T}} \\
&\leq 2 G_\infty \|g_{1:T-1,i}\|_2 + \sqrt{\frac{g_{T,i}^2}{T}} \\
&= 2 G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_T^2} + \sqrt{\frac{g_{T,i}^2}{T}}
\end{aligned}$$

From, $\|g_{1:T,i}\|_2^2 - g_{T,i}^2 + \frac{g_{T,i}^4}{4\|g_{1:T,i}\|_2^2} \geq \|g_{1:T,i}\|_2^2 - g_{T,i}^2$, we can take square root of both side and have,

$$\begin{aligned}
\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\|g_{1:T,i}\|_2} \\
&\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\sqrt{T G_\infty^2}}
\end{aligned}$$

Rearrange the inequality and substitute the $\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2}$ term,

$$G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_T^2} + \sqrt{\frac{g_{T,i}^2}{T}} \leq 2 G_\infty \|g_{1:T,i}\|_2$$

□

Lemma 10.4. *Let $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. For β_1, $\beta_2 \in [0, 1)$ that satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ and bounded g_t, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$, the following inequality holds*

$$\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} \leq \frac{2}{1-\gamma} \frac{1}{\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

Proof. Under the assumption, $\frac{\sqrt{1-\beta_2^t}}{(1-\beta_1^t)^2} \leq \frac{1}{(1-\beta_1)^2}$. We can expand the last term in the summation using the update rules in Algorithm 1,

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} &= \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(\sum_{k=1}^{T}(1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T\sum_{j=1}^{T}(1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^{T} \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T\sum_{j=1}^{T}(1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^{T} \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T(1-\beta_2)\beta_2^{T-k} g_{k,i}^2}} \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(1-\beta_1)^2}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^{T} T\left(\frac{\beta_1^2}{\sqrt{\beta_2}}\right)^{T-k} \|g_{k,i}\|_2 \\
&\leq \sum_{t=1}^{T-1} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} + \frac{T}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^{T} \gamma^{T-k} \|g_{k,i}\|_2
\end{aligned}$$

Similarly, we can upper bound the rest of the terms in the summation.

$$\begin{aligned}
\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} &\leq \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T-t} t\gamma^j \\
&\leq \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T} t\gamma^j
\end{aligned}$$

For $\gamma < 1$, using the upper bound on the arithmetic-geometric series, $\sum_t t\gamma^t < \frac{1}{(1-\gamma)^2}$:

$$\sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T} t\gamma^j \leq \frac{1}{(1-\gamma)^2\sqrt{1-\beta_2}} \sum_{t=1}^{T} \frac{\|g_{t,i}\|_2}{\sqrt{t}}$$

Apply Lemma 10.3,

$$\sum_{t=1}^{T} \frac{\widehat{m}_{t,i}^2}{\sqrt{t\widehat{v}_{t,i}}} \leq \frac{2G_\infty}{(1-\gamma)^2\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

□

To simplify the notation, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Intuitively, our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ, that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 10.5. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in R^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$,*

$\|\theta_m - \theta_n\|_\infty \leq D_\infty$ *for any* $m, n \in \{1, ..., T\}$*, and* β_1*,* $\beta_2 \in [0, 1)$ *satisfy* $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$*. Let* $\alpha_t = \frac{\alpha}{\sqrt{t}}$ *and* $\beta_{1,t} = \beta_1 \lambda^{t-1}, \lambda \in (0, 1)$*. Adam achieves the following guarantee, for all* $T \geq 1$*.*

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(\beta_1+1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Proof. Using Lemma 10.2, we have,

$$f_t(\theta_t) - f_t(\theta^*) \leq g_t^T(\theta_t - \theta^*) = \sum_{i=1}^{d} g_{t,i}(\theta_{t,i} - \theta_{,i}^*)$$

From the update rules presented in algorithm 1,

$$\begin{aligned}\theta_{t+1} &= \theta_t - \alpha_t \widehat{m}_t / \sqrt{\widehat{v}_t} \\ &= \theta_t - \frac{\alpha_t}{1-\beta_1^t}\left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_t}} m_{t-1} + \frac{(1-\beta_{1,t})}{\sqrt{\widehat{v}_t}} g_t\right)\end{aligned}$$

We focus on the i^{th} dimension of the parameter vector $\theta_t \in R^d$. Subtract the scalar $\theta_{,i}^*$ and square both sides of the above update rule, we have,

$$(\theta_{t+1,i} - \theta_{,i}^*)^2 = (\theta_{t,i} - \theta_{,i}^*)^2 - \frac{2\alpha_t}{1-\beta_1^t}\left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} m_{t-1,i} + \left(1 - \frac{\beta_{1,t})}{\sqrt{\widehat{v}_{t,i}}} g_{t,i}\right)(\theta_{t,i} - \theta_{,i}^*) + \alpha_t^2 \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}}\right)^2$$

We can rearrange the above equation and use Young's inequality, $ab \leq a^2/2 + b^2/2$. Also, it can be shown that $\sqrt{\widehat{v}_{t,i}} = \sqrt{\sum_{j=1}^{t}(1-\beta_2)\beta_2^{t-j} g_{j,i}^2} / \sqrt{1-\beta_2^t} \leq \|g_{1:t,i}\|_2$ and $\beta_{1,t} \leq \beta_1$. Then

$$\begin{aligned} g_{t,i}(\theta_{t,i} - \theta_{,i}^*) =& \frac{(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2\alpha_t(1-\beta_{1,t})}\left((\theta_{t,i} - \theta_{,t}^*)^2 - (\theta_{t+1,i} - \theta_{,i}^*)^2\right) \\ &+ \frac{\beta_{1,t}}{(1-\beta_{1,t})} \frac{\widehat{v}_{t-1,i}^{\frac{1}{4}}}{\sqrt{\alpha_{t-1}}} (\theta_{,i}^* - \theta_{t,i}) \sqrt{\alpha_{t-1}} \frac{m_{t-1,i}}{\widehat{v}_{t-1,i}^{\frac{1}{4}}} + \frac{\alpha_t(1-\beta_1^t)\sqrt{\widehat{v}_{t,i}}}{2(1-\beta_{1,t})} \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}}\right)^2 \\ \leq& \frac{1}{2\alpha_t(1-\beta_1)}\left((\theta_{t,i} - \theta_{,t}^*)^2 - (\theta_{t+1,i} - \theta_{,i}^*)^2\right)\sqrt{\widehat{v}_{t,i}} + \frac{\beta_{1,t}}{2\alpha_{t-1}(1-\beta_{1,t})}(\theta_{,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t-1,i}} \\ &+ \frac{\beta_1 \alpha_{t-1}}{2(1-\beta_1)} \frac{m_{t-1,i}^2}{\sqrt{\widehat{v}_{t-1,i}}} + \frac{\alpha_t}{2(1-\beta_1)} \frac{\widehat{m}_{t,i}^2}{\sqrt{\widehat{v}_{t,i}}} \end{aligned}$$

We apply Lemma 10.4 to the above inequality and derive the regret bound by summing across all the dimensions for $i \in 1, ..., d$ in the upper bound of $f_t(\theta_t) - f_t(\theta^*)$ and the sequence of convex functions for $t \in 1, ..., T$:

$$\begin{aligned} R(T) \leq& \sum_{i=1}^{d} \frac{1}{2\alpha_1(1-\beta_1)}(\theta_{1,i} - \theta_{,i}^*)^2 \sqrt{\widehat{v}_{1,i}} + \sum_{i=1}^{d}\sum_{t=2}^{T} \frac{1}{2(1-\beta_1)}(\theta_{t,i} - \theta_{,i}^*)^2 \left(\frac{\sqrt{\widehat{v}_{t,i}}}{\alpha_t} - \frac{\sqrt{\widehat{v}_{t-1,i}}}{\alpha_{t-1}}\right) \\ &+ \frac{\beta_1 \alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \frac{\alpha G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 \\ &+ \sum_{i=1}^{d}\sum_{t=1}^{T} \frac{\beta_{1,t}}{2\alpha_t(1-\beta_{1,t})}(\theta_{,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t,i}} \end{aligned}$$

From the assumption, $\|\theta_t - \theta^*\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$, we have:

$$\begin{aligned}
R(T) \leq & \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \frac{D_\infty^2}{2\alpha} \sum_{i=1}^{d} \sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t\widehat{v}_{t,i}} \\
\leq & \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 \\
& + \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha} \sum_{i=1}^{d} \sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t}
\end{aligned}$$

We can use arithmetic geometric series upper bound for the last term:

$$\begin{aligned}
\sum_{t=1}^{t} \frac{\beta_{1,t}}{(1-\beta_{1,t})} \sqrt{t} &\leq \sum_{t=1}^{t} \frac{1}{(1-\beta_1)} \lambda^{t-1} \sqrt{t} \\
&\leq \sum_{t=1}^{t} \frac{1}{(1-\beta_1)} \lambda^{t-1} t \\
&\leq \frac{1}{(1-\beta_1)(1-\lambda)^2}
\end{aligned}$$

Therefore, we have the following regret bound:

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^{d} \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^{d} \|g_{1:T,i}\|_2 + \sum_{i=1}^{d} \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha\beta_1(1-\lambda)^2}$$

□